OpenNeuro Dataset Analysis Technical Report

EEG Dataset Documentation

Datasets included: 242

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Preface

This report summarizes EEG datasets from OpenNeuro and includes 242 datasets. Each dataset is documented in its own section, covering paradigm, event structure, potential downstream tasks (including multi-task settings), and data-quality notes.

For retrieval and modeling, tasks are first grouped using the Category field in each dataset document: High-level cognitive and mental tasks (88); Complex applied and ecological tasks (56); Sensory stimulation and response tasks (51); Biological-state tasks (rest/sleep/clinical state) (30); Action-output and BCI tasks (12); Others (4); Motor-control tasks (1).

Based on a sample of 10 datasets, the task space spans cognition, sensory processing, motor control, complex applied paradigms, and biological-state analyses. The examples below provide short sample-level task summaries:

- openneuro_ds001784: Participants with treatment-resistant depression performed a modified Multi-Source Interference Task (MSIT) integrated with emotional sti...
- openneuro_ds003039: Participants walked two different outdoor routes, once while performing a self-paced button-pressing task, to study brain activity during...
- openneuro_ds003670: Participants performed a continuous tracking task (CTT) requiring them to maintain a ball at the center of a screen to assess vigilance a...
- openneuro_ds004017: A pre-post intervention study assessing literacy and orthographic processing in children through two tasks: a two-alternative forced choi...
- openneuro_ds004264: A ship-steering simulation where participants control a rotating arrow to maintain orientation against varying environmental and controll...
- openneuro_ds004504: Resting-state EEG recordings with eyes closed collected from healthy controls and patients with Alzheimer's disease (AD) and Frontotempor...
- openneuro_ds004718: Naturalistic listening task where older Cantonese-speaking adults (65+) listened to a Cantonese translation of "The Little Prince" divide...
- openneuro_ds005305: Resting-state EEG microstates were recorded to correlate spontaneous brain dynamics with performance on a nine-task battery of executive...
- openneuro_ds006018: A suite of standardized ERP tasks from the ERP CORE compendium (Kappenman et al.
- openneuro_ds007327: EEG was recorded during two sports-related tasks (ball passing and dribbling) to study motor intention and motor observation.

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001. openneuro_ds001784 - Effects of ON/OFF deep brain stimulation on cognitive control in treatment-resistant depression (EEG)

Dataset: `openneuro_ds001784` | Year: 2019 | URL: https://openneuro.org/datasets/ds001784
Category: High-level Mental | Subjects: N/A | Files (Completed): 38 | Channels: 65.0 | Sampling Rate: 1450.0 Hz

Paradigm

Participants with treatment-resistant depression performed a modified Multi-Source Interference Task (MSIT) integrated with emotional stimuli (IAPS images) while deep brain stimulation (DBS) was toggled ON or OFF. The study investigated how DBS of the internal capsule affects cognitive control and prefrontal cortex function. No trial-level event markers are present in the BIDS EEG files; the DBS condition (ON/OFF) is tracked as metadata rather than inline event triggers.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files; DBS condition tracked via metadata

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events; sliding-window segmentation
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level labels available)
- Notes: DBS ON vs OFF condition labels are subject-level metadata. If DBS condition is available from participants.tsv, it can be used as a subject-level binary label post-hoc.

Data Quality Notes

- No trial-level event markers detected in any EEG files; only continuous recordings available.
- DBS condition (ON/OFF) distinguishes experimental conditions but is not encoded as EEG events.
- High sampling rate (1450 Hz); consider downsampling for computational efficiency.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Deep Brain Stimulation of the Internal Capsule Enhances Human Cognitive Control and Prefrontal Cortex Function. Nature Communications.`

002. openneuro_ds001785 - Evidence accumulation relates to perceptual consciousness and monitoring

Dataset: openneuro_ds001785 | Year: 2021 | URL: https://openneuro.org/datasets/ds001785
Category: High-level Mental | Subjects: 21 | Files (Completed): 54 | Channels: 71.0 | Sampling Rate: 1024.0 Hz

Paradigm

Participants performed a threshold-level vibrotactile detection task in which a weak tactile stimulus was either present (threshold condition) or absent (adapt condition). After each trial, participants provided a confidence rating. The study investigated the neural mechanisms of evidence accumulation and metacognitive monitoring using signal detection theory outcomes (hit, miss, correct rejection, false alarm) as trial labels.

Event Structure

Event ID	Count (total)	Type	Meaning
stim-thr	large	short	Stimulus onset - threshold vibrotactile stimulus present
stim-adapt	large	short	Stimulus onset - adapt condition (no stimulus)
hit	large	short	Correct detection: target present, response given
miss	large	short	Miss: target present, no response
cr	large	short	Correct rejection: no target, no response
fa	large	short	False alarm: no target, but response given
conf	large	short	Confidence rating onset
conf-resp	large	short	Confidence rating button-press response

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (kernels/openneuro_specific/openneuro_ds001785.py); misc channels: confidence, response_time

Task 1: Signal Detection Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`hit`	0	Correct detection
`miss`	1	Missed target
`cr`	2	Correct rejection
`fa`	3	False alarm

- Notes: Four-class SDT labels enable analysis of detection sensitivity and response bias. `stim-thr` and `stim-adapt` are stimulus onset markers - use as alternative epoch triggers if SDT label counts are imbalanced. Regression target: `confidence` from misc channel.

Data Quality Notes

- Check class balance across hit/miss/cr/fa - miss and fa classes may be sparse for threshold stimuli.
- Confidence ratings stored in misc channel `confidence`; response times in `response_time`.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Evidence accumulation relates to perceptual consciousness and monitoring`

003. openneuro_ds001787 - EEG meditation study

Dataset: openneuro_ds001787 | Year: 2024 | URL: https://openneuro.org/datasets/ds001787
Category: Biological State | Subjects: 24 | Files (Completed): 40 | Channels: 80.0 | Sampling Rate: 256.0 Hz

Paradigm

Subjects engaged in focused meditation sessions that were interrupted every two minutes by a brief probe assessing concentration level and mind-wandering. EEG was recorded continuously with stimulus markers at probe onsets and response markers at button-press responses. The study examined neural correlates of sustained attention and mind-wandering during meditation using ERP and time-frequency analysis.

Event Structure

Event ID	Count (total)	Type	Meaning
response	40 (files)	short	Participant button-press response to attention probe
stimulus	40 (files)	short	Probe stimulus onset (attention check)
STATUS	5	admin	BioSemi system status marker - exclude

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (kernels/openneuro_specific/openneuro_ds001787.py); misc channels: response_time

Task 1: Stimulus-Response ERP (binary)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
stimulus	0	Probe stimulus onset
response	1	Button-press response

- Notes: STATUS is a BioSemi admin marker (count=5 files only) - excluded from target_events. Regression target: response_time from misc channel. The count column shows files-with-event, not total trial count.

Data Quality Notes

- STATUS events are BioSemi system markers, not task events - do not epoch on these.
- Multiple response events may occur per stimulus probe; consider windowing to first response only.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Arnaud Delorme (October 17, 2018; updated June 2024)

004. openneuro_ds001810 - EEG study of the attentional blink; before, during, and after transcranial Direct Current Stimulation (tDCS)

Dataset: `openneuro_ds001810` | Year: 2019 | URL: https://openneuro.org/datasets/ds001810
Category: Sensory/Response | Subjects: 40 | Files (Completed): 527 | Channels: 73.0 | Sampling Rate: 512.0 Hz

Paradigm

Participants performed an attentional blink (AB) task using a Rapid Serial Visual Presentation (RSVP) stream to identify two targets (T1 and T2) embedded in a stream of distractors. EEG was recorded before, during, and after receiving anodal or cathodal transcranial Direct Current Stimulation (tDCS) to study how cortical excitability modulates the attentional blink - the impaired detection of T2 when it follows T1 within ~500 ms. BioSemi STATUS channel encodes trial conditions as numeric codes.

Event Structure

Event ID	Count (total)	Type	Meaning
`T1_present_T2_present_lag3`	large	short	T1 and T2 both present, short lag (within AB window)
`T1_present_T2_absent_lag3`	large	short	T1 present, T2 absent, short lag
`T1_absent_T2_present`	large	short	T1 absent, T2 present
`T1_T2_lag8`	large	short	T1 and T2 at long lag (outside AB window)
`T1_onset`	large	short	T1 target onset
`T2_onset`	large	short	T2 target onset
`CM_baseline` (61440)	261 files	admin	BioSemi CM_in_range baseline - exclude
`CM_out_of_range` (61694)	263 files	admin	BioSemi CM status - exclude
`stimulus_onset` (61468)	large	short	General stimulus onset

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - BioSemi STATUS codes decoded via specific kernel
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds001810.py`); misc channels: `response_time`

Task 1: Attentional Blink Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)

- Target Events -> y:

Event	y	Meaning
`T1_present_T2_present_lag3`	0	T2 detected within AB window
`T1_present_T2_absent_lag3`	1	T2 missed within AB window
`T1_absent_T2_present`	2	T2 present without T1
`T1_T2_lag8`	3	T2 detected outside AB window (control)
`T1_onset`	4	T1 target onset
`T2_onset`	5	T2 target onset

- Notes: BioSemi STATUS codes decoded via specific kernel (TODO: verify exact code mapping from original paper doi.org/10.1111/j.1469-8986.2010.01081.x). `CM_baseline` (61440) and `CM_out_of_range` (61694) are admin markers excluded. Regression target: `response_time`.

Data Quality Notes

- BioSemi STATUS channel encodes all triggers numerically; specific kernel maps codes to human-readable labels.
- `61440` (CM_in_range) and `61694` (CM_out_of_range) are BioSemi hardware status codes - exclude from analysis.
- Large dataset (527 files, 40 subjects x pre/during/post x anodal/cathodal conditions).
- Verify that event code mapping is correct against the original paper before downstream use.

References

- Reference: `EEG study of the attentional blink; before, during, and after transcranial Direct Current Stimulation (tDCS)`

005. openneuro_ds001849 - RS_TMSEEG_Data

Dataset: `openneuro_ds001849` | Year: 2020 | URL: https://openneuro.org/datasets/ds001849
Category: Biological State | Subjects: N/A | Files (Completed): 120 | Channels: 30.0 | Sampling Rate: 5000.0 Hz

Paradigm

Resting-state EEG data recorded in conjunction with Transcranial Magnetic Stimulation (TMS). Subjects were at rest (no active task) while TMS pulses were delivered; the R128 marker in BrainVision format marks a system/response trigger present in all files but does not indicate a meaningful task event. The primary scientific interest is resting-state oscillatory dynamics and TMS-evoked potentials.

Event Structure

Event ID	Count (total)	Type	Meaning
`R128`	120 (files)	admin	BrainVision system/response trigger present in every file - not a task event

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no meaningful task events; sliding-window segmentation
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no task events; R128 is admin)
- Notes: R128 appears in all 120 files and is a BrainVision recording artifact marker. High sampling rate (5000 Hz); consider downsampling. TMS artifact rejection may be needed if TMS pulses are captured in recording.

Data Quality Notes

- R128 is a BrainVision system marker present in every file - it is not a stimulus or response event.
- High sampling rate (5000 Hz); significant downsampling required before most analyses.
- TMS artifacts may contaminate EEG within the first ~50 ms post-pulse; apply TMS artifact rejection if analyzing TMS-evoked potentials.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `RS_TMSEEG_Data`

006. openneuro_ds001971 - Audiocue walking study

Dataset: `openneuro_ds001971` | Year: 2019 | URL: https://openneuro.org/datasets/ds001971
Category: Complex Applied | Subjects: 18 | Files (Completed): 273 | Channels: 115.0 | Sampling Rate: 512.0 Hz

Paradigm

A mobile brain-body imaging (MoBI) study where participants walked on a treadmill and adapted their gait to auditory tone sequences. Participants walked at their preferred cadence, uncued, or responded to tempo shifts (delay or advance) in an auditory metronome. The study investigated the neural dynamics of gait adaptation and rhythm entrainment during real-world locomotion. Each walking condition is a sustained state lasting multiple seconds.

Event Structure

Event ID	Count (total)	Type	Meaning
`PreferredCadence`	234 (files)	long	Sustained walking at participant's preferred cadence
`UncuedWalking`	234 (files)	long	Sustained uncued walking (no auditory cue)
`DelayTempo`	232 (files)	long	Sustained walking to delayed tempo auditory cue
`AdvanceTempo`	230 (files)	long	Sustained walking to advanced tempo auditory cue
`STATUS`	39	admin	BioSemi system status marker - exclude

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds001971.py`); misc channels: `response_time`

Task 1: Gait Condition Classification (multiclass)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`PreferredCadence`	0	Walking at preferred cadence
`UncuedWalking`	1	Uncued walking
`DelayTempo`	2	Walking to delayed tempo cue

Event	y	Meaning
AdvanceTempo	3	Walking to advanced tempo cue

- Notes: Walking conditions are sustained states (multiple seconds each) - use long_event epoching. STATUS is a BioSemi admin marker - excluded. Misc channel response_time available for regression.

Data Quality Notes

- MoBI dataset with 115 channels including motion capture; verify EEG-only channel selection.
- STATUS events are BioSemi system markers - do not include in target_events.
- Walking produces motion artifacts; independent component analysis (ICA) for artifact removal is recommended.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Johanna Wagner (June 6, 2019)

007. openneuro_ds002034 - Real-time EEG feedback on alpha power lateralization leads to behavioral improvements in a covert attention task

Dataset: `openneuro_ds002034` | Year: 2022 | URL: https://openneuro.org/datasets/ds002034
Category: Action/Output | Subjects: 14 | Files (Completed): 167 | Channels: 82.0 | Sampling Rate: 512.0 Hz

Paradigm

Participants performed a covert spatial attention task where they were trained to lateralize their alpha-band EEG power using real-time neurofeedback. The study evaluated whether closed-loop neurofeedback of alpha lateralization improves behavioral performance in a spatial attention paradigm. No trial-level event markers are stored in the BIDS EEG files; the task condition is reflected continuously in the EEG signal itself.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files; continuous neurofeedback paradigm

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events; sliding-window segmentation
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level labels available)
- Notes: Continuous neurofeedback paradigm; alpha lateralization index may be computed from raw EEG as a continuous target if desired.

Data Quality Notes

- No trial-level event markers detected in any EEG files; only continuous recordings available.
- Real-time neurofeedback paradigm - EEG signal itself contains the task-relevant information.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Real-time EEG Feedback on Alpha Power Lateralization Leads to Behavioral Improvements in a Covert Attention Task`

008. openneuro_ds002158 - Disentangling the origins of confidence in speeded perceptual judgments through multimodal imaging

Dataset: openneuro_ds002158 | Year: 2020 | URL: https://openneuro.org/datasets/ds002158
Category: High-level Mental | Subjects: 119 | Files (Completed): 119
Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

Participants performed speeded perceptual judgments and subsequently reported their confidence in either their own decisions or in observed decisions made by another person. The study used multimodal neuroimaging (EEG + fMRI) to isolate the neural correlates of metacognitive confidence, distinguishing whether confidence reflects the quality of perceptual evidence or post-decisional processes. The confidence rating is stored as a misc channel for regression analysis. High sampling rate (5000 Hz) requires downsampling before epoching.

Event Structure

Event ID	Count (total)	Type	Meaning
unknown	119 (files)	short	Trial onset marker - actual trial type may be in events.tsv value column; used as epoch anchor

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds002158.py); misc channels: confidence, response_time

Task 1: Confidence Regression (regression)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:null(regression only; unknown serves as epoch trigger)
- Regression target:confidence(metacognitive rating stored as misc channel)
- Note: unknown event label is the trial onset marker - use as epoch anchor. Primary task is regression of confidence rating from EEG. High sampling rate (5000 Hz); downsample before epoching.

Data Quality Notes

- Event label unknown is a placeholder - actual trial type information may be in events.tsv value or trial_type columns.
- Confidence ratings stored in misc channel confidence; reaction times in response_time.
- High sampling rate (5000 Hz); significant downsampling required.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Disentangling the origins of confidence in speeded perceptual judgments through multimodal imaging

009. openneuro_ds002218 - Auditory and Visual Rhythm Omission EEG

Dataset: openneuro_ds002218 | Year: 2019 | URL: https://openneuro.org/datasets/ds002218
Category: Sensory/Response | Subjects: 18 | Files (Completed): 18
Channels: 32 | Sampling Rate: 256 Hz

Paradigm

Participants passively perceived rhythmic auditory tone sequences and visual flashing stimuli that occasionally omitted an expected beat, to study predictive neural activity (mismatch responses) versus sensory evoked responses. The paradigm investigates how the brain generates predictions based on temporal regularities and detects violations of expected rhythmic patterns. Only BioSemi STATUS admin markers are detectable in the locator metadata; no decoded trial events are available from the locator pipeline.

Event Structure

Event ID	Count (total)	Type	Meaning
STATUS	18 (files)	admin	BioSemi system status marker - exclude from epoch

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv) - only admin STATUS present in locator; sliding-window segmentation
Kernel:Specific (kernels/openneuro_specific/openneuro_ds002218.py); misc channels: response_time

Task 1: Resting / Continuous Segmentation (resting)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y:null(only STATUS admin events; no trial markers decoded)
- Note: STATUS is a BioSemi CM_in_range hardware marker. If events.tsv contains decoded rhythm onset/omission events (separate from the locator metadata), they could be used for ERP analysis - verify directly from raw events.tsv.

Data Quality Notes

- Only STATUS events found in locator metadata - BioSemi admin markers only.
- The actual auditory/visual rhythm and omission events may be encoded in the events.tsv files but were not decoded by the locator pipeline; check events.tsv directly.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Auditory and Visual Rhythm Omission EEG

010. openneuro_ds002336 - A multi-modal human neuroimaging dataset for data integration: simultaneous EEG and fMRI acquisition during a motor imagery neurofeedback task: XP1

Dataset: `openneuro_ds002336` | Year: 2021 | URL: https://openneuro.org/datasets/ds002336
Category: Action/Output | Subjects: 10 | Files (Completed): 107
Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

Simultaneous EEG-fMRI acquisition during a kinesthetic motor imagery neurofeedback task. Subjects performed right-hand motor imagery or motor execution during sustained task blocks alternating with rest blocks, while receiving unimodal (EEG or fMRI) or bimodal neurofeedback. Each task block (Task-NF: neurofeedback, Task-ME: motor execution, Task-MI: motor imagery) is a sustained mental state lasting several seconds. BrainVision format with MRI artifact markers (TPULSE, R128) that must be excluded.

Event Structure

Event ID	Count (total)	Type	Meaning
`Rest`	93 (files)	long	Rest / baseline block
`Task-NF`	60 (files)	long	Neurofeedback task block
`Task-ME`	19 (files)	long	Motor execution task block
`Task-MI`	14 (files)	long	Motor imagery task block
`Response/R128`	14	admin	BrainVision response trigger / MRI sync - exclude
`Stimulus/S 99`	14	admin	BrainVision stimulus code 99 - likely admin/sync
`Stimulus/S 2`	14	admin	BrainVision stimulus code 2 - likely admin/sync
`Pulse Artifact/R`	7	admin	MRI gradient pulse artifact marker - exclude
`Time 0/`	7	admin	MRI timing reference marker - exclude
`TPULSE/TSTART`	7	admin	MRI T-pulse start - exclude
`TPULSE/TPEAK`	7	admin	MRI T-pulse peak - exclude
`TPULSE/TEND`	7	admin	MRI T-pulse end - exclude

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Motor Imagery / Neurofeedback State Classification (multiclass)

- Epoch method: `epoch_by_long_event_hdf5`
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`Rest`	0	Rest / baseline block
`Task-NF`	1	Neurofeedback task block
`Task-ME`	2	Motor execution block
`Task-MI`	3	Motor imagery block

- Note: MRI artifact markers (TPULSE/*, Pulse Artifact/R, Time 0/, R128) are all admin markers - excluded. High sampling rate (5000 Hz); downsample before epoching. Task-ME and Task-MI are sparse (14-19 files).

Data Quality Notes

- MRI gradient artifact contamination present; TPULSE markers indicate artifact regions.
- High sampling rate (5000 Hz); downsampling to 250-1000 Hz recommended.
- Task-ME and Task-MI blocks are sparsely present (subset of sessions only).
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Lioi, G., et al. (2019). Simultaneous MRI-EEG during a motor imagery neurofeedback task: an open access brain imaging dataset for multi-modal data integration. BioRxiv.`

011. openneuro_ds002338 - A multi-modal human neuroimaging dataset for data integration: simultaneous EEG and fMRI acquisition during a motor imagery neurofeedback task: XP2

Dataset: `openneuro_ds002338` | Year: 2022 | URL: https://openneuro.org/datasets/ds002338
Category: Action/Output | Subjects: 16 | Files (Completed): 169 | Channels: 64.0 | Sampling Rate: 5000.0 Hz

Paradigm

Participants performed right-hand motor imagery and received neurofeedback (1D or 2D display) during simultaneous EEG-fMRI acquisition to learn to regulate sensorimotor rhythms. The task alternated between sustained Rest blocks and Task-NF (neurofeedback) blocks. MRI gradient artifacts are marked with TPULSE markers which must be excluded. This is XP2 of a two-experiment series; XP1 is ds002336.

Event Structure

Event ID	Count (total)	Type	Meaning
`Rest`	101 (files)	long	Rest / baseline block
`Task-NF`	101 (files)	long	Neurofeedback motor imagery task block
`Response/R128`	68	admin	BrainVision response trigger / MRI sync - exclude
`Stimulus/S 99`	68	admin	BrainVision admin code - exclude
`Stimulus/S 2`	68	admin	BrainVision admin code - exclude
`Stimulus/S101`	39	admin	BrainVision admin code - exclude
`Pulse Artifact/R`	34	admin	MRI gradient pulse artifact - exclude
`Time 0/`	34	admin	MRI timing reference - exclude
`TPULSE/TSTART`	34	admin	MRI T-pulse start - exclude
`TPULSE/TPEAK`	34	admin	MRI T-pulse peak - exclude
`TPULSE/TEND`	34	admin	MRI T-pulse end - exclude
`Stimulus/S 35`	2	admin	Sparse BrainVision code - exclude

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Motor Imagery Neurofeedback State Classification (binary)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`Rest`	0	Rest / baseline block
`Task-NF`	1	Motor imagery neurofeedback block

- Notes: All Stimulus/S, TPULSE/, Pulse Artifact/R, Time 0/, and R128 are MRI/system admin markers - excluded. High sampling rate (5000 Hz); downsample before epoching.

Data Quality Notes

- MRI gradient artifacts contaminate EEG; TPULSE markers indicate artifact windows.
- High sampling rate (5000 Hz); downsampling recommended.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Simultaneous MRI-EEG during a motor imagery neurofeedback task: an open access brain imaging dataset for multi-modal data integration`

012. openneuro_ds002578 - Visual Oddball Task (256 channels)

Dataset: `openneuro_ds002578` | Year: 2021 | URL: https://openneuro.org/datasets/ds002578
Category: Sensory/Response | Subjects: N/A | Files (Completed): 2 | Channels: 256.0 | Sampling Rate: 256.0 Hz

Paradigm

A visual oddball task designed to study selective attention using event-related potentials with high-density 256-channel electrode coverage. Despite the task description implying trial-level stimuli (standard vs. oddball), no event markers were detected in the BIDS EEG files. The dataset contains only 2 completed files and is a very small subset.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events; sliding-window segmentation
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002578.py`); misc channels: `response_time`

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level labels available)
- Notes: Only 2 completed files available - very small dataset. High-density 256-channel EEG provides spatial richness. Check events.tsv directly for potential trial markers not captured in locator pipeline.

Data Quality Notes

- Only 2 completed EEG files - extremely limited dataset size.
- No event markers detected despite visual oddball paradigm description.
- High-density 256-channel array; verify electrode positions from channel info files.

References

- Reference: `Visual Oddball Task (256 channels)`

013. openneuro_ds002680 - Go-nogo categorization and detection task

Dataset: `openneuro_ds002680` | Year: 2021 | URL: https://openneuro.org/datasets/ds002680
Category: Sensory/Response | Subjects: N/A | Files (Completed): 350 | Channels: 31.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants performed a visual go/no-go paradigm involving rapid animal categorization and image recognition tasks with brief 20 ms stimulus presentations. Go trials required a button-press response when an animal was detected (or a target category matched); no-go trials required withholding response. Each run contains approximately 100 stimulus events and ~50 response events, suggesting roughly 50% go rate. The paradigm studies rapid visual processing and response inhibition.

Event Structure

Event ID	Count (total)	Type	Meaning
`stimulus`	~35,000 (350x100)	short	Visual stimulus onset (20 ms image presentation)
`response`	~18,000 (350x52)	short	Button-press go response

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002680.py`); misc channels: `response_time`

Task 1: Stimulus-Locked ERP / Reaction Time Regression (regression)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to stimulus onset)
- Target Events -> y: null (regression only; stimulus serves as epoch anchor)
- Regression target: `response_time` (reaction time from stimulus to response, from misc channel)
- Notes: Go/no-go structure: ~100 stimuli per run, ~52 responses (go trials). Regression target is response_time on go trials. For classification, go vs no-go labels would need to be inferred from presence/absence of a response after each stimulus.

Data Quality Notes

- Large dataset: 350 runs; verify subject/session structure from BIDS metadata.
- ~50% go rate per run based on stimulus (100) vs response (52) counts.
- 1000 Hz sampling rate; consider downsampling to 250 Hz for efficiency.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Go-nogo categorization and detection task`

014. openneuro_ds002691 - Internal attention study

Dataset: openneuro_ds002691 | Year: 2021 | URL: https://openneuro.org/datasets/ds002691
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 32.0 | Sampling Rate: 250.0 Hz

Paradigm

A sustained internal attention task where participants were asked to either mentally concentrate on a target (concentrate condition) or relax (relax condition) for 15-second sustained blocks with eyes closed. Transitional cues (getready, wait) marked preparation periods between active blocks. The study investigated EEG signatures of sustained internal attention versus relaxation.

Event Structure

Event ID	Count (total)	Type	Meaning
concentrate	20 (files)	long	15-second sustained mental concentration block
relax	20 (files)	long	15-second sustained relaxation block
getready	20 (files)	admin	Preparation cue between blocks - transitional
wait	20 (files)	admin	Wait period between blocks - transitional

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (kernels/openneuro_specific/openneuro_ds002691.py); misc channels: response_time

Task 1: Attention State Classification (binary)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
concentrate	0	Sustained mental concentration (15 s blocks)
relax	1	Sustained relaxation (15 s blocks)

- Notes: getready and wait are transitional preparation markers - excluded from target_events. 15-second sustained blocks are appropriate for long_event epoching with 2 s windows and 0.5 overlap. Misc channel response_time available for regression.

Data Quality Notes

- Small dataset: 20 subjects, 1 file each.
- Eyes-closed recording throughout; no eye movement artifacts expected.

- The count column (20 files) reflects that every file has these events, not the per-subject trial count.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Internal attention study`

015. openneuro_ds002718 - Face processing EEG dataset for EEGLAB

Dataset: openneuro_ds002718 | Year: 2025 | URL: https://openneuro.org/datasets/ds002718
Category: High-level Mental | Subjects: 18 | Files (Completed): 18 | Channels: 74.0 | Sampling Rate: 250.0 Hz

Paradigm

Participants viewed famous, unfamiliar, and scrambled face images and performed a subjective spatial symmetry judgment task (yes/no button press). The primary scientific question concerns the neural basis of face recognition, with ERP components (N170, N250) distinguishing famous from unfamiliar and scrambled faces. This is an EEGLAB tutorial dataset based on the SPM multimodal dataset (ds000117); events encode face category, presentation order (first vs. second), and latency of second presentation (early vs. late).

Event Structure

Event ID	Count (total)	Type	Meaning
famous_new	18 (files)	short	Famous face - first presentation
unfamiliar_new	18 (files)	short	Unfamiliar face - first presentation
scrambled_new	18 (files)	short	Scrambled face - first presentation
famous_second_early	18 (files)	short	Famous face - second presentation (early lag)
famous_second_late	18 (files)	short	Famous face - second presentation (late lag)
unfamiliar_second_early	18 (files)	short	Unfamiliar face - second presentation (early lag)
unfamiliar_second_late	18 (files)	short	Unfamiliar face - second presentation (late lag)
scrambled_second_early	18 (files)	short	Scrambled face - second presentation (early lag)
scrambled_second_late	18 (files)	short	Scrambled face - second presentation (late lag)
right_sym	10 (files)	short	Right-hand response: face judged symmetric
left_nonsym	10 (files)	short	Left-hand response: face judged non-symmetric
left_sym	8 (files)	short	Left-hand response: face judged symmetric
right_nonsym	8 (files)	short	Right-hand response: face judged non-symmetric
boundary	18 (files)	admin	Segment boundary marker - exclude

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5

- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (kernels/openneuro_specific/openneuro_ds002718.py); misc channels: response_time

Task 1: Face Category Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
famous_new	0	Famous face - first presentation
unfamiliar_new	1	Unfamiliar face - first presentation
scrambled_new	2	Scrambled face - first presentation

- Notes: Three-class face categorization targets the primary N170/N250 ERP question. boundary is a segment boundary admin marker - excluded. Response events (right_sym, left_sym, etc.) are button-press responses, not stimulus events. Regression target: response_time.

Data Quality Notes

- Count column shows files-with-event (18 subjects), not individual trial counts - actual trial counts per category are much higher.
- boundary events are segment/epoch boundary markers - do not epoch on these.
- This is an EEGLAB tutorial dataset with well-documented preprocessing pipeline.
- Verify channel count consistency across subjects before pooling.

References

- Paper: Multimodal Integration of M/EEG and f/MRI Data in SPM12 (DOI/ref: ds000117)

016. openneuro_ds002720 - A dataset recorded during development of a tempo-based brain-computer music interface

Dataset: `openneuro_ds002720` | Year: 2020 | URL: https://openneuro.org/datasets/ds002720
Category: Action/Output | Subjects: 19 | Files (Completed): 165 | Channels: 19.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants used kinaesthetic motor imagery (imagining squeezing a ball) or relaxation to intentionally modulate the tempo of music in a continuous brain-computer music interface (BCMI) task. Event code 1 = motor_imagery (ball squeezing imagination) and code 2 = relaxation (rest), each lasting several seconds. The BCI decoded the EEG to adjust music tempo in real-time, creating a closed-loop feedback system.

Event Structure

Event ID	Count (total)	Type	Meaning
`motor_imagery` (code 1)	143 (files)	long	Sustained motor imagery - imagining squeezing a ball
`relaxation` (code 2)	143 (files)	long	Sustained relaxation period

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - codes decoded via specific kernel
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002720.py`); event_map: {"1": "motor_imagery", "2": "relaxation"}; misc channels: `response_time`

Task 1: Motor Imagery BCI Classification (binary)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.5 s, tmax = 4.0 s (motor imagery window)
- Target Events -> y:

Event	y	Meaning
`motor_imagery`	0	Motor imagery - imagining squeezing a ball
`relaxation`	1	Relaxation / rest period

- Notes: Motor imagery BCI task with multi-second imagination periods. Long 4-second epoch window captures the full motor imagery event. Specific kernel decodes raw codes (1->motor_imagery, 2->relaxation).

Data Quality Notes

- 19-channel EEG is low density but suitable for BCI motor imagery decoding.
- Count column (143 files) indicates files-with-event, not trial count per file.
- Verify channel count consistency across subjects before pooling.

References

- Reference: A dataset recorded during development of a tempo-based brain-computer music interface

017. openneuro_ds002721 - An EEG dataset recorded during affective music listening

Dataset: `openneuro_ds002721` | Year: 2024 | URL: https://openneuro.org/datasets/ds002721
Category: High-level Mental | Subjects: 31 | Files (Completed): 185 | Channels: 19.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40 music clips from film scores while their EEG was recorded, to investigate the neural correlates of music-induced emotional states. Each 40-second music clip is marked at onset with a numeric trigger code (800-908 range encode specific music tracks; 32768 and 257 are likely system markers). The study is part of a series with ds002722 (calibration) and ds002723/ds002724 (BCI sessions). Emotion ratings may be available in behavioral logs.

Event Structure

Event ID	Count (total)	Type	Meaning
`32768`	124 (files)	admin	System/hardware trigger - likely session marker
`1092`	124 (files)	admin	System code - likely hardware sync
`800-808` range	124 (files each)	long	Music clip onset - specific track identifiers
`833-840` range	111-123 (files)	long	Music clip onset - specific track identifiers
`902-908` range	118-123 (files)	long	Music clip onset - specific track identifiers
`786`, `788`	124 (files)	long	Music clip onset - specific track identifiers
`257`	120 (files)	admin	System code - likely hardware marker
`33575`, `33571`	112-119 (files)	admin	Large codes - likely system/admin markers

> Event Type:
> - `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
> - `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
> - `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Music Clip Onset Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
800	0	Music clip onset (track 800)
801	1	Music clip onset (track 801)
802	2	Music clip onset (track 802)
803	3	Music clip onset (track 803)
804	4	Music clip onset (track 804)
805	5	Music clip onset (track 805)
806	6	Music clip onset (track 806)
807	7	Music clip onset (track 807)
808	8	Music clip onset (track 808)
786	9	Music clip onset (track 786)
788	10	Music clip onset (track 788)
833	11	Music clip onset (track 833)
834	12	Music clip onset (track 834)
835	13	Music clip onset (track 835)
836	14	Music clip onset (track 836)
837	15	Music clip onset (track 837)
838	16	Music clip onset (track 838)
839	17	Music clip onset (track 839)
840	18	Music clip onset (track 840)
902	19	Music clip onset (track 902)
903	20	Music clip onset (track 903)
904	21	Music clip onset (track 904)
905	22	Music clip onset (track 905)
906	23	Music clip onset (track 906)
907	24	Music clip onset (track 907)
908	25	Music clip onset (track 908)

- Notes: Codes 32768, 1092, 257, 33575, 33571 are system/admin markers - excluded. Each music clip is ~40 seconds; these onset markers are short events marking start of a sustained music stimulus. Valence/arousal ratings from behavioral logs could serve as regression targets.

Data Quality Notes

- 30 distinct event types; codes 32768, 33575, 33571 are likely hardware sync markers.
- 19-channel EEG; low spatial resolution.
- Verify event code mapping from dataset README or paper (Scientific Data 2018).
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Neural and physiological data from participants listening to affective music (Scientific Data, 2018)`

018. openneuro_ds002722 - A dataset recorded during development of an affective brain-computer music interface: calibration session

Dataset: `openneuro_ds002722` | Year: 2020 | URL: https://openneuro.org/datasets/ds002722
Category: High-level Mental | Subjects: 19 | Files (Completed): 94 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40-second synthetic music clips designed to induce specific affective states (varying valence and arousal) as a calibration session for an affective brain-computer music interface (BCMI). EEG and physiological signals were recorded to build a mapping from neural/physiological features to targeted emotional states. Codes 1-9 are primary music condition labels; 100-200 range codes encode specific music clip identifiers.

Event Structure

Event ID	Count (total)	Type	Meaning
`9`	94 (files)	long	Music condition 9 onset (high arousalx)
`4`	90 (files)	long	Music condition 4 onset
`5`	89 (files)	long	Music condition 5 onset
`6`	89 (files)	long	Music condition 6 onset
`7`	89 (files)	long	Music condition 7 onset
`2`	89 (files)	long	Music condition 2 onset
`1`	60 (files)	long	Music condition 1 onset
`3`	59 (files)	long	Music condition 3 onset
`104-205` range	19 (files each)	long	Specific music clip identifiers

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Affective Music Condition Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to music clip onset)
- Target Events -> y:

Event	y	Meaning
`1`	0	Music condition 1

Event	y	Meaning
2	1	Music condition 2
3	2	Music condition 3
4	3	Music condition 4
5	4	Music condition 5
6	5	Music condition 6
7	6	Music condition 7
8	7	Music condition 8
9	8	Music condition 9

- Notes: Codes 1-9 are primary valence/arousal condition labels. 100-200 range codes are specific music clip IDs (22 clips per subject); these could alternatively be used as 22-class stimulus labels. Decode exact mapping from Daly et al. 2018. 40-second clips are long events; epoch on onset markers.

Data Quality Notes

- 30 distinct event types; codes 1-9 are the primary condition labels.
- Decode exact valence/arousal mapping for codes 1-9 from Daly et al. 2018 paper.
- 37-channel EEG includes physiological channels (ECG, GSR, etc.); identify EEG-only channels.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Daly, I., Nicolaou, N., Williams, D., Hwang, F., Kirke, A., Miranda, E., Nasuto, S.J., Neural and physiological data from participants listening to affective music, Scientific Data, 2018.`

019. openneuro_ds002723 - A dataset recorded during development of an affective brain-computer music interface: testing session

Dataset: `openneuro_ds002723` | Year: 2020 | URL: https://openneuro.org/datasets/ds002723
Category: Complex Applied | Subjects: 8 | Files (Completed): 44 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants interacted with a real-time affective brain-computer music interface (BCMI) that modified music on-the-fly based on detected emotional states, aiming to induce specific target affective trajectories. This is the testing session of the BCI system (see also ds002722 calibration and ds002724 training sessions). Codes 1-9 encode the target or detected affective music conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
2	44 (files)	long	Affective music condition 2
6	44 (files)	long	Affective music condition 6
9	42 (files)	long	Affective music condition 9
4	42 (files)	long	Affective music condition 4
5	39 (files)	long	Affective music condition 5
7	38 (files)	long	Affective music condition 7
1	29 (files)	long	Affective music condition 1
3	28 (files)	long	Affective music condition 3

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Affective Music Condition Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to music condition onset)
- Target Events -> y:

Event	y	Meaning
2	0	Music condition 2
6	1	Music condition 6
9	2	Music condition 9
4	3	Music condition 4
5	4	Music condition 5

Event	y	Meaning
7	5	Music condition 7
1	6	Music condition 1
3	7	Music condition 3

- Notes: 8 affective conditions; decode exact valence/arousal mapping from Daly et al. 2018. Small dataset (8 subjects, 44 files). Music conditions are sustained 40-second states; these onset markers epoch at transition points.

Data Quality Notes

- Small dataset: 8 subjects, 44 files.
- 37-channel array includes physiological channels; verify EEG-only channel selection.
- Decode code-to-condition mapping from Daly et al. 2018 (Scientific Data).
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Neural and physiological data from participants listening to affective music, Scientific Data, 2018`

020. openneuro_ds002724 - A dataset recorded during development of an affective brain-computer music interface: training sessions

Dataset: `openneuro_ds002724` | Year: 2020 | URL: https://openneuro.org/datasets/ds002724
Category: High-level Mental | Subjects: 17 | Files (Completed): 96 | Channels: 37.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to 40-second synthetic music clips designed to induce specific affective states (varying valence and arousal) during the training phase of an affective brain-computer music interface. This is the training session companion to ds002722 (calibration) and ds002723 (testing). Codes 1-9 encode the nine target affective music conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
6	96 (files)	long	Music condition 6 onset
4	96 (files)	long	Music condition 4 onset
5	96 (files)	long	Music condition 5 onset
2	96 (files)	long	Music condition 2 onset
1	95 (files)	long	Music condition 1 onset
8	95 (files)	long	Music condition 8 onset
7	95 (files)	long	Music condition 7 onset
9	94 (files)	long	Music condition 9 onset
3	94 (files)	long	Music condition 3 onset

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Affective Music Condition Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to music condition onset)
- Target Events -> y:

Event	y	Meaning
1	0	Music condition 1
2	1	Music condition 2
3	2	Music condition 3

Event	y	Meaning
4	3	Music condition 4
5	4	Music condition 5
6	5	Music condition 6
7	6	Music condition 7
8	7	Music condition 8
9	8	Music condition 9

- Notes: 9 affective music conditions; decode valence/arousal mapping from Daly et al. 2018. Well-balanced: all conditions present in ~95-96 files. 40-second sustained stimuli; epoch at onset. 37-channel array includes physiological channels.

Data Quality Notes

- Well-balanced event counts across all 9 conditions (94-96 files each).
- 37-channel array includes EEG plus physiological channels (ECG, GSR, etc.).
- Decode code-to-affective-state mapping from Daly et al. 2018 (Scientific Data).
- Verify channel count consistency across subjects before pooling.

References

- Reference: `Daly, I., Nicolaou, N., Williams, D., Hwang, F., Kirke, A., Miranda, E., Nasuto, S.J., Neural and physiological data from participants listening to affective music, Scientific Data, 2018.`

021. openneuro_ds002725 - A dataset recording joint EEG-fMRI during affective music listening

Dataset: openneuro_ds002725 | Year: 2020 | URL: https://openneuro.org/datasets/ds002725
Category: High-level Mental | Subjects: 21 | Files (Completed): 105 | Channels: 47.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants listened to music stimuli designed to elicit emotional responses (approach/withdrawal behavior) and self-reported their felt affective states during simultaneous EEG-fMRI recording. The study investigated how EEG reflects subcortical brain activity related to emotional responses to music. Codes 1283/1285 are likely BrainVision fMRI sync codes; 34053 may be a BrainVision hardware code; 47/10/1 may be music condition/block markers.

Event Structure

Event ID	Count (total)	Type	Meaning
47	105 (files)	long	Music block marker (likely condition or trial count)
10	105 (files)	long	Music onset marker (condition 10x)
1	105 (files)	long	Music onset marker (condition 1x)
1283	105 (files)	admin	BrainVision fMRI sync code - likely admin
1285	105 (files)	admin	BrainVision fMRI sync code - likely admin
34053	105 (files)	admin	Large BrainVision hardware code - admin
265	94 (files)	long	Music condition marker
768	84 (files)	long	Music condition marker

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Affective Music State Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
47	0	Music block/condition marker 47

Event	y	Meaning
10	1	Music block/condition marker 10
1	2	Music block/condition marker 1
265	3	Music condition marker 265
768	4	Music condition marker 768

- Notes: Codes 1283, 1285, 34053 are BrainVision system/fMRI sync markers - excluded. Decode exact music condition meaning from paper (subcortical approach/withdrawal). 47-channel EEG includes MRI-compatible electrodes. fMRI gradient artifact correction required.

Data Quality Notes

- EEG recorded in MRI scanner; MRI gradient artifacts require dedicated removal pipeline.
- Codes 1283, 1285, 34053 are BrainVision system/fMRI codes - exclude from analysis.
- 47-channel EEG; verify channel positions and identify MRI-compatible electrode set.
- Decode music condition codes from Daly et al. / Sharman et al. paper.

References

- Reference: `Electroencephalography reflects the activity of sub-cortical brain regions during approach-withdrawal behaviour while listening to music`

022. openneuro_ds002778 - UC San Diego Resting State EEG Data from Patients with Parkinson's Disease

Dataset: `openneuro_ds002778` | Year: 2021 | URL: https://openneuro.org/datasets/ds002778
Category: Biological State | Subjects: N/A | Files (Completed): 46 | Channels: 41.0 | Sampling Rate: 512.0 Hz

Paradigm

Spontaneous scalp EEG recorded from patients with Parkinson's disease (PD) and healthy controls during resting-state conditions. The study investigated abnormal beta-band synchrony and beta waveform shape features (phase amplitude coupling, non-sinusoidal characteristics) as biomarkers of PD. All events in the BioSemi recording are system status markers (65536 = CM_in_range baseline, 130817 = CM status); no task events exist.

Event Structure

Event ID	Count (total)	Type	Meaning
`1`	44 (files)	admin	BioSemi recording marker (onset of continuous record) - admin
`65536`	43 (files)	admin	BioSemi CM_in_range baseline status code - exclude
`2`	1 (file)	admin	BioSemi marker - sparse, likely admin
`130817`	1 (file)	admin	BioSemi CM status code - exclude

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - all events are BioSemi admin markers; sliding-window segmentation
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002778.py`)

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (all events are admin; PD vs HC labels from participants.tsv)
- Notes: 65536 is the BioSemi CM_in_range baseline status code - a hardware marker not a task event. PD vs healthy control group labels are in participants.tsv and can be used as subject-level binary labels. Specific kernel adds participant group annotation.

Data Quality Notes

- Pure resting-state dataset; all events are BioSemi hardware status codes.
- 65536 (CM_in_range) and 130817 are BioSemi system codes - do not epoch on these.
- PD vs HC group labels available in participants.tsv for disease classification.

- Verify channel count consistency across subjects before pooling.

References

- Paper: Characteristics of Waveform Shape in Parkinson's Disease Detected with Scalp Electroencephalography (DOI/ref: `10.1523/ENEURO.0151-19.2019`)

023. openneuro_ds002791 - DataSet1 - Visual Object Naming EEG

Dataset: `openneuro_ds002791` | Year: 2020 | URL: https://openneuro.org/datasets/ds002791
Category: High-level Mental | Subjects: N/A | Files (Completed): 92 | Channels: 256.0 | Sampling Rate: 1000.0 Hz

Paradigm

Participants viewed and named visual objects (pictures presented as bitmap images) during high-density 256-channel EEG recording. Each event label is the filename of the presented image (e.g., GLOVE.bmp, GORILLA.bmp), corresponding to 30 distinct visual object categories. The task likely involved overt or covert naming or picture-word matching. This is DataSet1 of a paired dataset series (see also DataSet2: ds002833).

Event Structure

Event ID	Count (total)	Type	Meaning
`GLOVE.bmp`	46 (files)	short	Visual object onset - glove image
`FEATHER.bmp`	46 (files)	short	Visual object onset - feather image
`GORILLA.bmp`	46 (files)	short	Visual object onset - gorilla image
`DUSTPAN.bmp`	46 (files)	short	Visual object onset - dustpan image
`COMPASS.bmp`	46 (files)	short	Visual object onset - compass image
`ACCORDIO.bmp`	46 (files)	short	Visual object onset - accordion image
`CAP.bmp`	46 (files)	short	Visual object onset - cap image
`BOTTLE.bmp`	46 (files)	short	Visual object onset - bottle image
`CROWN.bmp`	46 (files)	short	Visual object onset - crown image
`ANCHOR.bmp`	46 (files)	short	Visual object onset - anchor image
`CARROT.bmp`	46 (files)	short	Visual object onset - carrot image
`FORK.bmp`	46 (files)	short	Visual object onset - fork image
`DONKEY.bmp`	46 (files)	short	Visual object onset - donkey image
`EAR.bmp`	46 (files)	short	Visual object onset - ear image
`CLOUD.bmp`	46 (files)	short	Visual object onset - cloud image

Event ID	Count (total)	Type	Meaning
ASHTRAY.bmp	46 (files)	short	Visual object onset - ashtray image
frenchcr.bmp	46 (files)	short	Visual object onset - french curve image
FISH.bmp	46 (files)	short	Visual object onset - fish image
BIRDCAGE.bmp	46 (files)	short	Visual object onset - birdcage image
ASPARAGU.bmp	46 (files)	short	Visual object onset - asparagus image
DRESS.bmp	46 (files)	short	Visual object onset - dress image
BEE.bmp	46 (files)	short	Visual object onset - bee image
CHIMNEY.bmp	46 (files)	short	Visual object onset - chimney image
BOOT.bmp	46 (files)	short	Visual object onset - boot image
CUP.bmp	46 (files)	short	Visual object onset - cup image
APPLE.bmp	46 (files)	short	Visual object onset - apple image
CRAB.bmp	46 (files)	short	Visual object onset - crab image
CAR.bmp	46 (files)	short	Visual object onset - car image
ALLIGATO.bmp	46 (files)	short	Visual object onset - alligator image
HAMMOCK.bmp	46 (files)	short	Visual object onset - hammock image

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002791.py`); misc channels: `response_time`

Task 1: Visual Object ERP / Regression (regression)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to image onset)
- Target Events -> y: null (30 object categories; regression primary; classification possible but 30-class)
- Regression target: `response_time` (naming latency from misc channel)

- Notes: 30 visual object categories; for classification, consider grouping by semantic category (living/nonliving, animate/inanimate) rather than using all 30 individual object identities. High-density 256-channel EEG provides excellent spatial coverage.

Data Quality Notes

- 30 distinct object images - each present in 46 files (half of 92 total files).
- High-density 256-channel EGI/HydroCell array; verify electrode positions.
- Consider semantic category grouping (animate/inanimate, living/nonliving) for classification.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `DataSet1 - Visual Object Naming EEG`

024. openneuro_ds002814 - A Multimodal Neuroimaging Dataset to Study Spatiotemporal Dynamics of Visual Processing in Humans

Dataset: `openneuro_ds002814` | Year: 2021 | URL: https://openneuro.org/datasets/ds002814
Category: Sensory/Response | Subjects: N/A | Files (Completed): 168 | Channels: 68.0 | Sampling Rate: 1200.0 Hz

Paradigm

A multimodal neuroimaging study (EEG + MEG/MRI) investigating the spatiotemporal dynamics of visual processing in humans. Despite the dataset title implying an active visual processing task, no event markers are detectable in the BIDS EEG files. The visual stimulation protocol may be encoded in other modality files or events are not available in the EEG files processed by the locator pipeline.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files

Event Type:
- `short`: stimulus onset marker (<2 s per presentation), for `epoch_by_event_hdf5`
- `long`: sustained-state marker (>=2 s), for `epoch_by_long_event_hdf5`
- `rest`: resting-state segment marker, for `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events; sliding-window segmentation
Kernel: Specific (`kernels/openneuro_specific/openneuro_ds002814.py`); misc channels: `response_time`

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level labels available)
- Notes: No event markers found in EEG locator metadata. High sampling rate (1200 Hz); downsampling recommended. Part of a multimodal dataset - check MEG/MRI files for corresponding event timing.

Data Quality Notes

- No event markers detected in any of the 168 EEG files.
- High sampling rate (1200 Hz); downsampling required before most analyses.
- Part of multimodal dataset; MEG or MRI-derived event timing may be applicable if aligned.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `A Multimodal Neuroimaging Dataset to Study Spatiotemporal Dynamics of Visual Processing in Humans`

025. openneuro_ds002833 - DataSet2 - Visual Object Naming EEG (High-Density)

Dataset: openneuro_ds002833 | Year: 2020 | URL: https://openneuro.org/datasets/ds002833
Category: High-level Mental | Subjects: N/A | Files (Completed): 80 | Channels: 257.0 | Sampling Rate: 1000.0 Hz

Paradigm

High-density 257-channel EEG recorded during a visual object naming or picture viewing task, companion to DataSet1 (ds002791). The DIN1 marker is an EGI/Geodesic system digital input trigger marking stimulus onset. The unknown label in 20 files indicates undecodable trigger codes for those sessions. The study likely involves the same visual object paradigm as ds002791 (30 object categories), recorded with an EGI system.

Event Structure

Event ID	Count (total)	Type	Meaning
DIN1	60 (files)	short	EGI digital input trigger - stimulus onset marker
unknown	20 (files)	admin	Undecodable trigger code - treat as admin/uncertain

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Specific (kernels/openneuro_specific/openneuro_ds002833.py); misc channels: response_time

Task 1: Stimulus-Locked ERP / Regression (regression)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to DIN1 onset)
- Target Events -> y: null (regression only; DIN1 serves as epoch anchor; unknown excluded)
- Regression target: response_time from misc channel
- Notes: DIN1 is the EGI stimulus trigger; use as epoch anchor. unknown events are undecodable - exclude from target events. For classification, trial type information would need to come from events.tsv trial_type or value columns.

Data Quality Notes

- 257-channel EGI HydroCell array provides very high spatial density.
- unknown events in 20 files may indicate corrupted trigger codes - verify from events.tsv.
- DIN1 triggers mark stimulus onsets; actual stimulus identity requires events.tsv lookup.
- Verify channel count consistency across subjects before pooling.

References

- Reference: DataSet2 - Visual Object Naming EEG (High-Density)

026. openneuro_ds002893 - Auditory-Visual Shift Study

Dataset: openneuro_ds002893 | Year: 2022 | URL: https://openneuro.org/datasets/ds002893
Category: Sensory/Response | Subjects: 49 | Files (Completed): 52 | Channels: 36.0 | Sampling Rate: 250.0 Hz

Paradigm

An interleaved dual-oddball task in which subjects selectively attended to either auditory or visual stimuli while switching between modalities based on spoken and printed attention cue instructions. The study investigated modality-specificity of sensory aging in vision and audition using event-related potentials. Attention states (attend_auditory, attend_visual) are sustained conditions lasting multiple trials; shift_attention marks the transition cue between modalities.

Event Structure

Event ID	Count (total)	Type	Meaning
attend_auditory	51 (files)	long	Sustained auditory attention block
attend_visual	51 (files)	long	Sustained visual attention block
shift_attention	49 (files)	admin	Attention shift cue - transition marker between blocks

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Attention Modality Classification (binary)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
attend_auditory	0	Sustained auditory attention block
attend_visual	1	Sustained visual attention block

- Notes: shift_attention is a transition cue - it marks the moment of modality switching, not a sustained state; treat as admin. The attend_* events mark the onset of sustained attention blocks of multiple seconds.

Data Quality Notes

- shift_attention marks attention shift cues - not a sustained state; exclude from classification target.
- Count column reflects files-with-event (51 subjects out of 52 files), not per-session trial count.
- Older adult participants (aging study); verify age distribution from participants.tsv.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Modality-specificity of sensory aging in vision and audition: evidence from event-related potentials. Brain research, 1215, pp.53-68.

027. openneuro_ds003004 - Imagined Emotion Study

Dataset: openneuro_ds003004 | Year: 2022 | URL: https://openneuro.org/datasets/ds003004
Category: High-level Mental | Subjects: N/A | Files (Completed): 34 | Channels: 219.0 | Sampling Rate: 256.0 Hz

Paradigm

Participants used voice-guided imagery to induce and sustain 15 distinct emotional states, using button presses to indicate the onset and offset of each imagined emotion. Despite the paradigm involving discrete emotional state onsets (button presses), no event markers were detected in the BIDS EEG files by the locator pipeline. The button-press events may be stored in separate behavioral logs or were not encoded in the EEG trigger channel.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files; button-press onsets not captured in locator

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events in locator; sliding-window segmentation
Kernel: Specific (kernels/openneuro_specific/openneuro_ds003004.py); misc channels: response_time

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: epoch_by_segmentation_hdf5
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level event markers in locator; check events.tsv directly for button-press events)
- Notes: High-density 219-channel EEG. If button-press onset/offset events are available in events.tsv, 15 emotional states could be used as multiclass labels with long_event epoching.

Data Quality Notes

- No event markers detected in EEG files by locator pipeline - verify events.tsv directly.
- High-density 219-channel array; verify electrode positions and system (likely EGI/Geodesic).
- 15 distinct emotional states if properly annotated - valuable for affective BCI research.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Imagined Emotion Study

028. openneuro_ds003039 - Free walking study

Dataset: openneuro_ds003039 | Year: 2025 | URL: https://openneuro.org/datasets/ds003039
Category: Complex Applied | Subjects: 19 | Files (Completed): 19 | Channels: 67.0 | Sampling Rate: 500.0 Hz

Paradigm

Participants walked two different outdoor routes, once while performing a self-paced button-pressing task, to study brain activity during naturalistic free-walking gait. This is a mobile EEG study; no trial-level event markers are detectable in the EEG files. Button-press events from the self-paced task may be available in separate behavioral logs.

Event Structure

Event ID	Count (total)	Type	Meaning
(none)	-	-	No event markers detected in EEG files; naturalistic free-walking paradigm

Event Type:
- short: stimulus onset marker (<2 s per presentation), for epoch_by_event_hdf5
- long: sustained-state marker (>=2 s), for epoch_by_long_event_hdf5
- rest: resting-state segment marker, for epoch_by_segmentation_hdf5
- admin: system/boundary marker, exclude from epoch target_events

HDF5 Production Guide

Label Source: Annotation (events.tsv) - no trial events; sliding-window segmentation
Kernel: Specific (kernels/openneuro_specific/openneuro_ds003039.py); misc channels: response_time

Task 1: Resting / Continuous Segmentation (resting)

- Epoch Method: epoch_by_segmentation_hdf5
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null (no trial-level events in EEG files)
- Notes: Mobile EEG during naturalistic walking. Motion artifacts are expected; ICA-based artifact removal recommended. Route condition (route A vs route B) may be available as a subject-level metadata label.

Data Quality Notes

- Mobile EEG with significant motion artifacts from walking - ICA and motion artifact rejection required.
- No trial-level event markers in EEG files; button-press events may be in separate logs.
- 67-channel mobile EEG system; verify electrode positions.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Nadine Jacobsen (March, 2020)

029. openneuro_ds003061 - EEG data from an auditory oddball task

Dataset: openneuro_ds003061 | Year: 2022 | URL: https://openneuro.org/datasets/ds003061
Category: Sensory/Response | Subjects: 39 | Files (Completed): 39
Channels: 79 | Sampling Rate: 256 Hz

Paradigm

An auditory three-stimulus oddball task in which participants pressed a key in response to infrequent target tones among frequent standard tones and rare distractor tones. The paradigm is designed to elicit P300 event-related potentials. Event labels stimulus (tone onset) and response (keypress) are present in all 39 files; STATUS appears as a BioSemi admin marker in 2 files.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulus	39 (files)	short	Auditory tone stimulus onset
response	39 (files)	short	Participant keypress response
STATUS	2 (files)	admin	BioSemi system status marker - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003061.py); misc channels: response_time

Task 1: Stimulus-Locked Auditory ERP/Regression (binary + regression)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
stimulus	0	Auditory tone onset

- Regression target:response_time(reaction time from stimulus to response, from misc channel)
- Note: STATUS is a BioSemi CM_in_range admin marker - excluded. response marks keypresses; reaction time computed per trial. The events.tsv trial_type may further distinguish standard, target, and distractor tone types - check for sub-classification.

Data Quality Notes

- STATUS is a BioSemi hardware marker - not a task event; exclude from target_events.

- `response` events mark keypresses; sub-classification by tone type (standard/target/distractor) requires events.tsv lookup.
- 79-channel BioSemi EEG; check consistent montage across subjects.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG data from an auditory oddball task`

030. openneuro_ds003190 - Assessment of the visual stimuli properties in P300 paradigm

Dataset: openneuro_ds003190 | Year: 2020 | URL: https://openneuro.org/datasets/ds003190
Category: Sensory/Response | Subjects: 19 | Files (Completed): 384
Channels: 9 | Sampling Rate: 256 Hz

Paradigm

A P300 ERP experiment evaluating the effects of stimulus count and visual presentation styles (standard flash vs. cartoon face) on EEG brain activity. The experiment used a P300-based BCI paradigm with target and non-target stimulus flashes. Codes 1-9 represent different stimulus conditions or counts; codes 100-203 range encode specific stimulus configurations. The dataset has 19 subjects with many runs (384 files total).

Event Structure

Event ID	Count (total)	Type	Meaning
1	382 (files)	short	P300 stimulus condition 1 (non-target flashx)
2	382 (files)	short	P300 stimulus condition 2 (non-target flashx)
3	382 (files)	short	P300 stimulus condition 3
4	382 (files)	short	P300 stimulus condition 4
200	382 (files)	short	P300 stimulus configuration 200
201	382 (files)	short	P300 stimulus configuration 201
202	382 (files)	short	P300 stimulus configuration 202
203	382 (files)	short	P300 stimulus configuration 203
101	363 (files)	short	P300 target flash condition 101
102	354 (files)	short	P300 target flash condition 102
103	368 (files)	short	P300 target flash condition 103
104	370 (files)	short	P300 target flash condition 104
5	326 (files)	short	P300 stimulus condition 5
105	305 (files)	short	P300 target flash condition 105
6	224 (files)	short	P300 stimulus condition 6
106	212 (files)	short	P300 target flash condition 106
7	169 (files)	short	P300 stimulus condition 7

Event ID	Count (total)	Type	Meaning
107	161 (files)	short	P300 target flash condition 107
8	112 (files)	short	P300 stimulus condition 8
108	110 (files)	short	P300 target flash condition 108
9	56 (files)	short	P300 stimulus condition 9
109	47 (files)	short	P300 target flash condition 109

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003190.py); misc channels: response_time

Task 1: P300 Target vs Non-Target Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
target_flash	0	Target stimulus flash (P300 target)
nontarget_flash	1	Non-target stimulus flash

- Remarks: Specific kernel decodes numeric codes into semantic labels (target_flash, nontarget_flash). 9-channel EEG is very sparse; focus on occipital and parietal channels for P300. Regression target response_time available.

Data Quality Notes

- Very sparse 9-channel EEG; limited spatial coverage.
- 384 files across 19 subjects = ~20 runs/subject; verify session structure.
- Numeric trigger codes require specific kernel to decode semantic P300 target/non-target labels.
- Verify channel count consistency across subjects before pooling.

References

- Reference: Assessment of the visual stimuli properties in P300 paradigm

031. openneuro_ds003194 - Neuroepo multisession

Dataset: openneuro_ds003194 | Year: 2022 | URL: https://openneuro.org/datasets/ds003194
Category: Biological State | Subjects: 25 | Files (Completed): 29
Channels: 19 | Sampling Rate: 200 Hz

Paradigm

Resting-state EEG recorded before and after a six-month Neuroepo drug administration period to evaluate neuroprotective effects in Parkinson's disease patients. The protocol includes resting-state (eyes closed / eyes open), hyperventilation periods, and photic stimulation sections. This is a clinical neurological assessment battery recorded in Spanish (event labels in Spanish).

Event Structure

Event ID	Count (total)	Type	Meaning
ojos cerrados	29 (files)	long	Eyes-closed resting state
ojos abiertos	28 (files)	long	Eyes-open resting state
recuperacion	28 (files)	long	Recovery period after hyperventilation
hiperventilacion 1	27 (files)	long	Hyperventilation period 1
hiperventilacion 2	28 (files)	long	Hyperventilation period 2
hiperventilacion 3	28 (files)	long	Hyperventilation period 3
hiperventilacion 4	15 (files)	long	Hyperventilation period 4
photic stimulation	11 (files)	long	Photic stimulation block
fotoestimulacion	11 (files)	long	Photic stimulation (Spanish label)
discontinuity	9 (files)	admin	Recording discontinuity marker - exclude
HPVOJOS ABIERTOS	6 (files)	long	Eyes-open during hyperventilation
HEPERVENTILACION OJOS ABIERTOS	1 (file)	long	Hyperventilation eyes-open variant

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Brain State Classification (multiclass)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
ojos cerrados	0	Eyes-closed resting state
ojos abiertos	1	Eyes-open resting state
recuperacion	2	Recovery period
hiperventilacion 2	3	Hyperventilation period 2
hiperventilacion 3	4	Hyperventilation period 3
hiperventilacion 1	5	Hyperventilation period 1
hiperventilacion 4	6	Hyperventilation period 4
photic stimulation	7	Photic stimulation
fotoestimulacion	8	Photic stimulation (Spanish)
HPVOJOS ABIERTOS	9	Eyes-open during hyperventilation
HEPERVENTILACION OJOS ABIERTOS	10	Hyperventilation eyes-open variant

- Note: `discontinuity` is a recording discontinuity marker - excluded. All other events are sustained clinical states (>=2 s). `photic stimulation` and `fotoestimulacion` are equivalent labels (Spanish/English) for the same condition.

Data Quality Notes

- Clinical recording from Parkinson's patients; expect higher artifact rate than healthy subjects.
- `photic stimulation` / `fotoestimulacion` are duplicate labels for the same condition.
- `discontinuity` events mark recording gaps - exclude from epoch analysis.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Sci. data 6, 103 (2019)`

032. openneuro_ds003195 - Placebo Neuroepo multisession

Dataset: openneuro_ds003195 | Year: 2022 | URL: https://openneuro.org/datasets/ds003195
Category: Biological State | Subjects: 20 | Files (Completed): 20
Channels: 19 | Sampling Rate: 200 Hz

Paradigm

Resting-state EEG recorded from Parkinson's disease patients who received a placebo (not active Neuroepo) during a six-month drug evaluation trial. Recordings were taken before and after the administration period. This is the placebo control arm companion to ds003194 (Neuroepo treatment). Protocol includes resting-state (eyes open/closed), hyperventilation, and recovery periods.

Event Structure

Event ID	Count (total)	Type	Meaning
ojos cerrados	20 (files)	long	Eyes-closed resting state
ojos abiertos	19 (files)	long	Eyes-open resting state
hiperventilacion 1	19 (files)	long	Hyperventilation period 1
hiperventilacion 2	19 (files)	long	Hyperventilation period 2
hiperventilacion 3	19 (files)	long	Hyperventilation period 3
recuperacion	19 (files)	long	Recovery period after hyperventilation
hiperventilacion 4	12 (files)	long	Hyperventilation period 4
discontinuity	8 (files)	admin	Recording discontinuity marker - exclude
photic stimulation	3 (files)	long	Photic stimulation block
fotoestimulacion	3 (files)	long	Photic stimulation (Spanish label)
HPVOJOS ABIERTOS	2 (files)	long	Eyes-open during hyperventilation
hipervantilacion 4	1 (file)	long	Hyperventilation period 4 (variant spelling)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Brain State Classification (multiclass)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
ojos cerrados	0	Eyes-closed resting state
ojos abiertos	1	Eyes-open resting state
hiperventilacion 1	2	Hyperventilation period 1
hiperventilacion 2	3	Hyperventilation period 2
hiperventilacion 3	4	Hyperventilation period 3
recuperacion	5	Recovery period
hiperventilacion 4	6	Hyperventilation period 4
photic stimulation	7	Photic stimulation
fotoestimulacion	8	Photic stimulation (Spanish)
HPVOJOS ABIERTOS	9	Eyes-open during hyperventilation
hipervantilacion 4	10	Hyperventilation period 4 (variant spelling)

- Note: `discontinuity` is a recording gap marker - excluded. Sparse counts for photic stimulation (3 files) and HPVOJOS (2 files); consider merging equivalent Spanish/English labels. Placebo arm of Neuroepo trial (NCT04110678).

Data Quality Notes

- Clinical recording from Parkinson's patients; expect higher artifact rate.
- `photic stimulation` and `fotoestimulacion` are equivalent labels - merge if needed.
- `hipervantilacion 4` is a spelling variant of `hiperventilacion 4` - treat as same condition.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `NCT04110678`

033. openneuro_ds003420 - HD-EEGtask (Dataset 1)

Dataset: openneuro_ds003420 | Year: 2020 | URL: https://openneuro.org/datasets/ds003420
Category: High-level Mental | Subjects: 23 | Files (Completed): 92
Channels: 256 | Sampling Rate: 1000 Hz

Paradigm

Participants performed visual naming and spelling tasks of visually presented objects while high-density 256-channel EEG and speech onset data were recorded. This is Dataset 1 of the conneXion project. Trigger codes 0 and 1 mark stimulus events (likely object presentation onset and response/naming onset). Speech onset timing is also captured for reaction time analysis.

Event Structure

Event ID	Count (total)	Type	Meaning
1	92 (files)	short	Visual object stimulus onset (or naming task trigger)
0	91 (files)	short	Stimulus condition 0 (or response trigger)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003420.py); misc channels: response_time

Task 1: Stimulus-Locked ERP/Regression (regression)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:null(regression primary; code 1 serves as epoch anchor)
- Regression target:response_time(speech onset / naming latency from misc channel)
- Note: Codes 0 and 1 require decoding from events.tsv - likely stimulus onset and response. Specific kernel decodes numeric codes and extracts speech onset timing as response_time. High-density 256-channel EEG provides excellent spatial coverage for naming task analysis.

Data Quality Notes

- Trigger codes 0/1 require semantic decoding from events.tsv or conneXion project documentation.
- High-density 256-channel array; significant computational overhead.
- Speech onset timing recorded - useful for naming latency regression.
- Verify channel count consistency across subjects before pooling.

References

- Reference: conneXion

034. openneuro_ds003421 - HD-EEGtask (Dataset 2)

Dataset: openneuro_ds003421 | Year: 2020 | URL: https://openneuro.org/datasets/ds003421
Category: High-level Mental | Subjects: 20 | Files (Completed): 80
Channels: 257 | Sampling Rate: 1000 Hz

Paradigm

Participants completed a multi-condition experiment including resting state, visual and auditory object naming tasks, and a recognition-based working memory task, with high-density 257-channel EEG recording. This is Dataset 2 of the conneXion project (companion to ds003420). Trigger codes 0 and 1 mark stimulus onset and response events; the specific kernel provides speech onset timing as `response_time` for regression analysis.

Event Structure

Event ID	Count (total)	Type	Meaning
1	60 (files)	long	Sustained task or stimulus condition 1 onset
0	57 (files)	long	Sustained task or stimulus condition 0 onset

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel:Specific (`kernels/openneuro_specific/openneuro_ds003421.py`); misc channels: `response_time`

Task 1: Task State Classification / Regression (regression + long_event)

- Epoch method: `epoch_by_long_event_hdf5`
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:null(regression primary; codes require decoding)
- Regression target:`response_time`(speech onset / naming latency from misc channel)
- Note: Codes 0/1 mark multi-second sustained task states (resting, naming, memory) - long_event epoching appropriate. Specific kernel decodes numeric codes and extracts speech onset as `response_time`. Not all 80 files contain both events (60 vs 57).

Data Quality Notes

- Trigger codes 0/1 require semantic decoding from events.tsv or conneXion project documentation.
- High-density 257-channel EGI array; significant computational overhead.
- Multi-condition design includes resting state, naming, and working memory - verify condition assignments.
- Verify channel count consistency across subjects before pooling.

References

- Reference: HD-EEGtask (Dataset 2) - conneXion project

035. openneuro_ds003458 - EEG: Three armed bandit gambling task

Dataset: openneuro_ds003458 | Year: 2021 | URL: https://openneuro.org/datasets/ds003458
Category: High-level Mental | Subjects: 23 | Files (Completed): 23
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Participants performed a three-armed bandit gambling task with oscillating reinforcement probabilities to investigate reward processing and reward prediction error signals in EEG. Each trial includes a stimulus presentation (choice between Left, Right, or Up options), a button-press response, and outcome feedback (Win, Loss, or Null). The Fixation event marks trial onset. STATUS is a BioSemi admin marker.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus (Left	138 (files)	short	Stimulus onset with left option available
Right	138 (files)	short	Stimulus with right option available
Up)	138 (files)	short	Stimulus with up option available
X	92 (files)	short	Option marker X - decode from events.tsv
Y	92 (files)	short	Option marker Y - decode from events.tsv
Z	92 (files)	short	Option marker Z - decode from events.tsv
Response	69 (files)	short	Participant button-press choice response
Fixation	23 (files)	short	Trial-onset fixation cross
Feedback Win	23 (files)	short	Outcome: reward received
Feedback Loss	23 (files)	short	Outcome: no reward
Feedback Null	13 (files)	short	Outcome: null/neutral feedback
STATUS	23 (files)	admin	BioSemi system status marker - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003458.py); misc channels:

```
response_time
```

Task 1: Feedback Outcome Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`Feedback Win`	0	Win feedback (reward received)
`Feedback Loss`	1	Loss feedback (no reward)
`Feedback Null`	2	Null/neutral feedback

- Note: `STATUS` is a BioSemi admin marker - excluded. Event label `Stimulus (Left` contains a parenthesis fragment - this is a truncated label; the full label may span `Stimulus (Left, Right, Up)`. Feedback events are the primary scientific targets for reward prediction error analysis. `Response` marks button-press choices.

Data Quality Notes

- Event label `Stimulus (Left` is a truncated label fragment - verify full label from events.tsv.
- `STATUS` is BioSemi hardware marker - exclude from analysis.
- Small dataset: 23 subjects, 1 file each.
- Verify channel count consistency across subjects before pooling.

References

- Reference: `EEG: Three armed bandit gambling task`

036. openneuro_ds003478 - EEG: Depression rest

Dataset: openneuro_ds003478 | Year: 2021 | URL: https://openneuro.org/datasets/ds003478
Category: Biological State | Subjects: 122 | Files (Completed): 243
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recording featuring alternating one-minute intervals of eyes-open and eyes-closed conditions in participants with varying scores on the Beck Depression Inventory (BDI). The study aims to identify EEG biomarkers of depression severity. Each session includes multiple alternating eyes-open and eyes-closed blocks, with Start and Finish markers indicating session boundaries. STATUS is a BioSemi admin marker.

Event Structure

Event ID	Count (total)	Type	Meaning
Eyes Open	368 (files)	long	Eyes-open resting state block (~1 min)
Eyes Closed	368 (files)	long	Eyes-closed resting state block (~1 min)
STATUS	243 (files)	admin	BioSemi system status marker - exclude
Start and Finish	184 (files)	admin	Session start/end boundary marker - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003478.py); misc channels: response_time

Task 1: Eyes Open vs Eyes Closed Classification (binary)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Eyes Open	0	Eyes-open resting state (~1 min block)
Eyes Closed	1	Eyes-closed resting state (~1 min block)

- Note: STATUS is BioSemi admin marker; Start and Finish marks session boundaries - both excluded. Eyes-open vs eyes-closed is the primary binary classification. BDI scores in participants.tsv can serve as continuous regression targets for depression severity prediction.

Data Quality Notes

- STATUS is BioSemi hardware marker - exclude.
- Start and Finish marks session boundaries - admin marker, exclude.
- Large dataset: 122 subjects, 243 files; subject-level BDI scores available for depression prediction.
- Verify channel count consistency across subjects before pooling.

References

- Reference: EEG: Depression rest

037. openneuro_ds003490 - EEG: 3-Stim Auditory Oddball and Rest in Parkinson's

Dataset: openneuro_ds003490 | Year: 2021 | URL: https://openneuro.org/datasets/ds003490
Category: Sensory/Response | Subjects: 50 | Files (Completed): 75
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

A three-stimulus auditory oddball task (standard tones, target tones requiring button-press response, and novel distractor tones) combined with resting-state (eyes open and eyes closed) EEG recordings in Parkinson's disease patients and age-matched healthy controls. The study aims to identify auditory P300 and mismatch negativity differences between PD patients and controls. STATUS is a BioSemi admin marker.

Event Structure

Event ID	Count (total)	Type	Meaning
Standard Tone	75 (files)	short	Frequent standard auditory tone (non-target)
Target Tone	75 (files)	short	Infrequent target tone (requires button press)
Novel Tone	75 (files)	short	Rare novel distractor tone
Eyes Closed	75 (files)	long	Eyes-closed resting state block
Eyes Open	75 (files)	long	Eyes-open resting state block
STATUS	75 (files)	admin	BioSemi system status marker - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel:Specific (kernels/openneuro_specific/openneuro_ds003490.py); misc channels: response_time

Task 1: Auditory Tone Type Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 0.8 s (relative to tone onset, P300 window)
- Target Events -> y:

Event	y	Meaning
Standard Tone	0	Frequent standard tone (non-target)
Novel Tone	1	Rare novel distractor tone (P3a)
Target Tone	2	Infrequent target tone (P3b)

- Note: STATUS is BioSemi admin marker - excluded. Tone events are short stimuli (~50 ms), not sustained states; epoch_by_event_hdf5 is appropriate. Eyes-open/closed are separate resting blocks - do not mix with tone targets. PD vs HC group labels available in participants.tsv.

Task 2: Resting State Classification (binary)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, long_event_overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Eyes Open	0	Eyes-open resting block
Eyes Closed	1	Eyes-closed resting block

Data Quality Notes

- STATUS is BioSemi hardware marker - exclude.
- Dataset includes both Parkinson's patients and age-matched controls; group label from participants.tsv.
- Three-stimulus oddball design provides Standard/Target/Novel tone categories for ERP analysis.
- Verify channel count consistency across subjects before pooling.

References

- Reference: EEG: 3-Stim Auditory Oddball and Rest in Parkinson's

038. openneuro_ds003506 - EEG: Reinforcement Learning in Parkinson's

Dataset: openneuro_ds003506 | Year: 2021 | URL: https://openneuro.org/datasets/ds003506
Category: High-level Mental | Subjects: 56 | Files (Completed): 84
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

This study investigates the Reward Positivity (RewP) EEG component in Parkinson's disease (PD) patients compared to matched healthy controls. Participants completed a reinforcement learning task with both volitional and instructed choices, earning rewards or receiving no-reward feedback based on their selections. The RewP, a frontocentral ERP component, scales with reward prediction error and mirrors dopamine-driven learning. The study examines whether PD diminishes this signal, given dopaminergic dysfunction in PD.

Event Structure

Event ID	Count (total)	Type	Meaning
Imperative Stimulus	84	short	Cue stimulus requiring a response
Left Button Choice	84	short	Left-hand button press response
Right Button Choice	84	short	Right-hand button press response
Instr	168	short	Instruction marker (Match/Choose)
FB	252	short	Feedback stimulus (reward +1, no-reward 0)
STATUS	84	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Feedback type classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
FB: +1	1	Reward feedback (reward)
FB: 0	0	No-reward feedback (no-reward)

- Note: FB events encode reward vs. no-reward outcomes; roughly balanced across subjects. Exclude `FB: 'Too Slow'` and `FB: 'No Match'` as they are rare and task-specific.

Data Quality Notes

- 67 channels including VEOG and positional channels (X, Y, Z); exclude non-EEG channels before analysis.
- Two sessions per subject for some participants; verify session-level consistency.
- STATUS is a single system marker per file; exclude from epoching.

References

- Reference: `An EEG marker of reward processing is diminished in Parkinson's disease. Brain Research, 2019. DOI: 10.1016/j.brainres.2019.146541`

039. openneuro_ds003509 - EEG: Simon Conflict in Parkinson's

Dataset: `openneuro_ds003509` | Year: 2021 | URL: https://openneuro.org/datasets/ds003509
Category: High-level Mental | Subjects: 56 | Files (Completed): 84
Channels: 67 | Sampling Rate: 500 Hz

Paradigm

This study investigates mid-frontal theta deficits in Parkinson's disease (PD) patients during a modified Simon reaction-time task that requires cognitive control. Participants (high-performing PD patients and matched controls) performed congruent and incongruent Simon trials where stimulus position and response direction could conflict. The task included both a training phase with reinforcement feedback and a test phase without feedback. Mid-frontal theta (4-8 Hz) activity serves as an index of cognitive control and is modulated by cortical dopamine.

Event Structure

Event ID	Count (total)	Type	Meaning
`Trn Stim`	1344	short	Training phase stimulus (congruent/incongruent)
`Test Stim`	1008	short	Test phase stimulus (congruent/incongruent)
`correct`	336	short	Correct response marker
`incorrect`	334	short	Incorrect response / error marker
`Trn Resp`	335	short	Training phase button-press response
`Test Resp`	335	short	Test phase button-press response
`FB`	168	short	Feedback stimulus in training phase
`Trn No Response`	83	short	Missing response in training phase
`Test No Response`	54	short	Missing response in test phase
`STATUS`	84	admin	System status marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Correct/Incorrect Classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`correct`	1	Correct response
`incorrect`	0	Error response

- Note: Correct and incorrect counts are approximately balanced (~336 vs ~334 total). Epoch on the response marker; exclude no-response trials.

Data Quality Notes

- 67 channels including VEOG and positional channels (X, Y, Z); exclude non-EEG channels before analysis.
- Two sessions per subject for many participants; verify session consistency.
- `Trn Stim` and `Test Stim` have detailed subtypes in events.tsv (congruent/incongruent) that can support conflict classification.

References

- Reference: `Mid-frontal theta activity is diminished during cognitive control in Parkinson's disease. Neuropsychologia, 2018. DOI: 10.1016/j.neuropsychologia.2018.05.020`

040. openneuro_ds003516 - EEG: Attended Speaker Paradigm (Own Name in Ignored Stream)

Dataset: openneuro_ds003516 | Year: 2022 | URL: https://openneuro.org/datasets/ds003516
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 49 | Sampling Rate: 500 Hz

Paradigm

Participants performed a cocktail-party selective auditory attention task with two competing speakers. They listened to audio books from one speaker (attended stream) while their own name was occasionally presented in the ignored stream. This paradigm investigates the neural correlates of selective attention and the salience of self-referential stimuli (own name) in the unattended auditory channel. No discrete trial-level event markers were recorded in the EEG data.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; continuous EEG recording

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag Source: No event marker, use segmentation method
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no labels, for unsupervised or feature extraction)
- Note: No trial-level annotations available; treat as continuous listening EEG. Suitable for neural tracking analyzes or subject-level classification using demographic labels.

Data Quality Notes

- No event markers present; cannot epoch by stimulus onset.
- Continuous EEG during selective attention task; audio stimuli timing not embedded in EEG annotations.

References

- Reference: Bjorn Holtze (January, 2021). Own-name salience in selective auditory attention.

041. openneuro_ds003517 - EEG: Continuous Gameplay of an 8-Bit Style Video Game

Dataset: openneuro_ds003517 | Year: 2021 | URL: https://openneuro.org/datasets/ds003517
Category: Complex Applied | Subjects: 17 | Files (Completed): 34
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

Participants played a continuous 8-bit style video game while EEG was recorded to study canonical neural events during naturalistic gameplay. The experiment also included a visual oddball task and a gambling task embedded within the session. Game events (shooting, collecting stars, collisions) and oddball stimuli (standard and rare tones/targets) were time-locked to EEG recordings. The study aims to identify ERP components (P300, error-related negativity) in a real-world, engaging task context.

Event Structure

Event ID	Count (total)	Type	Meaning
ODDBALL STANDARD	17	short	Standard tone in oddball task (frequent)
ODDBALL RARE	17	short	Rare/target tone in oddball task (infrequent)
ODDBALL START	17	short	Oddball task block start marker
ODDBALL DONE	17	short	Oddball task block end marker
SHOOT_BUTTON	18	short	Player shoot button press
COLLECT_STAR	18	short	Player collected a star (reward)
PLAYER_CRASH_ENEMY	18	short	Player crashed into enemy
MISSILE_HIT_ENEMY	17	short	Missile hit enemy (success)
PLAYER_CRASH_WALL	17	short	Player crashed into wall
COLLECT_AMMO	17	short	Player collected ammunition
GAMBLING WIN	14	short	Gambling task win outcome
GAMBLING LOSS	14	short	Gambling task loss outcome
GAME START	16	short	Game session start
GAME OVER	16	short	Game session end
STATUS	34	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Mission 1: Oddball Standard/Rare Classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`ODDBALL STANDARD`	0	Standard frequent stimulation
`ODDBALL RARE`	1	Rare target stimulation

- Note: Classic oddball P300 paradigm embedded in gameplay. Only 17 files contain oddball events; some files may be game-only sessions.

Task 2: Gambling result classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`GAMBLING WIN`	1	Win
`GAMBLING LOSS`	0	Lose

- Remarks: Only 14 files contain gambling events; small sample size.

Data Quality Notes

- 65-channel EEG recorded during naturalistic, continuous task; higher artifact rate expected from movement and engagement.
- Some sessions may be game-only (no oddball/gambling); verify event presence per file.

References

- Reference: `Identification of canonical neural events during continuous gameplay of an 8-bit style video game. NeuroImage, 2016. DOI: 10.1016/j.neuroimage.2016.02.075`

042. openneuro_ds003518 - EEG: Simon Conflict with Reinforcement + Cabergoline Challenge

Dataset: openneuro_ds003518 | Year: 2021 | URL: https://openneuro.org/datasets/ds003518
Category: High-level Mental | Subjects: 110 | Files (Completed): 137
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

This study examines whether conflict acts as an implicit cost in reinforcement learning using a modified Simon task with pharmacological manipulation. Healthy participants performed a Simon conflict task across training and test phases, with a double-blind Cabergoline (dopamine D2/D3 agonist) challenge administered to examine dopaminergic modulation. The training phase included reinforcement feedback to shape behavior; the test phase probed learned conflict avoidance. Mid-frontal theta and error-related negativity are key EEG measures.

Event Structure

Event ID	Count (total)	Type	Meaning
Trn Stim	2192	short	Training phase stimulus (congruent/incongruent Simon)
Test Stim	1644	short	Test phase stimulus
correct	548	short	Correct response outcome
incorrect	547	short	Incorrect response / error outcome
Trn Resp	547	short	Training phase button-press response
Test Resp	548	short	Test phase button-press response
FB	274	short	Feedback stimulus in training phase
Trn No Response	136	short	Missing response in training phase
Test No Response	55	short	Missing response in test phase
STATUS	137	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Correct/Incorrect Classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`correct`	1	Correct response
`incorrect`	0	Error response

- Note: Correct and incorrect counts are closely balanced (~548 vs ~547 total). Largest dataset in the Simon-conflict series; includes drug condition variation (cabergoline vs placebo) - consider as a covariate.

Data Quality Notes

- 64-channel EEG with 110 subjects across cabergoline and placebo conditions.
- Drug condition information in participants.tsv; use as a grouping variable for stratified analysis.
- Some subjects may have two sessions; verify file-to-subject mapping.

References

- Reference: `Conflict acts as an implicit cost in reinforcement learning. Nature Communications, 2014. DOI: 10.1038/ncomms6394`

043. openneuro_ds003519 - EEG: Visual Working Memory + Cabergoline Challenge

Dataset: openneuro_ds003519 | Year: 2021 | URL: https://openneuro.org/datasets/ds003519
Category: High-level Mental | Subjects: 27 | Files (Completed): 54
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed a visuospatial working memory task under a double-blind Cabergoline (dopamine D2/D3 agonist) drug challenge to investigate pharmacological modulation of working memory processing. The task involved memorizing spatial locations of stimuli followed by a probe stimulus requiring a match/non-match judgment. Distractor stimuli were presented during the retention interval. The study examines individual differences in working memory capacity and dopaminergic modulation.

Event Structure

Event ID	Count (total)	Type	Meaning
Stim Onset	162	short	Study stimulus onset (to-be-remembered)
Question Mark Probe	108	short	Probe stimulus requiring match judgment
Response	108	short	Participant button-press response
STATUS	54	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus presentation ERP (single class, can be used for feature extraction)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stim Onset	0	Memory stimulus presentation
Question Mark Probe	1	Probe stimulus presentation

- Note: Study stimulus vs probe comparison captures encoding vs retrieval phases. Two sessions per subject (drug vs placebo); treat session as covariate.

Data Quality Notes

- 54 files for 27 subjects; two sessions per subject (drug/placebo crossover design).

- Drug condition information should be in participants.tsv or scans.tsv.
- Limited event types; primary utility is ERP characterization of encoding and retrieval phases.

References

- Reference: Dopamine D2 agonist affects visuospatial working memory distractor interference depending on individual differences in baseline working memory span. Cognitive, Affective, & Behavioral Neuroscience, 2018. DOI: 10.3758/s13415-018-0584-6

044. openneuro_ds003522 - EEG: Three-Stimulus Auditory Oddball and Rest in Acute and Chronic TBI

Dataset: openneuro_ds003522 | Year: 2021 | URL: https://openneuro.org/datasets/ds003522
Category: Sensory/Response | Subjects: 62 | Files (Completed): 196
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

This study investigates EEG markers in mild traumatic brain injury (mTBI) using a three-stimulus auditory oddball task and resting-state recordings. Participants (sub-acute mTBI patients, chronic mTBI patients, and healthy controls) passively listened to standard, novel, and target tones while performing a target detection task. Resting-state blocks with eyes open and eyes closed were also recorded. The P3a and P3b ERP components - reflecting bottom-up and top-down attentional processes - are the primary measures examined as TBI biomarkers.

Event Structure

Event ID	Count (total)	Type	Meaning
Standard Tone	194	short	Frequent standard auditory tone (non-target)
Novel Tone	194	short	Rare novel/distractor tone eliciting P3a
Target Tone	194	short	Rare target tone requiring button press (P3b)
Eyes Open	196	rest	Eyes-open resting-state block marker
Eyes Closed	196	rest	Eyes-closed resting-state block marker
STATUS	196	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Oddball three-category classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Standard Tone	0	Standard frequent tone
Novel Tone	1	Novel interference tone (P3a)

Event	y	Meaning
`Target Tone`	2	Target pitch (P3b)

- Note: Classic 3-stimulus oddball. Standard tones are ~80% frequent (~184/file), novel and target each ~10% (~38/file). Class imbalance: undersample standards or weight loss. Eyes-open/closed events are resting-state block markers lasting several minutes each and are not trial-level stimuli.

Data Quality Notes

- Eyes Open/Closed markers are resting block delimiters, not instantaneous trial events; do not epoch them as stimuli.
- Some files contain "Eyes Open: Every 1000 ms" style sub-event annotations; these are timing markers within rest blocks.
- Clinical groups (acute mTBI, chronic mTBI, controls) should be used as stratification variables.

References

- Reference: `ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. Neuropsychologia, 2019. DOI: 10.1016/j.neuropsychologia.2019.107125`

045. openneuro_ds003523 - EEG: Visual Working Memory in Acute TBI

Dataset: openneuro_ds003523 | Year: 2021 | URL: https://openneuro.org/datasets/ds003523
Category: High-level Mental | Subjects: 62 | Files (Completed): 219
Channels: 65 | Sampling Rate: 500 Hz

Paradigm

A visual working memory task with intermittent mind-wandering probes was administered longitudinally to healthy controls and patients with sub-acute mild traumatic brain injury (mTBI). Participants memorized visual stimuli during encoding, maintained them across a delay period, and responded to probe stimuli. Occasional mind-wandering probes assessed attentional state. The dataset spans multiple time points post-injury to track recovery of executive functioning, with ERP markers (P3b) serving as neurophysiological outcome measures.

Event Structure

Event ID	Count (total)	Type	Meaning
Stim Onset	657	short	Visual study stimulus onset
Question Mark Probe	434	short	Working memory probe stimulus
Response	434	short	Participant button-press response
Mind Wandering	406	short	Mind-wandering probe response marker
Mind Wandering Probe	217	short	Mind-wandering probe stimulus
STATUS	219	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus type classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stim Onset	0	Memory stimulus presentation (encoding phase)
Question Mark Probe	1	Probe stimulus presentation (retrieval stage)

- Note: Longitudinal dataset; multiple sessions per subject across recovery timepoints. Mind-wandering probes can support on-task vs off-task classification with careful labeling.

Data Quality Notes

- Longitudinal design: multiple sessions per subject at different post-injury timepoints.
- Clinical group and time-point information in participants.tsv; use as stratification variables.
- Mind Wandering Probe and Mind Wandering events can be used to classify attentional state.

References

- Reference: ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. Neuropsychologia, 2019. DOI: 10.1016/j.neuropsychologia.2019.107125

046. openneuro_ds003555 - Dataset of EEG Recordings of Pediatric Patients with Epilepsy Based on the 10-20 System

Dataset: `openneuro_ds003555` | Year: 2021 | URL: https://openneuro.org/datasets/ds003555
Category: Biological State | Subjects: 30 | Files (Completed): 257
Channels: 52 | Sampling Rate: 2000 Hz

Paradigm

Continuous sleep EEG recordings from pediatric patients with epilepsy, focused on identifying High-Frequency Oscillations (HFOs) as clinical biomarkers during N3 slow-wave sleep stages. The dataset was recorded using the standard 10-20 electrode system and is intended for automated HFO detection algorithm development and sleep stage analysis. No task-based paradigm was employed; recordings capture spontaneous brain activity during sleep.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; continuous sleep EEG recording

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag Source: No event marker, use segmentation method
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous sleep EEG segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no label, for unsupervised or HFO detection)
- Note: High sampling rate (2000 Hz) is appropriate for HFO detection (80-500 Hz). Sleep stage labels (if available from hypnogram) can serve as y labels for sleep classification tasks. No trial-level events present.

Data Quality Notes

- Very high sampling rate (2000 Hz); high memory requirements for processing.
- Pediatric epilepsy population; inter-individual variability in seizure history and medications.
- HFO detection requires high-frequency filtering (80-500 Hz); standard 0.5-40 Hz bandpass not appropriate for primary analysis goal.

References

- Reference: `Dataset of EEG recordings of pediatric patients with epilepsy based on the 10-20 system`

047. openneuro_ds003570 - EEG: Improvisation and Musical Structures (Auditory Oddball Chords)

Dataset: `openneuro_ds003570` | Year: 2021 | URL: https://openneuro.org/datasets/ds003570
Category: Sensory/Response | Subjects: 34 | Files (Completed): 34
Channels: 64 | Sampling Rate: 2048 Hz

Paradigm

Participants listened to sequences of three-chord musical progressions in an oddball paradigm designed to detect functional and exemplar deviants occurring within a stream of standard chord progressions. The study investigates neural responses (mismatch negativity, P600) to harmonic violations and musical structure deviants, probing the auditory system's predictive coding of musical sequences. Only STATUS system markers were detected in the EEG data; the chord-level event annotations may be stored externally.

Event Structure

Event ID	Count (total)	Type	Meaning
STATUS	34	admin	System status marker (only detected event)

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag Source: No trial-level event marker available, use segmented approach
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no labels, for unsupervised analysis)
- Note: Only STATUS admin markers detected; no chord-level event annotations in EEG triggers. High sampling rate (2048 Hz). If events.tsv contains chord onset information, epoching by event may be feasible with custom preprocessing.

Data Quality Notes

- Very high sampling rate (2048 Hz); high memory requirements.
- No usable task events detected in EEG triggers beyond STATUS admin marker.
- Chord-level event timing may require reconstruction from audio or behavioral logs.

References

- Reference: `EEG dataset on improvisation and musical structures. OpenNeuro ds003570.`

048. openneuro_ds003574 - Reward Biases Spontaneous Neural Reactivation During Sleep

Dataset: openneuro_ds003574 | Year: 2021 | URL: https://openneuro.org/datasets/ds003574
Category: Complex Applied | Subjects: 18 | Files (Completed): 17
Channels: 69 | Sampling Rate: 500 Hz

Paradigm

Participants played two rewarded and non-rewarded video games (FACE and MAZE) followed by a nap sleep session recorded with simultaneous EEG-fMRI. The study examines whether reward enhances spontaneous neural reactivation of game-related memory traces during sleep. Brain patterns from waking gameplay are decoded and compared to sleep activity to identify replay events. EEG event structure consists of BrainVision numeric S-code triggers from the game paradigm with limited per-file occurrence.

Event Structure

Event ID	Count (total)	Type	Meaning
S 1-S 27	1 each	short	BrainVision stimulus triggers (game/task events)
R128	1	short	BrainVision response trigger
Sync On	1	short	MRI/EEG synchronization marker
R	1	admin	Generic response marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: There are very few event markers (each type only appears 1 time/file), not suitable for epoch analysis
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous sleep EEG segmentation

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no label, suitable for unsupervised analysis)
- Note: Only 1 occurrence per event type per file - insufficient for epoch-based classification. Dataset is primarily suited for EEG-fMRI sleep replay analysis using pattern decoding, not standard ERP epoching. The 17 completed files (of 18 subjects) are sleep EEG recordings.

Data Quality Notes

- Very sparse event structure; each trigger type appears only once per recording.
- EEG-fMRI simultaneous recording; gradient artifact and MRI-related noise present.
- Primary analysis requires decoding waking neural patterns and cross-correlating with sleep activity.

References

- Reference: Reward biases spontaneous neural reactivation during sleep. Nature Communications, 2021.

049. openneuro_ds003602 - Childhood Sexual Abuse and Problem Drinking in Women: Neurobehavioral Mechanisms

Dataset: openneuro_ds003602 | Year: 2022 | URL: https://openneuro.org/datasets/ds003602
Category: High-level Mental | Subjects: 234 | Files (Completed): 699
Channels: 35 | Sampling Rate: 1000 Hz

Paradigm

This study investigates neurobehavioral mechanisms linking childhood sexual abuse (CSA) history and problem drinking in women using affective picture viewing and startle paradigms. Participants viewed negative, neutral, and positive images including alcohol-related and non-alcohol-related content while EEG, EMG (startle), and physiological measures were recorded. Resting-state blocks (eyes open/closed) were also included. The study examines how trauma history modulates emotional reactivity and appetitive motivation for alcohol cues.

Event Structure

Event ID	Count (total)	Type	Meaning
NegStim	230	short	Negative picture stimulus onset
NeuStim	230	short	Neutral picture stimulus onset
PosStim	230	short	Positive picture stimulus onset
NegAlcStimWithoutStartle	230	short	Negative alcohol picture (no startle probe)
NeuAlcStimWithoutStartle	230	short	Neutral alcohol picture (no startle probe)
PosAlcStimWithoutStartle	230	short	Positive alcohol picture (no startle probe)
NegNonAlcStimWithoutStartle	230	short	Negative non-alcohol picture (no startle probe)
NeuNonAlcStimWithoutStartle	230	short	Neutral non-alcohol picture (no startle probe)
PosNonAlcStimWithoutStartle	230	short	Positive non-alcohol picture (no startle probe)
NegAlcP1000/P2000/P3000	230 each	short	Negative alcohol picture with startle probe at 1/2/3 s
NeuAlcP1000/P2000/P3000	230 each	short	Neutral alcohol picture with startle probe at 1/2/3 s
PosAlcP1000/P2000/P3000	230 each	short	Positive alcohol picture with startle probe at 1/2/3 s
NegNonAlcP1000/P2000/P3000	230 each	short	Negative non-alcohol picture with startle probe
NeuNonAlcP1000/P2000/P3000	230 each	short	Neutral non-alcohol picture with startle probe
PosNonAlcP1000/P2000/P3000	230 each	short	Positive non-alcohol picture with startle probe

Event ID	Count (total)	Type	Meaning
`EyesOpen1` / `EyesOpen2`	234 each	rest	Eyes-open resting-state block marker
`EyesClose1` / `EyesClose2`	234 each	rest	Eyes-closed resting-state block marker
`EndOfRest`	234	admin	End of resting-state block
`ignore`	458	admin	Practice/ignored trial marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Emotional valence classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`NegStim`	0	Negative pictures
`NeuStim`	1	Neutral image
`PosStim`	2	Positive pictures

- Note: Balanced across valence conditions (~230 per category). Large dataset (699 files). Alcohol-related vs non-alcohol-related stimulus distinction available for secondary analysis.

Data Quality Notes

- Large dataset with 30+ distinct event types; group by valence (Neg/Neu/Pos) and content (Alc/NonAlc) for tractable analysis.
- Eyes Open/Closed markers are resting-state block delimiters, not trial-level stimuli.
- `ignore` and `EndOfRest` are administrative markers; exclude from epoching.

References

- Reference: `Childhood Sexual Abuse and problem drinking in women: Neurobehavioral mechanisms. OpenNeuro ds003602.`

050. openneuro_ds003620 - Runabout: A Mobile EEG Study of Auditory Oddball Processing in Laboratory and Real-World Conditions

Dataset: openneuro_ds003620 | Year: 2021 | URL: https://openneuro.org/datasets/ds003620
Category: Complex Applied | Subjects: 44 | Files (Completed): 271
Channels: 35 | Sampling Rate: 250 Hz

Paradigm

Participants performed an auditory oddball task under three levels of environmental complexity and physical activity: sitting in a laboratory, walking on a sports field (oval), and navigating a university campus. The paradigm examined how cognitive and motor demands of real-world locomotion affect P300 oddball responses. Target tones required mental counting ("count" condition) or were simply listened to ("don't count" condition), with both counted and uncounted targets interleaved across environments.

Event Structure

Event ID	Count (total)	Type	Meaning
t_count_lab	134	short	Target tone, count condition, laboratory
t_count_oval	134	short	Target tone, count condition, oval
t_count_campus	134	short	Target tone, count condition, campus
ntcount_lab	134	short	Non-target (standard) tone, count condition, laboratory
ntcount_oval	134	short	Non-target tone, count condition, oval
ntcount_campus	134	short	Non-target tone, count condition, campus
t_Count_lab	80	short	Target tone, Count (capital) condition, laboratory
t_Count_oval	80	short	Target tone, Count condition, oval
t_Count_campus	80	short	Target tone, Count condition, campus
ntCount_lab	80	short	Non-target tone, Count condition, laboratory
ntCount_oval	80	short	Non-target tone, Count condition, oval
ntCount_campus	80	short	Non-target tone, Count condition, campus
t_Dontcount_lab	208	short	Target tone, don't-count condition, laboratory
t_Dontcount_oval	214	short	Target tone, don't-count condition, oval
t_Dontcount_campus	214	short	Target tone, don't-count condition, campus

Event ID	Count (total)	Type	Meaning
ntDontcount_lab	208	short	Non-target tone, don't-count condition, laboratory
ntDontcount_oval	214	short	Non-target tone, don't-count condition, oval
ntDontcount_campus	214	short	Non-target tone, don't-count condition, campus
empty	226	admin	Empty/padding event marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Target/non-target classification (binary, cross-environment)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
t_count_lab, t_count_oval, t_count_campus, t_Count_lab, t_Count_oval, t_Count_campus, t_Dontcount_lab, t_Dontcount_oval, t_Dontcount_campus	1	Target tone (all target variants)
ntcount_lab, ntcount_oval, ntcount_campus, ntCount_lab, ntCount_oval, ntCount_campus, ntDontcount_lab, ntDontcount_oval, ntDontcount_campus	0	Non-target tone (all standard variants)

- Note: Event names encode environment (lab/oval/campus) and condition (count/Count/Dontcount). The Stimulus/ prefixed variants in some files are duplicates. Pool by target (t_) vs non-target (nt_) for a tractable binary task. Mobile EEG; motion artifacts expected during walking conditions.

Data Quality Notes

- Mobile EEG recording; significant motion artifacts during outdoor walking conditions (oval, campus).
- Lowercase vs uppercase variants (count/Count) may reflect different recording sessions or systems.
- empty event is an admin placeholder; exclude from epoching.
- Stimulus/-prefixed variants appear in a subset of files and are redundant with non-prefixed versions.

References

- Reference: Runabout: A mobile EEG study of auditory oddball processing in laboratory and real-world conditions. OpenNeuro ds003620.

051. openneuro_ds003626 - Inner Speech

Dataset: `openneuro_ds003626` | Year: 2022 | URL: https://openneuro.org/datasets/ds003626
Category: High-level Mental | Subjects: 10 | Files (Completed): 30
Channels: 137 | Sampling Rate: 1024 Hz

Paradigm

Participants performed inner speech (imagined speech), pronounced speech, and visualization tasks involving four directional commands (Up, Down, Right, Left) to explore the neural correlates of the 'inner voice' phenomenon for potential BCI applications. High-density EEG (137 channels) was recorded during all three modalities of speech production/imagery. No discrete event markers were captured in the EEG trigger channel; paradigm timing is encoded externally.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; continuous EEG recording

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag Source: No event marker, use segmentation method
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no tag, needs to be restored from events.tsv or external annotation)
- Note: No EEG trigger markers detected. If events.tsv contains trial onset annotations, epoching by event may be possible. High-density 137-channel EEG suitable for source imaging of inner speech networks. Only 10 subjects.

Data Quality Notes

- No EEG trigger markers present; trial timing must be reconstructed from events.tsv or behavioral logs.
- Small sample size (10 subjects, 30 files across 3 conditions).
- High-density 137-channel EEG; requires careful re-referencing and ICA for artifact removal.

References

- Reference: `Inner Speech Dataset. OpenNeuro ds003626.`

052. openneuro_ds003638 - EEG: Electrophysiological Biomarkers of Behavioral Dimensions from Cross-Species Paradigms

Dataset: `openneuro_ds003638` | Year: 2021 | URL: https://openneuro.org/datasets/ds003638
Category: High-level Mental | Subjects: 57 | Files (Completed): 57
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

This study employs three cross-species behavioral paradigms (5-Choice Continuous Performance Task [5CCPT], Probabilistic Reward/Punishment task [PRBP], and Pavlovian-to-Instrumental Transfer [PST]) to identify electrophysiological biomarkers associated with various behavioral dimensions including attention, impulsivity, and reward sensitivity. All 57 subjects completed all three paradigms in a single session. Event markers are encoded as numeric codes with paradigm suffix (e.g., "N 5CCPT", "N PRBP", "N PST").

Event Structure

Event ID	Count (total)	Type	Meaning
`1 5CCPT-80 5CCPT`	57 each	short	5-Choice CPT numeric trigger codes
`116 PRBP, 117 PRBP`	57 each	short	Probabilistic reward/punishment task trigger codes
`31 PST, 32 PST, 39 PST`	56-57 each	short	Pavlovian-to-Instrumental Transfer task trigger codes
`STATUS`	57	admin	System status marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: 5CCPT stimulus classification (need to decode event meaning)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`41 5CCPT`	0	5CCPT correct response (need to confirm from events.tsv)
`43 5CCPT`	1	5CCPT commission error (need to confirm from events.tsv)

- Note: Numeric trigger codes require decoding from events.tsv or dataset README. Events with suffix 5CCPT correspond to the 5-Choice CPT; PRBP to reward/punishment; PST to Pavlovian transfer. Exact meaning of each numeric code must be verified from dataset documentation.

Data Quality Notes

- 30+ distinct numeric trigger types; decode from events.tsv before epoching.
- All three paradigms recorded in a single 57-file session; paradigm blocks need to be separated by trigger suffix.
- STATUS is a single admin marker per file; exclude from epoching.

References

- Reference: Electrophysiological biomarkers of behavioral dimensions from cross-species paradigms. OpenNeuro ds003638.

053. openneuro_ds003645 - Face Processing MEEG Dataset with HED Annotation

Dataset: `openneuro_ds003645` | Year: 2023 | URL: https://openneuro.org/datasets/ds003645
Category: Sensory/Response | Subjects: 18 | Files (Completed): 114
Channels: 395 | Sampling Rate: 1100 Hz

Paradigm

Participants performed a simple perceptual task judging the spatial symmetry of famous, unfamiliar, and scrambled faces during simultaneous MEG and EEG (MEEG) recording. This is a replication of the Wakeman and Henson (2015) dataset with Hierarchical Event Descriptor (HED) annotations added. The study serves as a reference dataset for multimodal neuroimaging analysis methods, with face symmetry judgment as the overt task while examining neural responses to face identity and familiarity.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No EEG trigger markers detected; events encoded in HED annotations or MEEG-specific channels

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: HED annotation (needs to be parsed from events.tsv HED column); EEG trigger channel no event available
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (HED annotation requires additional parsing)
- Remarks: 395-channel MEEG dataset (MEG + EEG combined); only EEG channels are relevant here. Very high channel count. Face stimuli (famous/unfamiliar/scrambled) are annotated via HED; standard BIDS event reading should recover trial_type if events.tsv is properly formatted.

Data Quality Notes

- MEEG dataset; 395 channels include both MEG and EEG sensors. Isolate EEG channels before analysis.
- No EEG trigger events detected in the EEG .set files; events may be in MEEG-format files or HED sidecar.
- Very high sampling rate (1100 Hz) and channel count; high computational requirements.

References

- Reference: `Wakeman and Henson (2015) / Robbins et al. (2021). Face processing MEEG dataset with HED annotation.`

054. openneuro_ds003655 - VerbalWorkingMemory

Dataset: openneuro_ds003655 | Year: 2023 | URL: https://openneuro.org/datasets/ds003655
Category: High-level Mental | Subjects: 26 | Files (Completed): 156
Channels: 21 | Sampling Rate: 500 Hz

Paradigm

A modified Sternberg working memory task requiring either simple retention or mental manipulation (alphabetization) of letter sequences at three load levels (5, 6, or 7 letters). Participants encoded a letter sequence, maintained it across a delay period, then responded to a probe letter. The mental manipulation condition (alphabetization) places additional executive demands on working memory compared to simple retention. EEG measures during encoding, delay, and probe phases capture load-dependent neural activity.

Event Structure

Event ID	Count (total)	Type	Meaning
encoding	936	short	Letter sequence encoding phase onset
delay period	936	long	Working memory retention/manipulation delay period
probe	936	short	Probe letter presentation
start of the baseline period	936	short	Baseline period onset (pre-trial)
start of the trial	156	short	Trial start marker
presentation of the task instruction	156	short	Task instruction display onset
Response	1807	short	Participant button-press response
STATUS	156	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Delay period analysis - keep vs operation (binary, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
delay period: 5R / 6R / 7R	0	Simple retention (Retention) condition

Event	y	Meaning
`delay period: 5M / 6M / 7M`	1	Mental arithmetic operation (Manipulation) conditions

- Note: Delay periods typically last several seconds (>=2 s). Event subtypes encode both load level (5/6/7) and condition (R=Retention, M=Manipulation). Load level can be used as a regression target or ordinal label.

Task 2: Detect stimulus ERP (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`probe: 5R / 6R / 7R`	0	Keep condition detection
`probe: 5M / 6M / 7M`	1	Operating condition detection

- Note: Probe events are short (<1 s); suitable for ERP epoching. Match/non-match accuracy from Response events can be used as trial-level correctness labels.

Data Quality Notes

- 21-channel EEG; limited spatial resolution but sufficient for frontal working memory analyses.
- Event subtypes encoded in trial_type column (e.g., "delay period: 5M"); parse carefully.
- Multiple sessions per subject (156 files / ~26 subjects ≈ 6 files each).

References

- Reference: `VerbalWorkingMemory EEG dataset. OpenNeuro ds003655.`

055. openneuro_ds003670 - Dataset of Concurrent EEG, ECG, and Behavior with Multiple Doses of Transcranial Electrical Stimulation

Dataset: `openneuro_ds003670` | Year: 2021 | URL: https://openneuro.org/datasets/ds003670
Category: Complex Applied | Subjects: 19 | Files (Completed): 62
Channels: 35 | Sampling Rate: 2000 Hz

Paradigm

Participants performed a continuous tracking task (CTT) requiring them to maintain a ball at the center of a screen to assess vigilance and sustained alertness. Simultaneously, multi-site transcranial electrical stimulation (tES) was applied at varying dose combinations (sham, low, high) to modulate arousal and attention. EEG, ECG, and behavioral tracking data were recorded concurrently. The study aims to identify optimal tES parameters for enhancing cognitive performance.

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulation Start`	62	long	Transcranial stimulation onset (sustained state)
`Stimulation Stop`	62	long	Transcranial stimulation offset
`Block Start`	62	long	Task block onset marker
`STATUS`	62	admin	System status marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Stimulation period analysis (long event)

- Epoch method: `epoch_by_long_event_hdf5`
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`Stimulation Start`	1	Transcranial electrical stimulation activation period
`Block Start`	0	Task block (baseline/control)

- Note: Stimulation periods are sustained states (several minutes each); use long-event epoching. Dose condition (sham/low/high) should be in events.tsv or participants.tsv. Very high sampling rate (2000 Hz) increases memory requirements.

Data Quality Notes

- Very high sampling rate (2000 Hz); downsample to 250-500 Hz for standard EEG analyses.
- tES artifacts present during stimulation periods; independent component analysis required.
- ECG channels included; can be used for cardiac artifact removal.

References

- Reference: `Dataset of concurrent EEG, ECG, and behavior with multiple doses of transcranial Electrical Stimulation. OpenNeuro ds003670.`

056. openneuro_ds003690 - EEG, ECG and Pupil Data from Young and Older Adults: Rest and Auditory Cued Reaction Time Tasks

Dataset: `openneuro_ds003690` | Year: 2021 | URL: https://openneuro.org/datasets/ds003690
Category: Sensory/Response | Subjects: 75 | Files (Completed): 375
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Young and older adults performed auditory-cued simple reaction time and go/no-go tasks to investigate age-related differences in neural and physiological activity. An auditory cue preceded each trial; participants pressed a button to go stimuli and withheld responses to no-go stimuli. Error feedback tones were provided after commission errors. Resting-state EEG was also recorded. The study examines how aging affects preparatory neural activity, inhibitory control, and pupil dilation responses.

Event Structure

Event ID	Count (total)	Type	Meaning
`cue`	375	short	Auditory warning cue preceding stimulus
`go`	300	short	Go stimulus (respond with button press)
`no-go`	150	short	No-go stimulus (withhold response)
`button press`	300	short	Participant button-press response
`error of commission`	64	short	Commission error (response to no-go)
`error feedback tone`	179	short	Auditory error feedback tone
`task start`	320	short	Task block start marker
`task end`	375	short	Task block end marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Go/No-go classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
go	1	Go stimulation (reaction required)
no-go	0	No-go stimulation (requires suppression)

- Note: Go:No-go ratio is 2:1 (~300 go vs ~150 no-go); mild class imbalance. 375 files from 75 subjects x 5 sessions each. Age group (young/older) in participants.tsv can be used as a grouping variable.

Task 2: Correct/Error Suppression Classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
no-go (correct inhibition)	0	Correct inhibition no-go
`error of commission`	1	Error response (key pressed during no-go)

- Remarks: Commission errors (64 total) are rare compared to correct inhibitions; severe class imbalance requires oversampling or weighted loss.

Data Quality Notes

- 66-channel EEG; includes ECG and pupil data (non-EEG channels should be excluded).
- Longitudinal design with 5 sessions per subject; verify session consistency.
- Age group comparison (young vs older) is the primary experimental variable.

References

- Reference: `Age-related differences in EEG, ECG and pupilography during auditory cued reaction time tasks. OpenNeuro ds003690.`

057. openneuro_ds003702 - Social Memory cuing

Dataset: openneuro_ds003702 | Year: 2024 | URL: https://openneuro.org/datasets/ds003702
Category: Complex Applied | Subjects: N/A | Files (Completed): 47
Channels: 64.0 | Sampling Rate: 500.0 Hz

Paradigm

A visual working memory task performed in a virtual reality environment where participants recalled object locations and statuses following social (avatar) or non-social (stick) directional cues.

Event Structure

Event ID	Count (files)	Meaning
unknown	47	Event marker in working memory paradigm - see paradigm description for decoding
## Downstream Task Recommendations		

Classification Tasks

- Insufficient distinct event types for classification (only 1 event class found).

Regression Tasks

- No clear regression target identified from available event structure. Consider reaction time if behavioral logs are available.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

058. openneuro_ds003710 - APPLESEED Example Dataset

Dataset: `openneuro_ds003710` | Year: 2022 | URL: https://openneuro.org/datasets/ds003710
Category: Biological State | Subjects: 13 | Files (Completed): 48
Channels: 32 | Sampling Rate: 5000 Hz

Paradigm

Longitudinal spontaneous EEG recordings from infants at 4, 8, and 12 months of age during passive viewing of social and non-social stimuli (faces, speech, objects, water). The dataset validates the APPLESEED (Automated Preprocessing Pipe-Line for the Estimation of Scale-wise Entropy from EEG Data) pipeline for pediatric EEG preprocessing and entropy estimation. Stimuli include social (face, speech) and non-social (object, water) categories presented as short video clips, with attention-getter events to orient the infant.

Event Structure

Event ID	Count (total)	Type	Meaning
`face`	48	short	Face stimulus onset (social)
`speech`	48	short	Speech stimulus onset (social)
`object`	48	short	Object stimulus onset (non-social)
`water`	48	short	Water stimulus onset (non-social)
`attention-getter`	48	short	Attention-orienting stimulus before trial
`experimenter-pause`	45	admin	Experimenter pause marker
`New Segment`	48	admin	Recording segment boundary marker
`Comment`	48	admin	Experimenter annotation

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Social vs non-social stimulus classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`face`	1	Face stimulation (social)
`speech`	1	Verbal stimulation (social)

Event	y	Meaning
object	0	Object stimulus (non-social)
water	0	Water stimulation (non-social)

- Note: Very high sampling rate (5000 Hz); downsample to 250-500 Hz before epoching. Infant EEG; high artifact rates expected. Only 13 subjects (48 files across 3 age points). New Segment, Comment, and experimenter-pause are admin markers.

Data Quality Notes

- Extremely high sampling rate (5000 Hz); downsampling required before standard EEG analysis.
- Infant population; high motion artifacts and data loss expected.
- Small sample (13 subjects); longitudinal design with 3-4 files per subject across age points.
- New Segment and Comment markers should be excluded from epoching.

References

- Reference: The Automated Preprocessing Pipe-Line for the Estimation of Scale-wise Entropy from EEG Data (APPLESEED): Development and validation for use in pediatric populations. OpenNeuro ds003710.

059. openneuro_ds003739 - Perturbed beam-walking task

Dataset: `openneuro_ds003739` | Year: 2021 | URL: https://openneuro.org/datasets/ds003739
Category: Complex Applied | Subjects: N/A | Files (Completed): 120
Channels: 149.0 | Sampling Rate: 256.0 Hz

Paradigm

Participants performed walking on a treadmill-mounted balance beam while being subjected to sensorimotor perturbations via virtual reality visual rotations and physical waist pulls.

Event Structure

Event ID	Count (files)	Meaning
`stimulus`	120	Stimulus onset marker
## Downstream Task Recommendations		

Classification Tasks

- Insufficient distinct event types for classification (only 1 event class found).

Regression Tasks

- No clear regression target identified from available event structure. Consider reaction time if behavioral logs are available.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `eNeuro publication`

060. openneuro_ds003751 - Dataset on Emotion with Naturalistic Stimuli (DENS)

Dataset: `openneuro_ds003751` | Year: 2023 | URL: https://openneuro.org/datasets/ds003751
Category: High-level Mental | Subjects: 40 | Files (Completed): 40
Channels: 132 | Sampling Rate: 250 Hz

Paradigm

Participants viewed naturalistic multimedia video clips (film excerpts, documentaries) to elicit a range of emotional states while EEG, ECG, and EMG data were simultaneously recorded. After each video clip, participants provided continuous subjective ratings of valence, arousal, dominance, liking, familiarity, and emotional category. The dataset supports multimodal emotion recognition research using naturalistic (ecologically valid) stimuli rather than standardized posed expressions.

Event Structure

Event ID	Count (total)	Type	Meaning
`stm`	40	long	Stimulus (video clip) onset marker
`IBEG`	40	long	Image/video block beginning
`base`	40	long	Baseline period onset
`bend`	40	long	Baseline period end
`IEND`	39	long	Image/video block end
`vlnc`	40	short	Valence rating event
`arsl`	40	short	Arousal rating event
`dmns`	40	short	Dominance rating event
`lkng`	40	short	Liking rating event
`relv`	40	short	Relevance rating event
`fmrt`	40	short	Familiarity rating event
`cate`	40	short	Emotion category rating event
`clic`	40	short	Mouse click rating response
`trno`	40	short	Trial number marker
`fixL`	40	short	Fixation cross onset
`quiz`	15	short	Comprehension quiz question
`qdon`	14	short	Quiz done marker
`stop`	37	admin	Stop/pause marker
`puse`	34	admin	Pause marker
`rsrt`	32	admin	Restart marker
`net`	2	admin	Network marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Emotion analysis during video stimulation period (long event)

- Epoch method: `epoch_by_long_event_hdf5`
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`stm`	-	Video stimulation period (continuous emotion labels from vlnc/arsl scoring)

- Note: Video clips are naturalistic stimuli lasting many seconds; use long-event epoching. Continuous emotion ratings (valence, arousal) are the primary labels - treat as regression targets from behavioral data rather than discrete y values. Only 40 subjects and files.

Data Quality Notes

- 132-channel high-density EEG; requires careful preprocessing and ICA.
- Naturalistic stimuli with varying durations; trial lengths are not uniform.
- `stop`, `puse`, `rsrt`, `net` are administrative markers; exclude from epoching.
- Continuous emotion ratings stored separately; must be merged with EEG epochs for regression tasks.

References

- Reference: `Sudhakar Mishra and Md. Asif and Uma Shanker Tiwary and Narayanan Srinivasan (2023). Dataset on Emotion with Naturalistic Stimuli (DENS). OpenNeuro ds003751.`

061. openneuro_ds003753 - EEG: Probabilistic Learning with Affective Feedback: Exp #2

Dataset: openneuro_ds003753 | Year: 2021 | URL: https://openneuro.org/datasets/ds003753
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task involving probabilistic affective feedback to investigate the Reward Positivity (RewP) EEG component and its sensitivity to affective stimulus properties. Participants made binary choices (left/right button) between cue stimuli (CUE) associated with positive or negative affective feedback images. Following each choice, an affective feedback image (FB) was presented. The study examines whether the RewP tracks reward value beyond mere monetary reward, responding to affective liking of feedback stimuli.

Event Structure

Event ID	Count (total)	Type	Meaning
CUE	100	short	Cue stimulus requiring binary choice
FB	100	short	Affective feedback stimulus (liked/disliked image)
Left Button	25	short	Left button press response
Right Button	25	short	Right button press response
STATUS	25	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Feedback type classification (requires decoding FB subtypes from events.tsv)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
FB (liked/rewarding)	1	Liked emotional feedback (needs to be decoded from events.tsv)
FB (disliked/non-rewarding)	0	Disliked emotional feedback (needs to be decoded from events.tsv)

- Note: FB event subtypes (liked vs disliked) must be decoded from events.tsv `trial_type` column. 25 subjects only; small dataset. Sister dataset to ds003822 (Exp #1).

Data Quality Notes

- Small dataset (25 subjects, 25 files).
- FB event subtype labels (affective value) must be decoded from events.tsv metadata.
- STATUS is a single admin marker per file; exclude from epoching.

References

- Reference: `The Reward Positivity is sensitive to affective liking. OpenNeuro ds003753.`

062. openneuro_ds003766 - A Resource for Assessing Dynamic Binary Choices in the Adult Brain Using EEG and Mouse-Tracking

Dataset: `openneuro_ds003766` | Year: 2024 | URL: https://openneuro.org/datasets/ds003766
Category: High-level Mental | Subjects: 31 | Files (Completed): 217
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Participants performed dynamic decision-making tasks including food preference choices and semantic judgments while high-density EEG and continuous mouse-tracking were recorded simultaneously. The study captures the real-time neural dynamics of binary choice formation, including evidence accumulation, conflict, and response selection. Numeric trigger codes (0200-0506 series) encode stimulus, response, and condition information. Boundary markers appear in some files from EEGLAB preprocessing.

Event Structure

Event ID	Count (total)	Type	Meaning
`0400`	186	short	Trigger code 0400 (stimulus/task event - decode from events.tsv)
`0500`	186	short	Trigger code 0500 (stimulus/task event - decode from events.tsv)
`0401-0403`	124 each	short	Stimulus variant trigger codes
`0503`, `0505`, `0506`	124 each	short	Response/outcome trigger codes
`0501`	95	short	Trigger code 0501
`0202-0221`	93 each	short	Trial-phase trigger codes (0200 series)
`0210`	93	short	Trigger code 0210
`boundary`	97	admin	EEGLAB boundary marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Binary choice classification (need to decode event meaning)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
0400	0	Stimulus/Option A (decode from events.tsv)
0500	1	Stimulus/Option B (decode from events.tsv)

- Note: Numeric trigger codes must be decoded from events.tsv. The 0200-series codes likely encode trial-phase timing; 0400/0500-series likely encode stimulus presentation. 217 files across 31 subjects (≈7 sessions each). `boundary` is an EEGLAB admin marker; exclude from epoching.

Data Quality Notes

- 30+ distinct numeric event codes; decode mapping required from events.tsv or dataset paper.
- `boundary` events are EEGLAB data-continuity markers; must be excluded from epoching.
- High-density 129-channel EEG at 1000 Hz; high computational requirements.
- Multiple sessions per subject; verify session consistency before pooling.

References

- Reference: Chen, K., Wang, R., Huang, J., Gao, F., Yuan, Z., Qi, Y., & Wu, H. (2022). A resource for assessing dynamic binary choices in the adult brain using EEG and mouse-tracking. Scientific Data, 9(1), 416.

063. openneuro_ds003768 - Simultaneous EEG and fMRI Signals During Sleep from Humans

Dataset: openneuro_ds003768 | Year: 2026 | URL: https://openneuro.org/datasets/ds003768
Category: Biological State | Subjects: 33 | Files (Completed): 252
Channels: 32 | Sampling Rate: 5000 Hz

Paradigm

Simultaneous EEG-fMRI recordings were acquired during resting-state wakefulness and multiple sleep sessions (NREM, REM) to analyze spontaneous brain activity and sleep stage dynamics. Participants underwent extended sleep sessions in an MRI scanner while 32-channel EEG was recorded concurrently with fMRI. BrainVision numeric stimulus codes (S1, S2, S3) and response codes (R128) mark sleep stage transitions or scoring events. The dataset supports multimodal sleep neuroscience research.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 1	183	long	BrainVision S1 trigger (sleep stage marker)
Stimulus/S 2	193	long	BrainVision S2 trigger (sleep stage marker)
Stimulus/S 3	183	long	BrainVision S3 trigger (sleep stage marker)
Response/R128	252	admin	BrainVision R128 response trigger

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Sleep stage classification (multiclass, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Stimulus/S 1	0	Sleep stage 1 (need to confirm the specific sleep stage from events.tsv)
Stimulus/S 2	1	Sleep Stage 2
Stimulus/S 3	2	Sleep Stage 3

- Note: S1/S2/S3 likely encode sleep stage transitions (Wake, NREM1/2/3, REM); exact mapping must be confirmed from events.tsv or dataset documentation. Sleep stages are sustained states (>=2 s); use

long-event epoching. Very high sampling rate (5000 Hz); downsample to 250 Hz for standard sleep scoring.

Data Quality Notes

- Very high sampling rate (5000 Hz); downsample to 250-500 Hz before analysis.
- EEG-fMRI simultaneous recording; gradient artifacts present; requires MRI artifact removal.
- R128 is a BrainVision response/sync marker; treat as admin.
- 252 files from 33 subjects across multiple sleep sessions.

References

- Reference: `Albouy et al., Journal of Sleep Research, 2013. Simultaneous EEG and fMRI during sleep. OpenNeuro ds003768.`

064. openneuro_ds003774 - Music Listening Genre EEG Dataset (MUSIN-G)

Dataset: `openneuro_ds003774` | Year: 2022 | URL: https://openneuro.org/datasets/ds003774
Category: High-level Mental | Subjects: 20 | Files (Completed): 260
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Twenty participants listened to 12 music tracks spanning various genres (classical, jazz, rock, etc.) with eyes closed while high-density EEG was recorded. After each track, participants provided subjective ratings for familiarity and enjoyment. The dataset supports EEG-based music genre classification and exploration of neural correlates of music preference and familiarity. No discrete event markers were embedded in the EEG recording.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; continuous music listening EEG

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag Source: No event marker; music track/genre tags need to be parsed from events.tsv or filename
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (music genre tags need to be appended after parsing from events.tsv)
- Remarks: 260 files for 20 subjects (13 files each: 12 tracks + 1 baseline). Music genre labels are file-level metadata; apply as segment-level labels for genre classification. High-density 129-channel EEG at 1000 Hz.

Data Quality Notes

- No EEG trigger events present; music onset timing not embedded in EEG.
- 260 files likely correspond to individual music track recordings; genre label per file.
- High-density 129-channel EEG; high computational requirements.
- Eyes-closed listening; fewer motion artifacts but possible drowsiness.

References

- Reference: `Music Listening Genre EEG dataset (MUSIN-G). OpenNeuro ds003774.`

065. openneuro_ds003775 - SRM Resting-State EEG

Dataset: openneuro_ds003775 | Year: 2022 | URL: https://openneuro.org/datasets/ds003775
Category: Biological State | Subjects: 111 | Files (Completed): 153
Channels: 64 | Sampling Rate: 1024 Hz

Paradigm

Four minutes of uninterrupted eyes-closed resting-state EEG recorded from healthy adult participants as part of a larger experimental paradigm. This dataset was collected to provide a normative baseline for comparing resting EEG characteristics across studies. The resting-state recording provides a standardized, task-free EEG sample suitable for spectral power analysis, functional connectivity, and individual difference studies.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; pure resting-state EEG

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag Source: No event marker, use segmentation method
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Resting state EEG segmentation

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (no labels, used for unsupervised analysis or individual differences research)
- Remarks: Pure 4-minute eyes-closed resting-state; 111 subjects, 153 files (some subjects may have multiple sessions). Suitable for individual alpha frequency, spectral power, and connectivity analyses. Large normative dataset.

Data Quality Notes

- Purely resting-state; no task events or stimuli.
- 153 files from 111 subjects; some subjects have multiple files (repeat sessions).
- 64-channel EEG at 1024 Hz; standard configuration for resting-state analysis.

References

- Reference: Hatlestad-Hall, C., Rygvold, T. W., & Andersson, S. (2022). BIDS-structured resting-state electroencephalography (EEG) data extracted from an experimental paradigm. Data in Brief, 45, 108647.

066. openneuro_ds003800 - Auditory Gamma Entrainment

Dataset: openneuro_ds003800 | Year: 2021 | URL: https://openneuro.org/datasets/ds003800
Category: Sensory/Response | Subjects: 8 | Files (Completed): 24
Channels: 19 | Sampling Rate: 250 Hz

Paradigm

Elderly participants with dementia received 40 Hz amplitude-modulated (AM) auditory stimulation to induce gamma-band entrainment of brain oscillations. The paradigm alternated between active 40 Hz AM stimulation (Stimulus) periods and quiet rest periods. The study investigates whether auditory gamma entrainment can modulate neural oscillations in a dementia population, potentially as a therapeutic intervention. EEG was recorded using a standard 10-20 system.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus	13	long	40 Hz AM auditory stimulation period onset
Rest	13	long	Resting/quiet period

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulation period vs resting period (binary, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Stimulus	1	40 Hz AM auditory stimulation activation period
Rest	0	Rest/quiet period

- Note: Stimulation and rest periods are sustained states (likely 1-5 min each); use long-event epoching. Only 8 subjects and 24 files - very small dataset. 19-channel EEG (10-20 system).

Data Quality Notes

- Very small dataset (8 subjects, 24 files); results may not generalize.
- Elderly dementia population; data quality may be variable.
- 19-channel 10-20 EEG; limited spatial resolution.
- 250 Hz sampling rate; adequate for gamma entrainment analysis up to 125 Hz.

References

- Reference: Auditory Gamma Entrainment in Elderly with Dementia. OpenNeuro ds003800.

067. openneuro_ds003801 - Neural Tracking to Go

Dataset: openneuro_ds003801 | Year: 2021 | URL: https://openneuro.org/datasets/ds003801
Category: Complex Applied | Subjects: 20 | Files (Completed): 20
Channels: 24 | Sampling Rate: 250 Hz

Paradigm

A mobile EEG study where participants performed a dichotic listening task (two competing speakers) while either sitting in a quiet indoor environment or walking along a pre-defined indoor route with environmental distractions. The study investigates how locomotion and environmental complexity affect neural tracking of attended speech (measured by auditory EEG envelope tracking). No discrete trial-level event markers were embedded in the EEG recording.

Event Structure

Event ID	Count (total)	Type	Meaning
-	-	-	No event markers detected; continuous EEG during dichotic listening and walking

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: No event marker; conditional (sitting/walking) tags need to be obtained from events.tsv or file name
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous EEG segmentation

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: null (conditional label sit/walk needs to be obtained from file-level metadata)
- Remarks: Mobile EEG; motion artifacts expected during walking condition. 20 subjects, 20 files (1 per subject). Condition (sitting/walking) is file-level metadata. Neural tracking of speech envelope requires cross-correlation with audio stimulus waveform.

Data Quality Notes

- Mobile EEG recording; significant motion artifacts during walking condition.
- No trial-level event markers; condition label is file-level or block-level.
- Small dataset (20 subjects, 20 files); limited statistical power.
- 24-channel EEG at 250 Hz; modest spatial and temporal resolution.

References

- Reference: Neural Tracking to Go. OpenNeuro ds003801.

068. openneuro_ds003810 - Motor Imagery vs Rest - Low-Cost EEG System

Dataset: openneuro_ds003810 | Year: 2021 | URL: https://openneuro.org/datasets/ds003810
Category: Action/Output | Subjects: 10 | Files (Completed): 50
Channels: 15 | Sampling Rate: 125 Hz

Paradigm

Participants performed kinesthetic motor imagery of grasping movements with the dominant hand during EEG recording using a low-cost consumer-grade EEG system. Sessions included a preliminary run of actual hand movements followed by motor imagery blocks alternating with resting-state baselines. OpenViBE (OVTK) event triggers mark trial onset, cue type, feedback, and session boundaries. The study evaluates BCI performance using motor imagery and demonstrates applicability of affordable EEG hardware.

Event Structure

Event ID	Count (total)	Type	Meaning
OVTK_GDF_Right	50	long	Right-hand motor imagery cue onset
OVTK_GDF_Tongue	50	long	Tongue motor imagery cue onset (second class)
OVTK_GDF_Start_Of_Trial	50	short	Trial start marker
OVTK_GDF_Cross_On_Screen	50	short	Fixation cross display (inter-trial)
OVTK_GDF_Feedback_Continuous	50	short	Continuous BCI feedback onset
OVTK_GDF_End_Of_Trial	50	short	Trial end marker
OVTK_GDF_Correct	50	short	Correct classification feedback
OVTK_GDF_Incorrect	50	short	Incorrect classification feedback
OVTK_StimulationId_BaselineStart	50	rest	Resting baseline period start
OVTK_StimulationId_BaselineStop	50	rest	Resting baseline period end
OVTK_StimulationId_Beep	50	short	Auditory beep cue for trial start
OVTK_StimulationId_Train	48	short	Training phase marker
OVTK_StimulationId_ExperimentStart	46	admin	Experiment start marker
OVTK_StimulationId_ExperimentStop	48	admin	Experiment stop marker
OVTK_GDF_End_Of_Session	48	admin	Session end marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5

- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Motor imagery category classification (binary, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
OVTK_GDF_Right	1	Right hand motor imagination period
OVTK_GDF_Tongue	0	Tongue movement imagination period (control type)

- Note: Motor imagery periods are typically 4-6 s; use long-event epoching. 10 subjects with 5 files each (50 total). Low-cost 15-channel EEG at 125 Hz; limited spatial and temporal resolution compared to research-grade systems.

Data Quality Notes

- Low-cost EEG system; lower signal quality and spatial resolution than research-grade systems.
- Low sampling rate (125 Hz); limits temporal resolution and applicable frequency range.
- OVTK_StimulationId_ExperimentStart/Stop and OVTK_GDF_End_Of_Session are admin markers; exclude from epoching.

References

- Reference: Motor Imagery vs Rest using a low-cost EEG system. OpenNeuro ds003810.

069. openneuro_ds003822 - EEG: Probabilistic Learning with Affective Feedback: Exp #1

Dataset: openneuro_ds003822 | Year: 2021 | URL: https://openneuro.org/datasets/ds003822
Category: High-level Mental | Subjects: 25 | Files (Completed): 25
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task using probabilistic affective feedback to investigate the Reward Positivity (RewP) EEG component and its sensitivity to subjective reward value. Participants made binary choices (left/right button) between cue stimuli (CUE) associated with liked or disliked affective feedback images. Following each choice, a feedback cue (FB cue) predicted the affective image, then the full feedback (FB type) was presented. This is Experiment 1 in a two-experiment series (see ds003753 for Exp #2).

Event Structure

Event ID	Count (total)	Type	Meaning
CUE	100	short	Cue stimulus requiring binary choice
FB cue	100	short	Feedback cue (predicts upcoming affective image)
FB type	200	short	Full affective feedback stimulus (liked/disliked)
Response Accurate	25	short	Accurate/correct response marker
Response Inaccurate	25	short	Inaccurate/incorrect response marker
Left Button	25	short	Left button press
Right Button	25	short	Right button press
STATUS	25	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Feedback type classification (binary, subtype needs to be decoded from events.tsv)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`FB type` (liked/rewarding)	1	Liked emotional feedback (needs to decode subtype from events.tsv)
`FB type` (disliked/non-rewarding)	0	Disliked emotional feedback

- Note: `FB type` appears 200 times (2x more than CUE/FB cue = 100 each) suggesting FB type encodes both the feedback onset and a second event (e.g., image offset or rating). Decode subtype from events.tsv `trial_type` column. Sister dataset to ds003753 (Exp #2). 25 subjects only.

Data Quality Notes

- Small dataset (25 subjects, 25 files).
- `FB type` count (200) is 2x `CUE` count (100); likely encodes two event variants per trial; verify from events.tsv.
- `Response Accurate` and `Response Inaccurate` appear only 25 times each (once per file summary count); likely one per session.
- STATUS is a single admin marker per file; exclude from epoching.

References

- Reference: `The Reward Positivity is sensitive to affective liking. OpenNeuro ds003822.`

070. openneuro_ds003825 - Human electroencephalography recordings from 50 subjects for 22,248 images from 1,854 object concepts

Dataset: `openneuro_ds003825` | Year: 2022 | URL: https://openneuro.org/datasets/ds003825
Category: Sensory/Response | Subjects: 50 | Files (Completed): 100
Channels: 64 | Sampling Rate: 250 Hz

Paradigm

Participants viewed a large-scale set of 22,248 images spanning 1,854 object concepts presented in rapid serial visual presentation (RSVP) streams at 10 Hz while performing a fixation color change detection task. The study aimed to record high-resolution neural representations of everyday objects across a large stimulus set. Each image was presented for 100 ms in continuous streams, and participants responded to a concurrent fixation task to maintain attention.

Event Structure

Event ID	Count (total)	Type	Meaning
`E 1`	~1,237,200	short	Image onset (RSVP stimulus)
`E 2`	~1,237,200	short	Second trial-type marker (redundant or condition flag)
`E 3`	~4,200	short	Block/sequence boundary or target event

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: RSVP Image Viewing (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`E 1`	0	RSVP image onset (primary stimulus stream)
`E 2`	1	Secondary condition marker

- Note: Extremely large dataset (~1.2M epochs per event type across 50 subjects). E 1 and E 2 occur in equal counts per subject (~22,248 each); E 3 is rare (~84 per subject) and may indicate special events or fixation targets. Consider subsampling for practical use.

Data Quality Notes

- Data from derivatives/eeglab directory; preprocessed with EEGLAB
- Very high epoch count (~24,648 E 1 events per subject); memory management required during HDF5 generation
- Sub-06 has fewer events (~12,360), suggesting a partial recording

References

- Reference: `Grootswagers, T., Zhou, I., Robinson, A.K. et al. Human EEG recordings for 1,854 concepts presented in rapid serial visual presentation streams. Sci Data 9, 3 (2022).`

071. openneuro_ds003838 - EEG, pupillometry, ECG and photoplethysmography, and behavioral data in the digit span task

Dataset: `openneuro_ds003838` | Year: 2021 | URL: https://openneuro.org/datasets/ds003838/versions/1.0.0
Category: High-level Mental | Subjects: 86 | Files (Completed): 296
Channels: 2 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a classic digit span working memory task involving serial recall of digit sequences of 5, 9, or 13 digits, along with a separate eyes-closed resting state condition. Each digit in a sequence was marked with an event indicating the digit position, sequence length, and whether it was later correctly recalled or forgotten. The dataset also includes ECG/PPG physiological signals recorded simultaneously with EEG.

Event Structure

Event ID	Count (total)	Type	Meaning
`memory X/05 correct`	high	short	Digit X in 5-item sequence, correctly recalled
`memory X/09 correct`	high	short	Digit X in 9-item sequence, correctly recalled
`memory X/13 correct`	high	short	Digit X in 13-item sequence, correctly recalled
`memory X/05 error`	moderate	short	Digit X in 5-item sequence, forgotten
`memory X/09 error`	moderate	short	Digit X in 9-item sequence, forgotten
`memory X/13 error`	high	short	Digit X in 13-item sequence, forgotten
STATUS	148	admin	System/hardware status marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Correct vs Error Memory Encoding (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`memory .*/correct` (all correct events)	0	Digit correctly recalled later
`memory .*/error` (all error events)	1	Digit forgotten later

- Note: Events encode digit position, sequence length, and recall outcome. Note this dataset contains only 2-channel ECG/PPG data (not EEG cap channels); EEG data may be in a separate modality. STATUS is an admin marker and must be excluded. Aggregate correct vs error across all sequence lengths for binary classification.

Data Quality Notes

- Only 2 channels (PPG + ECG) - this is a physiological signal dataset, not a scalp EEG cap
- EEG data may be stored separately; verify file naming convention
- Large number of event subtypes (~30 distinct labels); grouping by outcome (correct/error) is necessary

References

- Reference: `EEG, pupillometry, ECG and photoplethysmography, and behavioral data in the digit span task`

072. openneuro_ds003846 - Prediction Error

Dataset: openneuro_ds003846 | Year: 2025 | URL: https://openneuro.org/datasets/ds003846
Category: Complex Applied | Subjects: 19 | Files (Completed): 50
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

A virtual reality haptic interaction study where participants reached out to touch spawned virtual boxes while receiving varied combinations of visual, vibrotactile, and electrical muscle stimulation (EMS) feedback. Trials were marked as "normal" (congruent feedback) or "conflict" (incongruent prediction error condition), allowing investigation of multisensory prediction error processing. Three feedback modality conditions were run in separate sessions: visual-only, vibrotactile-only, and combined EMS conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
box	559 files	short	Box spawned or box touched event
vibroFeedback	198 files	short	Vibrotactile feedback onset
block	79 files	short	Block start marker
visualFeedback	50 files	short	Visual feedback onset

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Normal vs Conflict Prediction Error (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
box:touched;...normal_or_conflict:normal	0	Normal (congruent) trial - box touched, expected feedback
box:touched;...normal_or_conflict:conflict	1	Conflict (prediction error) trial - incongruent feedback

- Note: Events are richly annotated with metadata (reaction time, trial number, ISI, cube identity). The normal/conflict label is embedded in the event string. Epoch on box:touched events and extract the normal_or_conflict field for the label. Reaction time is also available as a regression target.

Data Quality Notes

- Events contain full JSON-like annotation strings; label extraction requires parsing the event description

- Multiple sessions per subject (EMS, Vibro, Visual) with differing session lengths
- Sampling rate is ~500 Hz (exact value varies slightly across files)

References

- Reference: `Cyber-Physical Systems: Prediction Error`

073. openneuro_ds003885 - Capacity for movement is an organisational principle in object representations: EEG data from Experiment 1

Dataset: `openneuro_ds003885` | Year: 2023 | URL: https://openneuro.org/datasets/ds003885
Category: High-level Mental | Subjects: 24 | Files (Completed): 48
Channels: 127 | Sampling Rate: 250 Hz

Paradigm

Participants performed a semantic classification task (alive vs. not alive) and a passive viewing task using RSVP streams of images to investigate the neural representations of object movement and aliveness. Images spanned animate and inanimate categories with varying degrees of movement capacity, presented at ~3 Hz in RSVP. The study aimed to decode movement-related object properties from EEG representational geometry.

Event Structure

Event ID	Count (total)	Type	Meaning
`E 8`	~115,200	short	RSVP image onset - condition 8
`E 9`	~115,200	short	RSVP image onset - condition 9
`E 1`	~38,400	short	RSVP image onset - condition 1
`E 2`	~38,400	short	RSVP image onset - condition 2
`E 3`	~38,400	short	RSVP image onset - condition 3
`E 5`	~38,400	short	RSVP image onset - condition 5
`E 4`	~960	short	Block boundary or special event

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Movement Capacity Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
E 1	0	Object category 1
E 2	1	Object category 2
E 3	2	Object category 3
E 5	3	Object category 5
E 8	4	Object category 8 (most frequent - likely animate/movement-capable)
E 9	5	Object category 9 (most frequent - likely inanimate/no movement)

- Note: E 8 and E 9 together account for ~76% of all events (~4,800 each per subject vs ~1,600 for conditions 1-5). E 4 is rare (40/subject) and likely a block marker. Data from derivatives/eeglab with 128-channel cap. Consider binary alive/not-alive classification using E 8/E 9.

Data Quality Notes

- Data from derivatives/eeglab; preprocessed with EEGLAB
- 128 channels recorded but analysis.md reports 127 (one likely reference or EOG)
- Very consistent event counts across all 24 subjects

References

- Reference: Shatek, S. M., Robinson, A. K., Grootswagers, T., & Carlson, T. A. (2022). Capacity for movement is an organisational principle in object representations. NeuroImage, 261, 119517.

074. openneuro_ds003887 - Capacity for movement is an organisational principle in object representations: EEG data from Experiment 2

Dataset: `openneuro_ds003887` | Year: 2023 | URL: https://openneuro.org/datasets/ds003887
Category: High-level Mental | Subjects: 24 | Files (Completed): 48
Channels: 127 | Sampling Rate: 250 Hz

Paradigm

Participants performed time-pressured categorization of static images based on movement capacity and completed a passive viewing task using Rapid Serial Visual Presentation (RSVP). This second experiment extends Experiment 1 by testing whether movement capacity organization in neural object representations generalizes across different task contexts and classification criteria.

Event Structure

Event ID	Count (total)	Type	Meaning
`E 8`	~115,200	short	RSVP image onset - condition 8
`E 9`	~115,200	short	RSVP image onset - condition 9
`E 1`	~38,400	short	RSVP image onset - condition 1
`E 2`	~38,400	short	RSVP image onset - condition 2
`E 3`	~38,400	short	RSVP image onset - condition 3
`E 5`	~38,400	short	RSVP image onset - condition 5
`E 4`	~960	short	Block boundary or special event

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Movement Capacity Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
E 1	0	Object category 1
E 2	1	Object category 2
E 3	2	Object category 3
E 5	3	Object category 5
E 8	4	Object category 8 (movement-capable / animate)
E 9	5	Object category 9 (no movement / inanimate)

- Note: Identical event structure to ds003885 (Experiment 1). E 8 and E 9 are the dominant categories (~4,800 each per subject). E 4 is rare (40/subject) and likely marks block boundaries. Binary E 8 vs E 9 classification (movement-capable vs not) is the primary task.

Data Quality Notes

- Data from derivatives/eeglab; preprocessed with EEGLAB
- 128-channel cap, 127 usable channels after reference removal
- Consistent event counts across all 24 subjects

References

- Reference: `Capacity for movement is a major organisational principle in object representations`

075. openneuro_ds003944 - EEG: First Episode Psychosis vs. Control Resting Task 1

Dataset: openneuro_ds003944 | Year: 2022 | URL: https://openneuro.org/datasets/ds003944
Category: Biological State | Subjects: 82 | Files (Completed): 82
Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A 5-minute eyes-open resting-state EEG recording comparing spontaneous brain activity between individuals with First Episode Psychosis (FEP) and age-matched healthy controls. The dataset is designed to identify EEG biomarkers of psychosis onset. No trial-level stimuli are presented; the label of interest is the clinical group membership (FEP vs. control) stored in participants.tsv.

Event Structure

Event ID	Count (total)	Type	Meaning
unknown	82	admin	Unidentified system marker - not suitable for epoching

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Label source: participants.tsv (group label: FEP vs. control)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: FEP vs. Control Classification (binary, resting-state segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:None (label from participants.tsv group column)
- Remarks: Purely resting-state dataset; segment continuously and assign subject-level diagnosis labels (FEP=1, control=0). No task events are available for epoch_by_event. Group label must be extracted from participants metadata.

Data Quality Notes

- No meaningful event markers; unknown events should be excluded
- Subject count not specified in header; 82 files suggests ~82 subjects
- Verify FEP/control group balance in participants.tsv before training

References

- No paper reference available

076. openneuro_ds003947 - EEG: First Episode Psychosis vs. Control Resting Task 2

Dataset: openneuro_ds003947 | Year: 2022 | URL: https://openneuro.org/datasets/ds003947
Category: Biological State | Subjects: 61 | Files (Completed): 61
Channels: 64 | Sampling Rate: 3000 Hz

Paradigm

Five minutes of resting-state EEG recording used to compare spontaneous brain activity between individuals with First Episode Psychosis (FEP) and healthy controls. This is a second independent resting-state dataset from the same clinical study series as ds003944, recorded at a higher sampling rate (3000 Hz). Group membership (FEP vs. control) serves as the classification label.

Event Structure

Event ID	Count (total)	Type	Meaning
unknown	61	admin	Unidentified system marker - not suitable for epoching

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Label source: participants.tsv (group label: FEP vs. control)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: FEP vs. Control Classification (binary, resting-state segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:None (label from participants.tsv group column)
- Note: Purely resting-state dataset; segment continuously and assign subject-level diagnosis labels (FEP=1, control=0). High sampling rate (3000 Hz) - consider downsampling to 500 Hz before processing to reduce computational load.

Data Quality Notes

- No meaningful event markers; unknown events should be excluded
- Very high sampling rate (~3000 Hz) may require downsampling
- Companion dataset to ds003944 (Task 1); can be combined for larger resting FEP cohort

References

- No paper reference available

077. openneuro_ds003987 - EEG: Amphetamine trials 5CCPT and Probabilistic Learning

Dataset: openneuro_ds003987 | Year: 2022 | URL: https://openneuro.org/datasets/ds003987
Category: High-level Mental | Subjects: 23 | Files (Completed): 69
Channels: 71 | Sampling Rate: 500 Hz

Paradigm

A double-blind, crossover pharmacological study investigating the effects of d-amphetamine on cognitive performance using two tasks: the 5-Choice Continuous Performance Task (5CCPT), which assesses sustained attention and impulsivity, and a Probabilistic Selection Task (PST), which measures reinforcement learning. EEG was recorded across three drug sessions (placebo, low-dose, high-dose amphetamine).

Event Structure

Event ID	Count (total)	Type	Meaning
21 5CCPT	69 files	short	5CCPT event type 21 (likely stimulus onset)
22 5CCPT	69 files	short	5CCPT event type 22
23 5CCPT	69 files	short	5CCPT event type 23
25 5CCPT	69 files	short	5CCPT event type 25
31 PST	69 files	short	PST event type 31 (stimulus/feedback)
32 PST	69 files	short	PST event type 32
39 PST	69 files	short	PST event type 39
STATUS	69 files	admin	System hardware status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: 5CCPT Attention Task (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
21 5CCPT	0	5CCPT condition 21 (e.g., target stimulus)
22 5CCPT	1	5CCPT condition 22

Event	y	Meaning
`23 5CCPT`	2	5CCPT condition 23

- Note: STATUS must be excluded (admin event). Events are named with task suffix (5CCPT or PST). Full event decoding requires consulting the dataset README. 23 subjects x 3 sessions (drug doses) = 69 files. Drug dose can serve as a between-condition variable.

Data Quality Notes

- Large number of event types (30); full decoding requires dataset documentation
- Three recording sessions per subject (placebo/low/high amphetamine) - consider treating sessions independently
- STATUS is an admin marker and must be excluded

References

- Reference: `Electrophysiological biomarkers of behavioral dimensions from cross-species paradigms`

078. openneuro_ds004000 - Fribourg Ultimatum Game in Schizophrenia Study

Dataset: `openneuro_ds004000` | Year: 2022 | URL: https://openneuro.org/datasets/ds004000
Category: High-level Mental | Subjects: 43 | Files (Completed): 84
Channels: 132 | Sampling Rate: 2048 Hz

Paradigm

A social decision-making study in which patients with psychosis and healthy controls played the Ultimatum Game in both responder and proposer roles across multiple blocks. EEG was recorded while participants decided to accept or reject monetary offers from a human or computer partner, enabling investigation of fairness processing and social cognition differences in psychosis.

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulus/S 1`	42 files	short	Stimulus onset type 1
`Stimulus/S 2`	42 files	short	Stimulus onset type 2
`Stimulus/S 3`	42 files	short	Stimulus onset type 3
`Stimulus/S 4`	42 files	short	Stimulus onset type 4
`Stimulus/S 64`	42 files	short	Stimulus onset - offer/condition marker
`Stimulus/S 96`	42 files	short	Stimulus onset marker
`Stimulus/S128`	42 files	short	Stimulus onset marker
`Stimulus/S160`	42 files	short	Stimulus onset marker
`Stimulus/S193-S201`	28-42 files	short	Condition-specific stimulus markers
`Stimulus/S225-S233`	28-42 files	short	Condition-specific stimulus markers
`Stimulus/S195-S200`	33-41 files	short	Condition-specific stimulus markers
`UserDefined/Blink`	84 files	admin	Blink artifact marker
`Time 0/`	84 files	admin	Recording start time marker
`Response/R 1`	6 files	short	Button press response
`Response/R 3`	1 file	short	Button press response

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Ultimatum Game Stimulus Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`Stimulus/S 1`	0	Stimulus type 1 (e.g., fair offer)
`Stimulus/S 2`	1	Stimulus type 2
`Stimulus/S 3`	2	Stimulus type 3
`Stimulus/S 4`	3	Stimulus type 4 (e.g., unfair offer)

- Note: UserDefined/Blink and Time 0/ are admin markers and must be excluded. Full decoding of S-codes requires dataset documentation. Psychosis vs. control group label is available from participants.tsv for group-level classification.

Data Quality Notes

- Large number of distinct Stimulus/ codes; full mapping requires consulting the study protocol
- UserDefined/Blink and Time 0/ are administrative markers to exclude
- 43 subjects x ~2 sessions = 84 files; verify session structure

References

- No paper reference available

079. openneuro_ds004010 - MAVIS

Dataset: openneuro_ds004010 | Year: 2022 | URL: https://openneuro.org/datasets/ds004010
Category: Sensory/Response | Subjects: 24 | Files (Completed): 24
Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A multisensory detection task (MAVIS) in which participants identified auditory and visual targets embedded within simultaneous audio-visual noise streams to investigate selective attention and crossmodal sensory processing. The dataset contains continuous EEG recordings without discrete event markers, suggesting the stimuli were presented as ongoing streams rather than discrete trials.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: participants.tsv (subject-level metadata)
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous EEG Segmentation (resting/continuous)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:None (no trial-level events available)
- Note: No event markers present. Use continuous segmentation for downstream tasks such as subject identification or spectral analysis. Task condition labels (if any) must come from external behavioral logs or stimulus delivery system.

Data Quality Notes

- No trial-level event markers; only continuous EEG available
- Stimulus delivery information may be in auxiliary files outside the BIDS structure
- 64 channels at 1000 Hz provides good temporal resolution for auditory processing analysis

References

- Reference: MAVIS

080. openneuro_ds004015 - Attended speaker paradigm (cEEGrid data)

Dataset: openneuro_ds004015 | Year: 2022 | URL: https://openneuro.org/datasets/ds004015
Category: High-level Mental | Subjects: 36 | Files (Completed): 36
Channels: 18 | Sampling Rate: 500 Hz

Paradigm

A two-competing speaker auditory attention paradigm in which participants were instructed to selectively attend to one of two simultaneously presented audiobooks under varying spatial (same location vs. different locations) and loudness conditions. EEG was recorded using a cEEGrid around-the-ear electrode array. The dataset is intended for auditory attention decoding (AAD) research using ear-worn EEG.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations; stimulus delivery is continuous (audiobook streams).

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: participants.tsv (attended speaker condition)
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Auditory Attention Decoding - Continuous Segmentation

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:None (attended speaker label from task metadata)
- Note: cEEGrid only provides 18 channels around the ear; limited spatial coverage compared to standard scalp EEG. Auditory attention decoding typically requires aligning EEG segments with envelope of the attended speech stream. No discrete event markers; stimulus is continuous.

Data Quality Notes

- Only 18 around-the-ear channels (cEEGrid) - very limited spatial coverage
- No discrete event markers; continuous stimulus paradigm
- Attended speaker condition metadata needed from external files for supervised learning

References

- Reference: Holtze et al. (2021)

081. openneuro_ds004017 - Embodied Learning for Literacy EEG

Dataset: openneuro_ds004017 | Year: 2023 | URL: https://openneuro.org/datasets/ds004017
Category: High-level Mental | Subjects: 21 | Files (Completed): 126
Channels: 65 | Sampling Rate: 2048 Hz

Paradigm

A pre-post intervention study assessing literacy and orthographic processing in children through two tasks: a two-alternative forced choice letter discrimination task (b vs. d confusion) and a visual search task. EEG was recorded across multiple sessions to evaluate the neural effects of an embodied learning intervention on letter processing. Trigger codes 10/20 and 100/200 likely encode the two letter types or correct/incorrect responses.

Event Structure

Event ID	Count (total)	Type	Meaning
100.0	63 files	short	Stimulus onset - letter type B or condition 100
200.0	63 files	short	Stimulus onset - letter type D or condition 200
20.0	61 files	short	Stimulus onset - response or sub-condition 20
10.0	61 files	short	Stimulus onset - response or sub-condition 10

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Letter Discrimination (b vs d) (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
100.0	0	Letter/condition type 100 (e.g., letter b)
200.0	1	Letter/condition type 200 (e.g., letter d)

- Note: 21 subjects x 6 sessions = 126 files. Codes 10.0/20.0 may indicate sub-conditions or response events - verify against dataset README. Pre-post intervention design allows longitudinal analysis.

Data Quality Notes

- Numerical trigger codes; full decoding requires dataset documentation
- 21 subjects x 6 sessions - session-level structure must be considered
- High sampling rate (2048 Hz); consider downsampling for efficiency

References

- Reference: `Embodied Learning for Literacy EEG`

082. openneuro_ds004018 - EEG recordings for 200 object images presented in RSVP sequences at 5Hz or 20Hz

Dataset: `openneuro_ds004018` | Year: 2022 | URL: https://openneuro.org/datasets/ds004018
Category: Sensory/Response | Subjects: 2 | Files (Completed): 2
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Rapid Serial Visual Presentation (RSVP) of 200 object images at two presentation frequencies (5 Hz and 20 Hz) to investigate the temporal dynamics of visual object representations. Images were categorized as "stars" or "boats" (two superordinate categories), allowing investigation of how presentation rate affects category-level neural decoding. This is a very small dataset (2 subjects).

Event Structure

Event ID	Count (total)	Type	Meaning
`stars`	2 files	short	RSVP onset - star/celestial category image
`boats`	2 files	short	RSVP onset - boat/vehicle category image

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Mission 1: Stars vs Boats Category Decoding (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`stars`	0	Star/celestial object category
`boats`	1	Boat/vehicle category

- Note: Very small dataset (only 2 subjects) - not suitable for cross-subject generalization studies. Useful for single-subject or within-subject decoding analyses. RSVP rates of 5 Hz and 20 Hz may be in separate sessions; verify file structure.

Data Quality Notes

- Only 2 subjects - extremely limited statistical power for group-level analysis
- RSVP at 5 Hz and 20 Hz: at 20 Hz, images appear for only 50 ms (near threshold for conscious processing)
- Verify that `stars` and `boats` events correspond to onset markers and not sustained states

References

- Reference: The representational dynamics of visual objects in rapid serial visual processing streams. NeuroImage, 188, 668-679

083. openneuro_ds004019 - Effect of obesity on arithmetic processing in preteens with high and low math skills. An event-related potentials study

Dataset: `openneuro_ds004019` | Year: 2022 | URL: https://openneuro.org/datasets/ds004019
Category: High-level Mental | Subjects: 62 | Files (Completed): 62
Channels: 132 | Sampling Rate: 500 Hz

Paradigm

Obese and non-obese preteens (ages 9-10) performed a delayed-verification arithmetic task to evaluate the effects of weight status and math skill level on arithmetic processing. Participants viewed an arithmetic problem (prob), then a delay (bgin), then a verification digit (face) and responded (resp). Trigger codes 0 and 1 encode response correctness (0=incorrect, 1=correct). The study uses ERPs to investigate neural correlates of arithmetic cognition in the context of childhood obesity.

Event Structure

Event ID	Count (total)	Type	Meaning
`bgin`	~14,880	short	Trial beginning / delay period onset
`prob`	~14,880	short	Arithmetic problem onset
`ch++`	~14,880	short	Choice/comparison stimulus onset
`face`	~14,880	short	Verification digit (face stimulus) onset
`resp`	~14,600	short	Participant button-press response
`1`	~14,364	short	Correct response (1=correct)
`0`	~6,156	short	Incorrect response (0=incorrect)
`sync`	~310	admin	Hardware sync pulse
`CELL`	~248	admin	Cell/block marker
`SESS`	62	admin	Session start marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Correct vs Incorrect Arithmetic (binary)

- Epoch method: `epoch_by_event_hdf5`

- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`1`	1	Correct arithmetic response
`0`	0	Incorrect arithmetic response

- Note: sync, CELL, SESS are admin markers and must be excluded. The `1` and `0` events mark response correctness concurrent with `resp`. Class imbalance (~70% correct across subjects) should be addressed. Obesity group label is in participants.tsv.

Data Quality Notes

- 132-channel EGI cap (E1-E128 + ECG/Respiration/SP channels); remove non-EEG channels before analysis
- sync/CELL/SESS are admin markers to exclude
- Class imbalance: ~70% correct trials - apply undersampling or weighted loss

References

- Reference: `Effect of obesity on arithmetic processing in preteens with high and low math skills. An event-related potentials study.`

084. openneuro_ds004022 - Multimodal EEG and fNIRS Biosignal Acquisition during Motor Imagery Tasks in Patients with Orthopedic Impairment

Dataset: `openneuro_ds004022` | Year: 2022 | URL: https://openneuro.org/datasets/ds004022
Category: Action/Output | Subjects: 7 | Files (Completed): 21
Channels: 18 | Sampling Rate: 500 Hz

Paradigm

Participants with orthopedic impairment performed four distinct motor imagery tasks (Reach, Twist, Lift, and Grasp) guided by visual cues across three experimental sessions. EEG and fNIRS were recorded simultaneously to study motor imagery neural patterns in a clinical population for BCI development. Each trial began with a rest cue (S 1) followed by a task cue (S 2-S 10), with S 7-S 10 encoding the four MI conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
`S 2`	21 files	short	Cue onset (task start)
`S 7`	21 files	short	Motor imagery cue - task 1 (Reach)
`S 8`	21 files	short	Motor imagery cue - task 2 (Twist)
`S 9`	21 files	short	Motor imagery cue - task 3 (Lift)
`S 10`	21 files	short	Motor imagery cue - task 4 (Grasp)
`S 3`	21 files	short	Rest period cue
`S 4`	21 files	short	Rest period cue (variant)
`S 5`	21 files	short	Feedback/response marker
`S 6`	21 files	short	Feedback/response marker
`S 1`	21 files	short	Trial start / baseline cue
`condition 12`	21 files	admin	Condition block marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Four-Class Motor Imagery (multiclass)

- Epoch method: `epoch_by_event_hdf5`

- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`S 7`	0	Motor imagery - Reach
`S 8`	1	Motor imagery - Twist
`S 9`	2	Motor imagery - Lift
`S 10`	3	Motor imagery - Grasp

- Note: Only 7 subjects; small sample size limits generalization. `condition 12` is an admin/block marker to exclude. 3 sessions per subject (21 files total). Each session has 40 trials per MI class.

Data Quality Notes

- Only 7 subjects with orthopedic impairment - very small clinical sample
- 18-channel EEG with motor cortex coverage; suitable for BCI but limited spatial resolution
- `condition 12` is a block marker to exclude from epoching

References

- Reference: `Multimodal EEG and fNIRS Biosignal Acquisition during Motor Imagery Tasks in Patients with Orthopedic Impairment`

085. openneuro_ds004024 - TMS-EEG-MRI-fMRI-DWI data on paired associative stimulation and connectivity (Shirley Ryan AbilityLab, Chicago, IL)

Dataset: `openneuro_ds004024` | Year: 2022 | URL: https://openneuro.org/datasets/ds004024
Category: Sensory/Response | Subjects: 497 files | Files (Completed): 497
Channels: 69 | Sampling Rate: 20000 Hz

Paradigm

A multi-modal neuroimaging study using Paired Associative Stimulation (PAS) combined with TMS-EEG to investigate cortical connectivity and neuroplasticity. Participants received precisely timed TMS pulses paired with peripheral nerve stimulation, and EEG was recorded at very high sampling rate (20 kHz) to capture TMS-evoked potentials. The study also collected MRI, fMRI, and DWI data. This is a clinical/neurophysiology dataset focusing on brain connectivity assessment.

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulus/A`	345 files	short	TMS or PAS stimulus onset
`Out/A`	378 files	short	Stimulus output marker
`SB/A`	33 files	short	Sub-threshold baseline stimulus
`Video/B`	32 files	short	Video instruction marker
`New Segment/`	497 files	admin	Recording segment boundary - exclude

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: TMS-Evoked Potential Epoching (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`Stimulus/A`	0	TMS/PAS stimulus pulse - main evoked potential event
`SB/A`	1	Sub-threshold baseline TMS pulse

- Note: New Segment/ is an admin boundary marker and must be excluded. Very high sampling rate (20 kHz) - downsampling to 1-5 kHz strongly recommended before HDF5 generation. The TMS artifact (first

~10 ms post-pulse) requires specialized cleaning before analysis.

Data Quality Notes

- Very high sampling rate (20,000 Hz) - downsampling required
- TMS artifact contaminates first ~10 ms post-stimulus; specialized TMS-EEG artifact removal (e.g., SSP, ICA) is required
- New Segment/ is an admin boundary marker to exclude

References

- Paper: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Scientific Data, 6, 103`

086. openneuro_ds004040 - Trance channeling EEG study

Dataset: openneuro_ds004040 | Year: 2022 | URL: https://openneuro.org/datasets/ds004040
Category: High-level Mental | Subjects: 13 | Files (Completed): 26
Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants who are experienced trance channelers performed alternating 5-minute blocks of trance channeling (a mediumistic altered state of consciousness) and matched resting state conditions across two separate sessions. The study investigates neural correlates of voluntarily induced altered states of consciousness compared to wakeful rest. Labels for trance vs. rest blocks must be derived from the session/task structure rather than discrete event markers.

Event Structure

Event ID	Count (total)	Type	Meaning
STATUS	26 files	admin	System hardware status marker - not suitable for epoching

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: BIDS task name or participants.tsv (trance vs. rest condition)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Trance vs. Resting State (binary, segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:None (condition label from task/session structure)
- Note: STATUS is an admin marker and must be excluded. Block-level labels (trance vs. rest) should come from the BIDS task name or session metadata. 13 subjects x 2 sessions = 26 files. Small sample size; suitable for within-subject analysis.

Data Quality Notes

- Only STATUS admin marker present; no task-level event markers
- Small sample (13 specialized subjects who practice trance channeling)
- Two sessions per subject - verify session labeling in BIDS structure

References

- Reference: Trance channeling EEG study

087. openneuro_ds004043 - The time-course of feature-based attention effects dissociated from temporal expectation and target-related processes

Dataset: `openneuro_ds004043` | Year: 2022 | URL: https://openneuro.org/datasets/ds004043
Category: Sensory/Response | Subjects: 20 | Files (Completed): 40
Channels: 63 | Sampling Rate: 250 Hz

Paradigm

Participants viewed rapid sequences of overlaid oriented grating pairs (presented at ~10 Hz) and detected a target orientation while attention was cued to different feature dimensions (orientation or color). The study dissociates the time-course of feature-based attention effects from temporal expectation and target-detection processes using EEG frequency-tagging and ERP methods.

Event Structure

Event ID	Count (total)	Type	Meaning
`E 1`	20 files	short	Grating stimulus onset - condition 1 (e.g., attended feature)
`E 2`	20 files	short	Grating stimulus onset - condition 2
`E 3`	20 files	short	Grating stimulus onset - condition 3 (e.g., target/distractor)

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Attention Condition Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`E 1`	0	Attention condition 1
`E 2`	1	Attention condition 2
`E 3`	2	Attention condition 3 (target or distractor)

- Note: 20 subjects x 2 sessions = 40 files. E 1/E 2/E 3 are EEGLAB-style markers; full decoding requires consulting the original study events.tsv. Balanced across conditions per session.

Data Quality Notes

- Events encoded as generic E markers; semantic meaning requires events.tsv inspection
- 20 subjects x 2 files suggests two sessions or two task conditions per subject
- 250 Hz sampling rate; adequate for ERP/steady-state visual evoked potential analysis

References

- Reference: `Undivided attention: The temporal effects of attention dissociated from decision, memory, and expectation`

088. openneuro_ds004067 - Moral conviction and metacognitive ability shape multiple stages of information processing

Dataset: `openneuro_ds004067` | Year: 2022 | URL: https://openneuro.org/datasets/ds004067
Category: High-level Mental | Subjects: 80 | Files (Completed): 83
Channels: 63 | Sampling Rate: 2000 Hz

Paradigm

Participants evaluated their support for sociopolitical protest photographs following social influence cues from a fictitious community to assess how moral conviction and metacognitive ability modulate information processing. EEG was recorded while participants viewed protest photos (supporting or opposing social movements), received social feedback (high/low/medium agreement), and provided attitude ratings. The study examined how moral convictions and metacognitive ability shape early and late stages of social information processing.

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulus/PhotoSupport`	~7,290	short	Photo onset - support-framing protest image
`Stimulus/PhotoOppose`	~7,290	short	Photo onset - oppose-framing protest image
`Stimulus/SocLow`	~4,860	short	Social influence cue - low agreement
`Stimulus/SocHigh`	~4,860	short	Social influence cue - high agreement
`Stimulus/SocMed`	~4,860	short	Social influence cue - medium agreement
`Stimulus/Moral_Support`	~5,688	short	Moral attitude cue - support
`Stimulus/Moral_Oppose`	~5,688	short	Moral attitude cue - oppose
`Stimulus/Nonmoral_Oppose`	~5,400	short	Non-moral attitude cue - oppose
`Stimulus/Nonmoral_Neutral`	~4,968	short	Non-moral attitude cue - neutral
`Stimulus/Nonmoral_Support`	~4,560	short	Non-moral attitude cue - support
`Stimulus/Resp1-Resp5`	variable	short	Participant rating response (1-5 scale)
`Stimulus/ProtestStart`	~80	admin	Block start marker
`Stimulus/ProtestEnd`	~80	admin	Block end marker
`Stimulus/RestStart`	~79	rest	Resting block start
`Stimulus/RestEnd`	~79	rest	Resting block end

Event ID	Count (total)	Type	Meaning
Stimulus/ClipTaskStart	~79	admin	Clip task start
Stimulus/ClipTaskEnd	~79	admin	Clip task end
Stimulus/Social5/6start-end	~79	admin	Social clip block markers
Stimulus/Heroic6/7start-end	~79	admin	Heroic clip block markers
Stimulus/S 2	~79	admin	System/sync marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Support vs Oppose Photo Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stimulus/PhotoSupport	0	Support-framing sociopolitical protest photo
Stimulus/PhotoOppose	1	Oppose-framing sociopolitical protest photo

- Note: ProtestStart/End, ClipTaskStart/End, Socialstart/end, Heroicstart/end, RestStart/End, and S 2 are all admin/structural markers to exclude. Resp1-Resp5 encode 5-point rating responses and can be used for regression. Social cues (SocLow/Med/High) can be used for a 3-class social influence task.

Data Quality Notes

- Large number of event types; careful exclusion of admin markers required
- RestStart/RestEnd delimit resting-state blocks within the recording
- High sampling rate (2000 Hz); consider downsampling to 500-1000 Hz

References

- Reference: Moral conviction and metacognitive ability shape multiple stages of information processing

089. openneuro_ds004075 - what_are_we_talking_about

Dataset: openneuro_ds004075 | Year: 2022 | URL: https://openneuro.org/datasets/ds004075
Category: Others | Subjects: N/A | Files (Completed): 115
Channels: 64.0 | Sampling Rate: 1000.0 Hz

Paradigm

The provided title 'what_are_we_talking_about' without an accompanying README is too vague to determine a specific experimental paradigm.

Event Structure

Event ID	Count (files)	Meaning
unknown	115	Event marker unknown - meaning unclear from available metadata; check events.tsv
## Downstream Task Recommendations		

Classification Tasks

- Insufficient distinct event types for classification (only 1 event class found).

Regression Tasks

- No clear regression target identified from available event structure. Consider reaction time if behavioral logs are available.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

090. openneuro_ds004105 - BCIT Auditory Cueing

Dataset: openneuro_ds004105 | Year: 2022 | URL: https://openneuro.org/datasets/ds004105
Category: Complex Applied | Subjects: 34 | Files (Completed): 34
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Participants performed a long-duration simulated driving task involving lane maintenance with periodic lateral vehicle perturbations, while receiving auditory cues at varying intervals. EEG biomarkers of fatigue were investigated as performance and alertness declined over the driving session. The study is part of the BCIT (Brain Computer Interface and Technology) driving fatigue dataset series.

Event Structure

Event ID	Count (total)	Type	Meaning
4210	34 files	short	Driving event code 4210 (lateral perturbation onset)
4220	34 files	short	Driving event code 4220
4230	34 files	short	Driving event code 4230
4311	34 files	short	Lane deviation/correction event
4312	34 files	short	Lane deviation/correction event
3200	34 files	short	Speed/control event
3111	34 files	short	Perturbation/response event
3112	34 files	short	Perturbation/response event
3310	34 files	short	Performance metric event
1211	34 files	short	Auditory cue event
1212	34 files	short	Auditory cue event
2621	34 files	short	Behavioral response event
2622	34 files	short	Behavioral response event
1111-1322	17 files	short	Condition-specific events (half of subjects)
4411	1 file	short	Rare event

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Lateral Perturbation Response (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
4311	0	Lane perturbation - direction/type 1
4312	1	Lane perturbation - direction/type 2

- Note: Full BCIT event code decoding requires the dataset codebook. Part of the BCIT driving fatigue series alongside ds004106, ds004118, ds004119, ds004120, ds004121, ds004122, ds004123. Auditory cueing conditions are the primary manipulation.

Data Quality Notes

- Numerical event codes require codebook from BCIT documentation for full decoding
- 74-channel BioSemi ActiveTwo; high-density cap with 1024 Hz sampling
- Part of a larger BCIT driving series - consider combining with related datasets

References

- No paper reference available

091. openneuro_ds004106 - BCIT Advanced Guard Duty

Dataset: openneuro_ds004106 | Year: 2022 | URL: https://openneuro.org/datasets/ds004106
Category: Complex Applied | Subjects: 29 | Files (Completed): 29
Channels: 262 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a sustained vigilance task simulating advanced security guard duties by verifying multiple fields on replica ID cards against reference images, under varying workloads and stochastic arrival rates of cards. EEG was recorded to identify biomarkers of mental fatigue and vigilance decrement over extended duty periods. Part of the BCIT sustained attention / guard duty dataset series.

Event Structure

Event ID	Count (total)	Type	Meaning
1311	29 files	short	Card/stimulus arrival event - type 1311
1312	29 files	short	Card/stimulus arrival event - type 1312
1321	29 files	short	Verification response event
1322	29 files	short	Verification response event
1331-1362	27-29 files	short	Guard duty task events (various verification stages)
1341-1342	29 files	short	Guard duty task events
1351-1352	27-29 files	short	Guard duty task events
1221-1222	29 files	short	Guard duty task events
1211-1212	28 files	short	Guard duty task events
1231-1232	28 files	short	Guard duty task events
1241-1242	27 files	short	Guard duty task events
1251-1252	27 files	short	Guard duty task events
1121-1122	27 files	short	Guard duty task events
2110	29 files	short	Workload condition marker - high
2120	29 files	short	Workload condition marker - low
3111-3112	26 files	short	Block/session markers
3231-3232	2 files	short	Rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: High vs Low Workload (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`2110`	0	High workload condition marker
`2120`	1	Low workload condition marker

- Note: 262-channel BioSemi cap - very high-density EEG. Full BCIT event codebook required for precise decoding. Part of the BCIT guard duty series (see also ds004119 Basic Guard Duty).

Data Quality Notes

- 262 channels - largest EEG cap in the BCIT series; high memory requirements
- Full event decoding requires BCIT codebook documentation
- Very limited subjects (29) for the large channel count

References

- Reference: `BCIT Advanced Guard Duty`

092. openneuro_ds004117 - Sternberg Working Memory

Dataset: openneuro_ds004117 | Year: 2022 | URL: https://openneuro.org/datasets/ds004117
Category: High-level Mental | Subjects: 23 | Files (Completed): 85
Channels: 71 | Sampling Rate: 250 Hz

Paradigm

A modified Sternberg working memory task in which participants selectively memorized sets of 3, 5, or 7 target letters (marked with lowercase 'g' prefix, e.g., gH) while ignoring identical-font distractor letters. After a retention period, a probe letter was presented and participants indicated with auditory feedback whether it was in the memorized set. The study investigates frontal midline EEG dynamics (particularly theta oscillations) during working memory maintenance.

Event Structure

Event ID	Count (total)	Type	Meaning
WM	85 files	short	Working memory trial onset / encoding phase
nonWM	85 files	short	Non-working-memory (distractor) trial
gA-gZ (prefixed letters)	85 files	short	Working memory target letter onset (to be memorized)
A-Z (plain letters)	85 files	short	Distractor letter onset (to be ignored)
255	85 files	admin	System trigger (likely BrainVision marker 255 = all bits)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: WM vs Non-WM Letter Encoding (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
WM	1	Working memory encoding trial
nonWM	0	Non-working-memory (distractor) trial

- Note: 255 is an admin marker and must be excluded. Individual letter events (A-Z, gA-gZ) mark specific letter presentations and can be used for fine-grained analysis. 23 subjects x ~3.7 sessions on average = 85

files.

Data Quality Notes

- Trigger 255 (all-bits-high) is a system artifact and must be excluded
- Individual letter events provide fine-grained trial structure for ERP analysis
- 23 subjects x multiple sessions - verify session structure before pooling

References

- Reference: `Onton, J., Delorme, A. and Makeig, S., 2005. Frontal midline EEG dynamics during working memory. Neuroimage, 27(2), pp.341-356.`

093. openneuro_ds004118 - BCIT Calibration Driving

Dataset: openneuro_ds004118 | Year: 2022 | URL: https://openneuro.org/datasets/ds004118
Category: Complex Applied | Subjects: 247 | Files (Completed): 247
Channels: 266 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a 15-minute simulated driving calibration task on a straight road, maintaining lane position while correcting for periodic lateral perturbations. This calibration recording is designed to collect a baseline driving EEG profile for each subject to normalize fatigue prediction algorithms used in the broader BCIT driving fatigue study series. It is the largest file count in the BCIT series.

Event Structure

Event ID	Count (total)	Type	Meaning
3200	247 files	short	Speed/driving state marker
3111	247 files	short	Perturbation onset
3112	247 files	short	Perturbation offset/response
3310	247 files	short	Performance metric event
1211	247 files	short	Lane correction event
1212	247 files	short	Lane correction event
4311	246 files	short	Lateral perturbation type 1
4312	246 files	short	Lateral perturbation type 2
4230	246 files	short	Perturbation/direction event
4210	239 files	short	Perturbation onset event
4220	206 files	short	Perturbation event
1121	246 files	short	Driver response event
1122	246 files	short	Driver response event
1111	246 files	short	Driver response event
1112	246 files	short	Driver response event
2621	97 files	short	Auditory cue onset
2622	97 files	short	Auditory cue offset
4200-4612	2 files	short	Rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Lateral Perturbation Direction (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`4311`	0	Lateral perturbation - direction/type 1
`4312`	1	Lateral perturbation - direction/type 2

- Note: Largest file count in the BCIT series (247 subjects). 266-channel BioSemi cap - very high memory requirements. Calibration session is shorter (15 min) than fatigue driving sessions.

Data Quality Notes

- 266 channels - very high-density EEG; significant memory requirements for batch processing
- 247 files (subjects) - large cohort; suitable for cross-subject decoding
- Full event code decoding requires BCIT codebook

References

- No paper reference available

094. openneuro_ds004119 - BCIT Basic Guard Duty

Dataset: openneuro_ds004119 | Year: 2022 | URL: https://openneuro.org/datasets/ds004119
Category: Complex Applied | Subjects: 22 | Files (Completed): 22
Channels: 262 | Sampling Rate: 1024 Hz

Paradigm

A realistic sustained vigilance task in which subjects verified multiple identification card fields against reference images in a simulated security environment under varying card arrival rates. This basic guard duty variant uses a simpler verification protocol compared to the Advanced Guard Duty (ds004106), and is designed to study EEG correlates of sustained attention and vigilance decrement over extended duty periods.

Event Structure

Event ID	Count (total)	Type	Meaning
1311	22 files	short	Card arrival / verification stimulus onset
1312	22 files	short	Card arrival event
1321-1322	22 files	short	Verification response events
1331-1332	22 files	short	Guard duty task events
1341-1342	22 files	short	Guard duty task events
1351-1352	22 files	short	Guard duty task events
1361-1362	22 files	short	Guard duty task events
1231-1232	22 files	short	Guard duty task events
2110	22 files	short	High workload condition marker
2120	22 files	short	Low workload condition marker
3111-3112	20 files	short	Block/session markers
1111-1112	21 files	short	Guard duty task events
3241-3242	6 files	short	Rare condition events
3231-3232	1 file	short	Very rare condition events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: High vs Low Workload (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`2110`	0	High workload condition
`2120`	1	Low workload condition

- Note: Small cohort (22 subjects). 262-channel BioSemi cap. Paired with ds004106 (Advanced Guard Duty)
- can be combined for larger guard duty cohort.

Data Quality Notes

- 262 channels - very high-density EEG; high memory requirements
- Only 22 subjects - small sample size
- Full event code decoding requires BCIT codebook documentation

References

- Reference: `Gross, Matthews, Cohn, Kanade, & Baker, 2010`

095. openneuro_ds004120 - BCIT Baseline Driving

Dataset: openneuro_ds004120 | Year: 2022 | URL: https://openneuro.org/datasets/ds004120
Category: Complex Applied | Subjects: 131 | Files (Completed): 131
Channels: 266 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a 60-minute simulated driving task involving manual steering and speed control, requiring corrective responses to lateral vehicle perturbations to study fatigue-related EEG biomarkers over an extended session. This baseline driving condition serves as the primary long-duration fatigue paradigm in the BCIT series, without auditory cueing or traffic complexity manipulation.

Event Structure

Event ID	Count (total)	Type	Meaning
4311	131 files	short	Lateral perturbation - type 1
4312	131 files	short	Lateral perturbation - type 2
4210	130 files	short	Perturbation onset
4220	123 files	short	Perturbation event
4230	131 files	short	Perturbation/direction event
3200	131 files	short	Speed/driving state marker
3111	128 files	short	Perturbation onset
3112	125 files	short	Perturbation response
3310	131 files	short	Performance metric event
1211	131 files	short	Lane correction event
1212	130 files	short	Lane correction event
1121-1122	131 files	short	Driver response events
1111-1112	131 files	short	Driver response events
2621-2622	28 files	short	Auditory cue events (subset)
2611-2612	28 files	short	Auditory cue events (subset)
5241-5242	10 files	short	Rare condition events
4421-5232	2-5 files	short	Very rare events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Lateral Perturbation Direction (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`4311`	0	Lateral perturbation - direction/type 1
`4312`	1	Lateral perturbation - direction/type 2

- Remarks: 131 subjects with 60-minute sessions - large fatigue dataset. 266-channel BioSemi cap. Primary baseline driving condition for BCIT fatigue series comparison.

Data Quality Notes

- 266 channels - very high-density EEG; significant memory requirements
- 60-minute sessions provide extensive fatigue-related data per subject
- Full event code decoding requires BCIT codebook

References

- No paper reference available

096. openneuro_ds004121 - BCIT Mind Wandering

Dataset: openneuro_ds004121 | Year: 2022 | URL: https://openneuro.org/datasets/ds004121
Category: Complex Applied | Subjects: 60 | Files (Completed): 60
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a long-duration simulated driving task involving lane maintenance, speed control, and a police vehicle vigilance sub-task (detecting a police car among traffic), while exposed to lateral perturbations and varying background audio conditions (silence vs. music vs. podcast). The mind wandering manipulation used periodic thought probes to assess on-task vs. off-task mental state. EEG biomarkers of mind wandering during driving were investigated.

Event Structure

Event ID	Count (total)	Type	Meaning
4311	60 files	short	Lateral perturbation type 1
4312	59 files	short	Lateral perturbation type 2
4210	59 files	short	Perturbation onset
4220	59 files	short	Perturbation event
4230	59 files	short	Perturbation/direction event
4730	59 files	short	Special driving event
4411	60 files	short	Police vehicle event
4621-4622	60 files	short	Audio condition events
3200	60 files	short	Speed marker
3111	60 files	short	Perturbation onset
3310	60 files	short	Performance metric
1211	60 files	short	Lane correction event
1212	58 files	short	Lane correction event
1121-1122	60 files	short	Driver response
1111-1112	60 files	short	Driver response
2221-2222	59 files	short	Thought probe / mind wandering event
2241-2242	59 files	short	Thought probe response
2251-2252	59 files	short	Thought probe response
2621-2622	60 files	short	Behavioral response
2811-2812	59 files	short	Audio cue events
2631-2632	58 files	short	Behavioral response events

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5

- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Lateral Perturbation Direction (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`4311`	0	Lateral perturbation - type 1
`4312`	1	Lateral perturbation - type 2

- Note: Mind wandering thought probe events (2221/2241/2251) can be used to label on-task vs. off-task periods for attention state classification. Full event codebook required for precise decoding of all 30+ event types.

Data Quality Notes

- 30+ distinct event types; full decoding requires BCIT codebook
- Thought probe events provide mind-wandering labels for attention classification
- Audio condition (silence/music/podcast) provides additional between-condition variable

References

- No paper reference available

097. openneuro_ds004122 - BCIT Speed Control

Dataset: openneuro_ds004122 | Year: 2022 | URL: https://openneuro.org/datasets/ds004122
Category: Complex Applied | Subjects: 63 | Files (Completed): 63
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

A simulated driving task assessing driver fatigue and performance through a speed control manipulation: subjects drove under either manual speed control (maintaining speed without cruise control) or automated speed control (cruise control active), while also performing manual steering corrections for lateral vehicle perturbations. The study investigates how automation of one driving sub-task affects neural fatigue signatures.

Event Structure

Event ID	Count (total)	Type	Meaning
4210	63 files	short	Perturbation onset
4220	63 files	short	Perturbation event
4230	63 files	short	Perturbation/direction event
4311	63 files	short	Lateral perturbation type 1
4312	63 files	short	Lateral perturbation type 2
3200	63 files	short	Speed control state marker
3111	63 files	short	Perturbation onset
3310	63 files	short	Performance metric event
1211	63 files	short	Lane correction event
1212	59 files	short	Lane correction event
1121-1122	63 files	short	Driver response events
1111-1112	63 files	short	Driver response events
2621-2622	61 files	short	Behavioral response / auditory cue
3112	59 files	short	Perturbation response
4411	5 files	short	Rare event
4421	6 files	short	Rare event

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Lateral Perturbation Direction (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
4311	0	Lateral perturbation - type 1
4312	1	Lateral perturbation - type 2

- Note: Speed control condition (manual vs. automated) can serve as a between-subjects or within-session variable. Part of the BCIT driving fatigue series.

Data Quality Notes

- 74-channel BioSemi ActiveTwo; 1024 Hz sampling
- Speed control condition variable requires session metadata from participants.tsv
- Full event code decoding requires BCIT codebook

References

- No paper reference available

098. openneuro_ds004123 - BCIT Traffic Complexity

Dataset: openneuro_ds004123 | Year: 2022 | URL: https://openneuro.org/datasets/ds004123
Category: Complex Applied | Subjects: 30 | Files (Completed): 30
Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Subjects performed a simulated driving task with varying levels of visual traffic complexity (low, medium, high density of surrounding vehicles) while correcting for lateral vehicle perturbations. EEG was recorded to assess how increasing traffic complexity interacts with fatigue-related neural signatures. Part of the BCIT driving fatigue dataset series examining environmental complexity effects on driver workload.

Event Structure

Event ID	Count (total)	Type	Meaning
4311	30 files	short	Lateral perturbation type 1
4312	30 files	short	Lateral perturbation type 2
4210	30 files	short	Perturbation onset
4220	30 files	short	Perturbation event
4230	30 files	short	Perturbation/direction event
4411	30 files	short	Special event
3200	30 files	short	Speed marker
3111	30 files	short	Perturbation onset
3310	30 files	short	Performance metric
3210	30 files	short	Traffic complexity marker - low
3230	30 files	short	Traffic complexity marker - medium
3240	30 files	short	Traffic complexity marker - high
1211	30 files	short	Lane correction event
1212	30 files	short	Lane correction event
1121-1122	30 files	short	Driver response events
1111-1112	30 files	short	Driver response events
2221-2222	30 files	short	Behavioral response events
2241-2242	30 files	short	Behavioral response events
2251-2252	30 files	short	Behavioral response events
2621-2622	30 files	short	Auditory/behavioral events
2711-2712	30 files	short	Traffic event markers
2811-2812	30 files	short	Traffic event markers

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5

- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Traffic Complexity Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`3210`	0	Low traffic complexity
`3230`	1	Medium traffic complexity
`3240`	2	High traffic complexity

- Note: Traffic complexity conditions (3210/3230/3240) are the primary manipulation variable. Only 30 subjects - small cohort. Full event code decoding requires BCIT codebook.

Data Quality Notes

- 30 subjects - smallest subject count in the BCIT driving series
- 30 distinct event types; full decoding requires BCIT codebook documentation
- Traffic complexity codes (3210/3230/3240) are the key classification targets

References

- Reference: `BCIT Traffic Complexity`

099. openneuro_ds004147 - Average Task Value

Dataset: openneuro_ds004147 | Year: 2024 | URL: https://openneuro.org/datasets/ds004147
Category: High-level Mental | Subjects: 12 | Files (Completed): 12
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

A reinforcement learning task requiring participants to learn cue-response mappings for colored shapes under varying conditions of feedback validity (0.5 vs 0.8 probability). Participants learned which button press was rewarded for each cue, with the value of the correct response varying across conditions. EEG was recorded to investigate neural correlates of value-based learning and the average reward rate representation.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus	12 files	short	Stimulus onset (cue presentation)
New Segment	12 files	admin	Recording segment boundary - exclude
Comment	12 files	admin	Experimenter annotation - exclude

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus-Locked Reinforcement Learning (binary/multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stimulus	0/1	Cue onset; label from behavioral outcome (reward/no reward in events.tsv)

- Note: New Segment and Comment are admin markers and must be excluded. Only 12 subjects - very small dataset. The Stimulus event needs behavioral outcome (correct/incorrect or rewarded/not-rewarded) from events.tsv for supervised learning. Feedback validity condition (0.5 vs 0.8) is available as a task variable.

Data Quality Notes

- Only 12 subjects - very small sample; unsuitable for cross-subject generalization
- New Segment and Comment are admin markers to exclude

- Behavioral outcome labels must be extracted from events.tsv alongside Stimulus events

References

- Reference: `Average Task Value`

100. openneuro_ds004148 - A test-retest resting and cognitive state EEG dataset

Dataset: `openneuro_ds004148` | Year: 2022 | URL: https://openneuro.org/datasets/ds004148
Category: High-level Mental | Subjects: 60 | Files (Completed): 1797 | Channels: 61 | Sampling Rate: 500 Hz

Paradigm

A test-retest reliability study recording EEG during five cognitive/resting conditions: eyes closed (EC), eyes open (EO), mental arithmetic (Ma), memory task (Me), and music listening (Mu), repeated across three sessions. Each file corresponds to one 5-minute condition block, with a single condition marker embedded (S 33=EO, S 44=EC, S 55=Mu, S 66=Me, S 77=Ma). Subjects are healthy adults aged 18-45.

Event Structure

Event ID	Count (total)	Type	Meaning
`S 44`	360	long	Eyes-closed resting state block label
`S 77`	360	long	Mental arithmetic task block label
`S 66`	360	long	Memory task block label
`S 55`	360	long	Music listening block label
`S 33`	357	long	Eyes-open resting state block label
`Sync Off`	~270000	admin	Periodic sync signal (~150/file), not a trial event

> Event Type: short=stimulus onset <2s (epoch_by_event); long=sustained state >=2s (epoch_by_long_event); rest=resting marker (segmentation); admin=exclude from epochs

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Cognitive State Classification (multiclass)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`S 33`	0	Eyes-open resting state
`S 44`	1	Eyes-closed resting state
`S 55`	2	Music listening
`S 66`	3	Memory task
`S 77`	4	Mental arithmetic

- Notes: Each file contains exactly one condition marker; 1797 files x ~150s each. Sync Off is a periodic synchronization signal (150 per file), excluded.

Data Quality Notes

- Sync Off at ~150 events/file is hardware sync, not a task event
- Three test-retest sessions allow reliability analyses

References

- Reference: `A test-retest resting and cognitive state EEG dataset`

101. openneuro_ds004151 - Effect of obesity on inhibitory control in preadolescents during stop-signal task. An event-related potentials study

Dataset: `openneuro_ds004151` | Year: 2022 | URL: https://openneuro.org/datasets/ds004151
Category: Sensory/Response | Subjects: 57 | Files (Completed): 57
Channels: 128 | Sampling Rate: 500 Hz

Paradigm

Preadolescents (obese and normal-weight, ages 9-10) performed a stop-signal task to investigate the impact of obesity on reactive inhibitory control using ERPs. Participants respond to a go stimulus (trigger code 1) but must suppress the response when a stop signal follows (trigger code 2). The study tests whether obese children show electrophysiological differences in inhibition-related brain activity (N2, P3 components) compared to normal-weight peers.

Event Structure

Event ID	Count (total)	Type	Meaning
1	~14,364	short	Go trial stimulus onset
2	~6,156	short	Stop-signal trial (inhibition required)

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Mission 1: Go vs Stop-Signal (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
1	0	Go trial
2	1	Stop-signal trial

- Note: ~252 go trials and ~108 stop trials per subject. Class imbalance (~70% go trials); apply undersampling or weighted loss. Obesity group label available from participants.tsv for group-level analysis. 128-channel EGI cap.

Data Quality Notes

- Event IDs are numeric (1, 2); decoded from stop-signal paradigm description
- 128-channel EGI cap; verify consistent montage across subjects

- Class imbalance: ~70% go trials vs ~30% stop-signal trials

References

- Reference: `Effect of obesity on inhibitory control in preadolescents during stop-signal task (DOI: 10.1016/j.ijpsycho.2021.04.003)`

102. openneuro_ds004152 - Drum Trainer

Dataset: `openneuro_ds004152` | Year: 2022 | URL: https://openneuro.org/datasets/ds004152
Category: Complex Applied | Subjects: 21 | Files (Completed): 21 | Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

Participants learned and performed specific rhythmic drumming patterns at varying tempos using keyboard keys, receiving adaptive visual feedback on timing accuracy. EEG was recorded continuously during drumming training. Each key press generates a Stimulus event marker; the dataset contains thousands of drumming events per participant (2300-4700 Stimulus events per file).

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulus`	~65000	short	Individual drum key press / beat onset
`New Segment`	21	admin	Recording segment boundary
`Comment`	21	admin	Experimenter comment marker

> Event Type: short=stimulus onset <2s (epoch_by_event); long=sustained state >=2s (epoch_by_long_event); rest=resting marker (segmentation); admin=exclude from epochs

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Drum Beat Detection (single-class)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`Stimulus`	0	Drum key press / beat onset

- Notes: Only one event class. Useful for stimulus-locked ERP analysis. New Segment and Comment are admin events, excluded.

Data Quality Notes

- Very high event rate; check for event jitter in timing accuracy analysis
- Only 21 subjects; limited generalizability

References

- Reference: `Drum Trainer EEG dataset`

103. openneuro_ds004196 - Bimodal dataset on Inner speech

Dataset: `openneuro_ds004196` | Year: 2023 | URL: https://openneuro.org/datasets/ds004196
Category: High-level Mental | Subjects: 4 | Files (Completed): 4 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An inner-speech production task where participants mentally vocalized eight specific words from social and numeric categories during nonsimultaneous EEG and fMRI recording. Subjects imagined speaking words without producing overt speech or movement. No trial-level event markers were embedded in the EEG recordings; the data is treated as continuous.

Event Structure

No trial events detected. This dataset contains continuous EEG without trial-level annotations.

> Event Type: short=stimulus onset <2s (epoch_by_event); long=sustained state >=2s (epoch_by_long_event); rest=resting marker (segmentation); admin=exclude from epochs

HDF5 Production Guide

> Label Source: None (no events)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Inner Speech Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: null
- Notes: No event markers available. Sliding window segmentation only.

Data Quality Notes

- Very small dataset (N=4); not suitable for cross-subject generalization
- Consider using fMRI labels aligned to EEG timing if available

References

- Reference: `Bimodal dataset on Inner speech`

104. openneuro_ds004200 - Temporal Scaling

Dataset: openneuro_ds004200 | Year: 2022 | URL: https://openneuro.org/datasets/ds004200
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 37 | Sampling Rate: 1000 Hz

Paradigm

Participants learned to produce and judge short, medium, and long temporal intervals through interleaved production and perception tasks with visual feedback. Trigger codes encode condition combinations: tens digit indicates task type and interval length (1x=production, 2x-6x=other conditions), units digit indicates subcondition or phase. The study investigates how humans scale temporal interval representations.

Event Structure

Event ID	Count (total)	Type	Meaning
11	~20	short	Production short interval phase 1
12	~20	short	Production short interval phase 2
13	~20	short	Production short interval phase 3
21	~20	short	Condition 2, phase 1
22	~20	short	Condition 2, phase 2
23	~20	short	Condition 2, phase 3
31	~20	short	Condition 3, phase 1
32	~20	short	Condition 3, phase 2
33	~20	short	Condition 3, phase 3
41	~20	short	Condition 4, phase 1
42	~20	short	Condition 4, phase 2
43	~20	short	Condition 4, phase 3
51	~20	short	Condition 5, phase 1
52	~20	short	Condition 5, phase 2
53	~20	short	Condition 5, phase 3
61	~20	short	Condition 6, phase 1
62	~20	short	Condition 6, phase 2
63	~20	short	Condition 6, phase 3
17	~20	short	Feedback/response code
27	~20	short	Feedback/response code
47	~20	short	Feedback/response code
57	~20	short	Feedback/response code
67	~20	short	Feedback/response code
2	~20	admin	Block/session boundary marker

> Event Type: short=stimulus onset <2s (epoch_by_event); long=sustained state >=2s (epoch_by_long_event); rest=resting marker (segmentation); admin=exclude from epochs

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Temporal Condition Multiclass (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
11	0	Cond 1 phase 1
12	1	Cond 1 phase 2
13	2	Cond 1 phase 3
21	3	Cond 2 phase 1
22	4	Cond 2 phase 2
23	5	Cond 2 phase 3
31	6	Cond 3 phase 1
32	7	Cond 3 phase 2
33	8	Cond 3 phase 3
41	9	Cond 4 phase 1
42	10	Cond 4 phase 2
43	11	Cond 4 phase 3
51	12	Cond 5 phase 1
52	13	Cond 5 phase 2
53	14	Cond 5 phase 3
61	15	Cond 6 phase 1
62	16	Cond 6 phase 2
63	17	Cond 6 phase 3
17	18	Feedback 1
27	19	Feedback 2
47	20	Feedback 4
57	21	Feedback 5
67	22	Feedback 6

- Notes: Trigger code 2 is a block marker, excluded. Decode exact condition meanings from paper or events.tsv.

Data Quality Notes

- Numerical trigger codes require decode from paper
- Small dataset (N=20)

References

- Reference: `Temporal Scaling`

105. openneuro_ds004252 - Rotation-tolerant representations elucidate the time course of high-level object processing

Dataset: openneuro_ds004252 | Year: 2022 | URL: https://openneuro.org/datasets/ds004252
Category: Sensory/Response | Subjects: 1 | Files (Completed): 1
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

A single participant viewed visual stimuli of objects at various rotations to investigate the neural time course of rotation-tolerant, high-level object representations. A rapid serial visual presentation (RSVP) design was used with three event types encoding distinct stimulus categories or conditions. The study is a single-subject pilot or proof-of-concept recording.

Event Structure

Event ID	Count (total)	Type	Meaning
Event/E 1	14400	short	Stimulus condition 1 (object/rotation category A)
Event/E 2	14400	short	Stimulus condition 2 (object/rotation category B)
Event/E 3	120	short	Stimulus condition 3 (object/rotation category C)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Object Category Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Event/E 1	0	Object/rotation condition 1
Event/E 2	1	Object/rotation condition 2
Event/E 3	2	Object/rotation condition 3

- Remarks: Only 1 subject; not suitable for cross-subject generalization. Event/E 3 count (120) is far lower than E 1 and E 2 (14400 each) - severe class imbalance. Decode exact condition mapping from paper.

Data Quality Notes

- Single subject; results not generalizable across subjects
- 127 channels (1 reference electrode excluded)
- Severe class imbalance: E 1 and E 2 have 14400 trials each vs. E 3 with only 120

References

- Reference: `Rotation-tolerant representations elucidate the time course of high-level object processing`

106. openneuro_ds004256 - Encoding of Sound Source Elevation in Human Cortex

Dataset: openneuro_ds004256 | Year: 2022 | URL: https://openneuro.org/datasets/ds004256
Category: Sensory/Response | Subjects: 53 | Files (Completed): 53
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed an auditory deviant detection task and a one-back sound localization task to investigate the cortical encoding of sound source elevation. Stimuli consisted of sounds presented from different vertical elevations, with subjects indicating perceived location. The study uses a multiclass design with conditions likely corresponding to different elevation angles.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 1	53x320	short	Auditory stimulus condition 1 (elevation angle)
Stimulus/S 2	53x320	short	Auditory stimulus condition 2 (elevation angle)
Stimulus/S 3	53x320	short	Auditory stimulus condition 3 (elevation angle)
Stimulus/S 4	53x320	short	Auditory stimulus condition 4 (elevation angle)
Stimulus/S 5	23x320	short	Auditory stimulus condition 5 (elevation angle)
Stimulus/S 6	23x320	short	Auditory stimulus condition 6 (elevation angle)
Stimulus/S 7	23x96	short	Auditory stimulus condition 7 (deviant/rare)
New Segment/	53	admin	Recording segment boundary
BAD boundary	53	admin	Bad epoch boundary marker
EDGE boundary	53	admin	Edge boundary artifact marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Sound Elevation Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`Stimulus/S 1`	0	Auditory stimulus condition 1
`Stimulus/S 2`	1	Auditory stimulus condition 2
`Stimulus/S 3`	2	Auditory stimulus condition 3
`Stimulus/S 4`	3	Auditory stimulus condition 4
`Stimulus/S 5`	4	Auditory stimulus condition 5
`Stimulus/S 6`	5	Auditory stimulus condition 6
`Stimulus/S 7`	6	Auditory stimulus condition 7 (deviant)

- Note: S 5/S 6/S 7 only present in 23 of 53 subjects; consider two-task splits. BAD boundary and EDGE boundary are admin events - exclude from target_events.

Data Quality Notes

- S 5, S 6, S 7 appear in only 23 subjects (possibly a different task condition block)
- BAD boundary and EDGE boundary markers must not be used as target events
- Balance classes by undersampling; verify event counts per subject before pooling

References

- Reference: `Encoding of Sound Source Elevation in Human Cortex`

107. openneuro_ds004262 - Continuous Feedback Processing

Dataset: openneuro_ds004262 | Year: 2022 | URL: https://openneuro.org/datasets/ds004262
Category: High-level Mental | Subjects: 21 | Files (Completed): 21
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

Participants predicted the final height of an animated rising bar based on cues with varying predictability to study continuous feedback processing and prediction error. Each trial involved observing bar motion and registering a prediction, allowing examination of EEG correlates of reward prediction error and uncertainty-driven learning.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus	21x900	short	Stimulus onset marker (bar appearance)
New Segment	21	admin	Recording segment boundary
Comment	21	admin	Experimenter comment/annotation

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus Processing (single-class)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stimulus	0	Stimulus onset (bar presentation)

- Note: Only one event class available for epoching. New Segment and Comment are admin markers. Behavioral metadata (prediction, actual height) may be available in events.tsv for regression tasks.

Data Quality Notes

- Only one non-admin event type (Stimulus); classification requires additional label parsing from events.tsv metadata columns
- New Segment and Comment are system/admin events excluded from epoching

References

- Reference: `Continuous Feedback Processing`

108. openneuro_ds004264 - Steer the Ship

Dataset: openneuro_ds004264 | Year: 2022 | URL: https://openneuro.org/datasets/ds004264
Category: Complex Applied | Subjects: 21 | Files (Completed): 21
Channels: 31 | Sampling Rate: 1000 Hz

Paradigm

A ship-steering simulation where participants control a rotating arrow to maintain orientation against varying environmental and controller noise using unstable pole-and-cart physics. The task involves continuous manual control with varying noise levels, making it a real-world closed-loop BCI paradigm. EEG was recorded to study cognitive load and motor control under dynamic uncertainty.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus	21x1000+	short	Stimulus/condition onset marker
New Segment	21	admin	Recording segment boundary
Comment	21	admin	Experimenter comment/annotation

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus Processing (single-class)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Stimulus	0	Stimulus/condition onset

- Note: Only one event class available for epoching. New Segment and Comment are admin markers. Behavioral metadata (noise level, performance metrics) may be available in events.tsv for regression tasks.

Data Quality Notes

- Only one non-admin event type (Stimulus); classification requires additional label parsing from events.tsv metadata columns
- New Segment and Comment are system/admin events excluded from epoching
- Variable trial counts across subjects (1473-2373 per subject)

References

- Reference: `Steer the Ship`

109. openneuro_ds004279 - Large Spanish EEG

Dataset: openneuro_ds004279 | Year: 2024 | URL: https://openneuro.org/datasets/ds004279
Category: High-level Mental | Subjects: 56 | Files (Completed): 60
Channels: 69 | Sampling Rate: 1000 Hz

Paradigm

Participants performed speech perception and silent speech production tasks involving 30 daily Spanish sentences. Each sentence trial followed a fixed structure: a rest period, a preparation cue, a perception phase (hearing the sentence), and a production phase (silently producing the sentence). The study investigated EEG correlates of perceived and silently produced speech.

Event Structure

Event ID	Count (total)	Type	Meaning
perception_N (N=1-30)	60x7	long	Perception phase onset for sentence N
production_N (N=1-30)	60x7	long	Silent production phase onset for sentence N
preparation_N (N=1-30)	60x7	long	Preparation cue onset for sentence N
rest_N (N=1-30)	60x7	rest	Rest period onset for sentence N
bad_artifact	60xvariable	admin	Bad artifact annotation
Impedance_test	60x2	admin	Impedance check marker
bad_before_experiment	60x1	admin	Pre-experiment bad segment
bad_after_experiment	60x1	admin	Post-experiment bad segment

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Perception vs Production Classification (binary)

- Epoch method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5
- Target Events -> y:

Event	y	Meaning
perception_1...perception_30	0	Speech perception phase
production_1...production_30	1	Silent speech production phase

- Note: Group all perception_N events as class 0 and all production_N events as class 1. Each sentence has 7 repetitions per subject. bad_artifact, Impedance_test, and bad_* events are admin markers excluded from epoching.

Data Quality Notes

- 90+ distinct event types (perception/production/preparation/rest for 30 sentences)
- bad_artifact count varies by subject (31-39 per file)
- Group sentence-level events by phase type for practical classification

References

- Reference: `Valle, C., Mendez-Orellana, C., Herff, C., & Rodriguez-Fernandez, M. (2024). Identification of perceived sentences using deep neural networks in EEG. Journal of neural engineering, 21(5), 056044.`

110. openneuro_ds004284 - eeg-neuroforecasting

Dataset: `openneuro_ds004284` | Year: 2022 | URL: https://openneuro.org/datasets/ds004284
Category: High-level Mental | Subjects: N/A | Files (Completed): 18
Channels: 129.0 | Sampling Rate: 1000.0 Hz

Paradigm

A neuroforecasting task that utilizes EEG activity during decision-making or valuation to predict aggregate, population-level outcomes.

Event Structure

Event ID	Count (files)	Meaning
`image`	18	Visual stimulus
## Downstream Task Recommendations		

Classification Tasks

- Insufficient distinct event types for classification (only 1 event class found).

Regression Tasks

- No clear regression target identified from available event structure. Consider reaction time if behavioral logs are available.

Data Quality Notes

- Verify channel count consistency across subjects before pooling.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG-BIDS, an extension to the brain imaging data structure for electroencephalography`

111. openneuro_ds004295 - Reward gain and punishment avoidance reversal learning

Dataset: openneuro_ds004295 | Year: 2022 | URL: https://openneuro.org/datasets/ds004295
Category: High-level Mental | Subjects: 26 | Files (Completed): 26
Channels: 66 | Sampling Rate: 1024 Hz

Paradigm

A reversal learning task comparing brain activity during reward gain (monetary) versus punishment avoidance (white noise) reinforcement. Participants learned stimulus-response associations that reversed mid-task, requiring cognitive flexibility. EEG was recorded to investigate feedback-related neural signatures under different reinforcement conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
S 81	26x~164	short	Stimulus trigger - decode from events.tsv
S 61	26x~162	short	Stimulus trigger - decode from events.tsv
S 75	26x~153	short	Stimulus trigger - decode from events.tsv
S 71	26x~148	short	Stimulus trigger - decode from events.tsv
S 52	26x~141	short	Stimulus trigger - decode from events.tsv
S 51	26x~139	short	Stimulus trigger - decode from events.tsv
S 55	26x~137	short	Stimulus trigger - decode from events.tsv
S 72	26x~132	short	Stimulus trigger - decode from events.tsv
S 60	26x~118	short	Stimulus trigger - decode from events.tsv
S 80	26x~116	short	Stimulus trigger - decode from events.tsv
S100	26x~1	short	Sample stimulus (first in pair)
S200	26x~1	short	Probe stimulus (second in pair)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: dedicated kernel (`kernels/openneuro_ds004295.py`)-needs to decode BrainVision trigger codes

Task 1: Reversal Learning Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`stimulus_A_prereversalReward`	0	Stimulus A before reversal (reward condition)
`stimulus_B_prereversalPunishment`	1	Stimulus B before reversal (punishment condition)
`correct_response`	2	Correct response
`incorrect_response`	3	Incorrect response
`reversal_cue`	4	Reversal signal
`stimulus_A_postreversalPunishment`	5	Stimulus A after reversal (punishment condition)
`feedback_positive`	6	Positive feedback
`feedback_negative`	7	Negative feedback

- Note: Specific kernel required to decode BrainVision trigger codes (S 51-S 81 range) to semantic labels. Regression target: response_time from misc channels.

Data Quality Notes

- BrainVision trigger codes (S 51-S 81 range) require kernel-level decoding to semantic labels
- S150/S151 appear in only 22 subjects, possibly a separate task block

References

- Reference: `Reward gain and punishment avoidance reversal learning`

112. openneuro_ds004315 - Mood Manipulation and PST, Experiment 1

Dataset: openneuro_ds004315 | Year: 2022 | URL: https://openneuro.org/datasets/ds004315
Category: High-level Mental | Subjects: 50 | Files (Completed): 50
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task (Probabilistic Selection Task) consisting of training and testing phases following a sad or neutral mood manipulation. Participants learned to select stimuli associated with reward or punishment, and were subsequently tested on novel stimulus pairs. EEG was recorded to examine how mood state affects the neural correlates of reinforcement learning.

Event Structure

Event ID	Count (total)	Type	Meaning
STIM	50x~240	short	Stimulus onset (stimulus pair presented)
RESP	50x~230	short	Participant button-press response
FB	50x~240	short	Feedback onset (reward/punishment)
Correct	50x~200	short	Correct response outcome
Incorrect	50x~85	short	Incorrect response outcome
TESTING	50x~1700	short	Testing phase trial marker
STATUS	50	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Feedback Outcome Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Correct	0	Correct response
Incorrect	1	Incorrect response

- Note: STATUS is an admin marker (exclude). TESTING events have rich sub-labels in the actual events.tsv (e.g., "TESTING: Right Press, Correct") useful for further analysis. Class imbalance: Correct >> Incorrect.

Data Quality Notes

- STATUS is an admin/system marker and must not be included in target events
- TESTING events contain rich sub-labels in events.tsv; the trial_type column provides full label strings
- Class imbalance between Correct (~200) and Incorrect (~55 training) responses

References

- Reference: `Mood Manipulation and PST, Experiment 1`

113. openneuro_ds004317 - Mood Manipulation and PST, Experiment 2

Dataset: openneuro_ds004317 | Year: 2022 | URL: https://openneuro.org/datasets/ds004317
Category: High-level Mental | Subjects: 50 | Files (Completed): 50
Channels: 66 | Sampling Rate: 500 Hz

Paradigm

A reinforcement learning task (Probabilistic Selection Task) performed by participants following sad or happy mood manipulations. The paradigm mirrors Experiment 1 but compares sad and happy mood conditions rather than sad versus neutral. EEG was recorded to examine mood-dependent differences in reinforcement learning neural signatures.

Event Structure

Event ID	Count (total)	Type	Meaning
STIM	50x~240	short	Stimulus onset (stimulus pair presented)
RESP	50x~230	short	Participant button-press response
FB	50x~240	short	Feedback onset (reward/punishment)
Correct	50x~198	short	Correct response outcome
Incorrect	50x~100	short	Incorrect response outcome
TESTING	50x~1664	short	Testing phase trial marker
STATUS	50	admin	System status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Feedback Outcome Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Correct	0	Correct response
Incorrect	1	Incorrect response

- Note: STATUS is an admin marker (exclude). TESTING events contain rich sub-labels in events.tsv. Better class balance than Experiment 1 (~198 correct vs ~100 incorrect in training phase).

Data Quality Notes

- STATUS is an admin/system marker and must not be included in target events
- TESTING events contain rich sub-labels in events.tsv; the trial_type column provides full label strings
- Better class balance than Experiment 1

References

- Reference: `Mood Manipulation and PST, Experiment 2`

114. openneuro_ds004324 - ToonFaces

Dataset: openneuro_ds004324 | Year: 2023 | URL: https://openneuro.org/datasets/ds004324
Category: High-level Mental | Subjects: 26 | Files (Completed): 26
Channels: 38 | Sampling Rate: 500 Hz

Paradigm

Participants viewed rapid serial visual presentation (RSVP) streams of three types of face images - real photographs (img), stylized toon faces (toon), and schematic stick-figure faces (sf) - each shown at different emotional intensity levels (0-4). Subjects evaluated emotion detection and affective states, allowing ERP-based comparison across face types and intensity levels.

Event Structure

Event ID	Count (total)	Type	Meaning
img/0	26x45	short	Real face image, intensity 0 (neutral)
img/1	26x195	short	Real face image, intensity 1
img/2	26x15	short	Real face image, intensity 2
img/3	26x15	short	Real face image, intensity 3
img/4	26x30	short	Real face image, intensity 4 (most intense)
toon/0	26x45	short	Toon face, intensity 0 (neutral)
toon/1	26x195	short	Toon face, intensity 1
toon/2	26x15	short	Toon face, intensity 2
toon/3	26x15	short	Toon face, intensity 3
toon/4	26x30	short	Toon face, intensity 4
sf/0	26x45	short	Schematic face, intensity 0 (neutral)
sf/1	26x195	short	Schematic face, intensity 1
sf/2	26x15	short	Schematic face, intensity 2
sf/3	26x15	short	Schematic face, intensity 3
sf/4	26x30	short	Schematic face, intensity 4
BlockEnd	26x3	admin	End of block marker
block1	26	admin	Block 1 start marker
block2	26	admin	Block 2 start marker
block3	26	admin	Block 3 start marker
Intro	22	admin	Introduction screen marker
Instructions1	25	admin	Instructions screen 1
Instructions2	25	admin	Instructions screen 2
Instructions3	25	admin	Instructions screen 3
PracticeInstructions	25	admin	Practice instructions marker

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Face Type Classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`img/0...img/4`	0	Real photograph face
`toon/0...toon/4`	1	Stylized toon face
`sf/0...sf/4`	2	Schematic stick-figure face

- Note: Block markers (block1/2/3, BlockEnd) and instruction markers are admin events - exclude from target_events. Intensity sub-levels can be used for finer-grained multiclass analysis.

Data Quality Notes

- Block/instruction markers must be excluded from epoching
- Unequal counts across intensity levels (intensity 1: 195 vs intensity 2/3: 15 per face type)
- 26 subjects; consistent event structure across subjects

References

- Reference: `Images of stylized faces improve ERP features used for emotion detection`

115. openneuro_ds004347 - Symmetry perception and affective responses: a combined EEG/EMG study

Dataset: openneuro_ds004347 | Year: 2022 | URL: https://openneuro.org/datasets/ds004347
Category: Sensory/Response | Subjects: 24 | Files (Completed): 24
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

Participants performed a visual perception task identifying whether stimuli were regular (symmetric) or random (asymmetric) patterns to investigate symmetry perception and affective responses. EEG and EMG were simultaneously recorded to capture both neural and facial muscle activity. No trial-level event markers are present in the EEG files; the dataset is processed as continuous recordings.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (resting/segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: No trial markers in EEG files; use sliding window segmentation. Subject-level labels (symmetric vs. random condition blocks) may be available from experimental design metadata.

Data Quality Notes

- No trial-level event markers in the EEG files
- 73 channels includes EMG/EOG channels that should be excluded from EEG analysis
- Continuous recordings suitable for spectral or connectivity analysis

References

- Reference: Symmetry perception and affective responses: a combined EEG/EMG study.

116. openneuro_ds004348 - Ear-EEG Sleep Monitoring 2017 (EESM17)

Dataset: `openneuro_ds004348` | Year: 2024 | URL: https://openneuro.org/datasets/ds004348
Category: Biological State | Subjects: 9 | Files (Completed): 36
Channels: 35 | Sampling Rate: 200 Hz

Paradigm

Nightly sleep monitoring and pre-sleep resting recordings using ear-EEG and partial polysomnography in a naturalistic home environment. Subjects wore ear-mounted EEG devices during sleep, allowing investigation of sleep stage classification and comparison between ear-EEG and conventional PSG signals.

Event Structure

Event ID	Count (total)	Type	Meaning
`Lights Off`	9	long	Lights-off marker (sleep onset reference)
`Lights On`	9	long	Lights-on marker (sleep end reference)
`Awake`	1	long	Wakefulness annotation
`Artifact`	9	admin	Artifact annotation
`Arousal`	9	admin	Arousal annotation
`Beginning of Recording`	9	admin	Recording start marker
`End of Study`	9	admin	Recording end marker
`Montage`	9	admin	Montage configuration event
`Started Analyzer - Sleep Events`	9	admin	Analyzer start marker
`Started Analyzer - ECG`	9	admin	ECG analyzer start
`Stopped Analyzer - ECG`	9	admin	ECG analyzer stop
`Started Analyzer - Auto-Staging`	9	admin	Auto-staging start
`Stopped Analyzer - Auto-Staging`	9	admin	Auto-staging stop
`512`	9	admin	Numerical trigger code 512

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: N/A - no structured trial events; segmentation-based processing
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Sleep Staging (long event)

- Epoch method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`Lights Off`	0	Sleep period start
`Lights On`	1	Sleep period end

- Note: Sleep stage labels (N1/N2/N3/REM) are typically stored in annotation files, not event markers. Use Lights Off/On as block boundaries. The dataset is primarily suitable for sleep staging classification.

Data Quality Notes

- Sleep stage annotations are in PSG annotation files, not directly as trial events
- Artifact markers must be excluded from analysis windows
- 35 channels includes non-EEG channels (ECG, respiratory belt, etc.)
- Only 9 subjects; limited generalization

References

- Reference: `Ear-EEG Sleep Monitoring 2017 (EESM17)`

117. openneuro_ds004350 - Executive Functioning Study for Assessing the Effect of Neurofeedback

Dataset: openneuro_ds004350 | Year: 2025 | URL: https://openneuro.org/datasets/ds004350
Category: High-level Mental | Subjects: 24 | Files (Completed): 480
Channels: 64 | Sampling Rate: 256 Hz

Paradigm

Participants performed a battery of executive function tasks including a visual N-back working memory task (stimuli A/B/C/D), a Sustained Attention to Response Task (SART, target/nontarget), and a Local-Global hierarchical stimulus task (largeH/largeT x smallH/smallT). The study assessed the effect of neurofeedback training on executive function performance.

Event Structure

Event ID	Count (total)	Type	Meaning
target	44xx	short	Target stimulus (requires response)
nontarget	44xx	short	Non-target stimulus (no-go)
A	144xx	short	N-back stimulus A
B	144xx	short	N-back stimulus B
C	144xx	short	N-back stimulus C
D	144xx	short	N-back stimulus D
largeHSmallH	48xx	short	Large H / Small H (local-global congruent)
largeHSmallT	48xx	short	Large H / Small T (local-global incongruent)
largeTSmallH	48xx	short	Large T / Small H (local-global incongruent)
largeTSmallT	48xx	short	Large T / Small T (local-global congruent)
responseLargeH	48xx	short	Response: Large H selected
responseLargeT	48xx	short	Response: Large T selected
responseSmallH	48xx	short	Response: Small H selected
responseSmallT	48xx	short	Response: Small T selected
rt	188xx	short	Reaction time marker
255	2	admin	System trigger code 255

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: SART Target Detection (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`nontarget`	0	Non-target (go) stimulus
`target`	1	Target (no-go) stimulus

- Note: Numerical codes (1, 2, 4, 5, 6, 8, 9, 10, 11, 12, 128, 255) are likely system/artifact markers - exclude from target_events. Multiple tasks in one dataset; files are per-task. Separate HDF5 files per task recommended.

Data Quality Notes

- Multiple distinct tasks (SART, N-back, Local-Global) in same dataset; files are per-task
- Numerical trigger codes (1-255) are system markers, not task events
- 480 files across 24 subjects (multiple sessions/tasks per subject)

References

- Reference: `Executive Functionning Study for Assessing the Effect of Neurofeedback`

118. openneuro_ds004356 - Subcortical responses to music and speech are alike while cortical responses diverge

Dataset: openneuro_ds004356 | Year: 2024 | URL: https://openneuro.org/datasets/ds004356
Category: Sensory/Response | Subjects: 22 | Files (Completed): 24
Channels: 34 | Sampling Rate: 10000 Hz

Paradigm

The study measures subcortical Auditory Brainstem Responses (ABR) to continuous clicks, music genres, and speech stimuli using a deconvolution method. Participants passively listened to 12 different audio categories (Chinese speech, English speech, 8 music genres, acoustic guitar, and click trains) to compare subcortical and cortical neural encoding across stimulus types.

Event Structure

Event ID	Count (total)	Type	Meaning
click	22x10	long	Continuous click train stimulus
acoustic	22x40	long	Acoustic guitar stimulus
classical	22x40	long	Classical music stimulus
jazz	22x40	long	Jazz music stimulus
pop	22x40	long	Pop music stimulus
metal	22x40	long	Metal music stimulus
hiphop	22x40	long	Hip-hop music stimulus
chn_aud	22x40	long	Chinese speech stimulus
eng_aud	22x40	long	English speech stimulus
news	22x40	long	News speech stimulus
talk	22x40	long	Talk speech stimulus
interview	22x40	long	Interview speech stimulus
lecture	22x40	long	Lecture speech stimulus

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Music vs Speech Classification (binary)

- Epoch method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5
- Target Events -> y:

Event	y	Meaning
acoustic	0	Music stimulus
classical	0	Music stimulus
jazz	0	Music stimulus
pop	0	Music stimulus
metal	0	Music stimulus
hiphop	0	Music stimulus
chn_aud	1	Speech stimulus
eng_aud	1	Speech stimulus
news	1	Speech stimulus
talk	1	Speech stimulus
interview	1	Speech stimulus
lecture	1	Speech stimulus

- Note: Each stimulus block is a continuous audio segment (many seconds); use long event epoching. Sampling rate is 10 kHz (ABR recording). Consider subcortical ABR analysis vs. cortical EEG analysis separately.

Data Quality Notes

- Very high sampling rate (10 kHz) designed for subcortical ABR; bandpass filtering required for cortical analysis
- 2 subjects have 2 files each (24 files, 22 subjects)
- Balanced stimulus presentation (40 trials per condition for most categories)

References

- Reference: Subcortical responses to music and speech are alike while cortical responses diverge (Shan et al., 2024)

119. openneuro_ds004357 - Features-EEG

Dataset: openneuro_ds004357 | Year: 2024 | URL: https://openneuro.org/datasets/ds004357
Category: Sensory/Response | Subjects: 16 | Files (Completed): 16
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed rapid serial visual presentation (RSVP) streams of Gabor-like stimuli at 6.67 Hz and 20 Hz while performing an orthogonal fixation color change detection task. Three event types encode different feature or condition categories, enabling investigation of how visual features are dynamically encoded in EEG responses.

Event Structure

Event ID	Count (total)	Type	Meaning
Event/E 1	16x~16000	short	Stimulus feature condition 1
Event/E 2	16x~16000	short	Stimulus feature condition 2
Event/E 3	16x~60	short	Stimulus feature condition 3 (rare)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Visual Feature Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Event/E 1	0	Stimulus feature condition 1
Event/E 2	1	Stimulus feature condition 2
Event/E 3	2	Stimulus feature condition 3

- Note: Event/E 3 count (~60 per file) is far lower than E 1 and E 2 (~16000 each) - severe class imbalance. Decode exact condition mapping from paper. Similar structure to ds004252.

Data Quality Notes

- Severe class imbalance: E 1 and E 2 have ~16000 trials each vs. E 3 with ~60 per subject
- 127-channel high-density EEG
- RSVP paradigm at 6.67 Hz and 20 Hz; stimuli are very brief

References

- Reference: Mapping the Dynamics of Visual Feature Coding: Insights into Perception and Integration

120. openneuro_ds004367 - Meta-rdk: Raw EEG data

Dataset: openneuro_ds004367 | Year: 2022 | URL: https://openneuro.org/datasets/ds004367
Category: High-level Mental | Subjects: 42 | Files (Completed): 40
Channels: 68 | Sampling Rate: 1200 Hz

Paradigm

A visual motion discrimination task involving left/right direction decisions on random dot kinematogram (RDK) stimuli followed by a metacognitive confidence assessment using a visual analog scale. The numerical trigger codes encode stimulus direction, coherence level, response direction, confidence level, and feedback. The dataset supports both motion perception classification and confidence/metacognition regression.

Event Structure

Event ID	Count (total)	Type	Meaning
3072	40x300	short	Motion stimulus onset (specific direction/coherence)
3456	40x300	short	Motion stimulus onset (specific direction/coherence)
3584	40x300	short	Motion stimulus onset (specific direction/coherence)
3968	40x300	short	Motion stimulus onset (specific direction/coherence)
4096	40x300	short	Motion stimulus onset (specific direction/coherence)
4480	40x300	short	Motion stimulus onset (specific direction/coherence)
31744...25984	40x10 each	short	Response/confidence marker (numerical code)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Special kernel (kernels/openneuro_ds004367.py) - needs to decode digital trigger code

Task 1: Motion Direction Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)

- Target Events -> y:

Event	y	Meaning
`motion_stimulus_onset`	0	Motion stimulus onset
`left_response`	1	Left direction response
`right_response`	2	Right direction response
`confidence_low`	3	Low confidence rating
`confidence_medium`	4	Medium confidence rating
`confidence_high`	5	High confidence rating
`feedback_onset`	6	Feedback onset

- Note: Numerical trigger codes require specific kernel decoding. Regression target: response_time and confidence from misc channels.

Data Quality Notes

- All event IDs are numerical codes requiring kernel-level decoding
- 68 channels includes EOG/ECG channels
- 42 subjects but only 40 files completed

References

- Reference: `Meta-rdk: Raw EEG data`

121. openneuro_ds004369 - Blink-Pause-Relation (Competing Speaker Paradigm)

Dataset: openneuro_ds004369 | Year: 2023 | URL: https://openneuro.org/datasets/ds004369
Category: High-level Mental | Subjects: 41 | Files (Completed): 41
Channels: 7 | Sampling Rate: 500 Hz

Paradigm

A two-competing speaker paradigm where participants selectively attended to one of two concurrently presented audiobooks while ignoring the other. EEG was recorded to study the relationship between eye blinks and speech pauses during continuous auditory attention. No trial-level event markers are present in the EEG files.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (resting/segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: No trial markers in EEG files; use sliding window segmentation. Only 7 channels (very low-density). Subject-level labels (attended speaker, blink/pause occurrence) may be available from behavioral logs.

Data Quality Notes

- Very sparse channel setup (7 channels only)
- No trial-level event markers in the EEG files
- Continuous naturalistic listening paradigm (audiobook)

References

- Reference: Jaeger et al. (2020); Holtze et al. (2021)

122. openneuro_ds004381 - Intraoperative EEG dataset during medianus-tibialis stimulation with 8 different rates

Dataset: openneuro_ds004381 | Year: 2024 | URL: https://openneuro.org/datasets/ds004381
Category: Sensory/Response | Subjects: 18 | Files (Completed): 770
Channels: 4 | Sampling Rate: 20000 Hz

Paradigm

Intraoperative recording of Somatosensory Evoked Potentials (SEP) elicited by medianus and tibial nerve stimulation at eight different repetition rates to optimize signal-to-noise ratio for intraoperative monitoring. Recordings were made during neurosurgery using a minimal-channel invasive EEG setup with very high sampling rate for subcortical potential resolution.

Event Structure

No event markers detected (no_events). This dataset contains continuous SEP recordings without trial-level BIDS annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: No BIDS trial event markers; stimulation rate (8 levels) encoded in filename/metadata. SEP analysis requires specialized averaging pipeline rather than standard epoch classification.

Data Quality Notes

- Very high sampling rate (20 kHz) for subcortical SEP; downsampling required for standard EEG analysis
- Only 4 channels (intraoperative minimal setup)
- 770 files across 18 subjects (many runs per subject, 8 stimulation rates)

References

- Reference: Optimization of signal-to-noise ratio in short-duration SEP recordings by variation of stimulation rate

123. openneuro_ds004388 - Somatosensory evoked potentials in the human spinal cord to mixed nerve stimulation

Dataset: openneuro_ds004388 | Year: 2023 | URL: https://openneuro.org/datasets/ds004388
Category: Sensory/Response | Subjects: 40 | Files (Completed): 399
Channels: 115 | Sampling Rate: 10000 Hz

Paradigm

The dataset records somatosensory evoked potentials (SEPs) in response to mixed median and tibial nerve stimulation, alongside a resting state condition. Simultaneous EEG and epidural spinal cord recordings (ESG) were acquired to characterize the spinal and cortical processing of somatosensory signals.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulus	399x~500	long	Electrical nerve stimulation onset

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: SEP Stimulus Epoching (long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5
- Target Events -> y:

Event	y	Meaning
stimulus	0	Nerve stimulation onset

- Note: Only one event class; suitable for evoked potential analysis or regression on stimulation parameters. Multiple nerve sites and stimulation rates encoded in file naming conventions.

Data Quality Notes

- Very high sampling rate (10 kHz) for SEP/ESG; downsampling required for standard EEG analysis
- 115 channels includes spinal cord (ESG) and peripheral nerve (ENG/EMG) channels
- 399 files across 40 subjects (multiple runs/conditions per subject)

References

- Reference: Somatosensory evoked potentials in the human spinal cord to mixed nerve stimulation

124. openneuro_ds004389 - Somatosensory evoked potentials in the human spinal cord to mixed and sensory nerve stimulation

Dataset: openneuro_ds004389 | Year: 2023 | URL: https://openneuro.org/datasets/ds004389
Category: Sensory/Response | Subjects: 26 | Files (Completed): 260
Channels: 90 | Sampling Rate: 10000 Hz

Paradigm

Simultaneous EEG, epidural spinal cord (ESG), nerve conduction (ENG), and EMG recordings during resting state and various conditions of mixed and sensory nerve stimulation of the arms and legs to evoke somatosensory potentials. The study compares spinal cord vs. cortical potentials for different stimulation types and sites.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulus	260x~1500	long	Electrical nerve stimulation onset

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: SEP Stimulus Epoching (long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s (relative to event onset), overlap = 0.5
- Target Events -> y:

Event	y	Meaning
stimulus	0	Nerve stimulation onset

- Remarks: Only one event class; suitable for evoked potential analysis. Multiple nerve sites (median mixed, median sensory, tibial) and stimulation conditions encoded in file/run naming.

Data Quality Notes

- Very high sampling rate (10 kHz) for SEP/ESG; downsampling required for standard EEG analysis
- 90 channels includes ESG, ENG, EMG, EOG, ECG channels - filter to EEG channels only
- 260 files across 26 subjects (multiple runs/conditions per subject)

References

- Reference: Somatosensory evoked potentials in the human spinal cord to mixed and sensory nerve stimulation.

125. openneuro_ds004395 - Penn Electrophysiology of Encoding and Retrieval Study (PEERS)

Dataset: openneuro_ds004395 | Year: 2023 | URL: https://openneuro.org/datasets/ds004395
Category: High-level Mental | Subjects: 300 | Files (Completed): 6480
Channels: 129 | Sampling Rate: 500 Hz

Paradigm

A large-scale study investigating the electrophysiological correlates of memory encoding and retrieval using various free recall tasks. Participants encoded word lists and subsequently recalled them (free recall), also performing recognition memory tasks. The study provides a rich dataset for ERP analysis of encoding success, retrieval effort, and memory-related oscillations.

Event Structure

Event ID	Count (total)	Type	Meaning
WORD	6480x~256	short	Word presentation onset (encoding)
REC_WORD	6480x~100	short	Recalled word onset (retrieval)
RECOG_TARGET	3581x~160	short	Recognition target word onset
RECOG_LURE	3581x~80	short	Recognition lure word onset
RECOG_RESP	3581x~320	short	Recognition response
RECOG_CONF	3563x~320	short	Confidence rating
DISTRACTOR	2439xx	short	Distractor task stimulus
REC_WORD_VV	5001xx	short	Recalled word (voice-verified)
REST_REWET	5955x1	rest	Rest/reward-with-eye-tracking marker
SESS_START	6478	admin	Session start marker
SESS_END	6474	admin	Session end marker
REC_START	6480	admin	Recall period start
START	3668	admin	Start marker
STOP	3668	admin	Stop marker
BREAK_START	459	admin	Break start marker
BREAK_STOP	459	admin	Break stop marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Memory Encoding Classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`RECOG_TARGET`	0	Recognition target (previously seen)
`RECOG_LURE`	1	Recognition lure (not previously seen)

- Note: SESS_START/END, REC_START, START/STOP, BREAK_START/STOP are admin markers - exclude. REST_REWET is a rest marker. Very large dataset (300 subjects, 6480 files); suitable for large-scale ERP studies.

Data Quality Notes

- Very large dataset (300 subjects, 6480 files); processing time is substantial
- Session-level admin markers (SESS_START/END) must be excluded from epoching
- Multiple sub-tasks with different event structures; separate HDF5 per sub-task recommended

References

- Reference: `Penn Electrophysiology of Encoding and Retrieval Study (PEERS)`

126. openneuro_ds004408 - EEG responses to continuous naturalistic speech

Dataset: openneuro_ds004408 | Year: 2023 | URL: https://openneuro.org/datasets/ds004408
Category: High-level Mental | Subjects: 380 | Files (Completed): 380
Channels: 128 | Sampling Rate: 512 Hz

Paradigm

Participants listened to continuous segments of an audiobook version of 'The Old Man and the Sea' to record EEG responses to naturalistic narrative speech and language processing. The study investigates low-frequency cortical entrainment to speech and phoneme-level processing. No trial-level event markers are present; recordings are continuous.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings during naturalistic speech listening without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Remarks: No trial markers; use sliding window segmentation. Large dataset (380 files). Suitable for EEG-speech alignment analysis using stimulus audio envelope.

Data Quality Notes

- No trial-level event markers; continuous listening paradigm
- Very large dataset (380 files); batch processing required
- 128-channel EEG at 512 Hz

References

- Reference: Low-frequency cortical entrainment to speech reflects phoneme-level processing

127. openneuro_ds004444 - The BMI-HDEEG dataset 1

Dataset: openneuro_ds004444 | Year: 2023 | URL: https://openneuro.org/datasets/ds004444
Category: Action/Output | Subjects: 465 | Files (Completed): 465
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

Brain-computer interfacing task based on sensorimotor rhythms (SMR) using high-density scalp EEG. Subjects performed motor imagery of four limb movements (left hand, right hand, feet, tongue) following visual cues. This is the first of four BMI-HDEEG datasets, enabling classification of motor imagery across all four classes.

Event Structure

Event ID	Count (total)	Type	Meaning
1	465xx	long	Motor imagery cue - class 1 (left hand MI)
2	465xx	long	Motor imagery cue - class 2 (right hand MI)
3	465xx	long	Motor imagery cue - class 3 (feet MI)
4	465xx	long	Motor imagery cue - class 4 (tongue MI)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Specialized kernel (kernels/openneuro_ds004444.py) - decodes MI category tags

Task 1: Motor Imagery Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.5 s, tmax = 4.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
left_hand_MI	0	Left hand motor imagery
right_hand_MI	1	Right hand motor imagery
feet_MI	2	Feet motor imagery
tongue_MI	3	Tongue motor imagery

- Note: Numerical codes 1-4 decoded by specific kernel to semantic MI labels. Extended epoch window (-0.5 to 4.0 s) captures full motor imagery period. High-density 130-channel EEG.

Data Quality Notes

- Numerical event codes (1-4) require kernel-level decoding to semantic MI labels
- 130-channel high-density EEG with extended epochs
- Large dataset (465 files); batch processing required

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)`

128. openneuro_ds004446 - The BMI-HDEEG dataset 2

Dataset: openneuro_ds004446 | Year: 2023 | URL: https://openneuro.org/datasets/ds004446
Category: Action/Output | Subjects: 237 | Files (Completed): 237
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

A high-density EEG dataset focused on sensorimotor rhythm-based brain-computer interfacing (BCI) with three motor imagery classes (left hand, right hand, feet). This is the second of four BMI-HDEEG datasets, using the same high-density setup as dataset 1 but with a three-class motor imagery design.

Event Structure

Event ID	Count (total)	Type	Meaning
1	237xx	long	Motor imagery cue - class 1 (left hand MI)
2	237xx	long	Motor imagery cue - class 2 (right hand MI)
3	237xx	long	Motor imagery cue - class 3 (feet MI)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Specialized kernel (kernels/openneuro_ds004446.py) - decodes MI category tags

Task 1: Motor Imagery Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.5 s, tmax = 4.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
left_hand_MI	0	Left hand motor imagery
right_hand_MI	1	Right hand motor imagery
feet_MI	2	Feet motor imagery

- Note: Numerical codes 1-3 decoded by specific kernel to semantic MI labels. Same paradigm as dataset 1 (BMI-HDEEG) but 3-class design and different subjects.

Data Quality Notes

- Numerical event codes (1-3) require kernel-level decoding to semantic MI labels
- 130-channel high-density EEG with extended epochs
- 237 files; smaller than dataset 1

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)`

129. openneuro_ds004447 - The BMI-HDEEG dataset 3

Dataset: openneuro_ds004447 | Year: 2023 | URL: https://openneuro.org/datasets/ds004447
Category: Action/Output | Subjects: 418 | Files (Completed): 418
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

A brain-computer interface (BCI) task based on sensorimotor rhythms (SMR) recorded with high-density scalp EEG. Subjects performed motor imagery of four limb movements (left hand, right hand, feet, tongue) following visual cues. This is the third of four BMI-HDEEG datasets with a four-class design, though tongue and feet classes appear in fewer files.

Event Structure

Event ID	Count (total)	Type	Meaning
1	418xx	long	Motor imagery cue - class 1 (left hand MI)
2	418xx	long	Motor imagery cue - class 2 (right hand MI)
3	91xx	long	Motor imagery cue - class 3 (feet MI)
4	91xx	long	Motor imagery cue - class 4 (tongue MI)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Specialized kernel (kernels/openneuro_ds004447.py) - decodes MI category tags

Task 1: Motor Imagery Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.5 s, tmax = 4.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
left_hand_MI	0	Left hand motor imagery
right_hand_MI	1	Right hand motor imagery
feet_MI	2	Feet motor imagery
tongue_MI	3	Tongue motor imagery

- Note: Numerical codes 1-4 decoded by specific kernel. Classes 3 and 4 appear in only 91 files (subset of subjects). Binary left/right classification is the most balanced sub-task.

Data Quality Notes

- Numerical event codes (1-4) require kernel-level decoding
- Classes 3 and 4 (feet/tongue) only appear in 91 of 418 files
- 130-channel high-density EEG with extended epochs

References

- Reference: `High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023).`

130. openneuro_ds004448 - The BMI-HDEEG dataset 4

Dataset: openneuro_ds004448 | Year: 2023 | URL: https://openneuro.org/datasets/ds004448
Category: Action/Output | Subjects: 280 | Files (Completed): 280
Channels: 130 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a sensorimotor rhythm-based brain-computer interface task using high-density scalp EEG. This is the fourth BMI-HDEEG dataset with a two-class (binary) motor imagery design (left hand vs. right hand). Only one event code is present in the data, suggesting the class label is encoded differently.

Event Structure

Event ID	Count (total)	Type	Meaning
1	280xx	long	Motor imagery cue - class 1

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Specialized kernel (kernels/openneuro_ds004448.py) - decodes MI category tags

Task 1: Motor Imagery Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.5 s, tmax = 4.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
left_hand_MI	0	Left hand motor imagery
right_hand_MI	1	Right hand motor imagery

- Remarks: Only one numerical code (1) visible in locator_meta; actual binary class label likely encoded in events.tsv metadata columns or file-level metadata. Specific kernel required.

Data Quality Notes

- Only one event code in metadata; class label may be in events.tsv columns rather than trial_type
- 130-channel high-density EEG with extended epochs
- 280 files; binary MI design

References

- Reference: High-density scalp electroencephalogram dataset during sensorimotor rhythm-based brain-computer interfacing. Sci Data 10, 385 (2023)

131. openneuro_ds004460 - EEG and motion capture data set for a full-body/joystick rotation task

Dataset: openneuro_ds004460 | Year: 2024 | URL: https://openneuro.org/datasets/ds004460
Category: Complex Applied | Subjects: 20 | Files (Completed): 40
Channels: 160 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a spatial orientation task in a virtual environment, navigating via either active physical rotation of the body or joystick-controlled rotation on a flat screen. EEG and motion capture were simultaneously recorded to compare neural dynamics of active (full-body) vs. passive (joystick) spatial navigation and heading changes.

Event Structure

Event ID	Count (total)	Type	Meaning
ground_rotate	40	short	Ground/environment rotation onset
object_fadeInOut	40	short	Object appearance/disappearance
'/Event/Label/targ_color_change'	53	short	Target color change event
('/Event/Category/Participant response'	40	short	Participant response event
'/Event/Label/wii_button'	40	short	Wii/button press response
('/Event/Category/Experimental stimulus/Instruction/Attend/'	80	short	Attend instruction stimulus
('/Event/Category/Experimental stimulus/Instruction/Read/'	40	short	Read instruction stimulus
duration	40	admin	Duration annotation

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Navigation Mode Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
ground_rotate	0	Spatial rotation stimulus
object_fadeInOut	1	Object appearance/disappearance
'/Event/Label/targ_color_change'	2	Target color change

- Note: HED (Hierarchical Event Descriptor) tags are used as event labels - very verbose names. 160 channels includes motion capture channels. 40 files from 20 subjects (2 conditions: body vs. joystick rotation).

Data Quality Notes

- HED-format event labels are very long strings - ensure proper parsing
- 160 channels includes motion capture data channels in addition to EEG
- 40 files from 20 subjects (2 runs each: body rotation and joystick rotation)

References

- Reference: Human cortical dynamics during full-body heading changes. Scientific Reports, 11(1), 18186

132. openneuro_ds004475 - Mobile EEG split-belt walking study

Dataset: openneuro_ds004475 | Year: 2023 | URL: https://openneuro.org/datasets/ds004475
Category: Complex Applied | Subjects: 30 | Files (Completed): 30
Channels: 260 | Sampling Rate: 512 Hz

Paradigm

A mobile brain-body imaging experiment investigating brain activity during gait adaptation using abrupt and gradual split-belt treadmill walking. Participants walked on a split-belt treadmill with different belt speeds while EEG and gait events (heel strikes, toe-offs) were recorded to study locomotor adaptation and associated cortical dynamics.

Event Structure

Event ID	Count (total)	Type	Meaning
LHS	30xx	short	Left heel strike (gait event)
LTO	30xx	short	Left toe-off (gait event)
RHS	30xx	short	Right heel strike (gait event)
RTO	30xx	short	Right toe-off (gait event)
TRIG	30xx	short	External trigger marker
BUTTON	30xx	short	Button press event
SB1	30	long	Split-belt block 1 start (tied speed)
SB2	30	long	Split-belt block 2 start (split speed)
B1	30	long	Baseline block 1
B2	30	long	Baseline block 2
B3	30	long	Baseline block 3
P1	30	long	Perturbation block 1
P2	30	long	Perturbation block 2
End SB1	30	admin	End of split-belt block 1
End SB2	30	admin	End of split-belt block 2
End B1	30	admin	End of baseline block 1
End B2	30	admin	End of baseline block 2
End B3	30	admin	End of baseline block 3
End P1	30	admin	End of perturbation block 1
End P2	30	admin	End of perturbation block 2
boundary	8	admin	Segment boundary marker
9633791	30	admin	Numerical system trigger

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5

- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Gait Event Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
LHS	0	Left heel strike
LTO	1	Left toe-off
RHS	2	Right heel strike
RTO	3	Right toe-off

- Note: boundary and 9633791 are admin/system markers - exclude from target_events. Block markers (SB1/SB2/B1-3/P1-2) are experimental condition markers. End_* markers are admin events.

Data Quality Notes

- boundary and 9633791 must be excluded as admin markers
- 260 channels includes motion capture channels; filter to EEG only
- Gait event counts vary across walking conditions (tied vs. split-belt speeds)

References

- Reference: Mobile EEG split-belt walking study

133. openneuro_ds004477 - PES - Pandemic Emergency Scenario

Dataset: openneuro_ds004477 | Year: 2023 | URL: https://openneuro.org/datasets/ds004477
Category: Complex Applied | Subjects: 9 | Files (Completed): 9
Channels: 80 | Sampling Rate: 2048 Hz

Paradigm

A strategic pandemic simulation task where participants allocate limited resources to cities on a map and report their decision confidence while interacting with an artificial agent. EEG was recorded to investigate neural correlates of complex decision-making, confidence estimation, and strategy formation in a realistic multi-agent scenario.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulus	9xx	short	Decision stimulus onset
resp_hold	9xx	short	Response hold (button press start)
resp_release	9xx	short	Response release (button press end)
confidence	9xx	short	Confidence rating onset

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Decision Making (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
resp_hold	0	Response initiation
resp_release	1	Response release
stimulus	2	Decision stimulus
confidence	3	Confidence rating

- Note: Only 9 subjects; limited generalization. Small dataset suitable for within-subject analysis. Confidence ratings may be used as regression target.

Data Quality Notes

- Only 9 subjects; very small dataset
- 80 channels at 2048 Hz (high temporal resolution)
- Complex task with strategic decisions; behavioral data critical for labeling

References

- Reference: `PES - Pandemic Emergency Scenario`

134. openneuro_ds004502 - Anticipatory differences between Attention and Expectation

Dataset: openneuro_ds004502 | Year: 2023 | URL: https://openneuro.org/datasets/ds004502
Category: Sensory/Response | Subjects: 38 | Files (Completed): 38
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

An experimental task investigating neural mechanisms of anticipation by comparing stimulus-driven attention and predictive expectation. Participants attended to auditory or visual stimuli with varying validity cues to disentangle attention (stimulus-driven) from expectation (predictive) effects on neural anticipatory activity.

Event Structure

Event ID	Count (total)	Type	Meaning
S 17	38xx	short	BrainVision trigger - decode from events.tsv
S 18	38xx	short	BrainVision trigger - decode from events.tsv
S 19	38xx	short	BrainVision trigger - decode from events.tsv
S 20	38xx	short	BrainVision trigger - decode from events.tsv
S 21	38xx	short	BrainVision trigger - decode from events.tsv
S 22	38xx	short	BrainVision trigger - decode from events.tsv
S 23	38xx	short	BrainVision trigger - decode from events.tsv
S 24	38xx	short	BrainVision trigger - decode from events.tsv
S 33…S 57	38xx	short	BrainVision trigger codes (22 codes) - decode from events.tsv

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Attention vs Expectation Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)

- Target Events -> y:

Event	y	Meaning
S 17...S 24	0	Attention condition trials (S1x range)
S 33...S 57	1	Expectation condition trials (S3x-S5x range)

- Note: 30 distinct BrainVision trigger codes (S 17-S 57 range) require semantic decoding from events.tsv. Grouping into attention vs. expectation conditions requires understanding the trigger code mapping.

Data Quality Notes

- 30 distinct trigger codes require semantic decoding from events.tsv metadata
- All 30 codes appear in all 38 files with consistent counts
- 63 channels at 1000 Hz (BrainVision system)

References

- Reference: `Anticipatory differences between Attention and Expectation`

135. openneuro_ds004504 - A dataset of EEG recordings from: Alzheimer's disease, Frontotemporal dementia and Healthy subjects

Dataset: `openneuro_ds004504` | Year: 2024 | URL: https://openneuro.org/datasets/ds004504
Category: Biological State | Subjects: 88 | Files (Completed): 176
Channels: 19 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings with eyes closed collected from healthy controls and patients with Alzheimer's disease (AD) and Frontotemporal Dementia (FTD). The study provides a clinical EEG dataset for developing diagnostic biomarkers and machine learning classifiers to differentiate neurodegenerative conditions from healthy aging.

Event Structure

No event markers detected (`no_events`). This dataset contains resting-state EEG recordings without trial-level annotations.

> Event Type Description:
> - `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
> - `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
> - `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
> - `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

> Label source: N/A - resting-state; subject-level labels from participants.tsv
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Dementia Classification (multiclass)

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (subject-level labels from participants.tsv)
- Remarks: Subject-level diagnosis labels (HC/AD/FTD) in participants.tsv used as ground truth. 19-channel 10-20 system EEG; standard clinical montage. 2 sessions per subject (176 files / 88 subjects).

Data Quality Notes

- No trial-level event markers; pure resting-state dataset
- Subject diagnosis labels (HC/AD/FTD) available in participants.tsv
- Standard 10-20 montage (19 channels)
- 2 recordings per subject (176 files / 88 subjects)

References

- Reference: `A Dataset of Scalp EEG Recordings of Alzheimers Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG`

136. openneuro_ds004505 - Real World Table Tennis

Dataset: openneuro_ds004505 | Year: 2023 | URL: https://openneuro.org/datasets/ds004505
Category: Motor/Movement | Subjects: 19 | Files (Completed): 19 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants played real table tennis matches in two conditions: cooperative (rallying) and competitive (point-scoring). EEG was recorded during actual gameplay using a mobile EEG system. Events mark the condition type and specific ball-hit moments (moving_serve vs stationary_serve shots). The study investigates motor cortex dynamics during ecological real-world sport.

Event Structure

Event ID	Count (total)	Type	Meaning
moving_hit	~19xN	short	Ball hit during moving serve condition
stationary_hit	~19xN	short	Ball hit during stationary serve condition
cooperative	~19xN	long	Cooperative play condition block
competitive	~19xN	long	Competitive play condition block
moving_serve	~19xN	long	Moving serve condition block
stationary_serve	~19xN	long	Stationary serve condition block
STATUS	~19xN	admin	BrainVision hardware status marker
M 1	~19xN	admin	Miscellaneous hardware marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Hit Type Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
moving_hit	0	Ball hit in moving serve condition
stationary_hit	1	Ball hit in stationary serve condition

- Note: STATUS and M 1 are admin events - exclude from target_events. cooperative/competitive/moving_serve/stationary_serve are long condition blocks. Focus on discrete hit events for epoch-based analysis.

Data Quality Notes

- Real-world mobile EEG with significant motion artifacts expected
- STATUS and M 1 are hardware markers, not task events

References

- Reference: `Real World Table Tennis`

137. openneuro_ds004511 - Deception_data

Dataset: openneuro_ds004511 | Year: 2023 | URL: https://openneuro.org/datasets/ds004511
Category: High-level Mental | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed a deception task while EEG was recorded. The dataset contains continuous EEG recordings; only a single hardware synchronization marker (Sync) is present per recording rather than trial-level event markers. No meaningful trial annotations are available for epoch-based analysis.

Event Structure

No task event markers detected (no_events). Only hardware sync marker Sync(1) present - this is a system/hardware event, not a task event.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (resting/segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: Sync(1) is a hardware synchronization marker, not a task event. Use sliding window segmentation for feature extraction.

Data Quality Notes

- No trial-level event markers
- Sync(1) is a hardware-only event and should not be used for epoching

References

- Reference: Deception_data

138. openneuro_ds004514 - Simultaneous EEG and fNIRS recordings for semantic decoding of imagined animals and tools

Dataset: openneuro_ds004514 | Year: 2023 | URL: https://openneuro.org/datasets/ds004514
Category: High-level Mental | Subjects: 22 | Files (Completed): 22 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Participants imagined animals and tools in response to visual cues while simultaneous EEG and fNIRS were recorded. Events encode the specific animal or tool category being imagined across four modalities: visual imagery, silent naming, auditory imagery, and drawing imagery. The study investigates cross-modal semantic representations of living vs non-living concepts.

Event Structure

Event ID	Count (total)	Type	Meaning
animal_visual_imagery_task	~22xN	long	Visual imagery of animals
tool_visual_imagery_task	~22xN	long	Visual imagery of tools
animal_silent_naming_task	~22xN	long	Silent naming imagery of animals
tool_silent_naming_task	~22xN	long	Silent naming imagery of tools
animal_auditory_imagery_task	~22xN	long	Auditory imagery of animals
tool_auditory_imagery_task	~22xN	long	Auditory imagery of tools
animal_drawing_imagery_task	~22xN	long	Drawing imagery of animals
tool_drawing_imagery_task	~22xN	long	Drawing imagery of tools
rest	~22xN	rest	Rest period between imagery blocks
animal_visual_image	~22xN	short	Visual image of animal (cue)
tool_visual_image	~22xN	short	Visual image of tool (cue)
audio_animal	~22xN	short	Audio cue for animal
audio_tool	~22xN	short	Audio cue for tool
written_animal	~22xN	short	Written word cue for animal
written_tool	~22xN	short	Written word cue for tool
animal_drawing	~22xN	short	Drawing cue for animal
tool_drawing	~22xN	short	Drawing cue for tool
animal_text_cue	~22xN	short	Text cue for animal

Event ID	Count (total)	Type	Meaning
tool_text_cue	~22xN	short	Text cue for tool

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Animal vs Tool Semantic Decoding (binary, long_event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
animal_visual_imagery_task	0	Animal imagery (any modality)
animal_silent_naming_task	0	Animal imagery (any modality)
animal_auditory_imagery_task	0	Animal imagery (any modality)
animal_drawing_imagery_task	0	Animal imagery (any modality)
tool_visual_imagery_task	1	Tool imagery (any modality)
tool_silent_naming_task	1	Tool imagery (any modality)
tool_auditory_imagery_task	1	Tool imagery (any modality)
tool_drawing_imagery_task	1	Tool imagery (any modality)

- Note: Imagery tasks are sustained (>=2 s) - use long_event epoching. rest events are resting periods between imagery blocks. Short cue events (visual_image, audio, written, drawing, text_cue) mark the onset stimulus, not the imagery period.

Data Quality Notes

- Simultaneous EEG and fNIRS; only EEG channels used here
- 32 EEG channels (lower density than typical 64-ch systems)
- 4 imagery modalities x 2 categories = 8 sustained imagery conditions

References

- Reference: Simultaneous EEG and fNIRS recordings for semantic decoding of imagined animals and tools

139. openneuro_ds004515 - EEG: Alcohol imagery reinforcement learning task with light and heavy drinker participants

Dataset: openneuro_ds004515 | Year: 2023 | URL: https://openneuro.org/datasets/ds004515
Category: High-level Mental | Subjects: 42 | Files (Completed): 42 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants (light and heavy drinkers) performed a reinforcement learning task involving alcohol-related and neutral imagery cues. On each trial, participants chose between left and right buttons in response to cues, with feedback (FB) indicating correct/incorrect responses. The study investigates how reward learning interacts with alcohol cue reactivity across drinking groups.

Event Structure

Event ID	Count (total)	Type	Meaning
CUE	~42xN	short	Stimulus cue onset
Left Button	~42xN	short	Left button press response
Right Button	~42xN	short	Right button press response
FB	~42xN	short	Feedback onset
STATUS	~42xN	admin	BrainVision hardware status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Response Direction Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Left Button	0	Left button press
Right Button	1	Right button press

- Note: STATUS is an admin event - exclude from target_events. CUE and FB are trial markers but not the primary classification target. Left vs Right button press provides a binary motor/decision classification.

Data Quality Notes

- 42 subjects split between light and heavy drinkers

- STATUS markers are BrainVision hardware events, not task-related

References

- Reference: `EEG: Alcohol imagery reinforcement learning task with light and heavy drinker participants`

140. openneuro_ds004517 - EEG recordings for semantic decoding of imagined animals and tools during auditory imagery task

Dataset: openneuro_ds004517 | Year: 2023 | URL: https://openneuro.org/datasets/ds004517
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Participants imagined animals and tools in response to auditory cues while EEG was recorded. Each trial consisted of an audio cue followed by a sustained auditory imagery period (~5 s). The study investigates EEG correlates of semantic imagery for living (animals) vs non-living (tools) concepts in the auditory domain.

Event Structure

Event ID	Count (total)	Type	Meaning
animal_auditory_imagery_task	~20xN	long	Sustained auditory imagery of animals
tool_auditory_imagery_task	~20xN	long	Sustained auditory imagery of tools
audio_animal	~20xN	short	Auditory cue onset for animal
audio_tool	~20xN	short	Auditory cue onset for tool
rest	~20xN	rest	Rest period between imagery blocks

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Animal vs Tool Auditory Imagery (binary, long_event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
animal_auditory_imagery_task	0	Auditory imagery of animals
tool_auditory_imagery_task	1	Auditory imagery of tools

- Note: Imagery tasks are sustained (~5 s) - use long_event epoching. audio_animal/audio_tool are short cue onsets; rest events are inter-block resting periods.

Data Quality Notes

- 32 EEG channels
- Sustained imagery periods (~5 s) are suitable for long_event epoching
- 20 subjects with balanced animal/tool imagery conditions

References

- Reference: `EEG recordings for semantic decoding of imagined animals and tools during auditory imagery task`

141. openneuro_ds004532 - EEG: Probabilistic Selection Task (PST) + PST with Cabergoline Challenge

Dataset: openneuro_ds004532 | Year: 2023 | URL: https://openneuro.org/datasets/ds004532
Category: High-level Mental | Subjects: 40 | Files (Completed): 80 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants performed the Probabilistic Selection Task (PST) in two sessions: a placebo session and a session with cabergoline (dopamine D2/D3 agonist). The PST involves learning to choose symbols with higher reward probability through feedback. Events encode training stimuli (Trn Stim), test stimuli (Tst Stim), feedback (FB), and outcome (correct/incorrect). The study investigates dopaminergic modulation of reinforcement learning.

Event Structure

Event ID	Count (total)	Type	Meaning
Trn Stim	~80xN	short	Training stimulus onset
Tst Stim	~80xN	short	Test stimulus onset
FB	~80xN	short	Feedback onset
correct	~80xN	short	Correct response outcome
incorrect	~80xN	short	Incorrect response outcome
STATUS	~80xN	admin	BrainVision hardware status marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Outcome Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
correct	0	Correct response outcome
incorrect	1	Incorrect response outcome

- Note: STATUS is an admin event - exclude from target_events. Trn Stim and Tst Stim are stimulus onset events; FB is the feedback onset. correct/incorrect provide clear binary outcome labels.

Data Quality Notes

- 40 subjects x 2 sessions (placebo + cabergoline) = 80 files
- STATUS markers are BrainVision hardware events, not task-related
- Two sessions per subject enable within-subject drug effect analysis

References

- Reference: `EEG: Probabilistic Selection Task (PST) + PST with Cabergoline Challenge`

142. openneuro_ds004561 - Illusion of Agency over Electrically-Actuated Movements

Dataset: openneuro_ds004561 | Year: 2023 | URL: https://openneuro.org/datasets/ds004561
Category: High-level Mental | Subjects: 20 | Files (Completed): 20 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants experienced electrical muscle stimulation that actuated their hand movements in three conditions: stimulation (involuntary movement), pretest (voluntary movement before stimulation), and posttest (voluntary movement after stimulation). EEG was recorded to investigate illusory sense of agency over externally-driven movements. The study explores how prior voluntary movement influences agency attribution over stimulation-induced movements.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulation	~20xN	short	Electrical stimulation onset (involuntary movement)
pretest	~20xN	short	Pre-stimulation voluntary movement
posttest	~20xN	short	Post-stimulation voluntary movement

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Agency Condition Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
stimulation	0	Electrical stimulation (involuntary movement)
pretest	1	Pre-stimulation voluntary movement
posttest	2	Post-stimulation voluntary movement

- Note: 3-class classification of agency conditions. stimulation = externally actuated movement, pretest/posttest = voluntary movements before/after stimulation session.

Data Quality Notes

- 20 subjects with balanced condition structure

- 64 channels at 512 Hz provides good temporal resolution for movement-related potentials

References

- Reference: `Illusion of Agency over Electrically-Actuated Movements`

143. openneuro_ds004563 - Vicarious touch: overlapping neural patterns between seeing and feeling touch

Dataset: openneuro_ds004563 | Year: 2023 | URL: https://openneuro.org/datasets/ds004563
Category: Sensory/Response | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants either saw someone else being touched (vicarious touch, event 4) or felt touch themselves (direct touch, event 1). EEG was recorded to investigate overlapping neural patterns between observed and experienced tactile stimulation. The study tests whether vicarious touch engages somatosensory cortex similarly to directly felt touch.

Event Structure

Event ID	Count (total)	Type	Meaning
4	~29xN	short	Seeing touch (vicarious touch condition)
1	~29xN	short	Feeling touch (direct touch condition)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Mission 1: Vicarious vs Direct Touch (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
4	0	Seeing touch (vicarious touch)
1	1	Feeling touch (direct touch)

- Note: Binary classification of vicarious vs direct tactile stimulation. Clean 2-event structure per subject.

Data Quality Notes

- 29 subjects with binary condition structure
- 64 channels at 512 Hz

References

- Reference: Vicarious touch: overlapping neural patterns between seeing and feeling touch

144. openneuro_ds004574 - Cross-modal Oddball Task

Dataset: openneuro_ds004574 | Year: 2023 | URL: https://openneuro.org/datasets/ds004574
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed an oddball task with cross-modal stimuli (visual and auditory). Standard stimuli (S 1), deviant stimuli (S 2), and rare stimuli (S 3) were presented in mixed modality streams. The study investigates the mismatch negativity (MMN) and P300 components elicited by deviant stimuli across sensory modalities.

Event Structure

Event ID	Count (total)	Type	Meaning
S 1	~30xN	short	Standard stimulus
S 2	~30xN	short	Deviant stimulus
S 3	~30xN	short	Rare/target stimulus
boundary	~30xN	admin	Recording boundary marker
S255	~30xN	admin	BrainVision overflow/system marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Oddball Stimulus Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
S 1	0	Standard stimulus
S 2	1	Deviant stimulus
S 3	2	Rare/target stimulus

- Note: boundary and S255 are admin events - exclude from target_events. S255 is a BrainVision system marker (overflow/sync), not a task event.

Data Quality Notes

- boundary and S255 must be excluded from epoching
- S255 is a BrainVision system marker, not a stimulus event

- 30 subjects with standard oddball structure

References

- Reference: `Cross-modal Oddball Task`

145. openneuro_ds004577 - Dataset containing resting EEG for a sample of 103 normal infants in the first year of life

Dataset: openneuro_ds004577 | Year: 2023 | URL: https://openneuro.org/datasets/ds004577
Category: Biological State | Subjects: 103 | Files (Completed): 103 | Channels: 128 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings from 103 healthy infants (aged 1-12 months) during the first year of life. Recordings were made during natural infant rest/sleep states. Eyes-open and eyes-closed state markers are present but the primary data is continuous infant resting EEG. The dataset supports developmental neuroscience research on early brain maturation.

Event Structure

No task event markers. State markers (eyes_closed, eyes_open) and artifact markers (discontinuity) are present but the dataset is primarily resting-state continuous EEG.

Event ID	Count (total)	Type	Meaning
eyes_closed	~103xN	rest	Eyes-closed resting state
eyes_open	~103xN	rest	Eyes-open resting state
discontinuity	~103xN	admin	Recording discontinuity/artifact marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: N/A - resting-state infant EEG; segmentation-based processing
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (resting/segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: Infant resting EEG without trial structure. discontinuity markers indicate artifact segments. Use sliding window segmentation; consider excluding segments surrounding discontinuity markers.

Data Quality Notes

- 103 infants aged 1-12 months; high variability in developmental stage
- 128 channels; high-density pediatric EEG cap
- discontinuity markers indicate artifact/movement artifacts in infant recordings

References

- Reference: Dataset containing resting EEG for a sample of 103 normal infants in the first year of life

146. openneuro_ds004579 - Interval Timing Task

Dataset: openneuro_ds004579 | Year: 2023 | URL: https://openneuro.org/datasets/ds004579
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed an interval timing task where they estimated durations of different intervals. BrainVision S-codes (S 1-S 7) encode different timing conditions or interval lengths. Participants produced or reproduced time intervals by button press (R 3). The study investigates neural oscillations and temporal processing mechanisms.

Event Structure

Event ID	Count (total)	Type	Meaning
S 1	~30xN	short	Timing condition 1
S 2	~30xN	short	Timing condition 2
S 3	~30xN	short	Timing condition 3
S 4	~30xN	short	Timing condition 4
S 5	~30xN	short	Timing condition 5
S 6	~30xN	short	Timing condition 6
S 7	~30xN	short	Timing condition 7
R 3	~30xN	short	Button press response
boundary	~30xN	admin	Recording boundary marker
S255	~30xN	admin	BrainVision overflow/system marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Timing Condition Classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
S 1	0	Timing condition 1
S 2	1	Timing condition 2
S 3	2	Timing condition 3

Event	y	Meaning
S 4	3	Timing condition 4
S 5	4	Timing condition 5
S 6	5	Timing condition 6
S 7	6	Timing condition 7

- Note: boundary and S255 are admin events - exclude from target_events. R 3 is a response event - can be used as an additional classification target. S255 is a BrainVision system marker (overflow/sync), not a task event.

Data Quality Notes

- boundary and S255 must be excluded from epoching
- S255 is a BrainVision system marker, not a task event
- 30 subjects with 7 timing conditions

References

- Reference: `Interval Timing Task`

147. openneuro_ds004580 - Simon-conflict Task

Dataset: openneuro_ds004580 | Year: 2023 | URL: https://openneuro.org/datasets/ds004580
Category: Sensory/Response | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants performed the Simon conflict task in which stimulus location (left/right) could be congruent or incongruent with the required response direction. S 1 and S 2 encode congruent and incongruent trial types respectively; S 3 marks rare/catch trials. The study investigates conflict monitoring and executive control neural signatures (N2, P3 ERP components).

Event Structure

Event ID	Count (total)	Type	Meaning
S 1	30x384	short	Congruent trial (stimulus side matches response side)
S 2	30x383	short	Incongruent trial (stimulus side opposite to response side)
S 3	~30xN	short	Rare/catch trial
boundary	~30xN	admin	Recording boundary marker
S255	~30xN	admin	BrainVision overflow/system marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Congruent vs Incongruent Classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
S 1	0	Congruent trial
S 2	1	Incongruent trial

- Note: boundary and S255 are admin events - exclude from target_events. S255 is a BrainVision system marker (overflow/sync), not a task event. S 3 (rare trials) can be optionally included as a third class.

Data Quality Notes

- boundary and S255 must be excluded from epoching

- Balanced S 1 and S 2 counts (~384 vs ~383 per subject)
- 30 subjects with consistent congruent/incongruent structure

References

- Reference: `Simon-conflict Task`

148. openneuro_ds004582 - FakeFaceEmo_data

Dataset: openneuro_ds004582 | Year: 2023 | URL: https://openneuro.org/datasets/ds004582
Category: High-level Mental | Subjects: 30 | Files (Completed): 30 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Participants viewed fake face images with different emotional expressions while EEG was recorded. The dataset contains continuous EEG recordings; only hardware/metadata markers (New Segment, Comment) are present in the event files rather than trial-level stimulus markers. No trial-level event markers are available for epoch-based analysis.

Event Structure

No task event markers detected (no_events). Only hardware/metadata markers present.

Event ID	Count (total)	Type	Meaning
New Segment/	~30x1	admin	Recording segment boundary (hardware marker)

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: N/A - no trial events; segmentation-based processing
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Continuous Segmentation (resting/segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y:N/A (unsupervised segmentation)
- Note: New Segment/ is a BrainVision hardware recording marker, not a task event. Use sliding window segmentation for feature extraction.

Data Quality Notes

- No trial-level event markers
- New Segment/ is a hardware-only event and should not be used for epoching
- 30 subjects with continuous EEG recordings

References

- Reference: FakeFaceEmo_data

149. openneuro_ds004584 - Rest Eyes Open (Parkinson's Disease)

Dataset: `openneuro_ds004584` | Year: 2023 | URL: https://openneuro.org/datasets/ds004584
Category: Biological State | Subjects: 149 | Files (Completed): 149 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG was collected from 149 participants (100 with Parkinson's disease and 49 healthy controls) who sat quietly with eyes open for approximately two minutes. The study characterizes spontaneous neural oscillatory patterns to differentiate PD patients from controls. No active task was required; participants fixated on a cross and remained still.

Event Structure

Event ID	Count (total)	Type	Meaning
`boundary`	149	admin	Segment boundary marker inserted by recording software (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting State Segmentation (segmentation)

- Epoch Method: `segmentation`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None (no task events)
- Notes: Only `boundary` admin markers present - pure resting-state dataset. Use sliding-window segmentation. PD vs. control group labels available from participants.tsv for downstream classification.

Data Quality Notes

- Only boundary markers present; no trial events for epoch-locked classification.
- PD vs. control labels available in participants.tsv.

References

- Reference: `Rest eyes open EEG dataset - Parkinson's disease and healthy controls`

150. openneuro_ds004587 - IllusionGameEEG_data

Dataset: openneuro_ds004587 | Year: 2024 | URL: https://openneuro.org/datasets/ds004587
Category: Sensory/Response | Subjects: 103 | Files (Completed): 114 | Channels: 64 | Sampling Rate: 10000 Hz

Paradigm

Participants performed perceptual judgements on visual targets embedded within three classic visual illusions - Ebbinghaus, Müller-Lyer, and Vertical-Horizontal - to measure individual differences in illusion sensitivity and perceptual bias. The study investigates how prior experience and personality shape perceptual decision-making. EEG was recorded at 10 kHz; actual stimulus-event markers appear not to have been stored in events.tsv.

Event Structure

Event ID	Count (total)	Type	Meaning
New Segment/LostSamples: 2	370	admin	Lost-samples boundary marker (exclude)
New Segment/	115	admin	BrainVision segment boundary (exclude)
New Segment/LostSamples: 3	111	admin	Lost-samples boundary marker (exclude)
New Segment/LostSamples: 4	6	admin	Lost-samples boundary marker (exclude)
Comment/motion	9	admin	Motion artifact annotation (exclude)
Comment/ro post	6	admin	Post-recording comment (exclude)
Comment/error	3	admin	Error annotation (exclude)
Comment/rc pre	3	admin	Pre-recording comment (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Continuous Task EEG (segmentation)

- Epoch Method: segmentation
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: All events in events.tsv are administrative markers or artifact annotations. No stimulus/response trial events were captured. Trial structure may be recoverable from separate behavioral logs. Use sliding-window segmentation.

Data Quality Notes

- All events are admin; no stimulus-locked epochs possible from events.tsv alone.

- Very high sampling rate (10 kHz) - downsample before analysis.
- Paper reference (ARSQ 2.0) may be a metadata mismatch; verify README.

References

- Reference: `The ARSQ 2.0 reveals age and personality effects on mind-wandering experiences`

151. openneuro_ds004588 - Neuma (Neuromarketing)

Dataset: openneuro_ds004588 | Year: 2023 | URL: https://openneuro.org/datasets/ds004588
Category: Complex Applied | Subjects: 42 | Files (Completed): 84 | Channels: 6 | Sampling Rate: 300 Hz

Paradigm

A multimodal neuromarketing study where 42 participants browsed supermarket product brochures displayed on screen and selected items they intended to purchase by clicking the left mouse button. EEG was recorded alongside mouse interaction data to identify neural correlates of consumer purchase decisions during free-browsing. Each session involved scrolling through product leaflet images presented in a sequential format.

Event Structure

Event ID	Count (total)	Type	Meaning
MouseButtonLeft pressed	1734	short	Left mouse button press (product selection / page interaction)
MouseButtonLeft released	1712	short	Left mouse button release
Category:IMG=ID:FYLLADIO_1.tif=...	268	short	Product leaflet image 1 onset
Category:IMG=ID:FYLLADIO_2.tif=...	138	short	Product leaflet image 2 onset
MouseWheelDown120 pressed	210	admin	Scroll-down wheel event (exclude)
MouseWheelUp120 pressed	76	admin	Scroll-up wheel event (exclude)
fixation_cross	84	short	Fixation cross onset (trial start)
EOE	84	short	End-of-experiment marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Purchase Interaction Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
fixation_cross	0	Trial-start fixation
MouseButtonLeft pressed	1	Product purchase/page click

- Notes: Only 6 EEG channels. Category event labels embed full image filenames - aggregate by category type if needed. Scroll wheel events are admin.

Data Quality Notes

- Very limited spatial resolution: only 6 EEG channels.
- Category event labels contain full image paths; group by category type for analysis.

References

- Reference: `Neuma`

152. openneuro_ds004595 - EEG: RL Task (3-Armed Bandit) with Alcohol Cues in Hazardous Drinkers

Dataset: openneuro_ds004595 | Year: 2023 | URL: https://openneuro.org/datasets/ds004595
Category: High-level Mental | Subjects: 53 | Files (Completed): 53 | Channels: 66 | Sampling Rate: 500 Hz

Paradigm

Participants completed a reinforcement learning task using a 3-armed bandit paradigm incorporating alcohol-related and neutral beverage cues to study decision-making in hazardous drinkers versus controls. On each trial, a cue was presented followed by a three-arm choice and probabilistic win/loss feedback. The study examines electrophysiological markers of cue-driven exploration and aberrant reinforcement learning.

Event Structure

Event ID	Count (total)	Type	Meaning
Fixation	18550	short	Fixation cross onset (trial start)
FB: Win	9713	short	Win feedback delivered
FB: Loss	7254	short	Loss feedback delivered
Left Button	6276	short	Left arm choice response
Up Button	6188	short	Up arm choice response
Right Button	5553	short	Right arm choice response
STATUS	4271	admin	System status marker (exclude)
CUE: 123	2955	short	Cue onset - arm order permutation 1-2-3
CUE: 231	2934	short	Cue onset - arm order permutation 2-3-1
CUE: 132	2931	short	Cue onset - arm order permutation 1-3-2
CUE: 321	2919	short	Cue onset - arm order permutation 3-2-1
CUE: 312	2908	short	Cue onset - arm order permutation 3-1-2
CUE: 213	2853	short	Cue onset - arm order permutation 2-1-3
No Response	533	admin	No-response trial (exclude)
FB: No Resp	533	admin	No-response feedback (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Feedback Classification (binary)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`FB: Win`	0	Win feedback
`FB: Loss`	1	Loss feedback

- Notes: STATUS, No Response, and FB: No Resp are admin - exclude. CUE events encode arm-order permutations; aggregate to generic CUE if order-agnostic analysis desired.

Data Quality Notes

- STATUS markers must be excluded from training.
- CUE events encode arm order permutations - group to single CUE class if order irrelevant.

References

- Reference: `Electrophysiological markers of aberrant cue-specific exploration in hazardous drinkers`

153. openneuro_ds004602 - Registered Replication Report of ERN/Pe Psychometrics

Dataset: openneuro_ds004602 | Year: 2023 | URL: https://openneuro.org/datasets/ds004602
Category: Sensory/Response | Subjects: 182 | Files (Completed): 546 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

Participants completed Flanker, Stroop, and Go/no-go tasks to evaluate the psychometric reliability and convergent/divergent validity of error-related ERP components (ERN and Pe). The study tests whether error-related components are stable within individuals across tasks and sessions. Three task files per subject yield the 546-file count. cor/err mark correct/error responses; con/inc mark congruent/incongruent Flanker/Stroop trials; go/ng/neu mark Go/no-go trial types.

Event Structure

Event ID	Count (total)	Type	Meaning
cor	169236	short	Correct response (all tasks)
go	58251	short	Go trial (Go/no-go task)
con	54498	short	Congruent trial (Flanker/Stroop)
inc	47242	short	Incongruent trial (Flanker/Stroop)
err	24198	short	Error response (all tasks)
neu	20890	short	Neutral trial condition
ng	8636	short	No-go trial (Go/no-go task)
boundary	545	admin	Segment boundary marker (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Error vs Correct Classification (binary)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
cor	0	Correct response (ERN/Pe absent)
err	1	Error response (ERN/Pe present)

- Notes: boundary (545) is admin - exclude. cor (169,236) >> err (24,198) - significant class imbalance; undersample cor or use weighted loss. con/inc/go/ng/neu usable for separate task-level analyses.

Task 2: Congruent vs Incongruent (binary, Flanker/Stroop)

- Epoch Method: epoch_by_event_hdf5

- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`con`	0	Congruent trial
`inc`	1	Incongruent trial

- Notes: 54,498 vs 47,242 - reasonably balanced. Filter by Flanker/Stroop task files only.

Data Quality Notes

- boundary (545) is admin - exclude.
- 3 task files per subject (Flanker, Stroop, Go/no-go); filter by task for task-specific analyses.
- High channel count (129) may include non-EEG channels - verify channel types.

References

- Reference: `Registered Replication Report of ERN/Pe Psychometrics`

154. openneuro_ds004603 - Visual Attribute-Specific Contextual Trajectory Paradigm

Dataset: openneuro_ds004603 | Year: 2023 | URL: https://openneuro.org/datasets/ds004603
Category: Sensory/Response | Subjects: 37 | Files (Completed): 37 | Channels: 65 | Sampling Rate: 1024 Hz

Paradigm

Participants observed sequences of visual stimuli that either followed or violated an established contextual trajectory in one of three visual attributes - brightness (color), size, or orientation. The paradigm tests predictive coding by contrasting EEG responses to expected (step) stimuli against unexpected violations and a no-trajectory control condition. Each sequence comprised four ordered steps followed by a target or violation stimulus.

Event Structure

Event ID	Count (total)	Type	Meaning
step_1	31368	short	Sequence step 1 - first ordered stimulus in trajectory
step_2	31367	short	Sequence step 2
step_3	31367	short	Sequence step 3
step_4	31367	short	Sequence step 4 - final step before target/violation
control	14203	short	Control condition - no established trajectory
siz_vio	4736	short	Size trajectory violation
ori_vio	4735	short	Orientation trajectory violation
col_vio	4735	short	Color/brightness trajectory violation
catch	2958	short	Catch trial requiring button-press response
initalising_triggers	1	admin	System initialisation trigger (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Violation Detection (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
control	0	No-trajectory control

Event	y	Meaning
siz_vio	1	Size violation
ori_vio	2	Orientation violation
col_vio	3	Color violation

- Notes: initalising_triggers (1 total) is admin - exclude. Sequence steps (step_1-step_4) can be used for prediction build-up ERP analysis.

Data Quality Notes

- Exclude initalising_triggers from all analysis.
- Catch trials have a motor response component - treat separately or exclude from violation analysis.

References

- Reference: Visual Attribute-Specific Contextual Trajectory Paradigm

155. openneuro_ds004625 - Mind in Motion: Young Adults Walking Over Uneven Terrain

Dataset: openneuro_ds004625 | Year: 2023 | URL: https://openneuro.org/datasets/ds004625
Category: Complex Applied | Subjects: 32 | Files (Completed): 543 | Channels: 284 | Sampling Rate: 500 Hz

Paradigm

Participants walked over uneven terrain at different speeds while high-density EEG, EMG, and kinematic data were simultaneously recorded, with a seated resting condition also included. The study examines neural correlates of locomotor control and gait adaptation during ambulatory tasks. Gait events (heel-strikes, toe-offs) were synchronized with EEG via force-sensitive insoles.

Event Structure

Event ID	Count (total)	Type	Meaning
LHS	65124	short	Left heel strike
LTO	65105	short	Left toe off
RHS	63800	short	Right heel strike
RTO	63792	short	Right toe off
CGY-T 1	2786	short	Terrain change trigger
CGY-M 1	676	short	Gait measurement marker
spare-M 1	635	short	Spare measurement marker
boundary	543	admin	Segment boundary (exclude)
TrialStart	543	long	Trial start marker
TrialEnd	543	long	Trial end marker
LS-Sync	536	admin	Insole sync pulse (exclude)
CWR-M 1	269	short	Cobblestone walk marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Gait Event Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
LHS	0	Left heel strike
LTO	1	Left toe off
RHS	2	Right heel strike
RTO	3	Right toe off

- Notes: High channel count (284-channel HD-EEG). boundary and LS-Sync are admin - exclude. TrialStart/TrialEnd define walking bouts. Extensive motion artifact rejection required.

Data Quality Notes

- High-density EEG (284 channels) with significant motion artifacts from walking.
- boundary and LS-Sync markers are administrative - exclude.
- Mobile EEG data - apply ICA-based artifact correction.

References

- Reference: `Mind in Motion: Young Adults Walking Over Uneven Terrain`

156. openneuro_ds004626 - Can We Dissociate Hypervigilance to Social Threats from Altered Perceptual Decision-Making in Lonely Individualsx

Dataset: openneuro_ds004626 | Year: 2023 | URL: https://openneuro.org/datasets/ds004626
Category: High-level Mental | Subjects: 52 | Files (Completed): 52
Channels: 68 | Sampling Rate: 1000 Hz

Paradigm

A perceptual decision-making task investigating whether lonely individuals exhibit hypervigilance toward social threats, combining Drift Diffusion Modeling (DDM) with ERP analysis. Participants viewed social and non-social threat stimuli and made rapid perceptual judgements. ERP components (P1, N2pc, LPP) and computational decision parameters were compared across loneliness severity groups.

Event Structure

Event ID	Count (total)	Type	Meaning
208	36977	short	High-frequency stimulus/response trigger
998	35001	short	Secondary trigger (trial boundary/ITI)
236	6240	short	Stimulus trigger code
146	6239	short	Stimulus trigger code
206	6238	short	Stimulus trigger code
186	6238	short	Stimulus trigger code
142	6237	short	Stimulus trigger code
210	6233	short	Stimulus trigger code
999	2363	admin	Trial-end/admin marker (exclude)
248	1560	short	Trigger code
255	1560	admin	S255 admin sync trigger (exclude)
300	1528	short	Trigger code

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus classification (multiclass)

- Epoch method: epoch_by_event_hdf5

- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
208	0	Primary stimulus trigger
998	1	Secondary trigger
236	2	Trigger code
146	3	Trigger code
206	4	Trigger code
186	5	Trigger code
142	6	Trigger code
210	7	Trigger code

- Note: Trigger 255 is a known admin sync code - exclude. Trigger 999 is trial-end/admin - exclude. Full decode requires events.tsv inspection.

Data Quality Notes

- 30+ distinct numeric trigger codes - check events.tsv for full decode table.
- Codes 255 and 999 are administrative and must be excluded.

References

- Reference: Can we dissociate hypervigilance to social threats from altered perceptual decision-making processes in lonely individualsx DOI: 10.1111/psyp.14406

157. openneuro_ds004635 - Gaffrey Lab Infant EEG Microstates Reliability

Dataset: openneuro_ds004635 | Year: 2024 | URL: https://openneuro.org/datasets/ds004635
Category: Biological State | Subjects: 48 | Files (Completed): 1
Channels: 129 | Sampling Rate: 1000 Hz

Paradigm

Infants watched a series of engaging and relaxing videos to facilitate recording of spontaneous EEG activity. The dataset was collected to assess the test-retest reliability of EEG microstate metrics in infants. The single aggregated file contains data from 48 infants with recording-boundary markers; no active task was required.

Event Structure

Event ID	Count (total)	Type	Meaning
bgin	10	long	Beginning of a recording segment
TRSP	10	long	Transition / segment boundary marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Long-term classification of paragraph tags (binary)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
bgin	0	Recording segment start
TRSP	1	Transition segment marker

- Note: Only 10 events of each type - very limited for supervised classification. Primary scientific use is resting-state microstate analysis. Single aggregated file covers all 48 infants.

Data Quality Notes

- Single file aggregating 48 infants - individual subject effects not separable.
- Only 10 events of each type; insufficient for robust supervised classification.

References

- Reference: Gaffrey Lab Infant EEG Microstates Reliability

158. openneuro_ds004657 - Driving with Autonomous Aids

Dataset: openneuro_ds004657 | Year: 2023 | URL: https://openneuro.org/datasets/ds004657
Category: Complex Applied | Subjects: N/A | Files (Completed): 119 | Channels: 74 | Sampling Rate: 1024 Hz

Paradigm

Participants performed a semi-automated driving task involving lane maintenance and collision avoidance to evaluate human trust in automation across manual and varying autonomous assistance conditions. The 7-digit hierarchical trigger codes encode the driving condition, task phase, and condition variant. Codes 2310002/2310001 dominate with ~10,367 events each; 4111101 appears ~10,051 times. All other codes appear 1,081-1,819 times each - consistent with less-frequent driving scenarios or transitions.

Event Structure

Event ID	Count (total)	Type	Meaning
2310002	10367	long	Driving condition 231, variant 2 (dominant)
2310001	10367	long	Driving condition 231, variant 1 (dominant)
4111101	10051	long	Driving condition 411, variant 1
2320001	1819	long	Driving condition 232, variant 1
2320002	1819	long	Driving condition 232, variant 2
1410001	1383	long	Driving condition 141, variant 1
1410002	1383	long	Driving condition 141, variant 2
1420001	1226	long	Driving condition 142, variant 1
1420002	1226	long	Driving condition 142, variant 2
1320001	1175	long	Driving condition 132, variant 1
1320002	1175	long	Driving condition 132, variant 2
2541001	1174	long	Driving condition 254, variant 1
2781021	1094	long	Driving condition 278, variant 021
2781022	1094	long	Driving condition 278, variant 022
2561001	1081	long	Driving condition 256, variant 1

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Driving Condition Classification (multiclass)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`2310001`	0	Driving condition 231, variant 1
`2310002`	1	Driving condition 231, variant 2
`4111101`	2	Driving condition 411, variant 1

- Notes: 7-digit codes require mapping from dataset README. Codes 2310001/2310002/4111101 dominate (>10k each); all others are minority classes (1k-1.8k). Group by leading 3-4 digits for higher-level condition classification.

Data Quality Notes

- 7-digit hierarchical trigger codes require decode from README or paper.
- Codes 2310001/2310002/4111101 dominate (>10k each); other codes are minority classes.
- Subject count not specified in BIDS metadata.

References

- Reference: `Driving with Autonomous Aids`

159. openneuro_ds004660 - TNO (Military Cognitive Workload)

Dataset: `openneuro_ds004660` | Year: 2023 | URL: https://openneuro.org/datasets/ds004660
Category: Complex Applied | Subjects: 42 | Files (Completed): 42
Channels: 38 | Sampling Rate: 512 Hz

Paradigm

A military-relevant cognitive workload study conducted at TNO research institute, involving multiple hierarchically encoded task conditions represented by 5-digit numeric trigger codes. The paradigm likely involves surveillance, target detection, or dual-task conditions common in defense research. Event codes encode task type, difficulty level, and stimulus properties in a hierarchical scheme.

Event Structure

Event ID	Count (total)	Type	Meaning
`14211`	42922	short	Hierarchical task code - primary condition
`14212`	42922	short	Hierarchical task code - primary condition variant
`21111`	9228	short	Hierarchical task code
`14112`	9228	short	Hierarchical task code
`14111`	9228	short	Hierarchical task code
`13001`	3663	short	Hierarchical task code
`13002`	3663	short	Hierarchical task code
`21201`	3663	short	Hierarchical task code
`15521`	1840	short	Hierarchical task code
`15421`	1840	short	Hierarchical task code
`14121`	1680	short	Hierarchical task code
`14122`	1680	short	Hierarchical task code

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Conditional classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
14211	0	Primary task code
14212	1	Primary task code variant
21111	2	Task code
14112	3	Task code
14111	4	Task code
13001	5	Task code
13002	6	Task code
21201	7	Task code

- Note: All codes are hierarchically structured 5-digit identifiers. Full decode table required from TNO dataset README or associated paper.

Data Quality Notes

- All event codes are 5-digit hierarchical numeric identifiers - full decode needed.
- 30+ unique event types; group by high-level condition category for practical classification.

References

- Reference: TNO

160. openneuro_ds004661 - ANDI: Auditory N-back Dual-task EEG

Dataset: `openneuro_ds004661` | Year: 2024 | URL: https://openneuro.org/datasets/ds004661
Category: Complex Applied | Subjects: 17 | Files (Completed): 17 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

A dual-task paradigm combining a guided visual search task (tracking a red annulus) with an auditory N-back working memory task to study cognitive interference. Event code `10` marks the dominant condition (likely standard/no-target trials, 29,820 total) and `11` marks the minority condition (likely target/match trials, 2,948 total). The large imbalance between codes suggests `10` = non-target and `11` = N-back target detection.

Event Structure

Event ID	Count (total)	Type	Meaning
`10`	29820	short	Non-target trial (dominant condition)
`11`	2948	short	N-back target match (minority condition)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Target vs Non-target N-back Classification (binary)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`10`	0	Non-target trial
`11`	1	N-back target match

- Notes: Strong class imbalance (29,820 vs 2,948 ≈ 10:1); undersample class 0 or use weighted loss. Low sampling rate (128 Hz) limits spectral resolution. Verify code meanings from dataset README or Ries et al., 2016.

Data Quality Notes

- Only 2 event codes; binary classification.
- Strong class imbalance (~10:1); undersample majority class.
- Low sampling rate (128 Hz); limited for high-frequency analyses.

References

- Reference: `Ries, et al., 2016`

161. openneuro_ds004706 - Spatial Memory and Non-Invasive Closed-Loop Stimulus Timing

Dataset: openneuro_ds004706 | Year: 2023 | URL: https://openneuro.org/datasets/ds004706
Category: High-level Mental | Subjects: 149 | Files (Completed): 298
Channels: 137 | Sampling Rate: 2048 Hz

Paradigm

A hybrid spatial-navigation and free recall task where participants delivered items to locations in a virtual town environment, with EEG recorded during both encoding and retrieval phases. The paradigm incorporates closed-loop stimulus timing to investigate the relationship between memory encoding, theta oscillations, and stimulation-driven memory enhancement. Sessions included free recall and cued recall phases with video stimuli.

Event Structure

Event ID	Count (total)	Type	Meaning
FFR_REC_WORD	55013	short	Free-form recall word event
CUED_REC_STOP	44064	short	Cued recall stop marker
CUED_REC_CUE	44064	short	Cued recall cue onset
WORD	43260	short	Word stimulus onset (encoding)
REC_WORD	39061	short	Recalled word event
CUED_REC_WORD	34381	short	Cued recall word event
CLASSIFIER_WAIT	22890	short	Closed-loop classifier waiting
POINTER_ON	20872	short	Spatial navigation pointer onset
CLASSIFIER	19005	short	Closed-loop classifier event
EFR_MARK	11498	short	EFR (encoding-phase) marker
CUED_REC_WORD_VV	11328	short	Cued recall word (video verified)
TRIAL_END	2888	short	Trial end marker
TRIAL_START	2884	short	Trial start marker
REC_STOP	2952	short	Recording stop marker
REC_START	2952	short	Recording start marker

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Memory encoding/recall classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
`WORD`	0	Word encoding
`REC_WORD`	1	Recalled word
`CUED_REC_CUE`	2	Cued recall cue
`CUED_REC_WORD`	3	Cued recall response

- Note: High sampling rate (2048 Hz). BREAK_START/STOP, SESS_END are session markers. Closed-loop classifier events (CLASSIFIER, CLASSIFIER_WAIT) may interfere - treat as admin.

Data Quality Notes

- High sampling rate (2048 Hz) - downsample for efficiency.
- 30+ event types; focus on WORD/REC_WORD for memory encoding analysis.

References

- Reference: `Spatial memory and non-invasive closed-loop stimulus timing`

162. openneuro_ds004718 - Le Petit Prince Hong Kong: Naturalistic fMRI and EEG Dataset

Dataset: `openneuro_ds004718` | Year: 2025 | URL: https://openneuro.org/datasets/ds004718
Category: Complex Applied | Subjects: 52 | Files (Completed): 102 | Channels: 62 | Sampling Rate: 1000 Hz

Paradigm

Naturalistic listening task where older Cantonese-speaking adults (65+) listened to a Cantonese translation of "The Little Prince" divided into story sections, followed by periodic comprehension checks, during simultaneous EEG and fMRI recording. Numeric trigger codes 1-30+ mark story segments, comprehension questions, and experimental conditions. Code 14 is most frequent (1,053 total), followed by codes 15 (648), and many codes at ~258 each - consistent with a factorial design across 102 files with ~2-10 events per file per code.

Event Structure

Event ID	Count (total)	Type	Meaning
`14`	1053	long	Story section / condition code 14
`15`	648	long	Story section / condition code 15
`7`	270	long	Story section / condition code 7
`3`	268	long	Story section / condition code 3
`5`	266	long	Story section / condition code 5
`4`	263	long	Story section / condition code 4
`6`	262	long	Story section / condition code 6
`88`	259	long	Story section / condition code 88
`40-112`	~258 each	long	Additional story/condition codes
`1-30`	~257-258 each	long	Story section codes 1-30

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Story Section / Condition Classification (multiclass)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
1	0	Story/condition code 1
2	1	Story/condition code 2
3	2	Story/condition code 3
4	3	Story/condition code 4
5	4	Story/condition code 5
6	5	Story/condition code 6
7	6	Story/condition code 7

- Notes: Numeric codes require full decode from dataset README or events.tsv. Naturalistic listening paradigm; continuous EEG during extended story segments. Code 14 is dominant (1,053) vs many codes at ~258 - likely reflects story structure (code 14 = more frequent section). 30+ distinct numeric codes; consider grouping.

Data Quality Notes

- Numeric code meaning requires README or events.tsv decode.
- Naturalistic paradigm - EEG is continuous during extended listening blocks.
- Simultaneous 7T fMRI and EEG - verify gradient artifact removal.

References

- Reference: `Le Petit Prince Hong Kong: Naturalistic fMRI and EEG dataset from older Cantonese speakers`

163. openneuro_ds004745 - 8-Channel SSVEP EEG Dataset with Artifact Trials

Dataset: openneuro_ds004745 | Year: 2023 | URL: https://openneuro.org/datasets/ds004745
Category: Sensory/Response | Subjects: 6 | Files (Completed): 6
Channels: 8 | Sampling Rate: 1000 Hz

Paradigm

Participants performed Steady-State Visual Evoked Potential (SSVEP) tasks at stimulation frequencies of 2 Hz, 4 Hz, and 8 Hz, under both static artifact-free and voluntary movement artifact conditions. The dataset was created to test artifact removal algorithms for SSVEP-based BCIs under realistic conditions with deliberate motion contamination. Multiple trigger codes mark different frequency and artifact conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
10	44	short	Trigger code 10 - likely artifact or rest condition
4	38	short	4 Hz SSVEP frequency onset
2	36	short	2 Hz SSVEP frequency onset
1	36	short	Trigger code 1
3	36	short	Trigger code 3
5	36	short	Trigger code 5
6	36	short	Trigger code 6
7	36	short	Trigger code 7
8	38	short	8 Hz SSVEP frequency onset

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: SSVEP frequency condition classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
2	0	2 Hz SSVEP condition

Event	y	Meaning
4	1	4 Hz SSVEP condition
8	2	8 Hz SSVEP condition
10	3	Artifact/rest condition

- Note: Only 6 subjects - very small dataset. Full trigger decode needed from events.tsv. Deliberate motion artifacts present in some trials.

Data Quality Notes

- Only 6 subjects and 8 channels - limited for generalization.
- Deliberate motion artifacts present in some trials; note artifact condition labels.

References

- Reference: Kumaravel et al., (IEEE EMBC 2021)

164. openneuro_ds004752 - Dataset of Intracranial EEG, Scalp EEG and Beamforming Sources from Epilepsy Patients - Verbal Working Memory

Dataset: `openneuro_ds004752` | Year: 2023 | URL: https://openneuro.org/datasets/ds004752
Category: High-level Mental | Subjects: 15 | Files (Completed): 136
Channels: 64 | Sampling Rate: 2000 Hz

Paradigm

Epilepsy patients performed a modified Sternberg verbal working memory task consisting of encoding, maintenance, and recall phases while simultaneous scalp and intracranial EEG were recorded. The dataset includes iEEG from depth electrodes, scalp EEG, single-unit neuron spike times, and electrode localizations, enabling multi-level analysis of memory processes. No trial-level event annotations are available in the BIDS events.tsv.

Event Structure

Event ID	Count (total)	Type	Meaning
(no events)	-	-	No event markers detected in events.tsv

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Continuous Segmentation

- Epoch method: `epoch_by_segmentation_hdf5`
- Segment:segment_length = 2.0 s, overlap = 0.5
- Target Events -> y: None (no event tag)
- Note: No trial-level events available in events.tsv. EEG is continuous during task performance. Use sliding-window segmentation. Trial timing may be available in accompanying behavioral logs not captured in BIDS events.tsv.

Data Quality Notes

- No event markers - trial-epoch structure not available from events.tsv.
- Very high sampling rate (2000 Hz) - downsample before analysis.
- Epilepsy patients; ictal episodes should be identified and excluded.

References

- Reference: `Dataset of human medial temporal lobe neurons, scalp and intracranial EEG during a verbal working memory task. DOI: 10.1038/s41597-020-0364-3`

165. openneuro_ds004784 - Phantom EEG Dataset with Motion, Muscle, and Eye Artifacts

Dataset: openneuro_ds004784 | Year: 2023 | URL: https://openneuro.org/datasets/ds004784
Category: Others | Subjects: N/A | Files (Completed): 10 | Channels: 264 | Sampling Rate: 512 Hz

Paradigm

Technical validation study using an electrically conductive head phantom to simulate brain activity under various artifact conditions (motion, muscle, eye movements) for testing EEG cleaning algorithms. Code 65471 appears in all 10 files (20 total events); code 65343 appears in all 10 files (10 total events). These likely mark artifact-on vs artifact-off conditions or different artifact types.

Event Structure

Event ID	Count (total)	Type	Meaning
65471	20	short	Artifact condition marker A (e.g., artifact-on)
65343	10	short	Artifact condition marker B (e.g., clean condition)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Artifact Condition Classification (binary)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
65471	0	Artifact condition A
65343	1	Artifact condition B

- Notes: Very small dataset (10 files, 30 total events). Verify code meanings from dataset README. 264 channels - ultra-high density EEG phantom. Class imbalance 2:1 (20 vs 10).

Data Quality Notes

- Very small dataset (10 files, 30 events total) - insufficient for standard classification.
- 264-channel phantom EEG; not standard scalp EEG configuration.
- Designed for artifact algorithm validation, not neural decoding.

References

- Reference: Phantom EEG Dataset with Motion, Muscle, and Eye Artifacts and Example Scripts

166. openneuro_ds004796 - PEARL-Neuro Database (Alzheimer's Risk Genes, Lifestyle, Neuroimaging)

Dataset: openneuro_ds004796 | Year: 2025 | URL: https://openneuro.org/datasets/ds004796
Category: High-level Mental | Subjects: 79 | Files (Completed): 233
Channels: 127 | Sampling Rate: 1000 Hz

Paradigm

A multimodal neuroimaging database of healthy middle-aged adults (50-63 years) carrying Alzheimer's risk genes (APOE, PICALM). The EEG protocol includes resting-state recordings (eyes open and closed) and two active cognitive tasks: the Multi-Source Interference Task (MSIT) and Sternberg's memory task. Stimulus and button-press response events are recorded across task sessions.

Event Structure

Event ID	Count (total)	Type	Meaning
stimulus	27862	short	Stimulus onset (MSIT or Sternberg item)
response	24051	short	Participant button-press response

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Stimulus vs response classification (binary)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
stimulus	0	Stimulus onset
response	1	Button-press response

- Note: 233 files = 79 subjects x ~3 sessions. Resting-state sessions have no stimulus events; only task sessions contribute stimulus/response epochs.

Data Quality Notes

- Mixed resting-state and task sessions; filter by file task label.
- High channel count (127); may include non-EEG channels.

References

- Reference: PEARL-Neuro Database: EEG, fMRI, health and lifestyle data of middle-aged people at risk of dementia. DOI: 10.1038/s41597-024-03106-5

167. openneuro_ds004802 - Loneliness in the Brain: Distinguishing Between Hypersensitivity and Hyperalertness (Pilot)

Dataset: `openneuro_ds004802` | Year: 2023 | URL: https://openneuro.org/datasets/ds004802
Category: High-level Mental | Subjects: 79 | Files (Completed): 79
Channels: 73 | Sampling Rate: 512 Hz

Paradigm

A pilot EEG study investigating the neural underpinnings of loneliness by contrasting social hypersensitivity (amplified neural responses to social stimuli) and hyperalertness (heightened attention to potential social threats). Participants viewed social and non-social stimuli while EEG was recorded. Trigger codes encode stimulus type, valence, and response conditions. Code 20 dominates the event structure (49,726 total) and may represent a high-frequency refresh or fixation signal.

Event Structure

Event ID	Count (total)	Type	Meaning
`20`	49726	short	Dominant trigger (likely fixation/refresh signal)
`111`	4144	short	Trigger code 111
`25`	2823	short	Trigger code 25
`24`	2817	short	Trigger code 24
`21`	2747	short	Trigger code 21
`23`	2667	short	Trigger code 23
`22`	2666	short	Trigger code 22
`42`	2319	short	Trigger code 42
`44`	2316	short	Trigger code 44
`43`	2246	short	Trigger code 43
`61`	2221	short	Trigger code 61
`62`	2185	short	Trigger code 62
`65`	2182	short	Trigger code 65

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - no dedicated kernel required

Task 1: Stimulus classification (multiclass)

- Epoch method: `epoch_by_event_hdf5`

- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
20	0	Primary stimulus trigger
111	1	Trigger code 111
25	2	Trigger code 25
24	3	Trigger code 24
21	4	Trigger code 21
23	5	Trigger code 23
22	6	Trigger code 22

- Note: 30+ distinct trigger codes; full decode from events.tsv required. Code 20 dominates (49,726 total) - likely a high-frequency fixation or visual onset marker.

Data Quality Notes

- 30+ distinct trigger codes - check events.tsv decode table.
- Code 20 (49,726 total) is anomalously high - may be a fixation/refresh signal rather than stimulus.

References

- Reference: `Loneliness in the Brain: Distinguishing Between Hypersensitivity and Hyperalertness`

168. openneuro_ds004816 - EEG Attention RSVP Experiment 1

Dataset: openneuro_ds004816 | Year: 2023 | URL: https://openneuro.org/datasets/ds004816
Category: Sensory/Response | Subjects: N/A | Files (Completed): 40 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

A Rapid Serial Visual Presentation (RSVP) task investigating the neural dynamics of prioritizing task-relevant information using stimuli composed of small letters on large objects. Event codes `E 1` and `E 2` are the dominant condition markers (69,120 each total); codes `1` and `0` appear 31,104 times each; `E 3` appears 1,920 times. The `E 1/E 2` pair likely marks the two main stimulus streams (letter-prioritized vs object-prioritized). Codes `1/0` mark target-present vs target-absent trials. `Event/E x` variants are the same conditions from different recordings.

Event Structure

Event ID	Count (total)	Type	Meaning
`E 1`	69120	short	RSVP condition 1 (primary stream)
`E 2`	69120	short	RSVP condition 2 (secondary stream)
`1`	31104	short	Target-present marker
`0`	31104	short	Target-absent marker
`Event/E 1`	6912	short	RSVP condition 1 (alternate format)
`Event/E 2`	6912	short	RSVP condition 2 (alternate format)
`E 3`	1920	short	RSVP condition 3 (rare)
`Event/E 3`	192	short	RSVP condition 3 (alternate format, rare)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: RSVP Condition Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`E 1`	0	RSVP condition 1
`E 2`	1	RSVP condition 2
`1`	2	Target-present
`0`	3	Target-absent

- Notes: Event/E 1 and Event/E 2 are alternate-format versions of the same codes; merge with E 1/E 2 if needed. E 3 (1,920) and Event/E 3 (192) are rare - exclude or treat as separate class. 5 Hz RSVP; verify stimulus duration from events.tsv.

Data Quality Notes

- Event/E x variants likely from different recording format; merge with E x before epoching.
- E 3 and Event/E 3 are rare codes - may represent catch trials or errors.
- High-count RSVP paradigm; very large epoch counts per subject.

References

- Reference: Grootswagers T., Robinson A.K., Shatek S.M., Carlson T.A. (2021). The neural dynamics underlying prioritisation of task-relevant information. Neurons, Behaviour, Data Analysis, and Theory, 5(1)

169. openneuro_ds004817 - EEG-attention-rsvp-exp2

Dataset: openneuro_ds004817 | Year: 2023 | URL: https://openneuro.org/datasets/ds004817
Category: Sensory/Response | Subjects: 40 | Files (Completed): 40
Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

A Rapid Serial Visual Presentation (RSVP) task (Experiment 2) investigating the neural dynamics of prioritizing task-relevant information, using stimuli consisting of small objects superimposed on large letters. Participants performed target detection in rapid streams. Experiment 2 is a companion to Experiment 1 (ds004816) with different stimulus configurations (objects on letters rather than letters on objects).

Event Structure

Event ID	Count (total)	Type	Meaning
E 1	69137	short	RSVP event type 1 (task-relevant stimulus)
E 2	69137	short	RSVP event type 2 (task-irrelevant stimulus)
1	29376	short	Trigger code 1 (response or condition marker)
0	29376	short	Trigger code 0
Event/E 1	10385	short	RSVP event type 1 (alternate format)
Event/E 2	10385	short	RSVP event type 2 (alternate format)
E 3	1921	short	RSVP event type 3
Event/E 3	289	short	RSVP event type 3 (alternate format)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: RSVP target classification (multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
E 1	0	RSVP event type 1

Event	y	Meaning
E 2	1	RSVP event type 2
E 3	2	RSVP event type 3
1	3	Response/condition code 1
0	4	Response/condition code 0

- Note: Companion to ds004816 (Experiment 1). Event/E 1 and Event/E 2 are identical events in different naming format - treat as same class or aggregate.

Data Quality Notes

- E 1/Event/E 1 and E 2/Event/E 2 are duplicate event formats - aggregate for analysis.
- Companion dataset: ds004816 is Experiment 1.

References

- Reference: Grootswagers T., Robinson A.K., Shatek S.M., Carlson T.A. (2021). The neural dynamics underlying prioritisation of task-relevant information. Neurons, Behaviour, Data Analysis, and Theory, 5(1)

170. openneuro_ds004840 - EEG/ECG/EMG During Music Therapy in ICU Burn Patients

Dataset: `openneuro_ds004840` | Year: 2023 | URL: https://openneuro.org/datasets/ds004840
Category: Complex Applied | Subjects: 9 | Files (Completed): 68 | Channels: 10 | Sampling Rate: 1024 Hz

Paradigm

Electrophysiological recordings (EEG, ECG, EMG) of adult burn patients in a clinical ICU environment during Music-Assisted Relaxation (MAR) therapy sessions. No event markers are present; EEG is continuous throughout the therapy recording. The study investigates physiological responses to music therapy as a pain and anxiety management intervention.

Event Structure

Event ID	Count (total)	Type	Meaning
(no events)	-	-	No event markers detected in events.tsv

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Continuous Segmentation (segmentation)

- Epoch Method: `segmentation`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: No event markers. Use sliding-window segmentation. Condition labels (pre-therapy / during / post-therapy) may be available from session metadata or participants.tsv.

Data Quality Notes

- No event markers; continuous EEG during music therapy.
- Clinical ICU environment - subject motion, medical equipment interference expected.
- Only 10 channels including ECG and EMG - limited spatial resolution for EEG analysis.

References

- Reference: `Dataset of electrophysiological signals (EEG, ECG, EMG) during Music therapy with adult burn patients in the Intensive Care Unit.`

171. openneuro_ds004841 - TX14 (Semi-Autonomous Vehicle Monitoring)

Dataset: `openneuro_ds004841` | Year: 2023 | URL: https://openneuro.org/datasets/ds004841
Category: Complex Applied | Subjects: 11 | Files (Completed): 11 | Channels: 70 | Sampling Rate: 256 Hz

Paradigm

Participants monitored a semi-autonomous vehicle system and performed local situational awareness tasks within a simulated environment, with EEG recorded to characterize cognitive states during supervised automation. Event codes are hierarchically structured 7-digit identifiers encoding operational state and task type.

Event Structure

Event ID	Count (total)	Type	Meaning
`2210001`	998	long	Hierarchical operational state code
`2210002`	947	long	Hierarchical operational state code
`2110000`	220	long	Hierarchical operational state code
`4310000`	195	long	Hierarchical operational state code
`4100001`	195	long	Hierarchical operational state code
`4320000`	195	long	Hierarchical operational state code
`4300010`	107	long	Hierarchical operational state code
`4300020`	96	long	Hierarchical operational state code
`4300100`	80	long	Hierarchical operational state code
`1110002`	76	long	Hierarchical operational state code
`1110001`	76	long	Hierarchical operational state code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Operational State Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
2210001	0	Operational state code
2210002	1	Operational state code
2110000	2	Operational state code
4310000	3	Operational state code
4100001	4	Operational state code
4320000	5	Operational state code

- Notes: All codes are 7-digit hierarchical identifiers. Full decode table required from dataset README. Only 11 subjects - small dataset.

Data Quality Notes

- 7-digit hierarchical numeric codes - full decode needed.
- Only 11 subjects; limited sample size for generalization.

References

- Reference: TX14. ReadMe: TX14 dataset

172. openneuro_ds004843 - Visual Situational Awareness EEG (T16)

Dataset: openneuro_ds004843 | Year: 2023 | URL: https://openneuro.org/datasets/ds004843
Category: Others | Subjects: 7 | Files (Completed): 7 | Channels: 70 | Sampling Rate: 256 Hz

Paradigm

The VisualSituationalAwareness task investigates neural correlates of situational awareness during visual monitoring. Subjects viewed structured visual scenes with EEG recorded across multiple runs. The 5-digit hierarchical trigger codes encode stimulus category, load level, and response type. Codes 22301/22302 (3,384 each) and 22501/22502 (2,331/2,325) dominate; code 32000 (1,099) likely marks scene/trial boundaries. Codes ending in 01/02 mark paired conditions (onset/offset or left/right).

Event Structure

Event ID	Count (total)	Type	Meaning
22301	3384	short	Condition 2-2-3, variant 01
22302	3384	short	Condition 2-2-3, variant 02
22501	2331	short	Condition 2-2-5, variant 01
22502	2325	short	Condition 2-2-5, variant 02
22102	1327	short	Condition 2-2-1, variant 02
22101	1325	short	Condition 2-2-1, variant 01
32000	1099	admin	Scene/trial boundary marker (exclude)
22601	436	short	Condition 2-2-6, variant 01
22602	436	short	Condition 2-2-6, variant 02
42001	382	short	Condition 4-2-0, variant 01
42002	382	short	Condition 4-2-0, variant 02
22001	372	short	Condition 2-2-0, variant 01
22002	372	short	Condition 2-2-0, variant 02
12201	306	short	Condition 1-2-2, variant 01
12202	306	short	Condition 1-2-2, variant 02

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Visual Situational Awareness (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
22301	0	Condition 22301
22302	1	Condition 22302
22501	2	Condition 22501
22502	3	Condition 22502
22101	4	Condition 22101
22102	5	Condition 22102
22601	6	Condition 22601
22602	7	Condition 22602

- Notes: 32000 (1,099) is a scene/trial boundary - exclude. Very small dataset (7 files); balance classes by undersampling. 5-digit codes require README or paper decode.

Data Quality Notes

- Only 7 files - very small dataset for training.
- 5-digit hierarchical codes require decode from README or paper.
- 32000 is a boundary/transition marker - exclude from epoching.

References

- Reference: T16 - Visual Situational Awareness EEG

173. openneuro_ds004844 - T22 (Semi-Automated Driving Trust Paradigm)

Dataset: openneuro_ds004844 | Year: 2023 | URL: https://openneuro.org/datasets/ds004844
Category: Complex Applied | Subjects: 68 | Files (Completed): 68 | Channels: 72 | Sampling Rate: 1024 Hz

Paradigm

Simulated semi-automated driving in a leader-follower paradigm focused on predicting and influencing trust-based decisions regarding control authority hand-off and take-over between human and automation. EEG was recorded during sustained driving across multiple automation levels. Event codes are hierarchically structured 7-digit identifiers encoding driving state and automation mode.

Event Structure

Event ID	Count (total)	Type	Meaning
2310002	8978	long	Driving state code
2310001	8978	long	Driving state code
4111101	8716	long	Driving state code
2572001	3231	long	Driving state code
2562001	2880	long	Driving state code
2552001	1983	long	Driving state code
3110002	1884	long	Driving state code
2812002	1884	long	Driving state code
3110001	1884	long	Driving state code
2812001	1884	long	Driving state code
2714111	1591	long	Driving state code
1410001	1591	long	Driving state code
1410002	1591	long	Driving state code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Driving State Classification (multiclass/long)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
2310002	0	Driving state code
2310001	1	Driving state code
4111101	2	Driving state code

Event	y	Meaning
2572001	3	Driving state code
2562001	4	Driving state code
2552001	5	Driving state code
3110002	6	Driving state code
2812002	7	Driving state code

- Notes: All codes are 7-digit hierarchical identifiers encoding driving condition and automation level. Full decode table required from dataset README. Related dataset: ds004657.

Data Quality Notes

- All event codes are 7-digit hierarchical numeric identifiers - full decode needed.
- 40+ unique codes; group by condition category for practical classification.

References

- Reference: Predicting and influencing trust-based decisions about control authority hand-off and take-over during simulated, semi-automated driving in a leader-follower paradigm

174. openneuro_ds004849 - STRONG (Social-Cognitive EEG Dataset)

Dataset: openneuro_ds004849 | Year: 2023 | URL: https://openneuro.org/datasets/ds004849
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset recorded during a social-cognitive paradigm involving multiple hierarchically encoded stimulus conditions. The 5-digit numeric trigger codes encode stimulus type, valence, and response. The dataset contains a single aggregated file across participants; specific task details require decoding from events.tsv.

Event Structure

Event ID	Count (total)	Type	Meaning
2200	1627	short	Dominant trigger code (primary condition)
72011	338	short	Trigger code (category 7, type 2)
52011	332	short	Trigger code (category 5, type 2)
75011	320	short	Trigger code
55011	317	short	Trigger code
73011	316	short	Trigger code
74011	312	short	Trigger code
53011	308	short	Trigger code
54011	307	short	Trigger code
1	195	short	Trigger code 1 (sync/startx)
71011	174	short	Trigger code
2201	173	short	Trigger code
51011	170	short	Trigger code

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Stimulus Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
2200	0	Primary condition code

Event	y	Meaning
72011	1	Trigger code
52011	2	Trigger code
75011	3	Trigger code
55011	4	Trigger code
73011	5	Trigger code
74011	6	Trigger code
53011	7	Trigger code
54011	8	Trigger code
71011	9	Trigger code
2201	10	Trigger code
51011	11	Trigger code

- Notes: Full decode needed from events.tsv. Only 1 file (single aggregated file). Related datasets: ds004850, ds004852.

Data Quality Notes

- Single aggregated file; individual subject data not separable without additional metadata.
- Full trigger decode required from events.tsv.

References

- Reference: STRONG

175. openneuro_ds004850 - ODE (Social-Cognitive EEG Dataset)

Dataset: `openneuro_ds004850` | Year: 2023 | URL: https://openneuro.org/datasets/ds004850
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset (ODE) recorded during a social-cognitive paradigm with hierarchically encoded stimulus conditions. Shares the identical event structure with the STRONG (ds004849) and InsurgentCivilian (ds004852) datasets, suggesting these are companion recordings from the same experimental series. The single aggregated file requires events.tsv decoding for full interpretation.

Event Structure

Event ID	Count (total)	Type	Meaning
`2200`	1627	short	Dominant trigger code (primary condition)
`72011`	338	short	Trigger code (category 7, type 2)
`52011`	332	short	Trigger code
`75011`	320	short	Trigger code
`55011`	317	short	Trigger code
`73011`	316	short	Trigger code
`74011`	312	short	Trigger code
`53011`	308	short	Trigger code
`54011`	307	short	Trigger code
`1`	195	short	Trigger code 1
`71011`	174	short	Trigger code
`2201`	173	short	Trigger code
`51011`	170	short	Trigger code

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Stimulus Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`2200`	0	Primary condition code
`72011`	1	Trigger code

Event	y	Meaning
52011	2	Trigger code
75011	3	Trigger code
55011	4	Trigger code
73011	5	Trigger code
74011	6	Trigger code
53011	7	Trigger code
54011	8	Trigger code
71011	9	Trigger code
2201	10	Trigger code
51011	11	Trigger code

- Notes: Identical event structure to ds004849 (STRONG) and ds004852 (InsurgentCivilian). Single aggregated file.

Data Quality Notes

- Single aggregated file; individual subject data not separable.
- Identical event structure to ds004849 and ds004852 - likely companion recordings.

References

- Reference: ODE

176. openneuro_ds004851 - HID (High-Density EEG Operational Dataset)

Dataset: `openneuro_ds004851` | Year: 2024 | URL: https://openneuro.org/datasets/ds004851
Category: Complex Applied | Subjects: 65 | Files (Completed): 65 | Channels: 72 | Sampling Rate: 2048 Hz

Paradigm

High-density EEG recorded in an operational context involving multiple conditions encoded as 4-digit numeric trigger codes. The code structure suggests stimulus classes (1xxx: primary task events, 7xxx/8xxx: secondary stimulus categories) within a cognitive workload or situational awareness paradigm.

Event Structure

Event ID	Count (total)	Type	Meaning
`1010`	2181	short	Primary event onset (type A)
`1011`	2179	short	Primary event (type A variant)
`1020`	1970	short	Primary event (type B)
`1021`	1970	short	Primary event (type B variant)
`1040`	1912	short	Primary event (type D)
`1041`	1912	short	Primary event (type D variant)
`8100`	973	short	Secondary stimulus category 8, type 1
`8101`	973	short	Secondary stimulus category 8, type 1 variant
`1061`	474	short	Primary event (type F variant)
`1060`	474	short	Primary event (type F)
`8011`	195	short	Secondary stimulus
`8010`	195	short	Secondary stimulus
`8021`	194	short	Secondary stimulus
`8020`	194	short	Secondary stimulus
`7001-7019`	~65 each	short	Tertiary stimulus category (19 types)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Condition Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`

- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
1010	0	Primary event type A
1011	1	Primary event type A variant
1020	2	Primary event type B
1021	3	Primary event type B variant
1040	4	Primary event type D
1041	5	Primary event type D variant
8100	6	Secondary stimulus
8101	7	Secondary stimulus variant

- Notes: Very high sampling rate (2048 Hz) - downsample before analysis. Full decode table required from dataset README.

Data Quality Notes

- Very high sampling rate (2048 Hz) - downsample for efficiency.
- 30+ distinct codes; grouping by prefix category recommended.

References

- Reference: HID

177. openneuro_ds004852 - InsurgentCivilian (Social-Cognitive EEG Dataset)

Dataset: openneuro_ds004852 | Year: 2023 | URL: https://openneuro.org/datasets/ds004852
Category: Complex Applied | Subjects: 1 | Files (Completed): 1 | Channels: 64 | Sampling Rate: 128 Hz

Paradigm

An EEG dataset (InsurgentCivilian) recorded during a social-cognitive paradigm where participants were likely required to classify ambiguous human figures as insurgents or civilians, a common threat-detection paradigm used in military research. Shares the identical event structure with STRONG (ds004849) and ODE (ds004850), suggesting these are companion recordings from the same experimental series.

Event Structure

Event ID	Count (total)	Type	Meaning
2200	1627	short	Dominant trigger code (primary condition)
72011	338	short	Trigger code (category 7, type 2)
52011	332	short	Trigger code
75011	320	short	Trigger code
55011	317	short	Trigger code
73011	316	short	Trigger code
74011	312	short	Trigger code
53011	308	short	Trigger code
54011	307	short	Trigger code
1	195	short	Trigger code 1
71011	174	short	Trigger code
2201	173	short	Trigger code
51011	170	short	Trigger code

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Stimulus Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
2200	0	Primary condition code
72011	1	Trigger code

Event	y	Meaning
52011	2	Trigger code
75011	3	Trigger code
55011	4	Trigger code
73011	5	Trigger code
74011	6	Trigger code
53011	7	Trigger code
54011	8	Trigger code
71011	9	Trigger code
2201	10	Trigger code
51011	11	Trigger code

- Notes: Identical event structure to ds004849 (STRONG) and ds004850 (ODE). Single aggregated file.

Data Quality Notes

- Single aggregated file; individual subject data not separable.
- Identical event structure to ds004849 and ds004850 - companion recordings.

References

- Reference: `InsurgentCivilian`

178. openneuro_ds004902 - Resting-State EEG Dataset for Sleep Deprivation

Dataset: openneuro_ds004902 | Year: 2023 | URL: https://openneuro.org/datasets/ds004902
Category: Biological State | Subjects: 71 | Files (Completed): 218 | Channels: 61 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings with eyes open and eyes closed were conducted under two conditions: normal sleep and 24-hour sleep deprivation. The study investigates how sleep loss modulates spontaneous brain activity. Subjects participated in two sessions (normal sleep vs sleep-deprived). Files are split by condition (eyesopen/eyesclosed) with 5-minute continuous blocks. Event marker S 33 (60 total) appears as a resting-state session marker in a subset of files; all other S-codes are rare system markers.

Event Structure

Event ID	Count (total)	Type	Meaning
S 33	60	rest	Resting-state session marker (BrainVision)
S 45	7	admin	Rare system marker (exclude)
S 44	7	admin	Rare system marker (exclude)
S 55	7	admin	Rare system marker (exclude)
S119	7	admin	Rare system marker (exclude)
S 66	7	admin	Rare system marker (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Sleep Deprivation Resting-State Segmentation (segmentation)

- Epoch Method: segmentation
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: Majority of files have no event markers - purely continuous resting-state. Use filename-derived or participants.tsv condition labels (eyesopen/eyesclosed x normal/deprived) for downstream classification. S 44/45/55/66/119 are rare admin system markers - exclude.

Data Quality Notes

- Most files have no event markers; condition derived from filename (eyesopen/eyesclosed).
- Session condition (normal sleep / sleep deprivation) in participants.tsv.
- S 44/45/55/66/119 are rare admin BrainVision system markers - exclude.

References

- Reference: Xiang, C., Fan, X., Bai, D. et al. A resting-state EEG dataset for sleep deprivation. Sci Data 11, 427 (2024).

179. openneuro_ds004951 - Braille Letters EEG

Dataset: `openneuro_ds004951` | Year: 2024 | URL: https://openneuro.org/datasets/ds004951
Category: Sensory/Response | Subjects: 23 | Files (Completed): 23 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants identified Braille letters presented via tactile stimulation to left or right index fingers while EEG was recorded. The study investigates cortical plasticity by comparing sighted controls and congenitally blind individuals in processing Braille letter information. Stimulus codes S1-S16 mark individual Braille letter presentations (3,316-3,319 each); S101 and S222 (63,609/63,607) dominate and likely mark response-correct/response-incorrect feedback or block boundaries; S200 (10,535) is a block marker; S253 (9,211) and S254/S255 are admin/system markers.

Event Structure

Event ID	Count (total)	Type	Meaning
`Stimulus/S101`	63609	short	Response feedback / block marker A
`Stimulus/S222`	63607	short	Response feedback / block marker B
`Stimulus/S200`	10535	short	Block boundary / condition marker
`Stimulus/S253`	9211	admin	Admin / response marker (exclude)
`Stimulus/S 16`	3319	short	Braille letter 16 onset
`Stimulus/S 6`	3319	short	Braille letter 6 onset
`Stimulus/S 15`	3319	short	Braille letter 15 onset
`Stimulus/S 11`	3318	short	Braille letter 11 onset
`Stimulus/S 3`	3318	short	Braille letter 3 onset
`Stimulus/S 10`	3317	short	Braille letter 10 onset
`Stimulus/S 1-S 16`	~3316-3319 each	short	Individual Braille letter onsets (16 letters)
`Stimulus/S255`	~437	admin	BrainVision admin sync marker (exclude)
`New Segment/`	rare	admin	BrainVision segment boundary (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Braille Letter Identification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Stimulus/S 1	0	Braille letter 1
Stimulus/S 2	1	Braille letter 2
Stimulus/S 3	2	Braille letter 3
Stimulus/S 4	3	Braille letter 4
Stimulus/S 5	4	Braille letter 5
Stimulus/S 6	5	Braille letter 6
Stimulus/S 7	6	Braille letter 7
Stimulus/S 8	7	Braille letter 8
Stimulus/S 9	8	Braille letter 9
Stimulus/S 10	9	Braille letter 10
Stimulus/S 11	10	Braille letter 11
Stimulus/S 12	11	Braille letter 12
Stimulus/S 13	12	Braille letter 13
Stimulus/S 14	13	Braille letter 14
Stimulus/S 15	14	Braille letter 15
Stimulus/S 16	15	Braille letter 16

- Notes: New Segment/, Stimulus/S253, Stimulus/S254, Stimulus/S255 are admin - exclude. S101/S222 (63k each) are very high-frequency feedback/block markers - exclude from classification. S200 is a block boundary - exclude.

Data Quality Notes

- Stimulus/S101 and Stimulus/S222 are very high-frequency (63k each) - likely feedback or block markers, exclude from classification.
- New Segment/, Stimulus/S253, Stimulus/S254, Stimulus/S255 are admin/system - exclude.
- Dataset includes sighted controls and congenitally blind participants; group label in participants.tsv.

References

- Reference: The transformation of sensory to perceptual braille letter representations in the visually deprived brain

180. openneuro_ds004980 - EEG Data for an Architectural Affordances Task

Dataset: openneuro_ds004980 | Year: 2024 | URL: https://openneuro.org/datasets/ds004980
Category: Complex Applied | Subjects: 17 | Files (Completed): 17 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants navigated a virtual architectural environment using keyboard controls while EEG was recorded to investigate neural responses to architectural affordances - the action possibilities offered by built spaces. The study examines how the brain processes spatial features (doors, corridors, rooms) that afford specific navigational actions. SAM (Self-Assessment Manikin) rating events mark periodic subjective valence/arousal/dominance assessments.

Event Structure

Event ID	Count (total)	Type	Meaning
Key released: UpArrow (forward)	13072	short	Forward key release
Key pressed: UpArrow (forward)	13063	short	Forward key press
Key pressed: RightArrow (turn right)	8915	short	Right turn key press
Key released: RightArrow (turn right)	8911	short	Right turn key release
Key pressed: LeftArrow (turn left)	7360	short	Left turn key press
Key released: LeftArrow (turn left)	7356	short	Left turn key release
Player left StartSquare	6904	short	Participant exits start zone
Player entered SAM Activator	4584	short	SAM rating zone entered
SAM:start	4521	short	SAM rating starts
SAM:completed	4517	short	SAM rating completed
Player left SAM Activator	4567	short	SAM rating zone exited
Wall touched: trial failed!	2183	short	Navigation error
Door opened	2427	short	Door affordance activated
ColliderRoom1 has been entered	1528	short	Room 1 entered
ColliderRoom2 has been entered	1523	short	Room 2 entered

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Navigation Action Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`Key pressed: UpArrow (forward)`	0	Forward navigation
`Key pressed: RightArrow (turn right)`	1	Right turn
`Key pressed: LeftArrow (turn left)`	2	Left turn
`Door opened`	3	Door affordance activated
`ColliderRoom1 has been entered`	4	Room 1 entered
`ColliderRoom2 has been entered`	5	Room 2 entered

- Notes: SAM events (SAM:start, SAM:completed) mark subjective rating periods - treat separately or exclude. Reset/admin events (Reset insideActivator flag, player position reseted) are admin.

Data Quality Notes

- 30+ unique event types; focus on key presses and spatial boundary events.
- SAM rating events mark active rating periods - exclude from navigation classification.

References

- Reference: `EEG data set for an architectural affordances task`

181. openneuro_ds005087 - Rapid Hemifield Object EEG

Dataset: openneuro_ds005087 | Year: 2024 | URL: https://openneuro.org/datasets/ds005087
Category: Sensory/Response | Subjects: 80 | Files (Completed): 80 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed object and word stimuli presented at 5 Hz in rapid visual streams to the left hemifield, right hemifield, or central visual field while performing a red target detection task. Event codes E 1/E 2 (184,320 each) and E 3 (960) mark the three presentation positions (left/right/center hemifield). Numeric codes 2/3/1 (92,160/46,080/46,080) likely encode stimulus categories or object identities within each presentation stream.

Event Structure

Event ID	Count (total)	Type	Meaning
E 1	184320	short	Hemifield/position condition 1
E 2	184320	short	Hemifield/position condition 2
2	92160	short	Stimulus category/identity code 2
3	46080	short	Stimulus category/identity code 3
1	46080	short	Stimulus category/identity code 1
E 3	960	short	Hemifield/position condition 3 (rare)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Visual Hemifield Presentation Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
E 1	0	Hemifield/position condition 1
E 2	1	Hemifield/position condition 2
1	2	Stimulus category code 1
2	3	Stimulus category code 2
3	4	Stimulus category code 3

- Notes: E 3 (960) is rare - exclude or treat as separate class. 5 Hz RSVP streams; very short inter-stimulus interval. Verify code meanings from events.tsv (Robinson et al., 2025).

Data Quality Notes

- Verify full event decode from events.tsv or Robinson et al., 2025.
- 5 Hz RSVP - confirm exact stimulus duration from events.tsv.
- `E 3` (960) is rare relative to E 1/E 2 (184k each); exclude or treat separately.

References

- Reference: `Robinson A.K., Grootswagers T., Shatek S., Behrmann M., Carlson T.A. (2025). Dynamics of visual object coding within and across the hemispheres: Objects in the periphery. Science Advances, 11, eadq0889`

182. openneuro_ds005106 - Human Infant EEG for 200 Object Images in Rapid Visual Streams

Dataset: `openneuro_ds005106` | Year: 2024 | URL: https://openneuro.org/datasets/ds005106
Category: Sensory/Response | Subjects: 42 | Files (Completed): 42 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Infants viewed 200 distinct object images presented in rapid visual streams (RSVP) while EEG was recorded to study the development of visual object representations. Each numeric trigger code corresponds to one of 200 object IDs; code -1 is a baseline/inter-stimulus marker. The dataset enables decoding of object-specific neural representations in the infant brain.

Event Structure

Event ID	Count (total)	Type	Meaning
`-1`	43105	short	Baseline / inter-stimulus marker
`155`	220	short	Object image 155 onset
`123`	220	short	Object image 123 onset
`72`	219	short	Object image 72 onset
`147`	219	short	Object image 147 onset
`(~196 more object codes)`	~219 each	short	Individual object image onsets (IDs 1-200)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Object Image Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
(numeric codes 1-200)	0-199	Object image ID (200 classes)

- Notes: Code -1 is a baseline/ITI marker - exclude from classification. ~200 object classes with ~219 occurrences each. Representational similarity analysis (RSA) is a primary use case.

Data Quality Notes

- ~200 unique object trigger codes - each code represents one object image.
- Code -1 is a baseline/ITI marker and should be excluded.
- Infant EEG - expect high noise and movement artifacts.

References

- Reference: Human Infant EEG Recordings for 200 Object Images Presented in Rapid Visual Streams

183. openneuro_ds005114 - EEG: DPX Cognitive Control Task in Acute Mild TBI

Dataset: `openneuro_ds005114` | Year: 2024 | URL: https://openneuro.org/datasets/ds005114
Category: High-level Mental | Subjects: 62 | Files (Completed): 217 | Channels: 65 | Sampling Rate: 500 Hz

Paradigm

The Dot Pattern Expectancy (DPX) task, a version of the Continuous Performance Task, was used to assess proactive versus reactive cognitive control in sub-acute mild traumatic brain injury (mTBI) patients and healthy controls. Participants responded to cue-probe pairs (CueA/CueB x Probe aX/aY/bY), enabling decomposition of context-maintenance and response-inhibition processes via P3a/P3b ERP components.

Event Structure

Event ID	Count (total)	Type	Meaning
`Response: CueCorr`	50287	short	Correct response to cue
`Response: Probe_Corr`	46594	short	Correct response to probe
`CueAOnset`	43580	short	Cue A stimulus onset
`Probe_aX_onset`	35272	short	Probe aX onset (congruent with CueA)
`STATUS`	19463	admin	System status marker (exclude)
`Practice: filler trigger`	15787	admin	Practice trial filler (exclude)
`Probe_aY_onset`	6443	short	Probe aY onset
`Probe_bY_onset`	5428	short	Probe bY onset
`Response: Probe_Incorr`	3626	short	Incorrect probe response
`Feedback: ProbeErr`	3626	short	Error feedback
`CueBOnset`	2661	short	Cue B stimulus onset

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Cue and Probe Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`CueAOnset`	0	Cue A onset
`CueBOnset`	1	Cue B onset
`Probe_aX_onset`	2	Probe aX (AX trial - expected)

Event	y	Meaning
Probe_aY_onset	3	Probe aY (AY trial - unexpected)
Probe_bY_onset	4	Probe bY (BY trial)
Response: CueCorr	5	Correct cue response
Response: Probe_Corr	6	Correct probe response

- Notes: STATUS (19,463) and Practice: filler trigger (15,787) are admin - exclude. mTBI patients and healthy controls included; participant group labels in participants.tsv.

Data Quality Notes

- STATUS and Practice filler triggers must be excluded from training.
- Dataset includes mTBI patients and healthy controls - group label in participants.tsv.

References

- Reference: ERPs predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. DOI: 10.1016/j.neuropsychologia.2019.107125

184. openneuro_ds005169 - Intracranial EEG During Cortical Stimulation Evoking Visual Effects

Dataset: `openneuro_ds005169` | Year: 2024 | URL: https://openneuro.org/datasets/ds005169
Category: Sensory/Response | Subjects: 22 | Files (Completed): 112 | Channels: 82 | Sampling Rate: 4096 Hz

Paradigm

Intracranial EEG was recorded from drug-resistant epilepsy patients undergoing presurgical evaluation during direct electrical stimulation (DES) of cortical sites that evoke visual hallucinations and perceptual effects. `Comment/prestim` (112 total) marks pre-stimulation baseline periods and `Comment/poststim` (111) marks post-stimulation response windows. UUID-format Comment events (2-10 per file) are patient-specific or trial-specific annotations.

Event Structure

Event ID	Count (total)	Type	Meaning
`Comment/prestim`	112	long	Pre-stimulation baseline period
`Comment/poststim`	111	long	Post-stimulation response window
`Comment/[UUID]`	2-10/file	admin	Patient/trial-specific annotation (UUID format, exclude)
`Comment/[electrode] [mA] SYM`	~2/file	admin	Stimulation parameter annotation (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Pre vs Post Stimulation (binary)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`Comment/prestim`	0	Pre-stimulation baseline
`Comment/poststim`	1	Post-stimulation response

- Notes: UUID-format and electrode/mA Comment events are admin - exclude. Very high sampling rate (4096 Hz) - downsample before processing. iEEG data; stimulation artifact present in poststim windows.

Data Quality Notes

- Very high sampling rate (4096 Hz) - downsample to <=1000 Hz before analysis.
- Intracranial EEG - different noise profile from scalp EEG; stimulation artifacts expected.
- UUID-format Comment events are patient-specific annotations - exclude from epoching.

References

- Reference: Imaging the effective networks associated with cortical function through intracranial high-frequency stimulation. DOI: 10.1002/hbm.25749

185. openneuro_ds005178 - Ear-EEG Sleep Monitoring 2023 (EESM23)

Dataset: `openneuro_ds005178` | Year: 2024 | URL: https://openneuro.org/datasets/ds005178
Category: Biological State | Subjects: 10 | Files (Completed): 159 | Channels: 5 | Sampling Rate: 250 Hz

Paradigm

Naturalistic nightly sleep monitoring using ear-EEG (in-ear electrodes) and polysomnography (PSG) in an ecological home environment where participants self-applied the recording equipment. The study investigates the feasibility of home-based sleep staging using miniaturized ear-EEG compared to gold-standard PSG. The `trigger` event marks synchronization pulses between the ear-EEG and PSG systems.

Event Structure

Event ID	Count (total)	Type	Meaning
`trigger`	175	long	Synchronization trigger between ear-EEG and PSG

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Sleep Monitoring Segmentation (segmentation)

- Epoch Method: `segmentation`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: Only synchronization triggers present - no sleep stage labels in events.tsv. Sleep stage annotations from PSG should be used as external labels for downstream classification. Only 5 EEG channels (ear electrodes).

Data Quality Notes

- Only 5 channels (ear-EEG) - very limited spatial resolution.
- No sleep stage labels in events.tsv; PSG annotations needed for staging.
- Home-recorded data - subject-specific motion and environmental artifacts expected.

References

- Reference: `Ear-EEG Sleep Monitoring 2023 (EESM23)`

186. openneuro_ds005185 - Ear-EEG Sleep Monitoring 2019 (EESM19)

Dataset: `openneuro_ds005185` | Year: 2024 | URL: https://openneuro.org/datasets/ds005185
Category: Biological State | Subjects: 20 | Files (Completed): 516 | Channels: 25 | Sampling Rate: 500 Hz

Paradigm

Naturalistic multi-night sleep monitoring using ear-EEG and polysomnography (PSG) in a home environment collected in 2019. Participants self-applied in-ear EEG electrodes over multiple nights for sleep staging and feasibility assessment. Event markers are Danish-language protocol annotations (lights out, going to bed, waking up), synchronization pulses, and calibration periods. The dominant event is `trigger` (30,726 total) - a PSG synchronization pulse. Sleep stage labels are derived from PSG annotations rather than EEG event markers.

Event Structure

Event ID	Count (total)	Type	Meaning
`trigger`	30726	admin	Synchronization trigger between ear-EEG and PSG (exclude)
`Lights Out`	94	rest	Lights out - sleep onset marker
`Gik i seng`	80	rest	"Went to bed" (Danish)
`Biocalibrations`	79	admin	Biocalibration period (exclude)
`Sync`	77	admin	Synchronization marker (exclude)
`Start af måling`	75	admin	"Start of recording" (Danish, exclude)
`Stod op`	70	rest	"Got up" (Danish) - wake marker
`Biocalibrations start`	11	admin	Biocalibration start (exclude)
`Vågen / vågen`	13	rest	"Awake" (Danish)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Sleep Monitoring Segmentation (segmentation)

- Epoch Method: `segmentation`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: All event markers are Danish-language protocol annotations or sync triggers - not suitable as EEG classification targets. Sleep stage labels (N1/N2/N3/REM/Wake) should be obtained from PSG annotation files. `trigger` (30,726) is a high-frequency sync pulse - exclude.

Data Quality Notes

- `trigger` (30,726) is a sync pulse - exclude from classification.
- All event markers are protocol annotations, not neural response labels.
- 25-channel ear-EEG with limited spatial resolution.
- PSG reference annotations required for sleep staging labels.

References

- Reference: `Ear-EEG Sleep Monitoring 2019 (EESM19)`

187. openneuro_ds005262 - ArEEG: Arabic Inner Speech EEG Dataset

Dataset: `openneuro_ds005262` | Year: 2024 | URL: https://openneuro.org/datasets/ds005262
Category: High-level Mental | Subjects: 12 | Files (Completed): 186 | Channels: 8 | Sampling Rate: 250 Hz

Paradigm

EEG recordings of 12 subjects performing imagined (inner) speech tasks based on visual cues presented in Modern Standard Arabic. Subjects silently imagined saying directional words (Up, Down, Left, Right), a Select command, or remained at Rest. The study aims to develop Arabic-language BCI systems using inner speech decoding without overt vocalization.

Event Structure

Event ID	Count (total)	Type	Meaning
`Comment/Wait`	8949	short	Wait period (inter-trial interval)
`Comment/Rest`	4590	rest	Rest condition
`Comment/Up`	918	short	Inner speech - "Up" command
`Comment/Select`	918	short	Inner speech - "Select" command
`Comment/Right`	918	short	Inner speech - "Right" command
`Comment/Left`	917	short	Inner speech - "Left" command
`Comment/Down`	916	short	Inner speech - "Down" command
`Comment/WarmUp`	174	admin	Warm-up period (exclude)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Inner Speech Command Classification (multiclass)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`Comment/Rest`	0	Rest condition
`Comment/Up`	1	Inner speech "Up"
`Comment/Down`	2	Inner speech "Down"
`Comment/Left`	3	Inner speech "Left"
`Comment/Right`	4	Inner speech "Right"

Event	y	Meaning
Comment/Select	5	Inner speech "Select"

- Notes: Comment/Wait is a high-frequency inter-trial marker - exclude from classification. Comment/WarmUp is admin - exclude. Only 8 EEG channels.

Data Quality Notes

- Only 8 EEG channels - limited spatial resolution.
- Comment/Wait (8,949 total) is inter-trial interval - exclude from classification.
- Comment/WarmUp is admin - exclude.

References

- Reference: ArEEG: Arabic Inner Speech EEG dataset

188. openneuro_ds005305 - EEG Resting-State Microstates and Executive Functions

Dataset: openneuro_ds005305 | Year: 2024 | URL: https://openneuro.org/datasets/ds005305
Category: High-level Mental | Subjects: 165 | Files (Completed): 165 | Channels: 64 | Sampling Rate: 512 Hz

Paradigm

Resting-state EEG microstates were recorded to correlate spontaneous brain dynamics with performance on a nine-task battery of executive functions (dual N-back, Stroop, task-switching, etc.). Each file contains alternating eyes-closed (RS_EC, 825 total) and eyes-open (RS_EO, 660 total) resting blocks. The study investigates whether EEG microstate metrics predict individual differences in executive function. boundary (165) is a EEGLAB admin marker; key press events are incidental.

Event Structure

Event ID	Count (total)	Type	Meaning
RS_EC	825	rest	Resting-state eyes-closed block onset
RS_EO	660	rest	Resting-state eyes-open block onset
boundary	165	admin	EEGLAB boundary marker (exclude)
RIGHT pressed/RIGHT released	26	admin	Incidental key presses (exclude)
DOWN pressed/DOWN released	22	admin	Incidental key presses (exclude)
LEFT pressed/LEFT released	16	admin	Incidental key presses (exclude)
Other key events	rare	admin	Incidental key presses (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Eyes Open vs Eyes Closed (binary)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
RS_EC	0	Eyes-closed resting state
RS_EO	1	Eyes-open resting state

- Notes: boundary (165) and all key press events are admin - exclude. Each resting block is ~30-60 s; sliding windows capture microstate dynamics. RS_EC (825) > RS_EO (660) - slight imbalance.

Data Quality Notes

- `boundary` is EEGLAB admin marker - exclude.
- Key press events are incidental - exclude.
- Executive function scores in participants.tsv for regression analyses.

References

- Reference: `Investigating the relationship between Resting-State EEG Microstates and Executive Functions: A Null Finding`

189. openneuro_ds005345 - Le Petit Prince Multitalker: Naturalistic 7T fMRI and EEG Dataset

Dataset: openneuro_ds005345 | Year: 2025 | URL: https://openneuro.org/datasets/ds005345
Category: Complex Applied | Subjects: 25 | Files (Completed): 28
Channels: 64 | Sampling Rate: 500 Hz

Paradigm

Participants engaged in naturalistic listening to excerpts of "Le Petit Prince" (The Little Prince) under single-talker and multi-talker conditions during simultaneous EEG and 7T fMRI recording. The study investigates neural processing of speech under cocktail-party-style listening conditions. Stimulus/S 11 marks single-talker condition onsets; Stimulus/S 31-S 34 mark multi-talker condition variants. Comment/actiCAP Data On is an EEG system administrative marker.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 11	~19/file	long	Single-talker listening block onset
Stimulus/S 31	~6/file	long	Multi-talker condition variant 31 onset
Stimulus/S 32	~6/file	long	Multi-talker condition variant 32 onset
Stimulus/S 33	~6/file	long	Multi-talker condition variant 33 onset
Stimulus/S 34	~6/file	long	Multi-talker condition variant 34 onset
Comment/actiCAP Data On	25/file	admin	EEG recording system marker (exclude)

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Listening Condition Classification (binary/multiclass)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
Stimulus/S 11	0	Single-talker listening
Stimulus/S 31	1	Multi-talker condition 31

Event	y	Meaning
Stimulus/S 32	2	Multi-talker condition 32
Stimulus/S 33	3	Multi-talker condition 33
Stimulus/S 34	4	Multi-talker condition 34

- Remarks: Comment/actiCAP Data On is admin - exclude. Naturalistic listening paradigm with sustained EEG blocks. Very small dataset (28 files, 25 subjects).

Data Quality Notes

- Very small dataset (28 files, 25 subjects).
- Comment/actiCAP Data On is an admin system marker - exclude from epoching.
- Simultaneous 7T fMRI - MR gradient artifacts may require removal if present.

References

- Reference: Le Petit Prince Multitalker: Naturalistic 7T fMRI and EEG Dataset

190. openneuro_ds005363 - Object Recognition in Healthy Aging (ORHA) - EEG

Dataset: openneuro_ds005363 | Year: 2024 | URL: https://openneuro.org/datasets/ds005363
Category: Sensory/Response | Subjects: 43 | Files (Completed): 43 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed images across five visual categories (faces, animals, places, objects, and likely one additional) to investigate visual object representations and how they change with healthy aging. S65-S69 encode the five stimulus categories; S99 is a high-frequency event likely marking fixations or trial separators. The study uses representational similarity analysis to track age-related changes in neural object coding.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 99	21460	short	Fixation/trial separator (high-frequency onset)
Stimulus/S 66	4508	short	Visual category 2
Stimulus/S 67	4463	short	Visual category 3
Stimulus/S 65	4421	short	Visual category 1
Stimulus/S 69	4351	short	Visual category 5
Stimulus/S 68	4285	short	Visual category 4
Stimulus/S 4-Stimulus/S 57	~1377 each	short	Individual exemplar image onsets (60+ items)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Visual Category Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Stimulus/S 65	0	Visual category 1
Stimulus/S 66	1	Visual category 2
Stimulus/S 67	2	Visual category 3
Stimulus/S 68	3	Visual category 4
Stimulus/S 69	4	Visual category 5

- Notes: S99 (21,460 total) is a high-frequency fixation/separator marker - exclude. S4-S57 encode individual exemplar images (60+ codes, ~1377 each); use for RSA analysis.

Data Quality Notes

- S99 is a very high-frequency marker (21,460 total) - likely fixation onset, exclude from category decoding.
- 60+ exemplar codes (~1377 each) suitable for RSA.

References

- Reference: `Healthy aging delays and dedifferentiates high-level visual representations`

191. openneuro_ds005383 - TMNRED: A Chinese Language EEG Dataset for Fuzzy Semantic Target Identification in Natural Reading Environments

Dataset: openneuro_ds005383 | Year: 2024 | URL: https://openneuro.org/datasets/ds005383
Category: High-level Mental | Subjects: 30 | Files (Completed): 720
Channels: 30 | Sampling Rate: 200 Hz

Paradigm

Participants engaged in self-paced natural reading of Chinese sentences and news headlines to identify fuzzy semantic targets embedded in text. The study investigates neural correlates of semantic processing during naturalistic reading, recording EEG with 30 channels at 200 Hz. Numerical BioSemi trigger codes mark word/sentence onset events across three sessions per subject (3 x 240 files = 720 total). The trigger code structure encodes semantic conditions, target presence, and reading context.

Event Structure

Event ID	Count (total)	Type	Meaning
55	240/file	short	Stimulus condition code 55 (decode from events.tsv)
31	240/file	short	Stimulus condition code 31
56	240/file	short	Stimulus condition code 56
36	240/file	short	Stimulus condition code 36
41	240/file	short	Stimulus condition code 41
51	240/file	short	Stimulus condition code 51
16	240/file	short	Stimulus condition code 16
61	240/file	short	Stimulus condition code 61
26	240/file	short	Stimulus condition code 26
46	240/file	short	Stimulus condition code 46
21	240/file	short	Stimulus condition code 21
22	240/file	short	Stimulus condition code 22
32	240/file	short	Stimulus condition code 32
42	240/file	short	Stimulus condition code 42
52	240/file	short	Stimulus condition code 52

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Semantic Target Identification (multiclass)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
16	0	Condition code 16
21	1	Condition code 21
22	2	Condition code 22
26	3	Condition code 26
31	4	Condition code 31
32	5	Condition code 32
36	6	Condition code 36
41	7	Condition code 41
42	8	Condition code 42
46	9	Condition code 46
51	10	Condition code 51
52	11	Condition code 52
55	12	Condition code 55
56	13	Condition code 56
61	14	Condition code 61

- Remarks: 30 distinct numerical codes across all files; full decode from events.tsv or paper required. 720 files from 30 subjects x 24 sessions/files. Low sampling rate (200 Hz) limits high-frequency analysis.

Data Quality Notes

- Numerical trigger codes require decode from events.tsv or dataset paper for semantic interpretation.
- 30 distinct codes; group by semantic condition (target/non-target) when possible.
- Low sampling rate (200 Hz) - suitable for N400/P600 ERP components but not high-gamma analysis.

References

- Reference: `TMNRED, A Chinese Language EEG Dataset for Fuzzy Semantic Target Identification in Natural Reading Environments`

192. openneuro_ds005385 - Resting-State EEG Before and After Cognitive Activity Across the Adult Lifespan (5-Year Follow-Up)

Dataset: `openneuro_ds005385` | Year: 2024 | URL: https://openneuro.org/datasets/ds005385
Category: Biological State | Subjects: 608 | Files (Completed): 3264 | Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

Resting-state EEG was recorded with eyes open and eyes closed before and after a two-hour block of cognitive tasks (working memory, attention, processing speed) across a lifespan cohort of adults. A 5-year longitudinal follow-up is included. BrainVision S-codes (S1-S207+) encode the specific resting-state condition (pre/post x eyes open/closed) and session number.

Event Structure

Event ID	Count (total)	Type	Meaning
`boundary`	3375	admin	BrainVision segment boundary (exclude)
`S 4-S 7`	~207 each	long	Resting-state session markers (BrainVision S-codes)
`S 11-S 19`	~204-205 each	long	Resting-state condition codes
`S 14-S 16`	68 each	long	Resting-state condition (higher count subset)
`S196-S207`	~205-207 each	long	Session/condition index codes

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting Condition Segmentation (long event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
`S 4`	0	Resting condition S4
`S 5`	1	Resting condition S5
`S 6`	2	Resting condition S6
`S 7`	3	Resting condition S7

- Notes: `boundary` (3375) is admin - exclude. S-codes encode resting condition and session; full decode from README/events.tsv required. 3264 files = 608 subjects x multiple sessions.

Data Quality Notes

- boundary markers (3375) are admin - exclude.
- Full S-code decode needed from README to identify pre/post and eyes-open/closed conditions.
- Large dataset (3264 files, 608 subjects); high variability across sessions and age groups.

References

- Reference: `Gajewski et al., 2022, doi: 10.2196/32352`

193. openneuro_ds005403 - Delayed Auditory Feedback EEG/EGG

Dataset: openneuro_ds005403 | Year: 2024 | URL: https://openneuro.org/datasets/ds005403
Category: High-level Mental | Subjects: 66 | Files (Completed): 66
Channels: 66 | Sampling Rate: 1000 Hz

Paradigm

Participants performed a speech production task while receiving delayed auditory feedback (DAF) to study neural processing of self-generated speech and auditory-motor monitoring. Three DAF conditions were used: an adaptation condition (fixed delay) and two random delay conditions (random1, random2), compared against a baseline (no delay). EEG was recorded alongside electroglottography (EGG) to capture vocal fold activity simultaneously. The event markers label continuous speech production blocks.

Event Structure

Event ID	Count (total)	Type	Meaning
adaption	~64/file	long	Adaptation DAF condition block
random1	~64/file	long	Random delay condition 1 block
random2	~64/file	long	Random delay condition 2 block
baseline	~64/file	rest	Baseline (no delay) speech block

Event Type Description:
- short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
- long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
- rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
- admin: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: DAF Condition Classification (multiclass)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
baseline	0	Baseline (no delay)
adaption	1	Adaptation DAF condition
random1	2	Random delay condition 1
random2	3	Random delay condition 2

- Note: All four conditions are sustained speech production blocks (>=2 s). Simultaneous EGG channels included; verify channel naming in channel metadata.

Data Quality Notes

- Simultaneous EGG recording alongside EEG - verify channel type labels.
- Speech production task - EMG artifacts from jaw/throat movement expected.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Delayed Auditory Feedback EEG/EGG`

194. openneuro_ds005406 - EEG Frequency Tagging Reveals the Integration of Dissimilar Observed Actions

Dataset: openneuro_ds005406 | Year: 2024 | URL: https://openneuro.org/datasets/ds005406
Category: Sensory/Response | Subjects: 29 | Files (Completed): 29 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

A frequency-tagging EEG study measuring steady-state visual evoked potentials (SSVEP) as participants observed dissimilar human actions presented in rapid visual sequences. The study uses neural entrainment at different tagging frequencies to measure the integration of distinct action components (e.g., arm and leg movements). Four action-pair conditions are encoded as S-code pairs (S11/12, S21/22, S31/32, S41/42).

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 41	231	short	Action condition 4, sequence start
Stimulus/S 21	231	short	Action condition 2, sequence start
Stimulus/S 22	231	short	Action condition 2, sequence end
Stimulus/S 42	230	short	Action condition 4, sequence end
Stimulus/S 31	230	short	Action condition 3, sequence start
Stimulus/S 32	230	short	Action condition 3, sequence end
Stimulus/S 11	230	short	Action condition 1, sequence start
Stimulus/S 12	230	short	Action condition 1, sequence end
New Segment/	29	admin	BrainVision segment boundary (exclude)
Stimulus/S255	24	admin	S255 admin sync trigger (exclude)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Action Integration Condition Classification (multiclass)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Stimulus/S 11	0	Action condition 1 start
Stimulus/S 12	1	Action condition 1 end
Stimulus/S 21	2	Action condition 2 start
Stimulus/S 22	3	Action condition 2 end
Stimulus/S 31	4	Action condition 3 start
Stimulus/S 32	5	Action condition 3 end
Stimulus/S 41	6	Action condition 4 start
Stimulus/S 42	7	Action condition 4 end

- Notes: New Segment/ (29) and Stimulus/S255 (24) are admin - exclude. Four action conditions with start/end markers each.

Data Quality Notes

- New Segment/ and Stimulus/S255 are administrative - exclude.
- Frequency-tagging paradigm; SSVEP analysis may be more appropriate than traditional ERP epoching.

References

- Reference: EEG frequency tagging reveals the integration of dissimilar observed actions

195. openneuro_ds005416 - Fatigue Characterization of EEG under Mixed Reality Stereo Vision

Dataset: `openneuro_ds005416` | Year: 2024 | URL: https://openneuro.org/datasets/ds005416
Category: Complex Applied | Subjects: 23 | Files (Completed): 23 | Channels: 64 | Sampling Rate: 1000 Hz

Paradigm

Participants viewed resting and movement scenes in a Mixed Reality (MR) stereo vision environment to characterize EEG-based fatigue during periodic visual depth and velocity changes. The study investigates neural correlates of visual fatigue induced by MR headset viewing under different dynamic visual conditions. No trial events were captured in events.tsv; data is continuous.

Event Structure

Event ID	Count (total)	Type	Meaning
(no events)	-	-	No event markers detected in events.tsv

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Continuous Segmentation (segmentation)

- Epoch Method: `segmentation`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5
- Target Events -> y: None
- Notes: No event markers in events.tsv. EEG is continuous during MR viewing task. Use sliding-window segmentation. Condition labels (fatigue level, visual condition) may be available from session metadata.

Data Quality Notes

- No event markers; continuous EEG during MR task.
- High sampling rate (1000 Hz) adequate for spectral fatigue analysis.
- MR headset may introduce electrical interference artifacts.

References

- Reference: `Fatigue Characterization of EEG under Mixed Reality Stereo Vision`

196. openneuro_ds005420 - Resting State EEG with Closed Eyes and Open Eyes in Females from 60 to 80 Years Old

Dataset: openneuro_ds005420 | Year: 2024 | URL: https://openneuro.org/datasets/ds005420
Category: Biological State | Subjects: 72 | Files (Completed): 72
Channels: 20 | Sampling Rate: 500 Hz

Paradigm

Spontaneous resting-state EEG was recorded in a cohort of females aged 60-80 years during alternating eyes-open and eyes-closed conditions. The dataset investigates age-related changes in spontaneous brain activity in older women. No event markers were included in events.tsv; the EEG is stored as continuous recordings. Eyes-open vs. eyes-closed condition can be inferred from the file naming or participants.tsv.

Event Structure

No event markers detected (no_events). This dataset contains continuous resting-state EEG recordings without trial-level annotations.

> Event Type Description:
> - short: short stimulus onset marker (presentation duration <2 s), for epoch_by_event_hdf5
> - long: long-term status flag (retention period is usually >=2 s), used for epoch_by_long_event_hdf5
> - rest: Rest block tag, used for epoch_by_long_event_hdf5 or epoch_by_segmentation_hdf5
> - admin: system/boundary marker, not used for epoch

HDF5 Production Guide

> Tag source: Annotation (events.tsv trial_type column, Generic BIDS kernel directly readable)
> Kernel: Generic BIDS (kernels/bids.py) - no dedicated kernel required

Task 1: Resting-State Segmentation (segmentation)

- Epoch method: epoch_by_segmentation_hdf5
- Segment:segment_length = 2.0 s, overlap = 0.5 s
- Target Events -> y:null(no events; use file-level condition label for eyes-open/closed from filename or participants.tsv)
- Remarks: No event markers. Only 20 EEG channels. Female-only sample aged 60-80. Eyes-open/closed condition may be in filename.

Data Quality Notes

- No event markers - purely continuous resting-state recordings.
- 20-channel montage; limited spatial coverage.
- Female-only cohort aged 60-80; demographic info in participants.tsv.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: EEG-BIDS, an extension to the brain imaging data structure for electroencephalography

197. openneuro_ds005505 - Healthy Brain Network (HBN) EEG - Release 1

Dataset: openneuro_ds005505 | Year: 2025 | URL: https://openneuro.org/datasets/ds005505
Category: Complex Applied | Subjects: 136 | Files (Completed): 1342 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 1 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection (contrastTrial), surrounding suppression, sequence learning (seqLearning), and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	13278	short	Contrast detection stimulus onset
fixpoint_ON	13248	short	Fixation point onset (trial start cue)
contrastTrial_start	8884	long	Contrast detection trial block onset
left_buttonPress	5095	short	Participant button press - left hand
right_buttonPress	4634	short	Participant button press - right hand
right_target	4251	short	Right-side target stimulus
left_target	4243	short	Left-side target stimulus
break cnt	1488	admin	Break/counter marker between blocks
trialResponse	1247	short	Trial response event
instructed_toOpenEyes	816	long	Instruction to open eyes (resting block)
video_start	761	long	Movie clip onset
boundary	728	admin	Segment boundary marker
video_stop	723	long	Movie clip offset
instructed_toCloseEyes	680	long	Instruction to close eyes (resting block)
resting_start	136	long	Resting-state block onset
seqLearning_start	122	long	Sequence learning block onset
learningBlock_1	122	long	Sequence learning block 1 onset
9999	1136	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`break cnt`	0	Break/counter marker
`9999`	1	End-of-recording admin marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_buttonPress`	6	Left button press
`left_target`	7	Left-side target
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`contrastChangeB3_start`	11	Contrast change block 3 onset
`instructed_toOpenEyes`	12	Eyes-open instruction
`instructed_toCloseEyes`	13	Eyes-close instruction
`resting_start`	14	Resting block onset
`dot_no3_ON`	15	Dot stimulus 3 on
`dot_no2_ON`	16	Dot stimulus 2 on
`dot_no2_OFF`	17	Dot stimulus 2 off
`dot_no5_ON`	18	Dot stimulus 5 on
`dot_no3_OFF`	19	Dot stimulus 3 off
`dot_no4_OFF`	20	Dot stimulus 4 off
`dot_no4_ON`	21	Dot stimulus 4 on
`dot_no6_OFF`	22	Dot stimulus 6 off
`dot_no6_ON`	23	Dot stimulus 6 on
`dot_no1_OFF`	24	Dot stimulus 1 off
`dot_no1_ON`	25	Dot stimulus 1 on
`learningBlock_1`	26	Sequence learning block 1
`seqLearning_start`	27	Sequence learning start
`dot_no5_OFF`	28	Dot stimulus 5 off

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (136); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Alexander et al., Sci Data 2017`

198. openneuro_ds005506 - Healthy Brain Network (HBN) EEG - Release 2

Dataset: openneuro_ds005506 | Year: 2025 | URL: https://openneuro.org/datasets/ds005506
Category: Complex Applied | Subjects: 152 | Files (Completed): 1405 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 2 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	14116	short	Contrast detection stimulus onset
fixpoint_ON	14081	short	Fixation point onset (trial start cue)
contrastTrial_start	9310	long	Contrast detection trial block onset
left_buttonPress	5762	short	Participant button press - left hand
right_buttonPress	4970	short	Participant button press - right hand
right_target	4463	short	Right-side target stimulus
left_target	4439	short	Left-side target stimulus
break cnt	1577	admin	Break/counter marker between blocks
trialResponse	1372	short	Trial response event
instructed_toOpenEyes	883	long	Instruction to open eyes (resting block)
boundary	799	admin	Segment boundary marker
video_start	797	long	Movie clip onset
video_stop	765	long	Movie clip offset
instructed_toCloseEyes	736	long	Instruction to close eyes (resting block)
resting_start	150	long	Resting-state block onset
9999	1150	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`break cnt`	0	Break/counter marker
`9999`	1	End-of-recording admin marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`contrastChangeB3_start`	11	Contrast change block 3 onset
`resting_start`	12	Resting block onset
`instructed_toOpenEyes`	13	Eyes-open instruction
`instructed_toCloseEyes`	14	Eyes-close instruction
`dot_no2_ON`	15	Dot stimulus 2 on
`dot_no3_ON`	16	Dot stimulus 3 on
`dot_no2_OFF`	17	Dot stimulus 2 off
`dot_no5_OFF`	18	Dot stimulus 5 off
`dot_no1_ON`	19	Dot stimulus 1 on
`dot_no1_OFF`	20	Dot stimulus 1 off
`dot_no6_ON`	21	Dot stimulus 6 on
`dot_no6_OFF`	22	Dot stimulus 6 off
`dot_no4_ON`	23	Dot stimulus 4 on
`dot_no4_OFF`	24	Dot stimulus 4 off
`dot_no5_ON`	25	Dot stimulus 5 on
`dot_no3_OFF`	26	Dot stimulus 3 off
`seqLearning_start`	27	Sequence learning start
`learningBlock_1`	28	Sequence learning block 1

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.

- Large subject population (152); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Alexander et al., Sci Data 2017`

199. openneuro_ds005507 - Healthy Brain Network (HBN) EEG - Release 3

Dataset: openneuro_ds005507 | Year: 2025 | URL: https://openneuro.org/datasets/ds005507
Category: Complex Applied | Subjects: 183 | Files (Completed): 1812 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 3 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	18602	short	Contrast detection stimulus onset
fixpoint_ON	18564	short	Fixation point onset (trial start cue)
contrastTrial_start	12072	long	Contrast detection trial block onset
left_buttonPress	6995	short	Participant button press - left hand
right_buttonPress	6266	short	Participant button press - right hand
right_target	5800	short	Right-side target stimulus
left_target	5753	short	Left-side target stimulus
break cnt	1973	admin	Break/counter marker between blocks
trialResponse	1757	short	Trial response event
instructed_toOpenEyes	1101	long	Instruction to open eyes (resting block)
boundary	1074	admin	Segment boundary marker
video_start	1041	long	Movie clip onset
video_stop	1013	long	Movie clip offset
instructed_toCloseEyes	915	long	Instruction to close eyes (resting block)
resting_start	184	long	Resting-state block onset
9999	1446	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`contrastChangeB3_start`	11	Contrast change block 3 onset
`instructed_toOpenEyes`	12	Eyes-open instruction
`resting_start`	13	Resting block onset
`instructed_toCloseEyes`	14	Eyes-close instruction
`dot_no2_ON`	15	Dot stimulus 2 on
`dot_no3_ON`	16	Dot stimulus 3 on
`dot_no2_OFF`	17	Dot stimulus 2 off
`dot_no1_ON`	18	Dot stimulus 1 on
`dot_no1_OFF`	19	Dot stimulus 1 off
`dot_no6_ON`	20	Dot stimulus 6 on
`dot_no6_OFF`	21	Dot stimulus 6 off
`dot_no4_ON`	22	Dot stimulus 4 on
`dot_no4_OFF`	23	Dot stimulus 4 off
`dot_no3_OFF`	24	Dot stimulus 3 off
`dot_no5_ON`	25	Dot stimulus 5 on
`dot_no5_OFF`	26	Dot stimulus 5 off
`seqLearning_start`	27	Sequence learning start
`learningBlock_1`	28	Sequence learning block 1

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.

- Large subject population (183); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Alexander et al., Sci Data 2017`

200. openneuro_ds005508 - Healthy Brain Network (HBN) EEG - Release 4

Dataset: openneuro_ds005508 | Year: 2025 | URL: https://openneuro.org/datasets/ds005508
Category: Complex Applied | Subjects: 324 | Files (Completed): 3342 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 4 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	35171	short	Contrast detection stimulus onset
fixpoint_ON	35072	short	Fixation point onset (trial start cue)
contrastTrial_start	20646	long	Contrast detection trial block onset
left_buttonPress	11751	short	Participant button press - left hand
right_buttonPress	10656	short	Participant button press - right hand
right_target	9909	short	Right-side target stimulus
left_target	9806	short	Left-side target stimulus
break cnt	3452	admin	Break/counter marker between blocks
trialResponse	3286	short	Trial response event
instructed_toOpenEyes	1929	long	Instruction to open eyes (resting block)
boundary	1871	admin	Segment boundary marker
instructed_toCloseEyes	1608	long	Instruction to close eyes (resting block)
resting_start	322	long	Resting-state block onset
9999	2502	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_buttonPress`	6	Left button press
`left_target`	7	Left-side target
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`instructed_toOpenEyes`	11	Eyes-open instruction
`instructed_toCloseEyes`	12	Eyes-close instruction
`resting_start`	13	Resting block onset
`dot_no2_ON`	14	Dot stimulus 2 on
`dot_no3_ON`	15	Dot stimulus 3 on
`contrastChangeB3_start`	16	Contrast change block 3 onset
`dot_no2_OFF`	17	Dot stimulus 2 off
`dot_no1_OFF`	18	Dot stimulus 1 off
`dot_no6_ON`	19	Dot stimulus 6 on
`dot_no6_OFF`	20	Dot stimulus 6 off
`dot_no4_ON`	21	Dot stimulus 4 on
`dot_no4_OFF`	22	Dot stimulus 4 off
`dot_no1_ON`	23	Dot stimulus 1 on
`dot_no3_OFF`	24	Dot stimulus 3 off
`dot_no5_ON`	25	Dot stimulus 5 on
`dot_no5_OFF`	26	Dot stimulus 5 off
`seqLearning_start`	27	Sequence learning start
`learningBlock_1`	28	Sequence learning block 1

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (324); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

201. openneuro_ds005509 - Healthy Brain Network (HBN) EEG - Release 5

Dataset: openneuro_ds005509 | Year: 2025 | URL: https://openneuro.org/datasets/ds005509
Category: Complex Applied | Subjects: 330 | Files (Completed): 3326 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 5 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	35156	short	Contrast detection stimulus onset
fixpoint_ON	35136	short	Fixation point onset (trial start cue)
contrastTrial_start	18682	long	Contrast detection trial block onset
left_buttonPress	10192	short	Participant button press - left hand
right_buttonPress	9498	short	Participant button press - right hand
right_target	8977	short	Right-side target stimulus
left_target	8947	short	Left-side target stimulus
break cnt	3296	admin	Break/counter marker between blocks
trialResponse	3367	short	Trial response event
instructed_toOpenEyes	1969	long	Instruction to open eyes (resting block)
boundary	1839	admin	Segment boundary marker
instructed_toCloseEyes	1639	long	Instruction to close eyes (resting block)
resting_start	329	long	Resting-state block onset
9999	2339	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`instructed_toOpenEyes`	11	Eyes-open instruction
`resting_start`	12	Resting block onset
`instructed_toCloseEyes`	13	Eyes-close instruction
`dot_no2_ON`	14	Dot stimulus 2 on
`dot_no3_ON`	15	Dot stimulus 3 on
`dot_no2_OFF`	16	Dot stimulus 2 off
`dot_no1_ON`	17	Dot stimulus 1 on
`dot_no1_OFF`	18	Dot stimulus 1 off
`dot_no6_ON`	19	Dot stimulus 6 on
`dot_no6_OFF`	20	Dot stimulus 6 off
`dot_no4_ON`	21	Dot stimulus 4 on
`dot_no4_OFF`	22	Dot stimulus 4 off
`dot_no3_OFF`	23	Dot stimulus 3 off
`dot_no5_ON`	24	Dot stimulus 5 on
`dot_no5_OFF`	25	Dot stimulus 5 off
`seqLearning_start`	26	Sequence learning start
`learningBlock_1`	27	Sequence learning block 1
`learningBlock_3`	28	Sequence learning block 3

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (330); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Alexander et al., Sci Data 2017`

202. openneuro_ds005510 - Healthy Brain Network (HBN) EEG - Release 6

Dataset: openneuro_ds005510 | Year: 2025 | URL: https://openneuro.org/datasets/ds005510
Category: Complex Applied | Subjects: 3000 | Files (Completed): 1227 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 6 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	13706	short	Contrast detection stimulus onset
fixpoint_ON	13696	short	Fixation point onset (trial start cue)
contrastTrial_start	5817	long	Contrast detection trial block onset
left_buttonPress	3743	short	Participant button press - left hand
right_buttonPress	3249	short	Participant button press - right hand
right_target	2796	short	Right-side target stimulus
left_target	2793	short	Left-side target stimulus
break cnt	1242	admin	Break/counter marker between blocks
trialResponse	1483	short	Trial response event
instructed_toOpenEyes	810	long	Instruction to open eyes (resting block)
instructed_toCloseEyes	676	long	Instruction to close eyes (resting block)
boundary	666	admin	Segment boundary marker
resting_start	135	long	Resting-state block onset
9999	866	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
break cnt	0	Break/counter marker
9999	1	End-of-recording admin marker
video_start	2	Movie clip onset
video_stop	3	Movie clip offset
contrastTrial_start	4	Contrast detection trial onset
right_target	5	Right-side target
left_buttonPress	6	Left button press
left_target	7	Left-side target
right_buttonPress	8	Right button press
stim_ON	9	Contrast stimulus on
fixpoint_ON	10	Fixation point on
instructed_toOpenEyes	11	Eyes-open instruction
instructed_toCloseEyes	12	Eyes-close instruction
resting_start	13	Resting block onset
dot_no2_OFF	14	Dot stimulus 2 off
dot_no2_ON	15	Dot stimulus 2 on
dot_no1_OFF	16	Dot stimulus 1 off
dot_no6_ON	17	Dot stimulus 6 on
dot_no6_OFF	18	Dot stimulus 6 off
dot_no4_ON	19	Dot stimulus 4 on
dot_no4_OFF	20	Dot stimulus 4 off
dot_no1_ON	21	Dot stimulus 1 on
dot_no3_ON	22	Dot stimulus 3 on
dot_no3_OFF	23	Dot stimulus 3 off
dot_no5_ON	24	Dot stimulus 5 on
dot_no5_OFF	25	Dot stimulus 5 off
seqLearning_start	26	Sequence learning start
learningBlock_1	27	Sequence learning block 1
learningBlock_2	28	Sequence learning block 2

- Notes: Multi-task HBN battery. boundary and 9999 are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- boundary and 9999 are administrative markers and should be excluded from scientific analysis.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

203. openneuro_ds005511 - Healthy Brain Network (HBN) EEG - Release 7

Dataset: openneuro_ds005511 | Year: 2025 | URL: https://openneuro.org/datasets/ds005511
Category: Complex Applied | Subjects: 381 | Files (Completed): 3100 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 7 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	34708	short	Contrast detection stimulus onset
fixpoint_ON	34688	short	Fixation point onset (trial start cue)
contrastTrial_start	13272	long	Contrast detection trial block onset
left_buttonPress	7530	short	Participant button press - left hand
right_buttonPress	6992	short	Participant button press - right hand
right_target	6384	short	Right-side target stimulus
left_target	6361	short	Left-side target stimulus
trialResponse	4059	short	Trial response event
break cnt	3092	admin	Break/counter marker between blocks
instructed_toOpenEyes	2193	long	Instruction to open eyes (resting block)
instructed_toCloseEyes	1825	long	Instruction to close eyes (resting block)
boundary	1728	admin	Segment boundary marker
resting_start	371	long	Resting-state block onset
9999	2143	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`break cnt`	0	Break/counter marker
`9999`	1	End-of-recording admin marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`stim_ON`	5	Contrast stimulus on
`right_target`	6	Right-side target
`fixpoint_ON`	7	Fixation point on
`left_target`	8	Left-side target
`left_buttonPress`	9	Left button press
`right_buttonPress`	10	Right button press
`resting_start`	11	Resting block onset
`instructed_toOpenEyes`	12	Eyes-open instruction
`instructed_toCloseEyes`	13	Eyes-close instruction
`dot_no2_OFF`	14	Dot stimulus 2 off
`dot_no2_ON`	15	Dot stimulus 2 on
`dot_no3_ON`	16	Dot stimulus 3 on
`dot_no1_OFF`	17	Dot stimulus 1 off
`dot_no6_ON`	18	Dot stimulus 6 on
`dot_no6_OFF`	19	Dot stimulus 6 off
`dot_no4_ON`	20	Dot stimulus 4 on
`dot_no4_OFF`	21	Dot stimulus 4 off
`dot_no1_ON`	22	Dot stimulus 1 on
`dot_no3_OFF`	23	Dot stimulus 3 off
`dot_no5_ON`	24	Dot stimulus 5 on
`dot_no5_OFF`	25	Dot stimulus 5 off
`seqLearning_start`	26	Sequence learning start
`learningBlock_1`	27	Sequence learning block 1
`learningBlock_3`	28	Sequence learning block 3

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (381); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

204. openneuro_ds005512 - Healthy Brain Network (HBN) EEG - Release 8

Dataset: openneuro_ds005512 | Year: 2025 | URL: https://openneuro.org/datasets/ds005512
Category: Complex Applied | Subjects: 257 | Files (Completed): 2320 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 8 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	22981	short	Contrast detection stimulus onset
fixpoint_ON	22913	short	Fixation point onset (trial start cue)
contrastTrial_start	12963	long	Contrast detection trial block onset
left_buttonPress	6946	short	Participant button press - left hand
right_buttonPress	6875	short	Participant button press - right hand
right_target	6231	short	Right-side target stimulus
left_target	6151	short	Left-side target stimulus
trialResponse	2318	short	Trial response event
break cnt	2258	admin	Break/counter marker between blocks
instructed_toOpenEyes	1326	long	Instruction to open eyes (resting block)
boundary	1237	admin	Segment boundary marker
instructed_toCloseEyes	1104	long	Instruction to close eyes (resting block)
resting_start	223	long	Resting-state block onset
9999	1751	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`instructed_toOpenEyes`	11	Eyes-open instruction
`resting_start`	12	Resting block onset
`instructed_toCloseEyes`	13	Eyes-close instruction
`dot_no2_ON`	14	Dot stimulus 2 on
`dot_no3_ON`	15	Dot stimulus 3 on
`dot_no2_OFF`	16	Dot stimulus 2 off
`dot_no4_ON`	17	Dot stimulus 4 on
`seqLearning_start`	18	Sequence learning start
`dot_no1_OFF`	19	Dot stimulus 1 off
`dot_no6_ON`	20	Dot stimulus 6 on
`dot_no6_OFF`	21	Dot stimulus 6 off
`dot_no4_OFF`	22	Dot stimulus 4 off
`dot_no1_ON`	23	Dot stimulus 1 on
`dot_no3_OFF`	24	Dot stimulus 3 off
`dot_no5_ON`	25	Dot stimulus 5 on
`dot_no5_OFF`	26	Dot stimulus 5 off
`learningBlock_1`	27	Sequence learning block 1
`learningBlock_2`	28	Sequence learning block 2

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (257); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

205. openneuro_ds005514 - Healthy Brain Network (HBN) EEG - Release 9

Dataset: openneuro_ds005514 | Year: 2025 | URL: https://openneuro.org/datasets/ds005514
Category: Complex Applied | Subjects: 295 | Files (Completed): 2885 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 9 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, sequence learning, and symbol search tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	30059	short	Contrast detection stimulus onset
fixpoint_ON	30018	short	Fixation point onset (trial start cue)
contrastTrial_start	16449	long	Contrast detection trial block onset
left_buttonPress	10593	short	Participant button press - left hand
right_buttonPress	10002	short	Participant button press - right hand
right_target	7910	short	Right-side target stimulus
left_target	7857	short	Left-side target stimulus
break cnt	2807	admin	Break/counter marker between blocks
trialResponse	2569	short	Trial response event
instructed_toOpenEyes	1591	long	Instruction to open eyes (resting block)
boundary	1554	admin	Segment boundary marker
instructed_toCloseEyes	1325	long	Instruction to close eyes (resting block)
resting_start	269	long	Resting-state block onset
9999	2155	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`resting_start`	11	Resting block onset
`instructed_toOpenEyes`	12	Eyes-open instruction
`instructed_toCloseEyes`	13	Eyes-close instruction
`dot_no2_OFF`	14	Dot stimulus 2 off
`contrastChangeB2_start`	15	Contrast change block 2 onset
`dot_no2_ON`	16	Dot stimulus 2 on
`dot_no3_ON`	17	Dot stimulus 3 on
`dot_no1_OFF`	18	Dot stimulus 1 off
`dot_no6_ON`	19	Dot stimulus 6 on
`dot_no6_OFF`	20	Dot stimulus 6 off
`dot_no4_ON`	21	Dot stimulus 4 on
`dot_no4_OFF`	22	Dot stimulus 4 off
`dot_no1_ON`	23	Dot stimulus 1 on
`dot_no3_OFF`	24	Dot stimulus 3 off
`dot_no5_ON`	25	Dot stimulus 5 on
`dot_no5_OFF`	26	Dot stimulus 5 off
`seqLearning_start`	27	Sequence learning start
`learningBlock_1`	28	Sequence learning block 1

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (295); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

206. openneuro_ds005515 - Healthy Brain Network (HBN) EEG - Release 10

Dataset: openneuro_ds005515 | Year: 2025 | URL: https://openneuro.org/datasets/ds005515
Category: Complex Applied | Subjects: 533 | Files (Completed): 2515 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 10 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, symbol search, and sequence learning tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	10818	short	Contrast detection stimulus onset
fixpoint_ON	10562	short	Fixation point onset (trial start cue)
contrastTrial_start	5858	long	Contrast detection trial block onset
left_buttonPress	3689	short	Participant button press - left hand
right_buttonPress	3034	short	Participant button press - right hand
instructed_toOpenEyes	2862	long	Instruction to open eyes (resting block)
right_target	2812	short	Right-side target stimulus
left_target	2545	short	Left-side target stimulus
instructed_toCloseEyes	2365	long	Instruction to close eyes (resting block)
video_start	1627	long	Movie clip onset
video_stop	1610	long	Movie clip offset
trialResponse	1211	short	Trial response event
boundary	1385	admin	Segment boundary marker
resting_start	516	long	Resting-state block onset
9999	2105	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`video_start`	1	Movie clip onset
`video_stop`	2	Movie clip offset
`break cnt`	3	Break/counter marker
`instructed_toOpenEyes`	4	Eyes-open instruction
`resting_start`	5	Resting block onset
`instructed_toCloseEyes`	6	Eyes-close instruction
`contrastTrial_start`	7	Contrast detection trial onset
`right_target`	8	Right-side target
`left_target`	9	Left-side target
`left_buttonPress`	10	Left button press
`right_buttonPress`	11	Right button press
`stim_ON`	12	Contrast stimulus on
`fixpoint_ON`	13	Fixation point on
`contrastChangeB2_start`	14	Contrast change block 2 onset
`surroundSuppB1_start`	15	Surround suppression block 1 onset
`contrastChangeB3_start`	16	Contrast change block 3 onset
`contrastChangeB1_start`	17	Contrast change block 1 onset
`trialResponse`	18	Trial response
`newPage`	19	New page (symbol search)
`symbolSearch_start`	20	Symbol search block onset
`surroundSuppB2_start`	21	Surround suppression block 2 onset
`dot_no2_ON`	22	Dot stimulus 2 on
`dot_no3_ON`	23	Dot stimulus 3 on
`dot_no2_OFF`	24	Dot stimulus 2 off
`dot_no4_ON`	25	Dot stimulus 4 on
`dot_no5_ON`	26	Dot stimulus 5 on
`dot_no3_OFF`	27	Dot stimulus 3 off
`dot_no4_OFF`	28	Dot stimulus 4 off

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population (533); verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: Alexander et al., Sci Data 2017

207. openneuro_ds005516 - Healthy Brain Network (HBN) EEG - Release 11

Dataset: openneuro_ds005516 | Year: 2025 | URL: https://openneuro.org/datasets/ds005516
Category: Complex Applied | Subjects: 3000 | Files (Completed): 3396 | Channels: 129 | Sampling Rate: 500 Hz

Paradigm

The Healthy Brain Network (HBN) EEG Release 11 is a large-scale, multi-task developmental battery collected from children and adolescents. Participants completed naturalistic movie watching, eyes-open/closed resting state, sensory contrast detection, surrounding suppression, symbol search, and sequence learning tasks. The dataset aims to characterize neurodevelopmental and psychopathological profiles across a broad pediatric population.

Event Structure

Event ID	Count (total)	Type	Meaning
stim_ON	37825	short	Contrast detection stimulus onset
fixpoint_ON	37760	short	Fixation point onset (trial start cue)
contrastTrial_start	19366	long	Contrast detection trial block onset
left_buttonPress	14777	short	Participant button press - left hand
right_buttonPress	13758	short	Participant button press - right hand
right_target	9311	short	Right-side target stimulus
left_target	9226	short	Left-side target stimulus
trialResponse	4056	short	Trial response event
break cnt	3196	admin	Break/counter marker between blocks
video_start	1666	long	Movie clip onset
instructed_toOpenEyes	1644	long	Instruction to open eyes (resting block)
video_stop	1541	long	Movie clip offset
boundary	1462	admin	Segment boundary marker
instructed_toCloseEyes	1370	long	Instruction to close eyes (resting block)
resting_start	372	long	Resting-state block onset
9999	3038	admin	End-of-recording marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Multi-task Battery (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`9999`	0	End-of-recording admin marker
`break cnt`	1	Break/counter marker
`video_start`	2	Movie clip onset
`video_stop`	3	Movie clip offset
`contrastTrial_start`	4	Contrast detection trial onset
`right_target`	5	Right-side target
`left_target`	6	Left-side target
`left_buttonPress`	7	Left button press
`right_buttonPress`	8	Right button press
`stim_ON`	9	Contrast stimulus on
`fixpoint_ON`	10	Fixation point on
`contrastChangeB1_start`	11	Contrast change block 1 onset
`surroundSuppB1_start`	12	Surround suppression block 1 onset
`resting_start`	13	Resting block onset
`contrastChangeB2_start`	14	Contrast change block 2 onset
`instructed_toOpenEyes`	15	Eyes-open instruction
`instructed_toCloseEyes`	16	Eyes-close instruction
`surroundSuppB2_start`	17	Surround suppression block 2 onset
`contrastChangeB3_start`	18	Contrast change block 3 onset
`trialResponse`	19	Trial response
`newPage`	20	New page (symbol search)
`symbolSearch_start`	21	Symbol search block onset
`IEND`	22	Instruction end
`dot_no2_ON`	23	Dot stimulus 2 on
`IBEG`	24	Instruction begin
`dot_no2_OFF`	25	Dot stimulus 2 off
`dot_no3_ON`	26	Dot stimulus 3 on
`dot_no4_ON`	27	Dot stimulus 4 on
`dot_no5_ON`	28	Dot stimulus 5 on

- Notes: Multi-task HBN battery. `boundary` and `9999` are admin markers; left/right target classification is the primary task of interest.

Data Quality Notes

- `boundary` and `9999` are administrative markers and should be excluded from scientific analysis.
- Large subject population; verify channel count consistency across subjects.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Alexander et al., Sci Data 2017`

208. openneuro_ds005520 - Research data supporting 'EEG recording during playing MOBA game'

Dataset: `openneuro_ds005520` | Year: 2024 | URL: https://openneuro.org/datasets/ds005520
Category: Complex Applied | Subjects: 23 | Files (Completed): 23 | Channels: 60 | Sampling Rate: 256 Hz

Paradigm

Participants played a real MOBA (Multiplayer Online Battle Arena) game while 60-channel EEG was recorded alongside resting-state sessions. Trigger codes mark game events such as kills, deaths, and other in-game actions. The study investigates neural correlates of naturalistic gaming behavior.

Event Structure

Event ID	Count (total)	Type	Meaning
`13`	834	long	In-game event code 13 (decode from events.tsv)
`14`	520	long	In-game event code 14 (decode from events.tsv)
`66`	137	long	In-game event code 66 (decode from events.tsv)
`666`	89	long	In-game event code 666 (decode from events.tsv)
`444`	45	long	In-game event code 444 (decode from events.tsv)
`-255`	9	admin	System overflow / boundary marker
`4`	9	long	In-game event code 4 (decode from events.tsv)
`100008`	9	long	In-game event code 100008 (decode from events.tsv)
`164`	3	long	In-game event code 164 (decode from events.tsv)
`244`	1	long	In-game event code 244 (decode from events.tsv)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: MOBA Game Events (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`13`	0	Game event 13

Event	y	Meaning
14	1	Game event 14
66	2	Game event 66
666	3	Game event 666
444	4	Game event 444
-255	5	System marker
4	6	Game event 4
100008	7	Game event 100008
164	8	Game event 164
244	9	Game event 244

- Notes: Numerical trigger codes require mapping from events.tsv or dataset README. Very small dataset (23 subjects); low event counts for rare codes.

Data Quality Notes

- Event codes require decoding from dataset events.tsv or README.
- Very small dataset (23 subjects, 23 files); rare codes (164, 244, -255) have <10 occurrences total.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG recording during playing MOBA game`

209. openneuro_ds005586 - EEG responses to the number of objects in partially occluded and uncovered scenes

Dataset: openneuro_ds005586 | Year: 2024 | URL: https://openneuro.org/datasets/ds005586
Category: Sensory/Response | Subjects: 23 | Files (Completed): 23 | Channels: 63 | Sampling Rate: 1000 Hz

Paradigm

Participants passively viewed game boards depicting varying numbers of game pieces under partial occlusion and subsequent uncovering conditions. The study investigates EEG responses to numerosity perception under occlusion, measuring visual evoked potentials to quantify object tracking. Stimulus codes encode number of objects (1 or 2 digits) and condition (occluded vs uncovered).

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 95	14399	short	Catch / fixation stimulus
Stimulus/S 21	6884	short	Stimulus condition (2-digit code: object count x condition)
Stimulus/S 11	6875	short	Stimulus condition (2-digit code: object count x condition)
Stimulus/S 15	2241	short	Stimulus condition
Stimulus/S127	1413	short	Stimulus condition (3-digit code)
Stimulus/S223	1299	short	Stimulus condition (3-digit code)
Stimulus/S111	903	short	Stimulus condition (3-digit code)
Stimulus/S124	866	short	Stimulus condition (3-digit code)
Stimulus/S224	866	short	Stimulus condition (3-digit code)
Stimulus/S226	864	short	Stimulus condition (3-digit code)
Stimulus/S126	864	short	Stimulus condition (3-digit code)
Stimulus/S215	864	short	Stimulus condition (3-digit code)
Stimulus/S115	864	short	Stimulus condition (3-digit code)
Stimulus/S118	863	short	Stimulus condition (3-digit code)
Stimulus/S218	863	short	Stimulus condition (3-digit code)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Object Numerosity / Occlusion (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`Stimulus/S 95`	0	Catch/fixation
`Stimulus/S127`	1	Condition code 127
`Stimulus/S223`	2	Condition code 223
`Stimulus/S111`	3	Condition code 111
`Stimulus/S 21`	4	Condition code 21
`Stimulus/S 11`	5	Condition code 11
`Stimulus/S117`	6	Condition code 117
`Stimulus/S124`	7	Condition code 124
`Stimulus/S224`	8	Condition code 224
`Stimulus/S227`	9	Condition code 227
`Stimulus/S118`	10	Condition code 118
`Stimulus/S218`	11	Condition code 218
`Stimulus/S122`	12	Condition code 122
`Stimulus/S217`	13	Condition code 217
`Stimulus/S222`	14	Condition code 222
`Stimulus/S212`	15	Condition code 212
`Stimulus/S123`	16	Condition code 123
`Stimulus/S113`	17	Condition code 113
`Stimulus/S213`	18	Condition code 213
`Stimulus/S112`	19	Condition code 112
`Stimulus/S226`	20	Condition code 226
`Stimulus/S211`	21	Condition code 211
`Stimulus/S126`	22	Condition code 126
`Stimulus/S221`	23	Condition code 221
`Stimulus/S121`	24	Condition code 121
`Stimulus/S216`	25	Condition code 216
`Stimulus/S116`	26	Condition code 116
`Stimulus/S214`	27	Condition code 214
`Stimulus/S114`	28	Condition code 114

- Notes: Stimulus codes encode object count x occlusion condition; decode from events.tsv. `New Segment/` is admin and excluded.

Data Quality Notes

- Small dataset (23 subjects, 23 files); verify that all subjects completed the same number of trials.
- `New Segment/` is an admin event and should be excluded.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG responses to the number of objects in partially occluded and uncovered scenes`

210. openneuro_ds005589 - A multi-subject and multi-session EEG dataset for modelling human visual object recognition

Dataset: `openneuro_ds005589` | Year: 2024 | URL: https://openneuro.org/datasets/ds005589
Category: Sensory/Response | Subjects: 32 | Files (Completed): 1592 | Channels: 133 | Sampling Rate: 1000 Hz

Paradigm

Participants underwent Rapid Serial Visual Presentation (RSVP) at 5 Hz and a low-speed visual presentation task to record brain responses to visual object recognition across multiple image categories. Codes 1-20 correspond to object categories in the low-speed task; codes 21-27 to RSVP categories; code 99 is catch trials; codes 252/254 are feedback. Multiple sessions were recorded per subject.

Event Structure

Event ID	Count (total)	Type	Meaning
`252`	141374	short	Correct response feedback
`1-20`	~80332 each	short	Object category codes 1-20 (low-speed presentation)
`99`	~157200	short	Catch trial
`254`	~40000	short	Incorrect response feedback
`21-27`	~798 each	short	Object category codes 21-27 (RSVP task)

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Visual Object Recognition (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`1`	0	Object category 1
`2`	1	Object category 2
`3`	2	Object category 3
`4`	3	Object category 4
`5`	4	Object category 5
`6`	5	Object category 6
`7`	6	Object category 7
`8`	7	Object category 8

Event	y	Meaning
9	8	Object category 9
10	9	Object category 10
11	10	Object category 11
12	11	Object category 12
13	12	Object category 13
14	13	Object category 14
15	14	Object category 15
16	15	Object category 16
17	16	Object category 17
18	17	Object category 18
19	18	Object category 19
20	19	Object category 20
99	20	Catch trial
252	21	Correct feedback
254	22	Incorrect feedback
21	23	RSVP category 21
22	24	RSVP category 22
23	25	RSVP category 23
24	26	RSVP category 24
25	27	RSVP category 25
26	28	RSVP category 26
27	29	RSVP category 27

- Notes: 30 event classes. Primary task: decode object category from EEG (codes 1-20 or 21-27). Codes 252/254 and 99 may be used for filtering.

Data Quality Notes

- Large dataset (32 subjects x 1592 files); multi-session design - account for session effects.
- Apply bandpass filter (1-40 Hz) and baseline correction prior to analysis.

References

- Reference: `A multi-subject and multi-session EEG dataset for modelling human visual object recognition`

211. openneuro_ds005662 - A high-quality EEG dataset for studying visual touch perception

Dataset: `openneuro_ds005662` | Year: 2025 | URL: https://openneuro.org/datasets/ds005662
Category: Sensory/Response | Subjects: N/A | Files (Completed): 80 | Channels: 65 | Sampling Rate: 2048 Hz

Paradigm

Participants viewed rapid streams of videos depicting touch to a hand or an object to investigate visual touch perception. The experiment used target and non-target trials to measure brain responses to observed touch. No event markers are present in the BIDS metadata; the EEG is stored as continuous recordings requiring segmentation for downstream analysis.

Event Structure

No event markers detected (`no_events`). This dataset contains continuous EEG recordings without trial-level annotations in events.tsv.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv) - not available; use segmentation
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Continuous Segmentation (resting)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, segment_overlap = 0.5 s
- Target Events -> y: null (no events)
- Notes: No trial-level events found. Treat as continuous EEG; sliding window segmentation only.

Data Quality Notes

- No event markers available - cannot perform event-locked epoching without additional annotation files.
- High sampling rate (2048 Hz); downsample to 500 Hz before processing.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `Validated Touch-Video Database (Smit & Rich, 2025)`

212. openneuro_ds005787 - NOD-EEG

Dataset: openneuro_ds005787 | Year: 2025 | URL: https://openneuro.org/datasets/ds005787
Category: Others | Subjects: N/A | Files (Completed): 896 | Channels: 64 | Sampling Rate: 250 Hz

Paradigm

The NOD-EEG dataset contains 64-channel EEG recordings from a stimulus-response paradigm. Each trial includes a stimulus onset (stim_on), participant button-press response (resp), and block boundary markers (begin, end). The precise experimental task and stimuli are described in the dataset README.

Event Structure

Event ID	Count (total)	Type	Meaning
resp	2273060	short	Participant button-press response
stim_on	112000	short	Stimulus onset
begin	896	long	Block begin marker
end	896	long	Block end marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Stimulus-Response (epoch_by_event)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
resp	0	Button-press response
stim_on	1	Stimulus onset
begin	2	Block begin
end	3	Block end

- Notes: Very high count for resp (2.27M total); verify that events are correctly timestamped. begin/end mark block boundaries.

Data Quality Notes

- Very high total event count for resp; verify no duplicate events.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. Scientific Data, 6, 103

213. openneuro_ds005815 - YOTO (You Only Think Once): An EEG Dataset for Tracking Spontaneous Cognitive Transitions and Multimodal Stimuli Responses in Resting-State and Task Phases

Dataset: `openneuro_ds005815` | Year: 2025 | URL: https://openneuro.org/datasets/ds005815
Category: High-level Mental | Subjects: 20 | Files (Completed): 205 | Channels: 30 | Sampling Rate: 250 Hz

Paradigm

Participants engaged in resting-state (eyes open/closed) and task phases involving unimodal and multimodal stimuli (visual and auditory) and mental imagery, providing self-reported vividness ratings. The YOTO paradigm investigates spontaneous cognitive transitions and mental imagery across stimulus modalities with numeric condition codes and named state markers.

Event Structure

Event ID	Count (total)	Type	Meaning
`6-9`	6720 each	short	Stimulus condition codes (decode from events.tsv)
`TrialProc`	6528	long	Trial procedure marker
`23`	840	short	Stimulus/response code 23
`22`	420	short	Stimulus/response code 22
`21`	420	short	Stimulus/response code 21
`StartInstruction`	66	long	Instruction block onset
`Eyeclosed`	66	long	Eyes-closed resting onset
`FixationBreak`	66	short	Fixation break event
`EyeOpen`	66	long	Eyes-open resting onset
`EndInstruction`	66	long	Instruction block offset
`88`	64	short	Response/trigger code 88
`30-35`	280 each	short	Imagery/rating condition codes
`99`	101	admin	Admin/session marker
`1`	81	short	Condition code 1
`2-5`	68 each	short	Condition codes 2-5

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Multi-task Cognitive (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s

- Target Events -> y:

Event	y	Meaning
99	0	Admin/session marker
1	1	Condition code 1
2	2	Condition code 2
3	3	Condition code 3
5	4	Condition code 5
StartInstruction	5	Instruction onset
Eyeclosed	6	Eyes-closed onset
FixationBreak	7	Fixation break
EyeOpen	8	Eyes-open onset
EndInstruction	9	Instruction offset
88	10	Response code 88
6	11	Condition code 6
7	12	Condition code 7
8	13	Condition code 8
9	14	Condition code 9
23	15	Code 23
22	16	Code 22
21	17	Code 21
30	18	Imagery code 30
35	19	Imagery code 35
25	20	Imagery code 25
32	21	Imagery code 32
31	22	Imagery code 31
27	23	Imagery code 27
28	24	Imagery code 28
24	25	Imagery code 24
33	26	Imagery code 33
26	27	Imagery code 26
29	28	Imagery code 29
34	29	Imagery code 34

- Notes: Numerical codes require decoding from events.tsv. Eyes-open/closed markers (EyeOpen, Eyeclosed) can be used for resting-state analysis.

Data Quality Notes

- Numerical event codes require mapping from dataset events.tsv or README.
- Small dataset (20 subjects); low statistical power for rare conditions.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: YOTO (You Only Think Once): An EEG Dataset for Tracking Spontaneous Cognitive Transitions and Multimodal Stimuli Responses in Resting-State and Task Phases

214. openneuro_ds005841 - EEG Experiment measuring ERPs in VR

Dataset: `openneuro_ds005841` | Year: 2025 | URL: https://openneuro.org/datasets/ds005841
Category: Complex Applied | Subjects: 48 | Files (Completed): 288 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An investigation of event-related potentials (ERPs) during visual tasks performed within a virtual reality (VR) environment. Stimuli vary along two orthogonal dimensions: lighting (DARK vs LIGHT) and reference type (RAND = random/unpredictable vs REF = reference/predictable). The 2x2 design yields four event codes measuring visual ERP responses under prediction-error conditions.

Event Structure

Event ID	Count (total)	Type	Meaning
`RAND_DARK`	288	short	Random (unpredictable) stimulus - dark condition
`RAND_LIGHT`	288	short	Random (unpredictable) stimulus - light condition
`REF_DARK`	288	short	Reference (predictable) stimulus - dark condition
`REF_LIGHT`	288	short	Reference (predictable) stimulus - light condition

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: VR Stimulus Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`RAND_DARK`	0	Random dark stimulus
`RAND_LIGHT`	1	Random light stimulus
`REF_DARK`	2	Reference dark stimulus
`REF_LIGHT`	3	Reference light stimulus

- Notes: Perfectly balanced 4-class design (288 each); primary contrast is RAND vs REF (prediction error) and DARK vs LIGHT (luminance). Count per file reflects 1 file per subject.

Data Quality Notes

- VR environment may introduce motion artifacts; apply ICA for artifact removal.
- 48 subjects x 6 sessions = 288 files; verify session-level consistency.
- Apply bandpass filter (1-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

215. openneuro_ds006018 - Cognitive Electrophysiology in Socioeconomic Context in Adulthood: An EEG dataset

Dataset: openneuro_ds006018 | Year: 2025 | URL: https://openneuro.org/datasets/ds006018
Category: High-level Mental | Subjects: 127 | Files (Completed): 357 | Channels: 31 | Sampling Rate: 500 Hz

Paradigm

A suite of standardized ERP tasks from the ERP CORE compendium (Kappenman et al. 2021) measuring neural correlates of perception, cognition, and action in adults in relation to socioeconomic status and ADHD symptoms. Tasks include N170 (face perception), MMN (mismatch negativity), N2pc (visual attention), P3 (cognitive updating), N400 (semantic processing), ERN (error monitoring), and LRP (lateralized readiness potential).

Event Structure

Event ID	Count (total)	Type	Meaning
S201	40081	short	ERP CORE stimulus code 201
S 80	30199	short	ERP CORE stimulus code 80
S212	10400	short	ERP CORE stimulus code 212
S111	9971	short	ERP CORE stimulus code 111
S222	9017	short	ERP CORE stimulus code 222
S121	8449	short	ERP CORE stimulus code 121
S 22	8268	short	ERP CORE stimulus code 22
S 11	8263	short	ERP CORE stimulus code 11
S 12	8252	short	ERP CORE stimulus code 12
S 21	8244	short	ERP CORE stimulus code 21
S 70	7977	short	ERP CORE stimulus code 70
S202	6292	short	ERP CORE stimulus code 202
S221	5197	short	ERP CORE stimulus code 221
S122	4665	short	ERP CORE stimulus code 122
S211	3707	short	ERP CORE stimulus code 211

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: ERP CORE Multi-paradigm (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`S 12`	0	ERP code 12
`S 22`	1	ERP code 22
`S 21`	2	ERP code 21
`S 11`	3	ERP code 11
`S201`	4	ERP code 201
`S202`	5	ERP code 202
`S221`	6	ERP code 221
`S122`	7	ERP code 122
`S212`	8	ERP code 212
`S222`	9	ERP code 222
`S111`	10	ERP code 111
`S121`	11	ERP code 121
`S211`	12	ERP code 211
`S112`	13	ERP code 112
`S 80`	14	ERP code 80
`S 70`	15	ERP code 70
`S180`	16	ERP code 180
`S 1`	17	ERP code 1
`S 25`	18	ERP code 25
`S 23`	19	ERP code 23
`S 24`	20	ERP code 24
`S 53`	21	ERP code 53
`S 43`	22	ERP code 43
`S 45`	23	ERP code 45
`S 44`	24	ERP code 44
`S 41`	25	ERP code 41
`S 13`	26	ERP code 13
`S 14`	27	ERP code 14
`S 42`	28	ERP code 42

- Notes: Multi-paradigm ERP CORE dataset. `boundary` is admin and excluded. Codes encode task x condition combinations per ERP CORE documentation.

Data Quality Notes

- Multi-paradigm dataset - participants completed multiple ERP CORE tasks; segment by task if needed.

- `boundary` is an administrative marker, excluded from target_events.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection; baseline correct [-0.2, 0] s.

References

- Reference: `Isbell, E., Peters, A. N., Richardson, D. M., & Rodas De Len, N. E. (2025). Cognitive electrophysiology in socioeconomic context in adulthood. Scientific Data, 12(1), 19.`

216. openneuro_ds006040 - Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI

Dataset: `openneuro_ds006040` | Year: 2025 | URL: https://openneuro.org/datasets/ds006040
Category: High-level Mental | Subjects: 28 | Files (Completed): 1064 | Channels: 64 | Sampling Rate: 5000 Hz

Paradigm

The dataset uses the Gradual Continuous Performance Task (gradCPT) to measure sustained attention via simultaneous EEG and fMRI recording. Participants detect rare city scenes (targets, S245) within a stream of common mountain scenes (standards, S155). Additional tasks include resting state, visual checkerboard stimulation, and mental imagery. MRI synchronization pulses (TR, Sync On) and cardiac (qrs) signals are also recorded.

Event Structure

Event ID	Count (total)	Type	Meaning
`S155`	201922	short	Standard stimulus (mountain scene, frequent)
`S245`	201110	short	Target stimulus (city scene, rare)
`Sync On`	195242	admin	MRI sync pulse onset
`qrs`	170358	admin	Cardiac QRS complex
`TR`	106494	admin	MRI TR repetition marker
`T 1`	52960	admin	Timing marker
`S254`	813	short	Response / button press
`[]`	15	admin	Empty/null event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: gradCPT Sustained Attention (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
`Sync On`	0	MRI sync pulse
`S155`	1	Standard (mountain)
`TR`	2	MRI TR marker
`S245`	3	Target (city)
`qrs`	4	Cardiac QRS
`T 1`	5	Timing marker

Event	y	Meaning
S254	6	Button press/response
[]	7	Null event

- Notes: Primary scientific contrast is S155 (standard) vs S245 (target) for gradCPT attention decoding. Sync On, TR, qrs, T 1 are MRI/cardiac markers. Very high sampling rate (5000 Hz); downsample before processing.

Data Quality Notes

- Very high sampling rate (5000 Hz); downsample to <=1000 Hz for most analyses.
- Sync On, TR, qrs are MRI/cardiac markers and should be excluded from stimulus-locked analysis.
- Simultaneous EEG-fMRI data requires MRI artifact correction (template subtraction or similar).

References

- Reference: Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI

217. openneuro_ds006104 - EEG dataset for speech decoding

Dataset: `openneuro_ds006104` | Year: 2025 | URL: https://openneuro.org/datasets/ds006104
Category: High-level Mental | Subjects: 24 | Files (Completed): 80 | Channels: 62 | Sampling Rate: 2000 Hz

Paradigm

A phoneme discrimination task where participants listened to speech sounds - including single phonemes, phoneme pairs, and words - while undergoing transcranial magnetic stimulation (TMS) to test the role of speech motor cortex in auditory speech perception. Subjects provided button-press responses after each sound. The dataset supports decoding of speech perception from TMS-EEG.

Event Structure

Event ID	Count (total)	Type	Meaning
`TMS`	29260	short	TMS pulse delivery
`stimulus`	29260	short	Auditory speech stimulus onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: TMS vs Stimulus Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`TMS`	0	TMS pulse
`stimulus`	1	Speech stimulus onset

- Notes: Binary classification (TMS vs stimulus). TMS pulses produce large artifacts; apply TMS artifact rejection before epoching.

Data Quality Notes

- Very high sampling rate (2000 Hz); downsample to 500 Hz for most analyses.
- TMS pulses produce large EEG artifacts requiring specialized artifact removal.
- Apply bandpass filter (1-80 Hz) after TMS artifact correction.

References

- Reference: `Moreira et al. "An open-access EEG dataset for speech decoding: Exploring the role of articulation and coarticulation"`

218. openneuro_ds006142 - Essex EEG Movie Memory dataset

Dataset: openneuro_ds006142 | Year: 2025 | URL: https://openneuro.org/datasets/ds006142
Category: High-level Mental | Subjects: 27 | Files (Completed): 27 | Channels: 64 | Sampling Rate: 2048 Hz

Paradigm

Memory recognition task where participants watched 10-second movie clips and identified whether they remembered them from previous viewings, including time-stamping recognition instants. Clips were shown twice; the first watch (`startOfRememberedClipFirstWatch`, `startOfRecognisedClipFirstWatch`) established encoding, while the second watch (`startOfSecondWatch`) tested recognition. Participants clicked when they recognized clips (`firstRecognitionClick`, `recognitionClick`).

Event Structure

Event ID	Count (total)	Type	Meaning
`recognitionClick`	6675	short	Participant clicks upon recognition
`endOfClip`	6094	short	Movie clip ends
`trialStart`	4153	long	Trial onset
`trialEnd`	4153	long	Trial offset
`startOfNotRecognisedClip`	2212	long	Clip onset - not recognized
`startOfSecondWatch`	1941	long	Second viewing onset
`firstRecognitionClick`	1881	short	First recognition click in second watch
`startOfRememberedClipFirstWatch`	1426	long	First viewing onset - later remembered
`startOfRecognisedClipFirstWatch`	515	long	First viewing onset - later recognized

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Movie Memory Recognition (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`endOfClip`	0	Clip end
`trialStart`	1	Trial start
`trialEnd`	2	Trial end
`recognitionClick`	3	Recognition response

Event	y	Meaning
`startOfNotRecognisedClip`	4	Not-recognized clip onset
`startOfSecondWatch`	5	Second viewing onset
`firstRecognitionClick`	6	First recognition click
`startOfRememberedClipFirstWatch`	7	First watch - remembered
`startOfRecognisedClipFirstWatch`	8	First watch - recognized

- Notes: Key contrast is remembered vs not-recognized clips at first watch onset. High sampling rate (2048 Hz); consider downsampling.

Data Quality Notes

- High sampling rate (2048 Hz); downsample to 512 Hz for standard ERP analysis.
- Small dataset (27 subjects); limited statistical power.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection prior to epoching.

References

- Paper: `Annotating, Understanding, and Predicting Long-term Video Memorability`

219. openneuro_ds006159 - Implicit Learning EEG (BioSemi)

Dataset: openneuro_ds006159 | Year: 2025 | URL: https://openneuro.org/datasets/ds006159
Category: High-level Mental | Subjects: 35 | Files (Completed): 35 | Channels: 69 | Sampling Rate: 1024 Hz

Paradigm

An experimental study investigating the neural correlates of implicit learning using a BioSemi EEG system. Participants underwent repeated exposure to structured sequences without explicit instructions to learn the underlying patterns. No trial-level event annotations are present in the dataset; the EEG is recorded as a continuous stream during the implicit learning protocol.

Event Structure

No event markers detected (no_events). This dataset contains continuous EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Continuous Segmentation (segmentation)

- Epoch Method: epoch_by_segmentation_hdf5
- Segment: segment_length = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no event markers; segmentation only)
- Notes: No trial-level events available; purely continuous EEG. Labels must be derived from external metadata (e.g., block condition labels from participants.tsv if available).

Data Quality Notes

- No event markers present; continuous-only processing.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: MNE-BIDS: Organizing electrophysiological data into the BIDS format and facilitating their analysis

220. openneuro_ds006171 - EEG during Near-Threshold Visual Detection Tasks (No-Cue, 50%-Cue, 100%-Cue)

Dataset: openneuro_ds006171 | Year: 2025 | URL: https://openneuro.org/datasets/ds006171
Category: Sensory/Response | Subjects: 26 | Files (Completed): 104 | Channels: 144 | Sampling Rate: 1024 Hz

Paradigm

Participants performed near-threshold visual detection tasks across three cueing conditions: no-cue (0% validity), noninformative cue (50% validity), and informative cue (100% validity). The study examines how spatial attention modulates near-threshold detection performance and associated EEG signatures. Stimuli were brief visual Gabor patches presented at or near detection threshold, and participants reported detection via button press. Behavioral columns (gabor_contrast, subjective_r, objective_r, response_time) stored in events.tsv enable regression analyses.

Event Structure

Event ID	Count (total)	Type	Meaning
128	69653	admin	BioSemi status boundary marker
125	23250	short	100% validity cue - correct rejection (100pct_CR)
124	18239	short	100% validity cue - false alarm (100pct_FA)
2	13793	short	Response (detection report)
1	13790	short	Trial start
22	6299	short	50%-validity cue condition, stimulus B
21	6299	short	50%-validity cue condition, stimulus A
11	6297	short	No-cue condition, stimulus A
12	6295	short	No-cue condition, stimulus B
122	2753	short	50% validity cue - miss (50pct_miss)
121	2650	short	50% validity cue - hit (50pct_hit)
103	1530	short	No-cue - false alarm (nocue_FA)
101	1530	short	No-cue - hit (nocue_hit)
111	1529	short	50% validity cue - hit (50pct_hit)
104	1529	short	No-cue - correct rejection (nocue_CR)

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Specific (`kernels/openneuro_specific/openneuro_ds006171.py`)

Task 1: Near-Threshold Detection Outcome (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`catch_trial`	0	Catch trial (no stimulus)
`catch_response`	1	Response during catch trial
`nocue_hit`	2	No-cue condition - hit
`nocue_miss`	3	No-cue condition - miss
`nocue_FA`	4	No-cue condition - false alarm
`nocue_CR`	5	No-cue condition - correct rejection
`50pct_hit`	6	50%-validity cue - hit
`50pct_miss`	7	50%-validity cue - miss
`50pct_FA`	8	50%-validity cue - false alarm
`50pct_CR`	9	50%-validity cue - correct rejection
`100pct_hit`	10	100%-validity cue - hit
`100pct_miss`	11	100%-validity cue - miss
`100pct_FA`	12	100%-validity cue - false alarm
`100pct_CR`	13	100%-validity cue - correct rejection

- Notes: Specific kernel maps numeric BioSemi codes to semantic labels. `regression_target`: response_time. `continuous_cols`: response_time, gabor_contrast, subjective_r, objective_r. Code `128` is a BioSemi boundary marker (admin; exclude). Key contrast: hit vs miss across cueing conditions.

Data Quality Notes

- BioSemi 128/256-channel system; code `128` is a hardware boundary marker, not a stimulus.
- Continuous behavioral columns (gabor_contrast, subjective_r, objective_r, response_time) available for regression.
- Apply bandpass filter (0.1-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `EEG data during three near-threshold visual detection tasks: a no-cue task, a noninformative cue task (50% validity), and an informative cue task (100% validity)`

221. openneuro_ds006269 - Tethered EEG Recordings in Syngap1 Rats

Dataset: openneuro_ds006269 | Year: 2025 | URL: https://openneuro.org/datasets/ds006269
Category: Biological State | Subjects: 40 | Files (Completed): 40 | Channels: 33 | Sampling Rate: 1000 Hz

Paradigm

Continuous 6-hour tethered EEG recordings were collected from wildtype and Syngap1 mutant rats to monitor spontaneous brain activity and examine circadian-specific neural states. The Syngap1 gene encodes a synaptic GTPase-activating protein and its haploinsufficiency is a model of intellectual disability. The dataset examines spontaneous EEG patterns (sleep/wake states, oscillations) across genotypes.

Event Structure

Event ID	Count (total)	Type	Meaning
unknown	40	admin	Unknown system marker - not usable for classification

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Continuous Segmentation (long_event)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no usable event markers; target_events: null)
- Notes: unknown is an uninformative system marker. Use subject-level metadata (wildtype vs. Syngap1 mutant genotype) from participants.tsv as classification labels.

Data Quality Notes

- Only unknown events present - no meaningful trial-level annotation.
- Labels must be derived from external genotype metadata (participants.tsv).
- Animal (rat) EEG data; electrode placement and artifact characteristics differ from human EEG.

References

- Paper: Key roles of C2/GAP domains in SYNGAP1-related pathophysiology (DOI: 10.1016/j.celrep.2024.114733)

222. openneuro_ds006465 - 3M-CPSEED: EEG Dataset for Chinese Pinyin Production in Overt, Silent-Intended, and Imagined Speech

Dataset: `openneuro_ds006465` | Year: 2025 | URL: https://openneuro.org/datasets/ds006465
Category: High-level Mental | Subjects: 20 | Files (Completed): 84 | Channels: 33 | Sampling Rate: 500 Hz

Paradigm

Participants performed three modes of speech production - overt (spoken aloud), silent-intended (mouthed without sound), and imagined (mental speech without movement) - for specific Chinese Pinyin sounds including initials and finals. The dataset enables comparison of neural signatures across overt, silent, and imagined speech conditions. Each trial consisted of a visual cue (stimulus_presentation) followed by a production phase and a response confirmation. Numeric codes 1-10 encode specific phoneme or condition types.

Event Structure

Event ID	Count (total)	Type	Meaning
`stimulus_presentation`	42	short	Visual cue onset prompting speech production
`1`	302	short	Phoneme/condition code 1
`2`	200	short	Phoneme/condition code 2
`3`	200	short	Phoneme/condition code 3
`4`	200	short	Phoneme/condition code 4
`5`	200	short	Phoneme/condition code 5
`6`	200	short	Phoneme/condition code 6
`7`	200	short	Phoneme/condition code 7
`8`	200	short	Phoneme/condition code 8
`9`	200	short	Phoneme/condition code 9
`10`	200	short	Phoneme/condition code 10
`response`	84	short	Participant response / confirmation

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Speech Production Mode Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
stimulus_presentation	0	Visual cue for speech production
1	1	Phoneme/condition code 1
5	2	Phoneme/condition code 5
6	3	Phoneme/condition code 6
7	4	Phoneme/condition code 7
4	5	Phoneme/condition code 4
3	6	Phoneme/condition code 3
2	7	Phoneme/condition code 2
10	8	Phoneme/condition code 10
8	9	Phoneme/condition code 8
9	10	Phoneme/condition code 9

- Notes: Numeric codes 1-10 correspond to specific Pinyin sounds or speech modes; decode from events.tsv. response is a motor response event. Some files show only numeric triggers; mixed annotation format across subjects.

Data Quality Notes

- Some files show only numeric triggers while others include stimulus_presentation; mixed annotation format across sessions.
- Derivatives (preprocessed) files available alongside raw data.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection before epoching.

References

- Reference: 3M-CPSEED: An EEG-based Dataset for Chinese Pinyin Production in Overt, Silent-intended, and Imagined Speech

223. openneuro_ds006519 - Dataset of Intracranial EEG during Cortical Stimulations Evoking Negative Motor Responses

Dataset: `openneuro_ds006519` | Year: 2025 | URL: https://openneuro.org/datasets/ds006519
Category: Action/Output | Subjects: 23 | Files (Completed): 35 | Channels: 33 | Sampling Rate: 4096 Hz

Paradigm

Patients undergoing presurgical evaluation for drug-resistant epilepsy received intracranial electrical stimulation to map cortical function. The stimulation protocol targeted regions that, when stimulated, evoked negative motor responses (involuntary movement inhibition or arrest). EEG was recorded from intracranial electrodes during pre-stimulation (prestim) and post-stimulation (poststim) periods to characterize neural activity changes associated with negative motor mapping.

Event Structure

Event ID	Count (total)	Type	Meaning
`Comment/prestim`	35	short	Pre-stimulation baseline period onset
`Comment/poststim`	35	short	Post-stimulation period onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Pre- vs Post-Stimulation Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`Comment/prestim`	0	Pre-stimulation baseline period
`Comment/poststim`	1	Post-stimulation response period

- Notes: Binary classification of pre- vs post-stimulation neural states. Intracranial EEG from epilepsy patients; electrode locations vary across subjects. Very high sampling rate (4096 Hz); consider downsampling.

Data Quality Notes

- Intracranial EEG dataset; electrode montage is patient-specific and non-standard.
- Very high sampling rate (4096 Hz); downsampling to <=1000 Hz recommended.
- Small dataset (35 files, 23 subjects); limited sample size.

References

- Reference: Dataset of intracranial EEG during cortical stimulations evoking negative motor responses

224. openneuro_ds006525 - Resting EEG with Structural MRI

Dataset: openneuro_ds006525 | Year: 2025 | URL: https://openneuro.org/datasets/ds006525
Category: Biological State | Subjects: 34 | Files (Completed): 34 | Channels: 128 | Sampling Rate: 250 Hz

Paradigm

Spontaneous brain activity was recorded during resting-state EEG, supplemented by structural MRI scans. The dataset captures baseline neural dynamics in the absence of any task. Event markers include resting-state block onset (rsts), offset (rstf), and a session-begin marker (bgin), enabling segmentation of clean resting periods.

Event Structure

Event ID	Count (total)	Type	Meaning
bgin	20400	long	Session begin marker
rsts	20400	rest	Resting-state block start
rstf	34	rest	Resting-state block finish/end
NRST	34	admin	Non-resting state marker
DIN	2	admin	Digital input trigger

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Resting-State Segmentation (long_event)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
bgin	0	Session begin
rsts	1	Resting-state block start
rstf	2	Resting-state block end
NRST	3	Non-resting state period
DIN	4	Digital input trigger

- Notes: rsts and rstf mark the boundaries of resting blocks; segment within these boundaries. NRST may indicate periods outside resting state that should be excluded.

Data Quality Notes

- High channel count (128) with 250 Hz sampling rate; EGI/Geodesic system likely.
- NRST may indicate periods outside resting state that should be excluded from analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `Resting EEG`

225. openneuro_ds006547 - Visual EEG Study (BrainVision -> BIDS)

Dataset: openneuro_ds006547 | Year: 2025 | URL: https://openneuro.org/datasets/ds006547
Category: Sensory/Response | Subjects: 31 | Files (Completed): 31 | Channels: 67 | Sampling Rate: 500 Hz

Paradigm

High-density EEG recordings were collected during a visual stimulation task using a BrainVision system, subsequently converted to BIDS format. Participants viewed visual stimuli under multiple conditions encoded by numeric BrainVision trigger codes (S2 through S56). The 30 distinct stimulus codes suggest a factorial or multi-condition visual paradigm covering perceptual and attentional manipulations.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S 5	31	short	Visual stimulus condition S5
Stimulus/S 2	31	short	Visual stimulus condition S2
Stimulus/S 42	31	short	Visual stimulus condition S42
Stimulus/S 11	31	short	Visual stimulus condition S11
Stimulus/S 51	31	short	Visual stimulus condition S51
Stimulus/S 8	31	short	Visual stimulus condition S8
Stimulus/S 45	31	short	Visual stimulus condition S45
Stimulus/S 56	31	short	Visual stimulus condition S56
Stimulus/S 3	31	short	Visual stimulus condition S3
Stimulus/S 46	31	short	Visual stimulus condition S46
Stimulus/S 21	31	short	Visual stimulus condition S21
Stimulus/S 7	31	short	Visual stimulus condition S7
Stimulus/S 12	31	short	Visual stimulus condition S12
Stimulus/S 6	31	short	Visual stimulus condition S6
Stimulus/S 16	31	short	Visual stimulus condition S16
Stimulus/S 19	31	short	Visual stimulus condition S19

Event ID	Count (total)	Type	Meaning
Stimulus/S 44	31	short	Visual stimulus condition S44
Stimulus/S 10	31	short	Visual stimulus condition S10
Stimulus/S 55	31	short	Visual stimulus condition S55
Stimulus/S 17	31	short	Visual stimulus condition S17
Stimulus/S 41	31	short	Visual stimulus condition S41
Stimulus/S 53	31	short	Visual stimulus condition S53
Stimulus/S 50	31	short	Visual stimulus condition S50
Stimulus/S 22	31	short	Visual stimulus condition S22
Stimulus/S 20	31	short	Visual stimulus condition S20
Stimulus/S 52	31	short	Visual stimulus condition S52
Stimulus/S 14	31	short	Visual stimulus condition S14
Stimulus/S 13	31	short	Visual stimulus condition S13
Stimulus/S 54	31	short	Visual stimulus condition S54
Stimulus/S 4	31	short	Visual stimulus condition S4

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Visual Condition Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Stimulus/S 5	0	Visual condition S5
Stimulus/S 2	1	Visual condition S2
Stimulus/S 42	2	Visual condition S42
Stimulus/S 11	3	Visual condition S11
Stimulus/S 51	4	Visual condition S51

Event	y	Meaning
Stimulus/S 8	5	Visual condition S8
Stimulus/S 45	6	Visual condition S45
Stimulus/S 56	7	Visual condition S56
Stimulus/S 3	8	Visual condition S3
Stimulus/S 46	9	Visual condition S46
Stimulus/S 21	10	Visual condition S21
Stimulus/S 7	11	Visual condition S7
Stimulus/S 12	12	Visual condition S12
Stimulus/S 6	13	Visual condition S6
Stimulus/S 16	14	Visual condition S16
Stimulus/S 19	15	Visual condition S19
Stimulus/S 44	16	Visual condition S44
Stimulus/S 10	17	Visual condition S10
Stimulus/S 55	18	Visual condition S55
Stimulus/S 17	19	Visual condition S17
Stimulus/S 41	20	Visual condition S41
Stimulus/S 53	21	Visual condition S53
Stimulus/S 50	22	Visual condition S50
Stimulus/S 22	23	Visual condition S22
Stimulus/S 20	24	Visual condition S20
Stimulus/S 52	25	Visual condition S52
Stimulus/S 14	26	Visual condition S14
Stimulus/S 13	27	Visual condition S13
Stimulus/S 54	28	Visual condition S54
Stimulus/S 4	29	Visual condition S4

- Notes: 30 distinct stimulus codes; decode from events.tsv to determine condition grouping. All codes appear in all 31 files (perfectly balanced). BrainVision format; verify event alignment with EEG data.

Data Quality Notes

- BrainVision format converted to BIDS; verify event alignment with EEG data.
- Large number of distinct event types (30); group by experimental condition before classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

226. openneuro_ds006554 - Social Observation EEG Raw Data

Dataset: openneuro_ds006554 | Year: 2025 | URL: https://openneuro.org/datasets/ds006554
Category: Complex Applied | Subjects: 47 | Files (Completed): 47 | Channels: 64 | Sampling Rate: 500 Hz

Paradigm

EEG was recorded while participants engaged in social observation tasks, likely involving ecological social stimuli such as videos or images of human social interactions. The rich numeric trigger coding (30 distinct codes) suggests a multi-factorial design with combinations of social condition, stimulus type, or response category. The paradigm examines neural correlates of social perception and mentalizing.

Event Structure

Event ID	Count (total)	Type	Meaning
60	47	short	Social stimulus condition 60
61	47	short	Social stimulus condition 61
30	47	short	Social stimulus condition 30
31	47	short	Social stimulus condition 31
62	47	short	Social stimulus condition 62
63	47	short	Social stimulus condition 63
32	47	short	Social stimulus condition 32
33	47	short	Social stimulus condition 33
21	47	short	Social stimulus condition 21
25	47	short	Social stimulus condition 25
16	47	short	Social stimulus condition 16
13	47	short	Social stimulus condition 13
12	47	short	Social stimulus condition 12
24	47	short	Social stimulus condition 24
87	47	short	Social stimulus condition 87
68	47	short	Social stimulus condition 68

Event ID	Count (total)	Type	Meaning
11	47	short	Social stimulus condition 11
26	47	short	Social stimulus condition 26
14	47	short	Social stimulus condition 14
15	47	short	Social stimulus condition 15
22	47	short	Social stimulus condition 22
23	47	short	Social stimulus condition 23
42	47	short	Social stimulus condition 42
67	47	short	Social stimulus condition 67
88	47	short	Social stimulus condition 88
43	47	short	Social stimulus condition 43
36	47	short	Social stimulus condition 36
35	47	short	Social stimulus condition 35
48	47	short	Social stimulus condition 48
37	47	short	Social stimulus condition 37

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Social Condition Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
60	0	Social condition 60
61	1	Social condition 61
30	2	Social condition 30
31	3	Social condition 31
62	4	Social condition 62

Event	y	Meaning
63	5	Social condition 63
32	6	Social condition 32
33	7	Social condition 33
21	8	Social condition 21
25	9	Social condition 25
16	10	Social condition 16
13	11	Social condition 13
12	12	Social condition 12
24	13	Social condition 24
87	14	Social condition 87
68	15	Social condition 68
11	16	Social condition 11
26	17	Social condition 26
14	18	Social condition 14
15	19	Social condition 15
22	20	Social condition 22
23	21	Social condition 23
42	22	Social condition 42
67	23	Social condition 67
88	24	Social condition 88
43	25	Social condition 43
36	26	Social condition 36
35	27	Social condition 35
48	28	Social condition 48
37	29	Social condition 37

- Notes: Numeric trigger codes require decoding from events.tsv. All 30 event types appear in all 47 files (perfectly balanced). Group codes by tens-digit (10s, 20s, 30s, 60s, 80s) as likely condition groupings.

Data Quality Notes

- Numeric trigger codes need decoding from events.tsv or README.
- Large number of distinct event types (30); group codes before classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Social Observation EEG raw data`

227. openneuro_ds006647 - Poetry EEG Study

Dataset: openneuro_ds006647 | Year: 2025 | URL: https://openneuro.org/datasets/ds006647
Category: High-level Mental | Subjects: 51 | Files (Completed): 4 | Channels: 70 | Sampling Rate: 512 Hz

Paradigm

Participants read and evaluated 210 short poetic (Haiku, Senryu) and non-poetic control texts on five subjective dimensions: Aesthetic Appeal, Vivid Imagery, Being Moved, Originality, and Creativity. EEG was recorded during reading using a 64-channel BioSemi system. The study investigates neural correlates of aesthetic and creative cognition during poetry processing. BioSemi EDF trigger codes (65280-65288) encode stimulus onset and condition types.

Event Structure

Event ID	Count (total)	Type	Meaning
65284	852	short	BioSemi trigger - stimulus/condition onset
65281	848	short	BioSemi trigger - stimulus/condition onset
65282	848	short	BioSemi trigger - stimulus/condition onset
65283	848	short	BioSemi trigger - stimulus/condition onset
65280	848	short	BioSemi trigger - stimulus/condition onset
65285	848	short	BioSemi trigger - stimulus/condition onset
65286	848	short	BioSemi trigger - stimulus/condition onset
65287	848	short	BioSemi trigger - stimulus/condition onset
65288	848	short	BioSemi trigger - stimulus/condition onset
boundary	5	admin	Segment boundary marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Poetry Condition Classification (epoch_by_event)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
65284	0	BioSemi condition code 65284

Event	y	Meaning
65281	1	BioSemi condition code 65281
65282	2	BioSemi condition code 65282
65283	3	BioSemi condition code 65283
65280	4	BioSemi condition code 65280
65285	5	BioSemi condition code 65285
65286	6	BioSemi condition code 65286
65287	7	BioSemi condition code 65287
65288	8	BioSemi condition code 65288

- Notes: BioSemi trigger codes 65280-65288; decode from events.tsv to determine poetry vs. control mapping. `boundary` is an admin segment marker (exclude). Only 4 files present despite 51 subjects reported - dataset may be a subset or pilot.

Data Quality Notes

- Only 4 completed files despite 51 subjects reported; this appears to be a partial dataset.
- `boundary` is a segment boundary marker and must be excluded from analysis.
- BioSemi EDF trigger codes 65280-65288 need decoding from events.tsv.

References

- Reference: `Poetry EEG Study`

228. openneuro_ds006648 - Poetry Assessment EEG Dataset 1

Dataset: `openneuro_ds006648` | Year: 2025 | URL: https://openneuro.org/datasets/ds006648
Category: High-level Mental | Subjects: 47 | Files (Completed): 47 | Channels: 70 | Sampling Rate: 512 Hz

Paradigm

Participants read and evaluated 210 short English-language texts including Haikus and Senryus on five subjective dimensions related to aesthetic and creative cognition: Aesthetic Appeal, Vivid Imagery, Being Moved, Originality, and Creativity. EEG was recorded during reading using a 64-channel BioSemi system (70 channels total including references). This is the first of two poetry datasets (companion to ds006647) with full N=47 subject coverage.

Event Structure

Event ID	Count (total)	Type	Meaning
`65284`	9993	short	BioSemi trigger - stimulus/condition D onset
`65281`	9964	short	BioSemi trigger - stimulus/condition A onset
`65282`	9964	short	BioSemi trigger - stimulus/condition B onset
`65283`	9964	short	BioSemi trigger - stimulus/condition C onset
`65286`	47	short	BioSemi trigger - stimulus/condition F onset
`65287`	47	short	BioSemi trigger - stimulus/condition G onset
`65288`	47	short	BioSemi trigger - stimulus/condition H onset
`65285`	46	short	BioSemi trigger - stimulus/condition E onset
`65280`	18	short	BioSemi trigger - rare/practice condition

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Poetry Aesthetic Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`65284`	0	BioSemi condition code 65284

Event	y	Meaning
65281	1	BioSemi condition code 65281
65282	2	BioSemi condition code 65282
65283	3	BioSemi condition code 65283
65286	4	BioSemi condition code 65286
65287	5	BioSemi condition code 65287
65288	6	BioSemi condition code 65288
65285	7	BioSemi condition code 65285
65280	8	BioSemi condition code 65280

- Notes: BioSemi EDF trigger codes 65280-65288; decode from events.tsv for exact poetry/non-poetry mapping. Codes 65281-65284 are highly frequent (~9964-9993 each); codes 65285-65288 appear rarely (~46-47 total). Code 65280 appears in only 18 files.

Data Quality Notes

- Codes 65285-65288 and 65280 are rare events; consider excluding from classification due to imbalance.
- BioSemi trigger codes need decoding from events.tsv to determine poetry/non-poetry mapping.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Poetry Assessment EEG Dataset 1`

229. openneuro_ds006695 - Validation of Sleep Staging with Forehead EEG Patch

Dataset: `openneuro_ds006695` | Year: 2025 | URL: https://openneuro.org/datasets/ds006695
Category: Biological State | Subjects: 19 | Files (Completed): 19 | Channels: 3 | Sampling Rate: 500 Hz

Paradigm

This dataset validates a spectral sleep staging approach using a three-electrode forehead EEG patch against manual scoring from standard polysomnography (PSG). Overnight sleep recordings were collected using a minimally intrusive forehead device, and sleep stages (Wake, N1, N2, N3, REM) were scored both manually and algorithmically. The study examines whether simplified forehead EEG can reliably reproduce standard PSG hypnogram classifications.

Event Structure

No event markers detected (`no_events`). This dataset contains continuous overnight EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Continuous Segmentation (segmentation)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no event markers; sleep stage labels from external PSG scoring)
- Notes: No EEG event markers available. Sleep stage labels (Wake/N1/N2/N3/REM) must be derived from PSG hypnogram annotations stored as separate scoring files. Very few channels (3); limited spatial resolution for advanced decoding.

Data Quality Notes

- Only 3 channels (forehead placement); very limited spatial information.
- Overnight recordings are long (hours); memory-efficient processing needed.
- Sleep stage labels require alignment with PSG hypnogram files.

References

- Paper: `Validation of spectral sleep scoring with polysomnography using forehead EEG device` (DOI: 10.3389/frsle.2024.1349537)

230. openneuro_ds006768 - Multiple Object Monitoring EEG

Dataset: `openneuro_ds006768` | Year: 2025 | URL: https://openneuro.org/datasets/ds006768
Category: High-level Mental | Subjects: 30 | Files (Completed): 210 | Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

Subjects completed a Multiple Object Monitoring (MOM) task requiring sustained visual attention to track multiple moving objects simultaneously. In each trial, a subset of objects is designated as targets at the start, then all objects move during a tracking period, and subjects report which objects were targets. EEG captures neural correlates of visual attention load and tracking accuracy. Events mark trial onset, target designation, and distractor appearances.

Event Structure

Event ID	Count (total)	Type	Meaning
`event`	20384	short	Generic trial event marker
`target`	11004	short	Target object onset (object to be tracked)
`distractor`	11004	short	Distractor object onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Target vs Distractor Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`event`	0	Generic trial event marker
`target`	1	Target object to be tracked
`distractor`	2	Distractor object

- Notes: Well-balanced binary contrast: target vs. distractor (11004 each). `event` is a generic trial marker. 210 files from 30 subjects (7 runs/subject on average).

Data Quality Notes

- 210 files from 30 subjects suggests ~7 runs per subject.
- `event` is a generic trial marker; prefer specific `target`/`distractor` events for classification.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Multiple Object Monitoring (EEG)`

231. openneuro_ds006777 - SFARI AVSRT EEG

Dataset: openneuro_ds006777 | Year: 2025 | URL: https://openneuro.org/datasets/ds006777
Category: Sensory/Response | Subjects: 132 | Files (Completed): 132 | Channels: 73 | Sampling Rate: 512 Hz

Paradigm

An audiovisual simple reaction-time task (AVSRT) in which participants press a button as quickly as possible upon detecting auditory, visual, or simultaneous audiovisual stimuli. The paradigm measures unimodal vs. multimodal stimulus processing and associated reaction times. The dataset is from the SFARI (Simons Foundation Autism Research Initiative) collection, likely including participants with autism spectrum disorder and neurotypical controls.

Event Structure

Event ID	Count (total)	Type	Meaning
Response_button	46353	short	Participant button-press response
Visual_stim	16250	short	Visual stimulus onset
Audiovisual_stim	16130	short	Simultaneous audiovisual stimulus onset
Auditory_stim	15995	short	Auditory stimulus onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Stimulus Modality Classification (epoch_by_event)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Response_button	0	Button-press response
Audiovisual_stim	1	Simultaneous audiovisual stimulus
Auditory_stim	2	Auditory-only stimulus
Visual_stim	3	Visual-only stimulus

- Notes: Three balanced modality conditions (Auditory/Visual/Audiovisual ~16000 each) well-suited for multiclass decoding. Response_button is a motor response event; exclude from stimulus classification if needed.

Data Quality Notes

- SFARI autism cohort; likely includes participants with ASD and neurotypical controls.
- Response_button is a motor event marker, not a stimulus; consider excluding from stimulus decoding.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `SFARI AVSRT EEG`

232. openneuro_ds006801 - Resting-State EEG Before and After Different Study Methods

Dataset: openneuro_ds006801 | Year: 2025 | URL: https://openneuro.org/datasets/ds006801
Category: Biological State | Subjects: 21 | Files (Completed): 42 | Channels: 31 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG was recorded before and after an educational intervention in which participants engaged in active retrieval practice, passive review, or a control condition. The design examines how different study strategies affect resting-state neural oscillations. Each participant completed eyes-open and eyes-closed resting blocks. The 42 files correspond to pre- and post-intervention recordings for 21 subjects.

Event Structure

Event ID	Count (total)	Type	Meaning
New Segment/LostSamples	1	admin	Segment discontinuity marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (kernels/bids.py)

Task 1: Resting-State Segmentation (long_event)

- Epoch Method: epoch_by_long_event_hdf5
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no trial events; target_events: null)
- Notes: New Segment/LostSamples is an admin marker indicating signal discontinuity; never use as classification target. Labels for pre/post intervention and study condition must be derived from participants.tsv metadata.

Data Quality Notes

- New Segment/LostSamples is a signal discontinuity marker - exclude from all analyses.
- 42 files = 21 subjects x 2 sessions (pre/post intervention).
- Eyes-open vs eyes-closed blocks may be distinguished by duration or metadata.

References

- Reference: The RECAP-EEG: Retrieval with Feedback and Cognitive Adaptation EEG Dataset

233. openneuro_ds006839 - EEG Recordings during Sham Neurofeedback in Virtual Reality

Dataset: openneuro_ds006839 | Year: 2025 | URL: https://openneuro.org/datasets/ds006839
Category: Complex Applied | Subjects: 144 | Files (Completed): 144 | Channels: 29 | Sampling Rate: 1000 Hz

Paradigm

Participants performed an attentional task within a virtual reality environment while receiving sham neurofeedback with positive, negative, or control feedback valences. The study investigates alpha-band modulation in response to perceived performance feedback without actual neurofeedback. Numeric trigger codes mark trial onsets and feedback conditions. The paradigm examines placebo effects of neurofeedback on neural oscillations and attention.

Event Structure

Event ID	Count (total)	Type	Meaning
190	10264	short	Positive feedback condition marker
222	5389	short	Control (neutral) feedback condition
126	576	short	Trial/block onset marker
238	431	short	Negative feedback condition marker
0	98	admin	Null/reset event

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Neurofeedback Valence Classification (epoch_by_event)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
126	0	Trial/block onset marker
190	1	Positive sham neurofeedback
238	2	Negative sham neurofeedback
222	3	Control (neutral) feedback
0	4	Null/reset event

- Notes: Decode trigger codes from events.tsv. Code 0 is a null/reset event. Primary contrast is positive (190) vs negative (238) vs neutral (222) feedback valence. Classes are imbalanced (190: 10264 vs 238: 431).

Data Quality Notes

- Classes are imbalanced: code 190 (10264) >> 238 (431); balance before classification.
- VR environment may introduce motion artifact; check for movement-related noise.
- Numeric trigger codes require decoding from events.tsv or README.

References

- Reference: `EEG recordings during sham neurofeedback in virtual reality`

234. openneuro_ds006840 - IACKD: Intention Action Conflict EEG-Hand Kinematics Dataset

Dataset: `openneuro_ds006840` | Year: 2025 | URL: https://openneuro.org/datasets/ds006840
Category: High-level Mental | Subjects: 128 | Files (Completed): 128 | Channels: 29 | Sampling Rate: 1024 Hz

Paradigm

This dataset investigates the neural correlates and hand kinematics associated with conflicts between intended and executed actions. In an intention-action conflict paradigm, participants are cued to perform one action but must sometimes execute a competing action, creating conflict between motor intention and execution. EEG is recorded alongside hand kinematic data during pre-movement planning (`premove`), execution (`execution`), and successful hit trials (`hit`).

Event Structure

Event ID	Count (total)	Type	Meaning
`execution`	7032	short	Motor execution onset
`hit`	7032	short	Successful action completion marker
`premove`	7029	short	Pre-movement preparation period onset

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Motor Phase Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
`premove`	0	Pre-movement intention/preparation
`execution`	1	Motor execution onset
`hit`	2	Successful trial completion

- Notes: Well-balanced 3-class design (~7032 each). Hand kinematic data available alongside EEG for multimodal analysis. BrainVision format.

Data Quality Notes

- Consistent event structure (~7029-7032 events per type); well-controlled paradigm.
- Kinematic data channels (VEOG, HEOG, Trigger) should be excluded from EEG analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `IACKD: Intention Action Conflict EEG-Hand Kinematics Dataset`

235. openneuro_ds006848 - AlphaDirection1: EEG, ECG, PPG in the resting state and working memory for sequentially and simultaneously presented digits

Dataset: openneuro_ds006848 | Year: 2025 | URL: https://openneuro.org/datasets/ds006848/versions/1.0.0
Category: High-level Mental | Subjects: 30 | Files (Completed): 52
Channels: 65 | Sampling Rate: 1000 Hz

Paradigm

This dataset studies the directional characteristics of brain alpha waves during resting state and working memory tasks. Participants completed a verbal working memory digit span task in two encoding modes: serial presentation (one digit is displayed at a time) and simultaneous presentation (an entire set of digits are displayed at once), followed by serial recall; before the experiment, a resting state (eyes open/eyes closed alternating) recording was included, and cartoon clips were used to separate different experimental segments. Data is recorded simultaneously with EEG, ECG and PPG.

Event Structure

Event ID	Count (total)	Type	Meaning
Encoding_DigitValue_0	3163	short	Sequence encoding: number 0 presents onset
Encoding_DigitValue_1	3134	short	Sequence encoding: number 1 presents onset
Encoding_DigitValue_2	3077	short	Sequence encoding: number 2 presents onset
Encoding_DigitValue_3	3124	short	Sequence encoding: number 3 presents onset
Encoding_DigitValue_4	3164	short	Sequence encoding: number 4 presents onset
Encoding_DigitValue_5	3104	short	Sequence encoding: number 5 presents onset
Encoding_DigitValue_6	3103	short	Sequence encoding: number 6 presents onset
Encoding_DigitValue_7	3131	short	Sequence encoding: number 7 presents onset
Encoding_DigitValue_8	3117	short	Sequence encoding: number 8 rendering onset
Encoding_DigitValue_9	3131	short	Sequence encoding: number 9 presents onset
Encoding_Set_Simultaneous	1495	short	Simultaneous encoding: the entire set of numbers is presented at once onset
Digits_Retrieval	5960	short	The recall phase begins (the subject begins to report verbally)
Baseline_2s	5956	short	2 s before each trial, the beginning of the baseline segment

Event ID	Count (total)	Type	Meaning
Retention_Fast	1499	long	Fast delay hold period (sequence encoding, short delay)
Retention_FastDelay	1499	long	Fast delay additional waiting period (sequence encoding)
Retention_Slow	1466	long	Slow delay hold period (sequence encoding, long delay)
Retention_Simultaneous	1495	long	Retention period after simultaneous encoding
Rest	209	rest	Resting state segment start mark
Eyes_Closed	87	rest	Close your eyes and start to rest
Eyes_Opened	88	rest	Open your eyes and start to rest
Boundary	157	admin	BIDS file boundary mark
Experiment_End	29	admin	Experiment end mark
Start_Cartoon	22	admin	Cartoon short film starts (intersegment interference)
End_Cartoon	22	admin	End of cartoon short

Event Type Description:
- `short`: short stimulus onset marker (presentation duration <2 s), for `epoch_by_event_hdf5`
- `long`: long-term status flag (retention period is usually >=2 s), used for `epoch_by_long_event_hdf5`
- `rest`: Rest block tag, used for `epoch_by_long_event_hdf5` or `epoch_by_segmentation_hdf5`
- `admin`: system/boundary marker, not used for epoch

HDF5 Production Guide

Tag source: Annotation (events.tsv `trial_type` column, Generic BIDS kernel directly readable)
Kernel: Generic BIDS (`kernels/bids.py`) - No need for a dedicated kernel, event tags are already in human-readable form, no need to re-encode, no need for continuous variable injection

Task 1 (main task): coding mode two classification (binary)

- Epoch method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s (relative to event onset)
- Target Events -> y:

Event	y	Meaning
Encoding_DigitValue_0	0	Sequence code (number 0)
Encoding_DigitValue_1	0	Sequence code (number 1)
Encoding_DigitValue_2	0	Sequence code (number 2)
Encoding_DigitValue_3	0	Sequence code (number 3)
Encoding_DigitValue_4	0	Sequence code (number 4)
Encoding_DigitValue_5	0	Sequence code (number 5)

Event	y	Meaning
Encoding_DigitValue_6	0	Sequence code (number 6)
Encoding_DigitValue_7	0	Sequence code (number 7)
Encoding_DigitValue_8	0	Sequence code (number 8)
Encoding_DigitValue_9	0	Sequence code (number 9)
Encoding_Set_Simultaneous	1	Simultaneous encoding (whole group)

- Remarks: The categories are seriously unbalanced (sequence ~31000 times vs. simultaneous ~1495 times), downstream needs to downsample to 1:1

Task 2: Digital Identification (11-class multiclass)

- Epoch method: epoch_by_event_hdf5
- Segment:tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
Encoding_DigitValue_0	0	Number 0 (sequence condition)
Encoding_DigitValue_1	1	number 1
...	...	...
Encoding_DigitValue_9	9	Number 9
Encoding_Set_Simultaneous	10	Present the entire group simultaneously

- Remarks: Each number category ~3100 times, Encoding_Set_Simultaneous ~1495 times, the overall balance is relatively balanced; This is the actual configuration in task_config.json

Task 3: Retention period type classification (4-class, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Retention_Fast	0	Sequence encoding, short delay hold period
Retention_FastDelay	1	Sequence encoding, short delay and extra wait
Retention_Slow	2	Sequence encoding, long delay hold period
Retention_Simultaneous	3	Retention period after simultaneous encoding

- Note: This task needs to be configured separately (different from the epoch method of task 2). The current main entry of task_config.json does not contain this task.

Resting state: open/closed eyes classification (binary, long event)

- Epoch method: epoch_by_long_event_hdf5
- Segment:tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5
- Target Events -> y:

Event	y	Meaning
Eyes_Closed	0	Close eyes and rest
Eyes_Opened	1	Rest with eyes open

- Remarks: Each file has about 1-2 switches (87/88 times in total), the amount of data is limited; Rest is the segment start mark and is not directly used for epoch

Data Quality Notes

- The ratio of sequence encoding vs. simultaneous encoding categories is about 20:1, and downsampling is required to do a fair binary classification evaluation.
- Boundary is the BIDS file boundary, which will appear at the beginning of each file and must be excluded from the epoch target
- Some subjects' files lack the Eyes_Closed/Eyes_Opened mark (only 22/30 subjects have resting eye state records)
- Sampling rate 1000 Hz, stage5 will downsample to 200 Hz

References

- Reference: AlphaDirection1: EEG, ECG, PPG in the resting state and working memory for sequentially and simultaneously presented digits

236. openneuro_ds006921 - High-Density Resting-State EEG of Phantom Limb Pain and Controls

Dataset: `openneuro_ds006921` | Year: 2025 | URL: https://openneuro.org/datasets/ds006921
Category: Biological State | Subjects: 152 | Files (Completed): 152 | Channels: 128 | Sampling Rate: 2400 Hz

Paradigm

High-density resting-state EEG recordings were collected from three groups: amputees with phantom limb pain (PLP), amputees without phantom limb pain, and healthy controls. Recordings were made during eyes-open and eyes-closed conditions. The dataset enables investigation of neural correlates associated with phantom limb pain, cortical reorganization following amputation, and resting-state oscillatory differences across groups.

Event Structure

No event markers detected (`no_events`). This dataset contains continuous resting-state EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting-State Segmentation (segmentation)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no event markers; group labels from participants.tsv)
- Notes: No trial-level events. Group labels (PLP amputee / non-PLP amputee / healthy control) must be derived from participants.tsv metadata. Very high sampling rate (2400 Hz); downsample before analysis.

Data Quality Notes

- Very high sampling rate (2400 Hz); downsample to 500-1000 Hz for most analyses.
- 152 files may include multiple sessions per subject; verify session structure.
- High-density 128-channel system (likely EGI/Geodesic); non-standard montage.

References

- Reference: `High Density Resting State EEG of Phantom Limb Pain and Controls`

237. openneuro_ds007020 - EEG Mortality Dataset in Parkinson's Disease

Dataset: `openneuro_ds007020` | Year: 2025 | URL: https://openneuro.org/datasets/ds007020
Category: Biological State | Subjects: 94 | Files (Completed): 94 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

Resting-state EEG recordings (eyes open) were collected from Parkinson's disease (PD) patients and healthy controls to evaluate EEG-based biomarkers for mortality classification. The dataset enables investigation of spectral and connectivity biomarkers that predict mortality risk in PD. The sole event marker (`New Segment/`) indicates a recording segment boundary and carries no experimental meaning.

Event Structure

Event ID	Count (total)	Type	Meaning
`New Segment/`	94	admin	Recording segment boundary marker

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting-State Segmentation (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no trial events; target_events: null)
- Notes: `New Segment/` is a BrainVision segment boundary marker and must never be used as a classification target. Group labels (PD patient / healthy control, mortality outcome) must be derived from clinical metadata in participants.tsv.

Data Quality Notes

- `New Segment/` is a BrainVision format boundary marker; exclude from all analyses.
- Labels for mortality prediction must be derived from clinical metadata in participants.tsv.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Reference: `EEG Mortality Dataset in Parkinson's Disease`

238. openneuro_ds007172 - EEG-Asymmetries Dataset

Dataset: openneuro_ds007172 | Year: 2026 | URL: https://openneuro.org/datasets/ds007172
Category: High-level Mental | Subjects: 100 | Files (Completed): 501 | Channels: 32 | Sampling Rate: 500 Hz

Paradigm

Multi-task EEG battery investigating hemispheric asymmetries across resting state and four cognitive categorization tasks: emotion recognition (face stimuli with emotional valence), language comprehension, face perception (identity discrimination), and visual discrimination (house images). The dataset examines relationships between N1 lateralization and alpha asymmetry across task domains. BrainVision trigger codes (Stimulus/S1xx and S2xx families) encode stimulus conditions per task.

Event Structure

Event ID	Count (total)	Type	Meaning
Stimulus/S111	12800	long	Emotion/language task condition S111
Stimulus/S223	12800	long	Emotion/language task condition S223
Stimulus/S212	12800	long	Emotion/language task condition S212
Stimulus/S214	12800	long	Emotion/language task condition S214
Stimulus/S211	12800	long	Emotion/language task condition S211
Stimulus/S123	12800	long	Emotion/language task condition S123
Stimulus/S114	12800	long	Emotion/language task condition S114
Stimulus/S112	12800	long	Emotion/language task condition S112
Stimulus/S222	12000	long	Emotion/language task condition S222
Stimulus/S113	12000	long	Emotion/language task condition S113
Stimulus/S121	12000	long	Emotion/language task condition S121
Stimulus/S122	12000	long	Emotion/language task condition S122
Stimulus/S124	12000	long	Emotion/language task condition S124
Stimulus/S221	12000	long	Emotion/language task condition S221
Stimulus/S213	12000	long	Emotion/language task condition S213
Stimulus/S224	12000	long	Emotion/language task condition S224
Stimulus/S217	11998	long	Face task condition S217

Event ID	Count (total)	Type	Meaning
Stimulus/S218	11998	long	Face task condition S218
Stimulus/S118	11998	long	Face task condition S118
Stimulus/S117	11998	long	Face task condition S117
Stimulus/S227	11998	long	Face task condition S227
Stimulus/S228	11998	long	Face task condition S228
Stimulus/S127	11998	long	Face task condition S127
Stimulus/S128	11998	long	Face task condition S128
Stimulus/S 21	11998	long	Houses/visual task condition S21
Stimulus/S 11	11998	long	Houses/visual task condition S11
Stimulus/S 22	11998	long	Houses/visual task condition S22
New Segment/	501	admin	Segment boundary marker
Comment/actiCAP Data On	383	admin	Hardware status marker
Comment/actiCAP USB Power On	134	admin	Hardware status marker

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Multi-Condition EEG Asymmetry (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y:

Event	y	Meaning
Comment/actiCAP Data On	0	Hardware marker
Comment/actiCAP USB Power On	1	Hardware power marker
Stimulus/S111	2	Condition S111
Stimulus/S223	3	Condition S223
Stimulus/S212	4	Condition S212
Stimulus/S214	5	Condition S214
Stimulus/S211	6	Condition S211
Stimulus/S123	7	Condition S123
Stimulus/S114	8	Condition S114
Stimulus/S112	9	Condition S112
Stimulus/S222	10	Condition S222

Event	y	Meaning
Stimulus/S113	11	Condition S113
Stimulus/S121	12	Condition S121
Stimulus/S122	13	Condition S122
Stimulus/S124	14	Condition S124
Stimulus/S221	15	Condition S221
Stimulus/S213	16	Condition S213
Stimulus/S224	17	Condition S224
Stimulus/S217	18	Condition S217
Stimulus/S218	19	Condition S218
Stimulus/S118	20	Condition S118
Stimulus/S117	21	Condition S117
Stimulus/S227	22	Condition S227
Stimulus/S228	23	Condition S228
Stimulus/S127	24	Condition S127
Stimulus/S128	25	Condition S128
Stimulus/S 21	26	Condition S21
Stimulus/S 11	27	Condition S11
Stimulus/S 22	28	Condition S22

- Notes: 5 task files per subject (emotions, faces, houses, language, rest). `New Segment/`, `Comment/actiCAP Data On`, `Comment/actiCAP USB Power On` are hardware/admin markers; exclude from stimulus analysis.

Data Quality Notes

- 501 files from 100 subjects includes 5 task types per subject.
- `New Segment/` and all `Comment/` markers are administrative; never use as classification targets.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- Reference: `Hemispheric asymmetries in the EEG: Is there an association between N1 lateralization and alpha asymmetryx`

239. openneuro_ds007176 - Longitudinal EEG Test-Retest Reliability in Healthy Individuals

Dataset: `openneuro_ds007176` | Year: 2026 | URL: https://openneuro.org/datasets/ds007176
Category: Biological State | Subjects: 43 | Files (Completed): 300 | Channels: 60 | Sampling Rate: 1000 Hz

Paradigm

Longitudinal resting-state EEG recordings collected over four sessions spanning two years to evaluate test-retest reliability of quantitative EEG measures. Recordings are continuous resting-state without structured task events. The dataset enables assessment of EEG biomarker stability across sessions and time points.

Event Structure

Event ID	Count (total)	Type	Meaning
`0`	2	admin	Null trigger code - no experimental meaning

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Resting-State Segmentation (long_event)

- Epoch Method: `epoch_by_long_event_hdf5`
- Segment: tmin = 0.0 s, tmax = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no trial events; target_events: null)
- Notes: Only event code `0` present (2 total); this is a null trigger with no experimental meaning. Resting-state data; use segmentation across full recording. Session/time-point labels available from participants.tsv for longitudinal analysis.

Data Quality Notes

- No meaningful event markers; purely continuous resting-state EEG.
- 43 subjects x ~7 sessions/subject = 300 files; multi-session longitudinal design.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to segmentation.

References

- Paper: `Longitudinal test-retest reliability of quantitative EEG in healthy individuals using an automated preprocessing approach` (DOI: 10.1016/j.bspc.2026.109484)

240. openneuro_ds007180 - Exo-EEG Experiment

Dataset: openneuro_ds007180 | Year: 2026 | URL: https://openneuro.org/datasets/ds007180
Category: Complex Applied | Subjects: 25 | Files (Completed): 25 | Channels: 63 | Sampling Rate: 500 Hz

Paradigm

EEG was recorded during interaction with an exoskeleton in a gait or robotic-assisted movement task. Participants performed motor tasks with varying exoskeleton assistance levels and movement conditions. BrainVision trigger codes (S-prefixed) encode specific movement phases, assistance levels, and stimulus conditions across 30 categories.

Event Structure

Event ID	Count (total)	Type	Meaning
S21	1885	short	Exo stimulus condition S21
S22	1885	short	Exo stimulus condition S22
S12	1884	short	Exo stimulus condition S12
S11	1884	short	Exo stimulus condition S11
S32	1876	short	Exo stimulus condition S32
S31	1876	short	Exo stimulus condition S31
S41	1872	short	Exo stimulus condition S41
S42	1872	short	Exo stimulus condition S42
S62	1872	short	Exo stimulus condition S62
S61	1872	short	Exo stimulus condition S61
S52	1872	short	Exo stimulus condition S52
S51	1872	short	Exo stimulus condition S51
S72	1868	short	Exo stimulus condition S72
S71	1868	short	Exo stimulus condition S71
S81	1860	short	Exo stimulus condition S81
S82	1860	short	Exo stimulus condition S82
S13	1750	short	Exo stimulus condition S13
S43	1750	short	Exo stimulus condition S43
S73	1750	short	Exo stimulus condition S73
S63	1750	short	Exo stimulus condition S63
S53	1750	short	Exo stimulus condition S53
S33	1750	short	Exo stimulus condition S33
S23	1750	short	Exo stimulus condition S23
S83	1750	short	Exo stimulus condition S83
S102	1300	short	Exo stimulus condition S102
S101	1300	short	Exo stimulus condition S101

Event ID	Count (total)	Type	Meaning
S92	1300	short	Exo stimulus condition S92
S91	1300	short	Exo stimulus condition S91
S111	1275	short	Exo stimulus condition S111
S122	1275	short	Exo stimulus condition S122

Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

Label Source: Annotation (events.tsv)
Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Exoskeleton Condition Classification (epoch_by_event)

- Epoch Method: `epoch_by_event_hdf5`
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
S62	0	Exo condition S62
S61	1	Exo condition S61
S52	2	Exo condition S52
S51	3	Exo condition S51
S21	4	Exo condition S21
S22	5	Exo condition S22
S12	6	Exo condition S12
S11	7	Exo condition S11
S32	8	Exo condition S32
S41	9	Exo condition S41
S42	10	Exo condition S42
S31	11	Exo condition S31
S72	12	Exo condition S72
S71	13	Exo condition S71
S81	14	Exo condition S81
S82	15	Exo condition S82
S13	16	Exo condition S13
S43	17	Exo condition S43
S73	18	Exo condition S73
S63	19	Exo condition S63
S53	20	Exo condition S53
S33	21	Exo condition S33

Event	y	Meaning
S23	22	Exo condition S23
S83	23	Exo condition S83
S102	24	Exo condition S102
S101	25	Exo condition S101
S92	26	Exo condition S92
S91	27	Exo condition S91
S111	28	Exo condition S111
S122	29	Exo condition S122

- Notes: 30 distinct BrainVision stimulus codes; decode from events.tsv or README for exact condition meaning. S255 and start are admin markers in some files; exclude from analysis.

Data Quality Notes

- BrainVision format; 30 S-coded stimulus conditions require README decoding.
- S255 (saturation marker) and start events in some files; exclude from analysis.
- Apply bandpass filter (0.5-40 Hz) and artifact rejection prior to epoching.

References

- No paper reference available

241. openneuro_ds007181 - Structural MRI, Resting-State fMRI, and PSG/EEG Dataset of Zoster-Associated Neuralgia

Dataset: `openneuro_ds007181` | Year: 2026 | URL: https://openneuro.org/datasets/ds007181
Category: Biological State | Subjects: 59 | Files (Completed): 59 | Channels: 24 | Sampling Rate: 1024 Hz

Paradigm

Resting-state EEG (part of a PSG recording) was collected from patients with zoster-associated neuralgia and healthy controls, alongside structural MRI and resting-state fMRI. Manual sleep staging is available from PSG annotations. The dataset enables investigation of brain activity and sleep architecture differences between healthy controls and patients with herpes zoster-associated chronic pain.

Event Structure

No event markers detected (`no_events`). This dataset contains continuous PSG/EEG recordings without trial-level annotations.

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (`kernels/bids.py`)

Task 1: Continuous Segmentation (segmentation)

- Epoch Method: `epoch_by_segmentation_hdf5`
- Segment: segment_length = 2.0 s, overlap = 0.5 s
- Target Events -> y: N/A (no event markers; sleep stage labels from PSG scoring files)
- Notes: No EEG event markers available. Sleep stage labels (Wake/N1/N2/N3/REM) must be derived from PSG hypnogram annotations if present. Group labels (zoster neuralgia / healthy control) from participants.tsv.

Data Quality Notes

- No event markers present; continuous-only processing.
- PSG-derived sleep stage labels require alignment with EEG data if available.
- 24-channel PSG EEG system; non-standard montage compared to research-grade EEG.

References

- Reference: `Structural MRI, Resting-state fMRI, and PSG/EEG Dataset of Zoster-associated Neuralgia`

242. openneuro_ds007327 - EEG During Sport-Related Motor Imagery and Observation Tasks

Dataset: openneuro_ds007327 | Year: 2026 | URL: https://openneuro.org/datasets/ds007327
Category: Action/Output | Subjects: 35 | Files (Completed): 105 | Channels: 88 | Sampling Rate: 1000 Hz

Paradigm

EEG was recorded during two sports-related tasks (ball passing and dribbling) to study motor intention and motor observation. The paradigm uses a 2x2 design crossing motor intention (MI = motor intention, MO = motor observation) with movement execution modality, yielding four compound conditions: MIMI (imagine intention), MIMO (imagine observation), MOMI (observe intention), MOMO (observe observation). Additional single-factor conditions (MI, MO, MO_MI) and a static/rest baseline are included.

Event Structure

Event ID	Count (total)	Type	Meaning
MOMI	8396	short	Motor observation of intention trials
MIMO	8394	short	Motor imagination of observation trials
MOMO	8394	short	Motor observation of observation trials
MIMI	8335	short	Motor imagination of intention trials
static	4313	short	Static baseline condition
MO	4200	short	Motor observation condition
MO_MI	4198	short	Motor observation with motor intention
MI	4195	short	Motor intention condition
rest	4073	short	Resting/baseline period

> Event Type: short=stimulus onset <2s; long=sustained state >=2s; rest=resting marker; admin=exclude

HDF5 Production Guide

> Label Source: Annotation (events.tsv)
> Kernel: Generic BIDS (kernels/bids.py)

Task 1: Motor Intention/Observation Classification (epoch_by_event)

- Epoch Method: epoch_by_event_hdf5
- Segment: tmin = -0.2 s, tmax = 1.0 s
- Target Events -> y:

Event	y	Meaning
MIMI	0	Motor imagination x intention
MIMO	1	Motor imagination x observation
MOMI	2	Motor observation x intention

Event	y	Meaning
MOMO	3	Motor observation x observation
MI	4	Motor intention
MO_MI	5	Motor observation with intention
static	6	Static baseline
MO	7	Motor observation
rest	8	Rest baseline

- Notes: 9-class design; 2x2 factorial core (MIMI/MIMO/MOMI/MOMO ~8300 each) plus single-factor and baseline conditions (~4000 each). 105 files = 35 subjects x 3 task types (ballpass, dribble, passing).

Data Quality Notes

- High channel count (88) includes 24 BIP channels for EMG/kinematic data; exclude BIP channels from EEG analysis.
- 35 subjects x 3 task files = 105 files; verify task structure.
- Apply bandpass filter (0.5-40 Hz), remove BIP channels, and artifact reject prior to epoching.

References

- No paper reference available